



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
BHOPAL**



IIIRD SEMESTER

CATEGORY:-BSC

BE-31 - ENGINEERING MATHEMATICS III

TOTAL-60HOURS

COURSE OBJECTIVE:

To develop logical understanding of the subject.

To develop the knowledge, skills & attitudes necessary to solve problem pertaining to Electrical Engineering.

To make aware students about the importance & symbiosis between Mathematics & Engineering.

Unit -1

Functions of complex variables: Analytic functions, Harmonic Conjugate, Cauchy-Riemann Equations, Line Integral, Cauchy's Theorem, Cauchy's Integral Formula, Singular Points, Poles & Residues, Residue Theorem, Application of Residues theorem for evaluation of real integrals.

(12 HRS)

Unit -2

Errors & Approximations, Solution of Algebraic & Trancedental Equations (Regula Falsi Newton-Raphson, Iterative, Secant Method), Solution of simultaneous linear equations by Gauss Elimination, Gauss Jordan, Crout's methods , Jacobi's and Gauss-Siedel Iterative methods.

(12 HRS)

Unit -3

Difference Operators, Interpolation (Newton Forward & Backward Formulae, Central Interpolation Formulae, Lagrange's and divided difference formulae), Numerical Differentiation and Numerical Integration..

(12 HRS)

Unit -4

Solution of Ordinary Differential Equations(Taylor's Series, Picard's Method, Modified Euler's Method, Runge-Kutta Method, Milne's Predictor & Corrector method), Correlation and Regression, Curve Fitting (Method of Least Square).

(12 HRS)

Unit -5

Concept of Probability: Probability Mass function, Probability density function. Discrete Distribution: Binomial, Poisson's, Continuous Distribution: Normal Distribution, Exponential Distribution ,Gamma Distribution ,Beta Distribution ,Testing of Hypothesis :Students t-test, Fisher's z-test, Chi-Square Method.

(12 HRS)

COURSE OUTCOME

Apply basic knowledge of maths to solve real-world problems.

Able to generate sdolutions to unfamiliar problems.



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Reference:

1. Numerical Methods using Matlab by J.H.Mathews and K.D.Fink, P.H.I.
2. Numerical Methods for Scientific and Engg. Computation by MKJain, Iyengar and RK Jain, New Age International Publication
3. Mathematical Methods by KV Suryanarayan Rao, SCITECH Publications
4. Numerical Methods using Matlab by Yang, Wiley India
5. Probability and Statistics by Ravichandran, Wiley India
Mathematical Statistics by George R., Springer Grading IVth Semester w.e.f.2011-12



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
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IIIRD SEMESTER

CATEGORY:-DC

BE-32- BIOLOGY

TOTAL -60HOURS

COURSE OBJECTIVE:

To convey that Biology is as important a scientific discipline as Mathematics.
To develop understanding about Biology and its different parts.

Unit -1

Darwinian evolution, molecular perspective and classification, Phylogenetic trees, study of inter- and intra- species relationships, developmental Biology.

(12 HRS)

Unit -2

Cellular structure and function, cellular assembly and central dogma, of molecular Biology
Organismal physiology - Energy and energetic constraints.

(12 HRS)

Unit -3

3D structure and function of large biological molecules 5 Techniques in biophysics and biochemistry 6 Immunology - Self vs. Non-self, pathogens, human immune system, antigen-antibody reactions, Infectious disease Biology and vaccines 4.

(12 HRS)

Unit -4

Cancer biology, gene regulation, aging, apoptosis and stem cell Biology 9 Environmental issues, bioresources, biodiversity.

(12 HRS)

Unit -5

Drug design, Engineering designs inspired by examples in Biology, Engineering aspects of some Nobel Prizes in Physiology and Medicine & Chemistry 7 recent advances in Biology.

(12 HRS)

COURSE OUTCOME

Ability has been developed to describe how biological observations of 18th century that lead to major discoveries. Understanding is developed about Biology & its different parts..

Reference Books:

1. Watson JD, Baker, TA, Bell SP, Gann A, Levin M, Losick R, Molecular Biology of the Gene, Pearson Education, 2004.
2. Dawkins, R. The Greatest Show on Earth: The Evidence For Evolution, Bantam Press,
3. Transworld Publishers, Random House Group, London, 2009.
4. Dawkins, R. The Blind Watchmaker, W. W. Norton & Co., NY, 1996.



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5. Watson, J. D. The Double Helix: A Personal Account of the Discovery of the Structure of DNA (Copyright © 1968, 1996 by James D. Watson), Simon & Schuster Inc., first Touchstone Edition (2001).



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IIIRD SEMESTER

CATEGORY:-DC

EE-31- ELECTRICAL MEASUREMENT & MEASURING INSTRUMENTS

TOTAL-60HOURS

COURSE OBJECTIVE:

The primary objective of the course is to introduce operation principles of instruments, terminology related to measurements and to have an adequate knowledge in measurement techniques for voltage, current, power and energy

Unit -1

Measurement and Error- Accuracy and precision, sensitivity resolution, Error & Error analysis, Effect of temperature, Internal friction, Stray field, Hysteresis and Frequency variation & method of minimizing them, Loading effects, due to shunt connected and series connected instruments, calibration curve, Testing & calibration of instruments. Galvanometers – Theory & operation of ballistic galvanometer, D'Arsonal galvanometer, galvanometer motion & damping, Sensitivity, Flux meter, Vibration galvanometer, Spot deflection galvanometer. Definition of analog & digital instruments, Classification of analog instruments, their operating principle, Operating force, Types of supports, Damping, Controlling.

(12 HRS)

Unit -2

Different types of Ammeter & Voltmeter- PMMC, MI, Electrodynamometer, Hotwire, Electrostatic, Induction, Rectifier, Ferro dynamic & Electro-thermic, Expression for control & deflection torque, their advantages, disadvantages & error, Extension of range of instruments using shunt & multiplier.

(12 HRS)

Unit -3

Instrument Transformers-Potential and current transformers, ratio and phase angle errors, testing of instrument transformers, Difference between CT and PT, errors and reduction of errors. Measurement of power: Power in AC and DC Circuit, Electrodynamometer type of wattmeter, Construction, theory, operation & error, Low power factor & UPF wattmeter, Double element and three element dynamometer wattmeter, Measurement of power in three phase circuit, one, two & three wattmeter method, Measurement of reactive power by single wattmeter, Measurement of power using CTs & PTs.

(12 HRS)

Unit -4

Measurement of Energy, Single phase induction type energy meter – construction & operation – driving and braking torques –errors & compensations – Testing by phantom loading and using R.S.S. meter- Three phase energy meter – Tri-vector meter – Maximum demand meter, Ampere hour meter. Potentiometer – DC potentiometer standardization – Lab type Crompton's potentiometer, application of DC potentiometer, AC polar type and coordinate type potentiometer, their construction and applications.

(12 HRS)



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Unit -5

Miscellaneous Instruments & Measurements- Power factor meter, Single phase and three phase Electro-dynamometer type & moving iron type. Frequency meter – Vibrating reed, Resonance type & Weston type, Synchronoscope, Ohmmeter – series & stunt type, Multi-meter, Megger & Ratio meter. Resistance Measurement – Classification of low, medium & high resistance – Voltmeter, Ammeter, Wheatstone Bridge, Kelvin's double bridge & loss of charge methods for resistance measurement, Earth resistance measurement. Magnetic Measurement – B-H Curve, Hysteresis Loop determination, Power loss in sheet metal – Lloyd Fischer square for measurement of power loss.

(12 HRS)

COURSE OUTCOME:

After successful completion of course, Students are expected to possess an in-depth understanding and Knowledge of the concepts and principles of measurement of electrical and non electrical viz. physical quantities and instruments.

Reference Books:

1. E W Golding & F C Widdis; Electrical Measurement & Measuring Instruments; Wheeler Pub.
2. A.K. Sawhney; Electrical & Electronic Measurements & Instrument; Dhanpat Rai & Sons Pub.
3. Buckingham & Price; Electrical Measurements; Prentice Hall

List of Experiments:

1. Measurement of low resistance using Kelvin's Double bridge
2. Measurement of medium resistance using Whetstone's bridge
3. Measurement of high resistance by loss of charge method
4. Measurement of Insulation resistance using Megger
5. Measurement of earth resistance by fall of potential method and verification by using earth tester.
6. Measurement of power in a single phase ac circuit by 3 voltmeter/ 3 Ammeter methods.
7. Calibration of a dynamometer type of wattmeter with respect to a standard/Sub Standard wattmeter.
8. Calibration of an induction type single phase energy meter.
9. Calibration of a dynamometer type of wattmeter by Phantom Loading method.
10. Measurements using Instrument Transformers.
11. Study of various types of Indicating Instruments.
12. Measurement of Power in three phase circuit by one, two & three Wattmeter's.



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IIIRD SEMESTER

CATEGORY:-DC

EE-32 - ANALOG ELECTRONICS

TOTAL-60HOURS

COURSE OBJECTIVE:

The primary objective of this course is to develop an in-depth understanding of the design principles and applications of integrated analog circuits.

Unit -1

Semiconductor device, theory of P-N junction, temperature dependence and break down characteristics, junction capacitances, Zener diode, Varactor diode, PIN diode, LED, Photo diode, Transistors BJT, FET, MOSFET, types, working principle, characteristics, and region of operation, load line biasing methods, transistor as an amplifier, gain, bandwidth, frequency response, Various applications of diode and special diodes.

(12 HRS)

Unit- 2

Small signal analysis of transistor (low frequency) using h-parameters, thermal runaway and thermal stability.

(12 HRS)

Unit- 3

Feedback amplifier, negative feedback, voltage-series, voltage shunt, current series and current shunt feedback, Sinusoidal oscillators, L-C (Hartley-Colpitts) oscillators, RC phase shift, Wien bridge, and Crystal oscillators. Power amplifiers, class A, class B, class A B, C amplifiers, their efficiency and power Dissipation, Pushpull and complementary pushpull amplifier.

(12 HRS)

Unit- 4

Switching characteristics of diode and transistor, turn ON, OFF time, reverse recovery time, transistor as switch, Multivibrators, Bistable, Monostable, Astable multivibrators. Clippers and clamping, Differential amplifier, calculation of differential, common mode gain and CMRR using h-parameters, Darlington pair, Bootstrapping technique. Cascade and cascade amplifier.

(12 HRS)

Unit-5

Operational amplifier characteristics, slew rate, bandwidth, offset voltage, basic current, application inverting, non inverting amplifier, summer, average, differentiator, integrator, differential amplifier, instrumentation amplifier, log and antilog amplifier, voltage to current and current to voltage converters, comparators Schmitt trigger, active filters, 555 timer and its application.

(12 HRS)



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COURSE OUTCOME:

After successful completion of course, Students are expected to be able in applying theory and realize analog filter circuits, Understand the circuit operation of the 555 timer IC and regulator IC and identifying the faulty components within a circuit.

References:

1. Nashelsky & Boysted; Electronic Devices and Circuits; PHI
2. Millman Halkias; Electronic Devices and Circuits; McGraw- Hill
3. Achuthan MA and Bhatt KN; Fundamentals of semiconductor devices; TMH
4. Neamen Donald; Semiconductor Physics and devices
5. Millman & Grabel; Micro Electronics; McGraw-Hill
6. Bogart; Electronic Devices and Circuits; Universal Book Stall, NDelhi
7. Millman & Halkias; Integrated Electronics; McGraw- Hill.
8. Tobbey; OP- Amps their design and Application
9. R.A. Gaikward; OP- Amp and linear Integrated circuit; PHI
10. D. Raychowdhary and Shail Jain; Linear Integrated Circuits
11. Botkar; Integrated Circuits; Khanna
12. Clayton; Applications of linear Integrated circuits

List of experiments (Expandable):

1. V-I Characteristics of different types of Diodes.
2. Applications of diodes and Design of various clipping and clamping circuits.
3. Design half & full wave rectifier.
4. Design & Analysis of transistor amplifier in CE, CB & CC configuration.
5. Design & Analysis of JFET Amplifier.
6. Design & Analysis of MOSFET Amplifier.
7. To study and construct power amplifiers of various classes.
8. Study of various oscillators.
9. Char. of Op-Amp (input offset voltage, slew rate CMRR, BW, Input bias current)
10. Linear application of OP-Amp (voltage follower, inverting and non-inverting amplifier and their frequency response adder subtractor differential amplifier, integrator and differential frequency response) .
11. Study of Op-Amp as a comparator.
12. Design of Schmitt trigger.
13. Design of monoastable & astable multivibrator.

NOTE- All experiments (wherever applicable) should be performed through the following steps.

Step 1: Circuit should be designed/ drafted on paper.



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- Step 2:** Where ever applicable the designed/drafted circuit should be simulated using Simulation S/W (TINA-V7/ PSPICE/ Labview/ CIRCUIT MAKER etc.).
- Step 3:** The designed/drafted circuit should be tested on the bread board and compare the results with the simulated results.
- Step 4:** Where ever required the bread board circuit should be fabricated



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IIIRD SEMESTER

CATEGORY:-DC

EE-33 - NETWORK ANALYSIS

TOTAL-60HOURS

COURSE OBJECTIVE :

This course will develop students' knowledge in/on

1. To analyze the Circuits in time and frequency domain
2. To study network Topology, network Functions, two port network.
3. To synthesize passive network by various methods.

Unit I

Introduction to circuit elements R,L,C and their characteristics in terms of linearity & time dependant nature, voltage & current sources controlled & uncontrolled sources KCL and KVL analysis, Nodal & mesh analysis, analysis of magnetically coupled circuits, Transient analysis :- Transients in RL, RC&RLC Circuits, initial conditions, time constants. Steady state analysis- Concept of phasor & vector, impedance & admittance, Network topology, concept of Network graph, Tree, Tree branch & link, Incidence matrix, cut set and tie set matrices, dual networks, and Dot convention, coupling coefficient, tuned circuits, Series & parallel resonance.

(12 HRS)

Unit II

Network Theorems for AC & DC circuits- Thevenin's & Norton's, Superposition's, Reciprocity, Compensation, Substitution, Maximum power transfer, and Millman's theorem, Tellegen's theorem, problems with dependent & independent sources.

(12 HRS)

Unit III

Frequency domain analysis – Laplace transform solution of Integro-differential equations, transform of waveform synthesized with step ramp, Gate and sinusoidal functions, Initial & final value theorem, Network Theorems in transform domain.

(12 HRS)

Unit IV

Concept of signal spectra, Fourier series co-efficient of a periodic waveform, symmetries as related to Fourier coefficients, Trigonometric & Exponential form of Fourier series.

(12 HRS)

Unit V

Network function & Two port networks – concept of complex frequency, Network & Transfer functions for one port & two ports, poles and zeros, Necessary condition for driving point & transfer function. Two port parameters – Z, Y, ABCD, Hybrid parameters, their inverse & image parameters, relationship between parameters, Interconnection of two ports networks, terminated two port networks.

(12 HRS)



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COURSE OUTCOME:

After successful completion of the course student will be able to:

1. Apply their knowledge in analyzing Circuits by using network theorems.
2. Apply the time and frequency method of analysis.
3. Find the various parameters of two port network.
4. Apply network topology for analyzing the circuit.
5. Synthesize the network using passive elements

Reference Books:

1. M.E. Van Valkenburg, Network Analysis, (PHI)
2. F.F.Kuo, Network Analysis.
3. Mittal GK; Network Analysis; Khanna Publisher
4. Mesereau and Jackson; Circuit Analysis- A system Approach; Pearson.
5. Sudhakar & Pillai; Circuit & Networks- Analysis and Synthesis; TMH
6. Hayt W.H. & J.E. Kemmerly; Engineering Circuit Analysis; TMH
7. Decarlo lin; Linear circuit Analysis; Oxford
8. William D Stanley : Network Analysis with Applications, Pearson Education
9. Roy Choudhary D; Network and systems; New Age Pub
10. Charles K. Alexander & Matthew N.O. Sadiku: Electrical Circuits :TMH
11. Chakraborti :Circuit theory: Dhanpat Rai
12. B.Chattopadhyay & P.C.Rakshit; Fundamental of Electrical circuit theory; S Chand

List of Experiments:

1. To Verify Thevenin's Theorem.
2. To Verify Superposition Theorem.
3. To Verify Reciprocity Theorem.
4. To Verify Maximum Power Transfer Theorem.
5. To Verify Millman's Theorem.
6. To Determine Open Circuit parameters of a Two Port Network.
7. To Determine Short Circuit parameters of a Two Port Network.
8. To Determine A,B, C, D parameters of a Two Port Network
9. To Determine h parameters of a Two Port Network
10. To Find Frequency Response of RLC Series Circuit.
11. To Find Frequency Response of RLC parallel Circuit.

NOTE - All experiments (wherever applicable) should be performed through the following steps.

Step 1: Circuit should be designed/ drafted on paper.

Step 2: Where ever applicable the designed/drafted circuit should be simulated using Simulation S/W (TINA-V7/ PSPICE/ Lab view/ CIRCUIT MAKER etc.).



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Step 3: The designed/drafted circuit should be tested on the bread board and compare the results with the simulated results.

Step 4: Where ever required the bread board circuit should be fabricated on PCB.



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IIIRD SEMESTER

CATEGORY:-DLC

EE-34 - ELECTRICAL WORKSHOP

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

1. To understand the importance of workshop practice in Engineering
- 2: to acquire proper understanding of various manufacturing processes
- 3: to identify the significance and application of various tools and equipment used in workshop

Experiment List:

1. To prepare a wiring such that one lamp is controlled by one switch with one independent plug point and to prepare estimate for the same time.
2. To prepare an estimation and wire up a circuit for mercury vapour lamp and CFL.
3. To prepare an estimate and wire up the circuit to control two lamps from two switches such that one lamp is ON at a time.
4. To connect two lamps in series and parallel and to prepare the estimate for the same.
5. To connect two switches in order to control one lamp so that it can be controlled by each one independently. Also prepare the estimate for the same.
6. To prepare an estimate and wire up the circuit to control two lamps with two switches (one way switch and two way switch) so as to get both dim, both bright and only one bright.
7. To prepare wiring of distribution board including power plug using isolator, MCB, ELCB and to prepare the estimation for the same.
8. To measure the resistance of insulation using insulation megger.
9. To measure the resistance of earth using earth megger.
10. (a). Identification of electronic components.
(b). To practice soldering and prepare the estimation for the same.
(c). To solder an IC base and wires in a PCB.
(d). Insert a logic gate in the IC base and verify its function using IC tester.
11. To prepare an estimate and to setup a half wave and full wave rectifier with and without capacitor filter.

COURSE OUTCOME:

After completion of the course, the student will be able to

- 1: know and understand the types of trades in engineering
- 2: improve their practical skills to develop new products



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IIIRD SEMESTER

CATEGORY:-PDFS

BE-33 PROFESSIONAL DEVELOPMENT FINISHING SCHOOL

LEVEL-I

TOTAL - 36 HOURS

COURSE OBJECTIVE

The students are to be groomed with respect to personality development. In this regard, an effort is made to improve the knowledge with respect to basic in English, mathematics, aptitude and reasoning.

UNIT-I

Conversational English:

Grammar mainly Tenses, 100 small sentences of daily use tense wise, Letter Writing, Standard Format for CV writing.

(12 HRS)

UNIT-II

Basic Mathematics:

Arithmetic, Algebra, Unit Conversions.

Arithmetic

Number system, Decimals, Fractions, Simplification, HCF and LCM. Ratio and proportion, percentage, partnership, Average, profit and Losses, Simple Interest and Compound Interest, Mensuration, Time and work ,Time and Distance, Data Interpretation , Trigonometry Basics ,etc

Algebra

Basics Algebraic Formulae, Linear Equations, quadratics Eqations,Logarithms,Functions, Permutation and Combination, Binomial Theorem , Series (AP,GP,HP).Unit conversion SI,FPS,MKS,CGS .

(12 –HOURS)



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UNIT-III

Aptitude / Reasoning

Quantitative Aptitude and Logical

Reasoning- Level-1 Problem solving on.

Number System, problems on Ages, Number Theory, Algebra, Clocks and Calendars.

Alphabet Test, Series Completion, Coding- Decoding, Logical Sequence,
Insert missing figures.

(12 –HOURS)

OUTCOME

The students have gained confidence after improving their English, Math, and Aptitude and reasoning abilities.



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
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NATIONAL CADET CORPS



GROOMING TOMORROW'S LEADERS

**OFFERING NCC A GENERAL GENERIC ELECTIVE CREDIT
COURSE IN UNIVERSITIES UNDER CHOICE BASED CREDIT
SYSTEM TO ALIGN WITH NEW EDUCATION POLICY 2020**



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CONTENTS

1. Section I : NCC Credit Course Design
2. Section II : NCC Credit Course Rules & Regulations aligned to UGC.

NATIONAL CADET CORPS

SECTION I: NCC CREDIT COURSE DESIGN DOCUMENT

UNDER CHOICE BASED CREDIT SYSTEM AS GENERAL ELECTIVE FOR SENIOR DIVISION / SENIOR WING

1. **Preamble.** The National Cadet Corps (NCC) is governed by NCC Act 1948 and attendant NCC Rules. It functions under the Ministry of Defence and is headed by DGNCC. It is organised into 17 State Directorates each headed by an Additional/Deputy Director General. The aims of NCC are:-

- (a) To develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.
- (b) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regard less of which career they choose.
- (c) To provide a conducive environment to motivate young Indians to choose the Armed Forces as a career.

2. **Purpose.** Currently NCC training is imparted as extra-curricular activity to volunteer students from recognized schools and colleges who enroll as cadets. NCC as a Credit Course is designed with an intent to transform NCC training into a curricular activity from an extra-curricular thereby providing academic credits to students undergoing NCC training along with other attended advantages to the cadets in the college/ university.

3. **Introduction to NCC Credit Course Design.** Institutional Training is the mainstay of NCC training and it is conducted at colleges and universities by Associate NCC Officers and Armed Forces personnel. The application of knowledge gained through institutional training is further honed or developed to a higher degree in NCC Camps. The Institutional Training syllabus comprises Common Subjects and Specialised Subjects (military component). NCC Credit Course is designed to offer Institutional Training of Senior Wing /Division is over six semesters (three years), comprising 300 periods (excluding Camp), of which 120 periods are meant for theory with 08 credits and 180 periods for practical with 6 credits. Each period is counted as hour. The ratio between theory and practical in terms of number of hours of training is 5:6, but in terms of credits is 5:3, since as per CBCS two hours of practical is counted towards one period of training as against one hour for theory. In addition two separate courses have been designed for two Camps normally referred to as Annual Training Camps (ATC).



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Training schedules planned for cadets ensure that the optimum benefits of the NCC organization reach maximum number of cadets. The main emphasis is on practical training which in consonance with theory is made to facilitate active participation of learner, better assimilation of knowledge, and proper development of various skills, strengthening of mind and body which is the bedrock of NCC training.

NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	01	01	-	02	
Semester - II	01	01	-	02	
Semester – III	01	01	05	07	Credits of 1 st Camp merged with 3 rd Sem
Semester – IV	02	01	-	03	
Semester – V	01	01	05	07	Credits of 2 nd Camp merged with 5 th Sem
Semester - VI	02	01	-	03	
Total	08	06	10	24	Twenty-Four Credits

INSTITUTIONAL TRG SYLLABUS

COMMON SUBJECTS				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	NCC General	06	-	06
2	National Integration	04		04
3	Drill	-	45	45
4	Weapon Training	-	25	25
5	Personality Development	25		25
6	Leadership	12	-	12
7	Disaster Management	13		13
8	Social Service & Community Development	08	39	47
9	Health & Hygiene	-	10	10
10	Adventure	01		01
11	Environmental awareness & conservation	03		03
12	Obstacle Training	-	09	09
13	General Awareness	04		04
14	Border & Coastal Areas	06		06
TOTAL HOURS COMMON SUBJECTS(a)		82	128	210

SPECIALSED SUBJECTS (ARMY)				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practical	
1	Armed Forces	09	-	09
2	Map Reading	-	24	24
3	Communications	03	03	06
4	Infantry Weapons	03	03	06
5	Field Craft & Battle Craft		22	22
6	Military History	23	-	23
Total Hours		38	52	90

**BHOPAL****SPECIALISED SUBJECTS (NAVY)**

Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	Naval Orientation	12	-	12
2	Naval Communication	02	18	20
3	Navigation	02	03	05
4	Seamanship	15	18	33
5	Fire Fighting and Damage Control	04	03	07
6	Ship and Boat Modelling	03	10	13
Total hours		38	52	90

SPECIALISED SUBJECTS (AIR FORCE)

Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	General Service Knowledge	08	-	08
2	Air Campaign	06	02	08
3	Principles of flight	06	06	12
4	Airmanship	01	07	08
5	Navigation	05	-	05
6	Aeroengines	06	-	06
7	Basic flight Instruments	03	03	06
8	Aero modelling	03	34	37
Total Hours		38	52	90

INSTITUTIONAL TRAINING: TOTAL HOURS & CREDITS

INSTITUTIONAL TRAINING: TOTAL HOURS & CREDITS			
ITEM	Periods (1 hour duration each)		Total
	Lectures/Tutorials	Practicals	
TOTAL HOURS COMMON SUBJECTS	82	128	210
TOTAL HOURS SPECIALISED SUBJECTS (ARMY/NAVY/AIR FORCE)	38	52	90
TOTAL HOURS INSTITUTIONAL TRAINING	120	180	300
TOTAL CREDITS INSTITUTIONAL TRAINING	08 CREDITS (15 HOUR THEORY = 1 CREDIT POINT)	6 CREDITS (30 HOURS PRACTICAL TRAINING = 1 CREDIT POINT)	



BHOPAL
NCC CAMP TRAINING SYLLABUS

COMMON SUBJECTS				
S No.	Subjects	Periods		Total
		L/T	P	
1.	Physical Training	-	18	18
2.	Drill	-	32	32
3.	Weapon Training	08	28	36
4.	National Integration and Awareness	08	-	08
5.	Personality Development	08	12	20
6.	Leadership	08	-	08
7.	Disaster Management	08	-	08
8. .	Social Service and Community Development	-	08	08
9.	Health & Hygiene	08	-	08
10.	Obstacle Training	-	04	04
11.	Military History	04	-	04
12.	Communication	04	-	04
13.	Games	-	18	18
14.	Culture	-	18	18
	TOTAL	56	138	194
<u>SPECIALISED SUBJECTS</u>				
1.	Map Reading	-	24	24
2.	Infantry Weapons	04	02	06
3.	Field Craft & Battle Craft	-	16	16
	TOTAL	04	42	46
	GRAND TOTAL	60 (4 credit)	180 (6 credit)	240 (10 credit)

NCC CAMP TRAINING SYLLABUS (FOR THEORY)

Ser No	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Weapon Training	-	-	04	-	04	-	08
2.	National Integration & Awareness	-	-	04	-	04	-	08
3.	Personality Development	-	-	04	-	04	-	08
4.	Leadership	-	-	04	-	04	-	08
5.	Disaster Management	-	-	04	-	04	-	08
6.	Health & Hygiene	-	-	04	-	04	-	08
7.	Military History			02		02		04
8.	Communication			02		02		04
9.	Infantry Weapons	-	-	02	-	02	-	04
	TOTAL	-	-	30	-	30	-	60
	TOTAL Credit	-	-	2	-	2	-	4

NCC CAMP TRAINING SYLLABUS (FOR PRACTICAL)

Ser No	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Physical Training	-	-	09	-	09	-	18
2.	Drill	-	-	16	-	16	-	32
3.	Weapon Training	-	-	14	-	14	-	28
4.	Personality Development	-	-	06	-	06	-	12
5.	Social Service and Community Development	-	-	04	-	04	-	08
6.	Obstacle Training	-	-	02	-	02	-	04
7.	Games			09		09		18
8.	Culture			09		09		18
9.	Map Reading	-	-	12	-	12	-	224
10.	Infantry Weapons	-	-	01	-	01	-	02
11.	Field Craft & Battle Craft	-	-	08	-	08	-	16
	TOTAL			90		90		180
	TOTAL CREDIT			03		03		06



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
BHOPAL**



SEMESTER WISE COURSE DESIGN

ARMY CADETS

NATIONAL CADET CORPS

INSTITUTIONAL TRAINING: SEMESTER WISE DISTRIBUTION OF NCC SYLLABUS FOR THEORY (ARMY CADETS)

S. NO.	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	NCC General	06	-	-	-	-	-	06
2.	National Integration	04	-	-	-	-	-	04
3.	Personality Development	02	05	05	04	06	04	25
4.	Leadership	-	05	04	03	-	-	12
5.	Disaster Management	-	-	03	10	-	-	13
6.	Social Service & Community Development	03	05	-	-	-	-	08
7.	Adventure	-	-	01	-	-	-	01
8.	Environmental Awareness & Conservation	-	-	-	03	-	-	03
9.	General Awareness	-	-	-	04	-	-	04
10.	Border & Coastal Areas	-	-	02	-	02	02	06
11.	Armed Forces	-	-	-	06	-	03	09
12.	Infantry Weapons	-	-	-	-	3	-	3
13.	Communication	-	-	-	-	-	03	03
14.	Military Hospital	-	-	-	-	04	19	23
	TOTAL	15	15	15	30	15	30	120
	TOTAL Credit	1	1	1	2	1	2	08

INSTITUTIONAL TRAINING: SEMESTER WISE DISTRIBUTION OF NCC SYLLABUS FOR PRACTICAL (ARMY CADETS)

S. NO.	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Drill	12	12	08	07	03	03	45
2.	Field Craft & Battle Craft	03	04	04	04	04	03	22
3.	Map Reading	03	05	04	04	04	04	24
4.	Weapons Training	05	04	04	04	04	04	25
5.	Communication	-	-	-	-	-	03	03
6.	Infantry Weapons	-	-	-	-	-	03	03
7.	Social Service & Community Development	07	05	05	06	06	10	39
8.	Health & Hygiene	-	-	-	05	05	-	10
9.	Operation Training	-	-	05	-	04	-	09
	TOTAL	30	30	30	30	30	30	180
	TOTAL Credit	01	01	01	01	01	01	06



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INSTITUTIONAL TRAINING: SEMESTER WISE THEORY DETAILED SYLLABUS (ARMY CADETS)

<u>SEMESTER I</u>					
S.No	Subject	Periods	Chapter	Lesson	Hours
1	NCC General	6	NCC-I	Aims, Objectives and Org of NCC	1
			NCC-II	Incentives	2
			NCC-III	Duties of NCC Cadets	1
			NCC-IV	NCC Camps: Types and Conduct	2
2	National Integration and Awareness	4	NI-I	National Integration: Importance and Necessity	1
			NI-II	Factors affecting National Integration	1
			NI-III	Unity in Diversity	1
			NI-IV	Threats to National Security	1
3	Personality Development	2	PD - I	Factors Self-Awareness Empathy Critical and Creative Thinking Decision Making and Problem Solving	2
4	Social Service and Community Development	3	SSCD - I	Basics of Social Service Rural Development Programmes NGO's Contribution of Youth	3
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER II					
S.No	Subject	Periods	Chapter	Lesson	Hours
5	Personality Development	5	PD-II	Communication Skills	3
			PD-III	Group Discussion -Coping with Stress and Emotions	2
6	Leadership	5	L-I	Leadership Capsule Traits Indicators Motivation Moral Values Honour Code	3
			L-II	Case Studies Shivaji, Jhansi Ki Rani,	2
7	Social Service and Community Development	5	SS-IV	Protection of Children & Women Safety	1
			SS-V	Road/Rail Travel Safety	1
			SS-VI	New Initiatives	2
			SS-VII	Cyber and Mobile Security Awareness	1
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER III					
S.No	Subject	hours	Chapter	Lesson	HOURS
8	Personality Development	5	PD-III	Group Discussions - Change your Mindset	2
			PD-V	Public Speaking	3
9	Leadership	4	L-II	Case Studies – APJ Abdul Kalam, Deepa Malik, Maharana Pratap, N Narayan Murthy	4
10	Disaster Management	3	DM-I	Disaster Management Capsule Organisation Types of Disasters Essential Services Assistance Civil Defence Organisation	3
11	Adventure	1	AD-I	Adventure activities	1
12	Border & Coastal Areas	2	BCA-I	History, Geography & Topography of Border/ Coastal Areas	2
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER IV					
S.No	Subject	hours	Chapter	Lesson	HOURS
13	Personality Development	4	PD-III	Group Discussions - Time Management, Social Skills	4
14	Leadership	3	L-II	Case Studies – Ratan Tata, Rabindra Nath Tagore, Role of NCC cadets in 1965 war	3
15	Disaster Management	9	DM-II	Initiative Trg, Organising Skills, Dos and Don'ts Natural Disasters Man Made Disasters	9
		1	DM-III	Fire Services and Fire Fighting	1
16	Environmental Awareness	3	EA-I	Environmental Awareness and Conservation	3
17	General Awareness	4	GA-I	General Awareness	4
18	Armed Forces	6	AF-1	Army, Navy, Air Force and Central Armed Police Forces	6
TOTAL HOURS					30
TOTAL CREDITS					2



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SEMESTER V

S.No	Subject	hours	Chapter	Lesson	HOURS
19	Personality Development	6	PD-III	Group Discussions - Team Work	2
			PD-V	Public Speaking	4
20	Border & Coastal Areas	2	BCA-II	Security Setup and Border/Coastal management in the area	2
21	Introduction to Infantry Battalion and its Equipments	3	INF-1	Organisation of Infantry Battalion & its weapons	3
22	Military History	4	MH-3	Study of Battles of Indo-Pak Wars 1965 & 1971	4
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER VI

S.No	Subject	hours	Chapter	Lesson	HOURS
25	Personality Development	3	PD-IV	Career Counselling, SSB Procedure and Interview Skills	3
27	Border & Coastal Areas	2	BCA-III	Security Challenges & Role of cadets in Border management	2
28	Armed Forces	3	AF-2	Modes of Entry into Army, Police and CAPF	3
29	Military History	19	MH-1	Biographies of Renowned Generals	6
			MH-2	War Heroes : Param Veer Chakra Awardees	3
			MH-3	Study of Battles of Kargil	2
			MH-4	War Movies	8
30	Communication	3	C-1	Introduction to Communication & Latest Trends	3
TOTAL HOURS					30
TOTAL CREDITS					2

SIX SEMESTER NCC COURSE SYLLABUS

Training Objectives: Institutional Training

1. Institutional training includes basic military training of the cadets as part of the curriculum with its long-standing effort to mould young volunteers into disciplined and responsible citizens of India. NCC course is aimed to achieve following learning objectives:-

- (a) Develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.
- (b) To create interest in cadets by including and laying emphasis on those aspects of Institutional Training which attract young cadets into the NCC and provides them an element of thrill and excitement.
- (c) To inculcate defence Services work ethos that is characterized by hard work, sincerity of purpose, honesty, ideal of selfless service, dignity of labour, secular outlook, comradeship, spirit of adventure and sportsmanship.
- (d) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regardless of which career they choose.
- (e) To provide conducive environment to motivate young Indians to choose the Armed Forces as a career.



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SEMESTER I COURSE MODULE : NATIONAL CADET CORPS I

National Cadet Corps : Course Details			
Course Title: National Cadet Corps I			
Course Code	BNCC01GE03	Credits	1(Thr)+ 1(Pr) = 03
L / T + P	15+30	Course Duration	1 Semester
Semester	I (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e, 25% internal assessment and 75% end term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

2. Course Objectives: Cadets will be able to: -

- Know about the history of NCC, its organization, and incentives of NCC for their career prospects.
- Acquire knowledge of duties and conduct of ncc cadets.
- Understand about different NCC camps and their conducts.
- Understand the concept of national integration and its importance.
- Understand the concept of self-awareness and emotional intelligence.
- Understand the concept of critical & creative thinking.
- Understand the process of decision making & problem solving.
- Understand the concept of team and its functioning.
- Understand the concept and importance of Social service.

3. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Imbibe the conduct of NCC cadets.
- (b) Respect the diversity of different Indian culture.
- (c) Practice togetherness and empathy in all walks of their life.
- (d) Do their own self analysis and will workout to overcome their weakness for better performance in all aspects of life.
- (e) Understand creative thinking & its components.
- (f) Think divergently and will try to break functional fixedness.
- (g) Make a team and will work together for achieving the common goals.
- (h) Do the social services on different occasions.

4.Course Content Part (I) Theory

- (a) **Unit 1- NCC General (N) (Contact Hrs. 06).** Introduction of NCC, History, Aims, Objective of NCC & NCC as Organization, Incentives of NCC, Duties of NCC Cadet. NCC Camps: Types & Conduct.
- (b) **Unit 2- National Integration & Awareness (NI) (Contact Hrs. 04) .** National Integration: Importance & Necessity, Factors Affecting National Integration, Unity in Diversity & Role of NCC in Nation Building, Threats to National Security.
- (c) **Unit 3- Personality Development (Contact Hrs. 3).** Intra & Interpersonal skills - Self-Awareness- & Analysis, Empathy, Critical & creative thinking, Decision making and problem solving.
- (d) **Unit 4- Social Service and Community Development(Contact Hrs. 02).** Basics of social service and its need, Types of social service activities, Objectives of rural development programs and its importance, NGO's and their contribution in social welfare, contribution of youth and NCC in Social welfare.

Course Content Part (II) Practical

5. **Course Objectives:** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of foot drill.
- (c) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.



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- (d) Develop awareness about different types of terrain and how it is used in battle craft.
 - (e) Develop the concept of various markings on the map and how they are co-related to the ground features.
 - (f) Understand the various social issues and their impact on social life.
 - (g) Develop the sense of self-less social service for better social & community life.
6. **Expected Learning Outcomes:** After completing this course, the cadets will be able to: -
- (a) Perform foot drill and follow the different word of command.
 - (b) Fire a weapon effectively with fair degree of marksmanship.
 - (c) Undertake point to point navigation and take part in route marches by day and night.
 - (d) Perform the social services on various occasions for better community & social life.
7. **Course Content Part (II) Practical**
- (a) **Unit 1. Drill (Contact Hrs. 12).** Foot Drill- Drill ki Aam Hidayaten, Word ki Command, Savdhan, Vishram, Aram Se, Murdna, Kadvar Sizing, Teen Line Banana, Khuli Line, Nikat Line, Khade Khade Salute Karna Parade Par, Visarjan, Line Tod, Tej Chal, Tham aur Dhire Chal, Tham.
 - (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 05).** Introduction & Characteristics of .22 rifle, Handling of .22 rifle.
 - (c) **Unit 3. Map Reading (MR) (Contact Hrs. 03).** Definition of Map, Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Magnetic Variation and Grid Convergence.
 - (d) **Unit 4. Field Craft & Battle Craft (FC & BC) (Contact Hrs. 03).** Introduction of Field Craft & Battle craft, Judging Distance, Method of Judging Distance.
 - (e) **Unit 5. Social Service and Community Development (SSCD)(Contact Hrs.07).** Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc.

SEMESTER II COURSE MODULE : NATIONAL CADET CORPS II

Course Title: National Cadet Corps II			
Course Code	BNCC02GE03	Credits	1(Thr)+ 1(Pr)=02
L / T + P	15+30	Course Duration	1 Semester
Semester	II (Even)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

8. Course Objectives: Cadets will be able to: -

- (a) Understand the thinking & reasoning process.
- (b) Understand the process to cope with Stress & emotions.
- (c) Understand the importance of improving communication skills.
- (d) Identify the leadership traits.
- (e) Admire the qualities of great leaders.
- (f) Know about different legal provisions for children & women safety and protection.
- (g) Understand the various rules & measures to be taken to ensure Road/Rail safety.
- (h) Understand & spread awareness about latest Government initiatives for welfare of citizens and contribute towards Nation building.
- (i) Understand concepts of cyber and mobile security.



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9. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Define thinking, reasoning, critical thinking and creative thinking.
- (b) To think critically about different life related issues.
- (c) Think divergently and will try to break functional fixedness.
- (d) Creatively in their real-life problems.
- (e) Understand the organizations related to disaster management and their functioning.
- (f) Appreciate the role of NCC cadets in disaster management.

10. **Course Content Part (I) Theory**

(a) Unit 1. Personality Development (Contact Hrs.5)

- (i) Thinking- Meaning and Concept of thinking, Reasoning, Process of thinking.
- (ii) Critical Thinking- Meaning & concept of critical thinking, Features of critical thinking, Process of critical thinking.
- (iii) Creative thinking- Meaning & concept of creative thinking, Features of creative thinking, Process of creative thinking, levels of Creativity, Characteristics of creative person.

(b) Unit 2. Leadership Development (Contact Hrs.5)

- (i) Leadership capsule.
- (ii) Important Leadership traits, Indicators of leadership and evaluation.
- (iii) Motivation- Meaning & concept, Types of motivation. Factors affecting motivation.
- (iv) Ethics and Honor codes.

(c) Unit 3. Social Service and Community Development (Contact Hrs. 5)

- (i) Protection of Children & Women Safety.
- (ii) Road/Rail Safety.
- (iii) New Government Initiatives.
- (iv) Cyber and mobile Security Awareness.

Course Content Part (II) Practical

11. **Course Objectives.** Cadets will be able to: -
- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
 - (b) Appreciate grace and dignity in the performance of foot drill.
 - (c) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
 - (d) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
12. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Perform foot drill gracefully.
 - (b) Give and follow the different word of command.
 - (c) Fire a weapon effectively with fair degree of marksmanship.
 - (d) Use of bearing and service protractor and locate the places and objects on the ground.
 - (e) Do the social service and feel connected with social problems.
13. **Course Content Part (II) Practical**
- (a) **Unit 1. Drill (Contact Hrs. 12)**
 - (i) Foot Drill Dahine, Baen, Aageaur Piche Kadam Lena.
 - (ii) Tej Chal se Murdna, Tej Chal se Salute Karna, Tej Kadam Taal aur Tham, Tej Kadam Taal se Kadam Badalna.
 - (iii) Teeno Teen se Ek File aur ek file se Teeno Teen Banana
 - (b) **Unit 2. Weapon Training (Contact Hrs. 04)**
 - (i) Range procedure & Theory of group.
 - (ii) Short Range firing.
 - (c) **Unit 3. Map Reading (Contact Hrs. 05)**
 - (i) Protractor Bearing and its conversion methods.
 - (ii) Service protractor and its uses.
 - (iii) Prismatic compass and its uses and GPS.
 - (iv) Navigation by compass and GPS.



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(d) Unit 4. Field Craft & Battle Craft (Contact Hrs. 04)

- (i) Indications of landmarks and Targets.
- (ii) Intro, Definitions, Types of Ground, Indication of Landmarks, Methods of identification of targets, difficult targets.

(e) Unit 5. Social Service and Community Development (Contact Hrs. 05)

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc. as per the requirement and similar announced days- National and state level.

NATIONAL CADET CORPS

SEMESTER III COURSE MODULE : NATIONAL CADET CORPS III

<u>COURSE TITLE: NATIONAL CADET CORPS III</u>			
Course Code	BNCC03GE02	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15 +30	Course Duration	1 Semester
Semester	III (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

14. **Course Objectives.** Cadets will be able to: -

- (a) Understand the life history and leadership qualities of great leaders, sportspersons & entrepreneurs.
- (b) Understand the various aspects of types of mindset.
- (c) Understand public speaking methods & qualities.
- (d) Understand the organizations related to disaster management and their functioning.
- (e) Understand the role of NCC cadets in disaster management.
- (f) Understand the various types of adventure activities.
- (g) Understand the History, Geography & Topography of Border/ Coastal Areas.



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15. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Admire and get inspired from the accomplishments of leaders from various walks of life.
- (b) Develop public speaking skills.
- (c) Understand the importance of positive mindset and optimistic attitude in life.
- (d) Appreciate the need & requirement for disaster management and his role in disaster management activities.
- (e) Know the history & geographical peculiarity of our borders & coastal regions.

16. **Course Content Part (I) Theory**

(a) **Unit 1. Personality Development (Contact Hrs.5)**

- (i) Group Discussions - Change your Mindset
- (ii) Public Speaking.

(b) **Unit 2. Leadership Development (Contact Hrs.4).** Case Studies – APJ Abdul Kalam, Deepa Malik, Maharana Pratap, N Narayan Murthy.

(c) **Unit 3. Disaster management(Contact Hrs. 3)**

- (i) Disaster Management Capsule.
- (ii) Organisation.
- (iii) Types of Disasters.
- (iv) Essential Services.
- (v) Assistance.
- (vi) Civil Defence Organisation.

(d) **Adventure (Contact Hrs. 1).** Adventure activities.

(e) **Border & Coastal Areas(Contact Hrs. 2).** History, Geography & Topography of Border/ Coastal Areas.

Course Content Part (II) Practical

17. **Course Objectives.** Cadets will be able to : -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal
- (b) Appreciate grace and dignity in the performance of arm drill
- (c) Understand the concept and importance of social service.
- (d) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (e) Actively participate in social service and community development activities.

18. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform arm drill gracefully.
- (b) Give and follow the different word of command.
- (c) Fire a weapon effectively with fair degree of marksmanship.
- (d) Different positioning for fire and aiming.
- (e) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (f) Observe surroundings in better way.
- (g) Develop the qualities of patience and confidence and become better individuals.
- (h) Will develop physical as well as mental fitness.

19. **Course Content Part (II) Practical**

(a) **Unit 1. Drill(Contact Hrs. 08)**

- (i) Arm Drill.
- (ii) Rifle ke saath Savdhan, Vishram aur Aram se.
- (iii) Rifle ke saath Parade Par aur Saj, Rifle ke saath Visarjan, Line Tod.
- (iv) Bhumi Shastra aur Uthao Shastra, Bagal Shastra aur Baju Shastra.

(b) **Unit 2. Weapon Training(Contact Hrs. 04).** Short Range firing.

(c) **Unit 3. Map Reading (Contact Hrs. 04).**

- (i) Setting of Map.
- (ii) Findings North and Own Position.



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(d) Unit 4. Field Craft & Battle Craft (Contact Hrs. 04)

- (i) Observation.
- (ii) Camouflage.
- (iii) Concealment.

(e) Unit 5. Social Service and Community Development (Contact Hrs. 05).

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) Unit 6. Obstacle Training(Contact Hrs. 05)

- (i) Obstacle training - Introduction, Safety-measures, Benefits.
- (ii) Obstacle Course- Straight balance, Clear Jump, Gate Vault, Zig- Zag Balance, High Wall.

NATIONAL CADET CORPS

SEMESTER IV COURSE MODULE : NATIONAL CADET CORPS IV

Course Title: National Cadet Corps IV			
Course Code	BNCC04GE03	Credits	2(Thr)+ 1(Pr)=03
L /T + P	30+30	Course Duration	1 Semester
Semester	IV (Even)	Contact Hours	30(Thr)+30(Pr)=60Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

20. **Course Objectives.** Cadets will be able to: -

- (a) Develop a sense of time management and social skills.
- (b) Understand the life history & leadership qualities of personalities who have contributed in Nation Building and Literature.
- (c) Understand the role of NCC cadets as 2nd line Defence in 1965 War.
- (d) Develop awareness about various types of Natural and manmade disasters.
- (e) Know about life saving tips during disasters.
- (f) acquainted about Fire Services.
- (g) Understand importance of Environmental Awareness & conservation.
- (g) Understand importance of General Awareness.
- (h) Know about Armed Forces.



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(b) **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (i) Effectively Manage time.
- (ii) Develop the qualities of social skills.
- (iii) Imbibe leadership qualities.
- (iv) Do group discussions effectively.
- (v) Be motivated to serve the nation by joining Armed forces.
- (vi) Contribute in environmental awareness and conservation activities.
- (vii) Keep abreast of current affairs & general awareness.
- (viii) Effectively contribute in managing disaster relief tasks.

21. Course Content Part (I) Theory

- (a) **Unit 1. Personality Development (Contact Hrs.4).** Group Discussions – Social Skills & Time management.
- (b) **Unit 2. Leadership Development (Contact Hrs.3).** Case Studies – Case Studies – Ratan Tata, Rabindra Nath Tagore, Role of NCC cadets in 1965 war.
- (c) **Unit 3. Disaster management(Contact Hrs. 10)**
 - (i) Initiative Trg, Organising Skills.
 - (ii) Dos and Don'ts.
 - (iii) Natural Disasters.
 - (iv) Man Made Disasters.
 - (v) Fire Services and Fire Fighting.
- (d) **Environmental Awareness (Contact Hrs. 3).** Adventure Environmental Awareness and Conservation.
- (e) **General Awareness (Contact Hrs. 4).** General Awareness.
- (f) **Armed Forces(Contact Hrs. 6).** Army, Navy, Air Force and Central Armed Police Forces.

Course Content Part (II) Practical

22. **Course Objectives.** Cadets will be able to: -
- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
 - (b) Understand various signals to convey messages in the army.
 - (c) Get acquainted various section formations.
 - (d) Understand the basics of personal and public hygiene.
 - (e) Get acquainted with the procedure to treat the wounds and fractures during emergencies.
23. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Perform weapon drill gracefully.
 - (b) Give and follow the different word of command.
 - (c) Appreciate grace and dignity in the performance of foot drill.
 - (d) Apply signals in there day to day functioning.
 - (e) Provide first aid during the emergencies.
 - (f) Navigate to the given location on ground using compass and GPS.
 - (g) Practice healthy practices for the personal sanitation and hygiene.
24. **Course Content Part (II) Practical**
- (a) **Unit 1. Drill (Contact Hrs. 08)**
 - (i) Arm Drill.
 - (ii) Salami Shastra.
 - (iii) Squad Drill with Arms.
 - (b) **Unit 2. Weapon Training (Contact Hrs. 04).** Short Range firing
 - (c) **Unit 3. Map Reading(Contact Hrs. 04)**
 - (i) Map to Ground.
 - (ii) Ground to Map.
 - (d) **Unit 4. Field Craft & Battle Craft(Contact Hrs. 04)**
 - (i) Fire and Move Capsule.
 - (ii) Field signal- with hand, with Weapons, Signal with Whistle.
 - (iii) Field signals as means of giving orders.



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(iv) Field signals by day, Field signals by night.

(v) Section Formation.

(e) **Unit 5. Social Service and Community Development(Contact Hrs. 05)**

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) **Unit 6. Health & Hygiene(Contact Hrs. 05)**

(i) Hygiene & Sanitation (Hygiene- Personal & Camp Hygiene).

(ii) First Aid in common medical emergencies.

(iii) Treatment & Care of Wounds.

NATIONAL CADET CORPS

SEMESTER V COURSE MODULE : NATIONAL CADET CORPS V

Course Title: National Cadet Corps V			
Course Code	BNCC05GE02	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15 +30	Course Duration	1 Semester
Semester	V (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

25. **Course Objectives.** Cadets will be able to: -
- (a) Understand the concept of Team and its functioning.
 - (b) Hone Public speaking skills.
 - (c) Understand the security set up and management of Border/Coastal areas.
 - (d) Acquire knowledge about an Infantry Battalion organisation and its weapons.
 - (e) Acquire knowledge about Indo-Pak Wars fought in 1965 & 1971.
26. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Participate in team building exercise and value team work.
 - (b) Improve communication skills by public speaking activities.
 - (c) Understand the security mechanism and management of Border/Coastal areas.
 - (d) Get motivated to join armed forces.



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27. **Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (Contact Hrs.6).**
 - (i) Group Discussions –Team work.
 - (ii) Public speaking.
- (b) **Unit 2. Border & Coastal Areas(Contact Hrs.2).** Security Setup and Border/Coastal management in the area.
- (c) **Unit 3. Introduction to Infantry Battalion and its Equipment(Contact Hrs. 3).** Organisation of Infantry Battalion & its weapons
- (d) **Military History(Contact Hrs. 4).** Study of Battles of Indo-Pak Wars 1965 & 1971.

Course Content Part (II) Practical

28. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of ceremonial drill.
- (c) Use the compass and GPS to locate places on the ground and map.

29. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform ceremonial drill and follow the different word of command.
- (b) Do the social service on various occasions and get connected with the community.
- (c) Do all the asana and gain the physical& mental fitness.

30. **Course Content Part (II) practical**

- (a) **Unit 1. Drill(Contact Hrs. 03)**
 - (i) Ceremonial Drill.
 - (ii) Guard Mounting.
- (b) **Unit 2. Field Craft & Battle Craft(Contact Hrs. 04)**
 - (i) Fire control orders.
 - (ii) Types of fire control orders.

(iii) Fire and Movement- when to use fire and movements tactics, Basic considerations, Appreciation of ground cover, Types of cover, Dead ground, Common Mistakes, Map and air photography, Selection of Fire position and fire control.

(c) **Unit 3. Map Reading(Contact Hrs. 04)**.Google Maps & applications

(d) **Unit 4. Weapon Training(Contact Hrs. 04)**.Short Range firing

(e) **Unit 5. Social Service and Community Development (Contact Hrs. 05)**

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc. as per the requirement and similar announced days- National and State level.

(f) **Unit 6. Health & Hygiene(Contact Hrs. 05)**

(i) Yoga- Introduction, Definition, Purpose, Benefits.

(ii) Asanas-Padamsana, Siddhasana, Gyan Mudra, Surya Namaskar, Shavasana, Vajrasana, Dhanurasana, Chakrasana, Sarvaangasana, Halasana etc.

(f) **Unit 7. Obstacle Training(Contact Hrs. 05)**

(i) Obstacle training – Intro, Safety measures, Benefits.

(ii) Obstacle Course- Straight balance, Clear Jump, Gate Vault, Zig- Zag Balance, High Wall etc.



Course Title: National Cadet Corps VI			
Course Code	BNCC06GE03	Credits	2(Thr)+ 1(Pr)=03
L / T + P	30 +30	Course Duration	1 Semester
Semester	VI (Even)	Contact Hours	30(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

31. **Course Objectives.** Cadets will be able to: -

- (a) Get acquainted about counselling process its need and importance.
- (b) Know about SSB procedure and different tasks and tests.
- (c) Know about the conduction during the interview.
- (d) Understand the security challenges & role of cadets in Border Areas.
- (e) Know about the modes of entry in Armed forces, CAPF & police.
- (f) Understand the life history & leadership qualities of great generals.
- (g) Learn about 1999 Kargil war.
- (h) Acquire the knowledge about various wars and their heroes.
- (i) Know about various components of communication process.

32. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Get motivated to join Armed forces, police & CAPF.
- (b) Write their CV effective and appealing.
- (c) Face SSB interview effectively in their future.
- (d) Understand individual responsibilities & role in meeting the security challenges on Border/Coastal areas.
- (e) Imbibe the feeling of patriotism.
- (f) Communicate more effectively.

33. **Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (Contact Hrs.3).**
 - (i) Career Counselling.
 - (ii) SSB Procedure.
 - (iii) Interview Skills.
- (b) **Unit 2. Border & Coastal Areas(Contact Hrs.2).** Security Challenges & Role of cadets in Border management.
- (c) **Unit 3. Armed Forces(Contact Hrs. 3).** Modes of Entry into Army, Police and CAPF.
- (d) **Military History(Contact Hrs. 19).**
 - (i) Biographies of Renowned Generals.
 - (ii) War Heroes : Param Veer Chakra Awardees.
 - (iii) Study of Battles of Kargil.
 - (iv) War Movies.
- (e) **Communication(Contact Hrs. 3).** Introduction to Communication & Latest Trends.



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Course Content Part (II) Practical

34. **Course Objectives.** Cadets will be able to: -
- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
 - (b) Appreciate grace and dignity in the performance of ceremonial drill.
 - (c) Know about various knots and lashing used in soldiering.
 - (d) Acquire awareness about the basic weapon system in use in the Armed Forces.
35. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Perform foot drill and follow the different word of command.
 - (b) Aiming range and figure targets.
 - (c) Use the different knots and lashing in day-to-day life for different purposes.
 - (d) Develop the feeling of altruism.
36. **Course Content Part (II) Practical.**
- (a) **Unit 1. Drill (Contact Hrs. 03).**
 - (i) Ceremonial Drill.
 - (ii) Guard of Honour.
 - (b) **Unit 2. Weapon Training(WT) (Contact Hrs. 04).** Short Range firing.
 - (c) **Unit 3. Map Reading(MR) (Contact Hrs. 04).** Google maps and Applications.
 - (d) **Unit 4. Field Craft & Battle Craft(FCBC) (Contact Hrs. 03).** Knots, Lashing and Stretchers.
 - (e) **Unit 5. Social Service and Community Development(SSCD) (Contact Hrs. 05).** Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.
 - (f) **Unit 6 Introduction of Infantry Weapons & Equipment(INF) (Contact Hrs.03).** Characteristics of 5.56MM INSAS Rifle, Ammunition, Fire Power, Stripping, Assembling & Cleaning Practice.
 - (g) **Unit 7. Communication (COM) (Contact Hrs. 03).**
 - (i) Basic Radio Telephony (RT) Procedure.
 - (ii) Introduction, Advantages, Disadvantages, Need for standard procedures.

(iii) Types of Radio telephony communication.

(iv) Radio telephony procedure, Documentation.

NATIONAL CADET CORPS



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COURSE MODULE: NATIONAL CADET CORPS CAMP -I

Course Title: National Cadet Corps Camp I			
Course Code	BNCCCAMP03GE05	Credits	2(Thr)+ 3(Pr)=05
L /T + P	30+90	Course Duration	10 Days (24 hours each)
Semester	III (Odd)	Contact Hours	30(Thr)+90(Pr)=120Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, individual and group tasks, team work, field-based assignments, Physical Training, endurance building and skill development practices		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

37. **Course Objectives**. Cadets will be able to: -

- (a) Acquire knowledge about the various aspects of personality development.
- (b) Understand the concept of leadership traits, moral values and character traits.
- (c) Develop awareness about the various types of natural disasters.
- (d) Develop sensitivity to the changing environment and understand the importance of conservation.
- (e) Understand the importance of hygiene and sanitation and common first aid procedures.
- (f) Acquire awareness about various types of weapon systems in the Armed Forces.

38. **Expected Learning Outcomes**. After completing this course, the cadets will be able to: -

- (a) Acquire adequate skill sets to overcome their weakness and reshape their personality.
- (b) Imbibe good moral values and character traits in their daily life.
- (c) Become useful members of the society and form part of disaster response team, if need arises.
- (d) Respect and make efforts to conserve natural resources
- (e) Follow good personal hygiene practices and provide first aid in emergencies.
- (f) Be motivated to join the armed forces.

39. **NCC Camp-I : Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (PD) (Contact Hrs. 04)**. Introduction to Personality Development, Factors influencing/shaping personality, Time Management and Interview Skills.
- (b) **Unit 2. Leadership (LDR) (Contact Hrs. 04)**. Leadership Traits, Moral Values and Character Traits.
- (c) **Unit 3. Disaster Management (DM) (Contact Hrs. 04)**. Assistance during natural disasters, Do's and Don'ts for NCC Cadets performing Disaster Management Duties
- (d) **Unit 4. National Integration and Awareness (NIA)(Contact Hrs. 04)**.
Water Conservation and Rain Harvesting, Waste Management and Energy Conservation
- (e) **Unit 5. Health and Hygiene (H&H)(Contact Hrs. 04)**. Hygiene and Sanitation, First Aid in Common Medical Emergencies.
- (f) **Unit 6. Infantry Weapons (IW) (Contact Hrs. 02)**. Characteristics of Company Support Weapons.
- (g) **Unit 7. Weapon Training (WT) (Contact Hrs. 04)**. Characteristics of Point 22 Rifle and its Ammunition, Range Procedure and Safety Precautions.
- (h) **Unit 8. Military History (MH) (Contact Hrs. 04)**. Guest lectures by War Veterans/decorated soldiers/veterans.
- (i) **Unit 9. Communication (COM) (Contact Hrs. 04)**. Basics of communication.



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NCC Camp-I : Course Content Part (II) Practical

40. **Course Objectives.** Cadets will be able to: -
- (a) Understand that drill is the foundation of discipline and command a group for a common goal.
 - (b) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
 - (c) Develop awareness about different types of terrain and how it is used in Battle Craft.
 - (d) Develop the concept of various markings on the map and how they are co-related to the ground features.
 - (e) Acquire awareness about the various types of weapon systems in the Armed Forces.
 - (f) Understand the concept and importance of social service.
 - (g) Understand the various nuances of Personality Development.
 - (h) Understand the concept and importance of Physical Training in everyone's life.
 - (i) Acquire skill sets about various games and understand the importance of team work.
 - (j) Develop awareness about different cultures and different modes of its projection in artistic forms.
41. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Perform foot drill, arms drill, ceremonial drill and will be able to give out different words of command.
 - (b) Fire a weapon effectively with fair degree of marksmanship.
 - (c) Undertake point to point navigation and take part in route marches by day and night.
 - (d) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
 - (e) Be motivated to join the armed forces.
 - (f) Acquire adequate skill sets to overcome their weakness and enhance their personality.
 - (g) Gain adequate physical and mental endurance capabilities.
 - (h) Play team games and be able to communicate and coordinate effectively in group events or situations.

- (i) Respect the diversity of Indian culture and develop pride by showcasing their own culture to others.

42. NCC Camp-I : Course Content Part (II) Practical

- (a) **Unit 1. Drill (Drill)(Contact Hrs. 16)**. Drill ki Aam Hidayaten aur Words of Command, Savdhan, Vishram, Aram Se aur Mudna, Khuli Line aur Nikat Line mein march, Salute Karna Parade Par, Visarjan aur Line Tod, Tej Chal, Tham aur Dhire Chal, Tham, Dahine, Baen, Aage aur Piche Kadam lena, Tejchaal se Mudna, Tejchaal se Salute karna, Tej kadamtaal aur Tham, Tej Kadamtaal se kadam badhana, Teenon Teen se ek file Banana aur ek file se Teenon Teen Banana, Rifle Ke Saath Saavdhan, Vishram aur Aaram se, Rifle ke saath Parade par aur saaj, Rifle Ke saath visarjan aur line tod, Bhumi Sashttra aur Uthao Sashttra, Bagal Sashttra aur Baaju Shastra.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 14)**. Stripping, Assembling, Cleaning of Point 22 rifle, Sight Setting and Sight Picture of Point 22 Rifle, Loading, Cocking and Unloading, Lying Position, Holding and Aiming of Point 22 rifle, Trigger Control and Firing of Shot, Theory of Group, Short-Range Aiming and Firing, Firing Practice I to VII.
- (c) **Unit 3. Field Craft & Battle Craft (FC/BC) (Contact Hrs. 06)**. Introduction of Field Craft & Battle craft, Judging Distance, Indication of Landmarks and Targets, Observation, Camouflage and Concealment, Field Signals, Section formations.
- (d) **Unit 4. Map Reading (MR) (Contact Hrs. 12)**. Introduction to Map and Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Types of Bearing and use of Service Protector, Prismatic Compass and its use, setting of a map, Finding North and own Position, Map to Ground and Ground to map, Point to Point march, Route March – I, Route March -II.
- (e) **Unit 5. Infantry Weapons (IW) (Contact Hrs. 01)**. Characteristics of Battalion Support Weapons.
- (f) **Unit 6. Social Service and Community Development (SSCD) (Contact Hrs. 04)**. Basics of Social Service and its need, Rural Development Programme, Civic Responsibilities: Cadets will participate in various activities throughout the camp e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc., Road /Rail Travel Safety
- (g) **Unit 7. Personality Development (PD) (Contact Hrs. 06)**. Self-Awareness, Empathy, Critical and Creative Thinking, Decision making and problem Solving, Coping with Stress and Emotions, Time Management.
- (h) **Unit 8. Obstacle Training (OT) (Contact Hrs. 02)**. OT Practice – I:- Untimed, Cadets will be familiarized with all the obstacles in the Obstacle Course and briefed about the correct method to do them, OT Practice -II: Timed practice for all the cadets and record to be maintained.



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- (i) **Unit 9. Physical Training (PT) (Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in morning hours. Training has to be progressive in degree of difficulty to improve individual stamina and endurance. Training to include warming up, running, exercises to strengthen upper body, lower body and core muscles. Two period each to be devoted to route march by day and night respectively and one period will be earmarked for trekking expedition as part of Adventure Activity.
- (j) **Unit 10. Games Training (G)(Contact Hrs. 09).** Games Training will be carried out on each day of the camp, except on last day, in evening hours. Training has to be progressive in degree of difficulty to improve individual skills, coordination, team work and desire to excel. Training to ensure that each and every boy and girl cadets participate in at least one game activity everyday.
- (k) **Unit 11. Cultural Activity (C)(Contact Hrs. 09).** Cultural Activity will be carried out on each day of the camp, except on last day, in evening hours. Cadets have to divided in Nine Groups consisting of a mix of boy and girl cadets and preferably belonging to the same geographical area. Each group has to present the unique culture, custom, tradition, folk lore, songs, drama, paintings and cuisine during one hour allotted. There will be a prize for the best group to encourage participation and to develop pride in their unique culture. This training activity should ensure that each and every boy and girl cadet participate in at least one game activity every-day.
- (l) **Unit 12. Spare (S)(Contact Hrs. 02).** Two periods in each camp will be earmarked as spare to cover disruptions in training activity due to weather or other administrative reasons.

COURSE MODULE : NATIONAL CADET CORPS CAMP – II

Course Title: National Cadet Corps Camp II			
Course Code	BNCCCAMP05GE05	Credits	2(Thr)+ 3(Pr)=05
L /T + P	30+90	Course Duration	10 Days (24 hours each)
Semester	V (Odd)	Contact Hours	30(Thr)+90(Pr)=120Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, individual and group tasks, team work, field-based assignments, Physical Training, endurance building and skill development practices		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (II) Theory

43. **Course Objectives.** Cadets will be able to: -

- (a) Acquire the concept self-awareness, emotional intelligence, critical and creative thinking, decision making and problem solving.
- (b) Learn about various indicators of good leadership and get an insight on principle of leadership and motivation.
- (c) Develop awareness about the various types of natural disasters and disaster management organization in our country.
- (d) Familiarize with natural resources, changing environment and understand the importance of conservation and waste management.
- (e) Value the importance of Physical and Mental health and understand how to deal with wounds of various types.
- (f) Acquire awareness about organization and role of an Infantry Battalion in the Armed Forces.



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44. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Develop a sense of responsibility, smartness in appearance and improve self-confidence, inculcate importance of empathizing with others, improve their deep-thinking ability and apply ideas and be able to face problems in a constructive manner with solutions.
- (b) Imbibe good leadership traits and apply them in practical life and appreciate the visible outcome of leadership and motivation.
- (c) Appreciate role of the org during emergency and become useful members of disaster response team, if need arises.
- (d) Learn about the various natural resources, their utilization and practice method of conservation of these resources in daily life.
- (e) Appreciate value of physical and mental health in daily life and spread awareness about treatment and care of wounds in their society.
- (f) Be motivated to join the armed forces.

45. **NCC Camp-II : Course Content Part (I) Theory.**

- (a) **Unit 1. Personality Development (PD) (Contact Hrs. 04).** Self-Awareness, Emotional intelligence, Critical and Creative Thinking, Decision-Making and Problem Solving.
- (b) **Unit 2. Leadership (LDR) (Contact Hrs. 02).** Indicators of Good Leadership, Leadership and Motivation.
- (c) **Unit 3. Disaster Management (DM) (Contact Hrs. 02).** Disaster Management Organization NDMA and NDRF, Types of Disasters.
- (d) **Unit 4. Environmental Awareness and Conservation (EAC) (Contact Hrs. 02).** Natural Resources, Conservation and Management, Water Conservation, Waste Management, Energy Conservation.
- (e) **Unit 5. Health and Hygiene (H&H) (Contact Hrs. 02).** Physical and Mental Health, Treatment and Care of Wounds.
- (f) **Unit 6. Infantry Weapons (IW) (Contact Hrs. 01).** Organization of Infantry Battalion.
- (g) **Unit 7. Weapon Training (WT) (Contact Hrs. 02).** Characteristics of Point 22 Rifle and its Ammunition, Range Procedure and Safety Precautions.
- (h) **Unit 8. Military History (MH) (Contact Hrs. 04).** Guest lectures by War Veterans/decorated soldiers/veterans.
- (i) **Unit 9. Communication (COM) (Contact Hrs. 04).** Latest trends in communication.

NCC Camp-II : Course Content Part (II) Practical

46. **Course Objectives.** Cadets will be able to: -

- (a) Inculcate spirit of discipline and follow command as a group for a common goal.
- (b) Fire a weapon with adequate safety precautions necessary for safe firing.
- (c) Understand the lay of the ground and use it skillfully towards own objective.
- (d) Understand and use the map, satellite imagery and GPS effectively.
- (e) Identify and be well versed with the primary weapon systems used in the Armed Forces.
- (f) Lead a life of selflessness and provide service towards society development and nation building.
- (g) Understand the importance of changing mindset, team work, social skills etiquettes and manners, interview skills and importance of effective communication in daily life.
- (h) Learn the importance of physical fitness and nuances of physical training.
- (i) Inculcate esprit-de-corps through team games.
- (j) Have knowledge about cultural diversity of India and learn ways and means to adopt them.

47. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Practice problem solving, critical thinking in real life situations.
- (b) Practice leadership of small teams and groups under challenging environment.
- (c) Develop a positive attitude, have manners and etiquettes in social life, develop a sense of cooperation for group or team work, participate in an interview with confidence and inculcate verbal and non-verbal communication skills.
- (d) Develop adequate physical and mental endurance capabilities.
- (e) Fire a weapon effectively with fair degree of marksmanship.
- (f) Undertake point to point navigation and take part in endurance marches by day and night.
- (g) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (h) Be motivated to join the Armed Forces.
- (i) Play team games and be able to communicate and coordinate effectively in group events or situations.



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- (j) Perform foot drill, arms drill, ceremonial drill and will be able to take part in ceremonial parade and events.
- (k) Respect the diversity of indian culture and develop pride by showcasing their own culture to others.

48. NCC Camp-II : Course Content Part (II) Practical

- (a) **Unit 1. Drill (Drill) (Contact Hrs. 16).** Tejchaal se Mudna, Tejchaal se Salute karna, Tej kadamtaal aur Tham, Tej Kadamtaal se kadam badhana, Teenon Teen se ek file Banana aur ek file se Teenon Teen Banana, Rifle Ke Saath Saavdhan, Aaram se, Rifle ke saath Parade par aur saaj, Rifle Ke saath visarjanaur line tod, Bhumi Sashttra aur Uthao Sashttra, Bagal Sashttra aur Baaju Shastra, Salami Sashttra, Squad Drill, Guard Mounting, Guard of Honour, Platoon / Company Drill, Word of Command and Instructional Practice.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 14).** Stripping, Assembling, Cleaning of Point 22 rifle, Sight Setting and Sight Picture of Point 22 Rifle, Loading, Cocking and Unloading, Lying Position, Holding and Aiming of Point 22 rifle, Trigger Control and Firing of Shot, Theory of Group, Short-Range Aiming and Firing, Musketry Training, Firing Practice I to VII.
- (c) **Unit 3. Field Craft & Battle Craft (FC/BC) (Contact Hrs. 06).** Observation, Camouflage and Concealment, Field Signals, Section formations, Fire Control Orders, Fire and Movement, Knots and Lashings.
- (d) **Unit 4. Map Reading (MR) (Contact Hrs. 12).** Introduction to Map and Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Types of Bearing and use of Service Protector, Prismatic Compass and its use, setting of a map, Finding North and own Position, Map to Ground and Ground to map, Point to Point march, Endurance March – I (10 KM), Endurance March -II (20 KM).
- (e) **Unit 5. Infantry Weapons (IW) (Contact Hrs. 01).** Characteristics of Infantry Company support weapons and 5.56 MM INSAS Rifle.
- (f) **Unit 6. Social Service and Community Development (SSCD)(Contact Hrs. 04).** Contribution of Youth Towards Social Welfare: Cadets will participate in various activities throughout the camp e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc., Social Evils: Female Feticide, Dowry, Child Abuse, Trafficking and Corruption, Drug Abuse and Drug Trafficking, Protection of Children and POCSO Act 2012.
- (g) **Unit 7. Personality Development (PD)(Contact Hrs. 06).** Change Your Mindset, Team Work and Team Building, Social Skills, Etiquettes and Manners, Interview Skills, Communication Skills–I, Communication Skills -II

(h) **Unit 8. Obstacle Training (OT)(Contact Hrs. 02).** OT Practice – I: Untimed, Cadets will be familiarized with all the obstacles in the Obstacle Course and briefed about the correct method to do them, OT Practice -II: Timed practice for all the cadets and record to be maintained.

(i) **Unit 9. Physical Training (PT) (Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in morning hours. Training has to be progressive in degree of difficulty to improve individual stamina and endurance. Training to include warming up, running, exercises to strengthen upper body, lower body and core muscles. Two period each to be devoted to route march by day and night respectively and one period will be earmarked for trekking expedition as part of Adventure Activity.

(j) **Unit 10. Games Training (G)(Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in evening hours. Training has to be progressive in degree of difficulty to improve individual skills, coordination, team work and desire to excel. Training to ensure that each and every boy and girl cadets participate in at least one game activity everyday

(k) **Unit 11. Cultural Activity (C) (Contact Hrs. 09).** Cultural Activity will be carried out on each day of the camp, except on last day, in evening hours. Cadets have to divided in Nine Groups consisting of a mix of boy and girl cadets and preferably belonging to the same geographical area. Each group has to present the unique culture, custom, tradition, folk lore, songs, drama, paintings and cuisine during one hour allotted. There will be a prize for the best group to encourage participation and to develop pride in their unique culture. This training activity should ensure that each and every boy and girl cadets participate in at least one game activity every day (Contact Hrs. 09)

(l) **Unit 12. Spare (S) (Contact Hrs. 02).** Two periods in each camp will be earmarked as spare to cover disruptions in training activity due to weather or other administrative reasons.



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SECTION II : RULES AND REGULATIONS

**GOVERNING NCC CREDIT COURSE UNDER CHOICE BASED CREDIT SYSTEM AS
GENERIC ELECTIVE FOR SENIOR DIVISION/WING**

RULE 1 :Definitions of Key Terms

General Definitions

‘Choice Based Credit System’ (CBCS).The CBCS provides choice for the student to select courses from the prescribed courses (Elective or Soft – Skill courses). It provides a ‘Cafeteria’ approach in which the students can take courses of their choice, learn at their ownpace, study additional courses and acquire more than the minimum required credits, and adopt an inter-disciplinary approach.

‘Academic Year’.Two consecutive (one odd + one even) semesters shall constitute oneacademic year.

‘Credit Course’.Course, usually referred to as paper having specific title and code number, is a component of a programme. It consists of a list of topics/concepts/theories/principles/activities/tasks etc. which a student has to learn during the programme of study. Each course has some credits according to the nature and load ofcontent. Each course should define the learning objectives/learning outcomes. A course may be designed to be delivered through lectures/tutorials/laboratory work/field work/out reach activities/project work / vocational training / physical training /viva / seminars /term papers / assignments / presentations / self-study work etc., or a combination of some of these.

‘Course Instructor/Teacher’.The course instructor generally will be a teaching faculty who has taken up the responsibility of teaching it and evaluating the performance of the students in that course. NCC course will be imparted by the ANO (Associate NCC Officer) and PI (Permanent Instructor) / Girl Cadet Instructor (GCI) staff together according to their area of specialization. Certain specific topics and training activity is imparted by Military Officers and Whole Time Lady (WTLO).

‘Credit’.A unit by which the course work is measured. It determines the number of hours of instructions required per week. One credit is equivalent to one hour of teaching (lecture or tutorial) or two hours of practical work per week. Thus, in each semester’s NCC course, credits are assigned on the basis of the number of lecture/tutorial/field work/physical training/excursions and other forms of learning required for completing the contents in a 15-18 week schedule. 2 hours of laboratory work/field work is generally considered equivalent to 1 hour of lecture.

- i. 1 credit = 1 hour of instruction per week (1 credit course = 15 contact hours of instruction per semester)

- ii. 4 credit = 4 hour of instruction per week (4 credit course = 60 contact hours of instruction per semester)
- iii. 1 credit = 2 hour of practical per week (1 credit course = 30 contact hours of instruction per semester)
- iv. 4 credit = 8 hour of practical per week (4 credit course = 120 contact hours of instruction per semester)

Number(s) of credit(s) assigned to a particular course are mentioned in the detailed syllabus of the courses.

‘Credit Point’. It is the product of the grade point and the number of credits for a course.

‘Letter Grade’. It is an index of the performance of students in a said course. Grades are denoted by letters O, A+, A, B+, B, C, P, and F. A letter grade is assigned to a student on the basis of evaluation of her/his performance in a course on a ten-point scale.

‘Grade Point’. It is a numerical weight allotted to each letter grade on a 10 -point scale.

Letter Grade	Grade Point
O	9-10
A+	8-9
A	7-8
B+	6-7
B	5-6
C	4-5
P	4
F	0
Ab	0

Note : University may use the above said criteria for providing the grades to the students or may adopt the same criteria which they are practicing for providing the letter grade and grade point for other subjects.

‘Programme’. An educational programme leading to the award of degree, Diploma or Certificate course. NCC course shall be offered only at under graduate level programmes for any stream or type of programme for example – Nonprofessional courses BA, B.Sc. B. Com etc. professional courses – B.A., LLB, B.A./B.Sc., B.Ed., BCA, BBA, B. Tech, MBBS etc.

‘Credit – Based Semester System (CBSS)’. Under the CBSS, the requirement of awarding a degree or diploma or certificate is prescribed in terms of number of credits to be completed by the students.



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'Semester' Each semester shall consist of 15 to 16 weeks of academic work equivalent to 90 actual teaching days. The odd semester may be scheduled from July to December and even semester from January to June. The Credit-based semester system provides flexibility in designing curriculum and assessing credits based on the course content and hrs of teaching.

'Semester Grade-Point Average (SGPA)' Semester Grade Point Average or SGPA, is an average grade point earned by the student at the end of an academic session i.e. semester at college. The formula for calculation of SGPA is the sum of all the credit points awarded for the subjects divided by total credits allotted to that semester. It shall be expressed up to two decimal places.

'Cumulative Grade Point (CGPA)' It is a measure of overall cumulative performance of a student over all semesters. The CGPA is the ratio of total credit points secured by a student in various courses in all the semesters. It shall be expressed up to two decimal places.

'Transcript/ Grade card or certificate' Based on the grades earned, a grade certificate shall be issued to all the registered students after every semester. The grade certificate will display the course details (code, title, number of credits, grade and / or marks secured) along with SGPA of semester. Overall Grade Certificate will be issued on completion of the course showing semester wise SGPA & CGPA.

'The University/ College/ Institution' The University/ College/ Institution in present document means the any recognized central/ state/ Deemed university or institution meant for higher education.

'NCC Course' In the present document 'NCC Course' means the course designed for imparting NCC curriculum in educational institutions as elaborated in this document under Choice Based Credit System as a General Elective Course for Senior Division/ Senior Wing.

Definitions Specific to NCC

'Institutional Training' Implies training conducted for NCC cadets as per Training Manuals and Cadet Hand Book issued by DG NCC, Ministry of Defence.

'Common Subjects' Implies those subjects specifically taught in NCC curriculum which are common to Army, Navy and Air Force and general training that can be imparted by Associate NCC Officers or Military staff or a suitably qualified person.

'Specialised Subjects' Implies subjects specifically taught in NCC curriculum by military instructors comprising specialised topics for Army, Navy and Air Force Cadets respectively.

'NCC Camps and Centralised Training Events' Collective training events conducted usually for 10 days with large number of cadets living under field conditions in selected places away from home. The training camp comprises of focused physical and mental training routines of different types as per syllabus and curriculum. Some training like route marches may happen overnight. Camps include, adventure camps, national integration camps,

Republic Day Parade Training Camps, ThalSainik, VayuSainik and NauSainik camps and other outdoor training activities as described in DG NCC Training Manuals.

NCC 'B' and 'C' Certificate Examinations. These are defined in Special National Cadet Corps Order 2020 issued by DG NCC, Ministry of Defence.

'Training Faculty'. Persons suitably trained & responsible for imparting training of different types and nature to students.

'Military Officers'. They are regular commissioned officers of Indian Armed Forces who serve in the NCC and render command, administrative and instructional functions for NCC.

'Whole Time Lady Officers (WTLO)'. They are women officers commissioned directly into the NCC.

'Associate NCC Officer (ANO)'. ANO will be a university/ college/ school faculty who are qualified in the PRCN (Pre-commission Course of NCC) conducted by DGNCC and are commissioned as Associate Officers in NCC as defined in NCC Act 1948 and NCC Rules. They have the eligibility to impart certain component of NCC Course and undertake training of cadets.

'Permanent Instructor (PI)'. PI Staff are Junior Commissioned Officers (JCO) and Non-Commissioned Officers (NCO) on deputation from Armed Forces to NCC as governed by NCC Act 1948. Retired PI Staff may be hired by a college as a substitute for ANO with prior concurrence of DGNCC.

'Girl Cadet Instructors (GCI)'. GCI are lady instructors' equivalent to PI Staff for specifically imparting instructions to women NCC cadets of Senior Wing.

'NCC Organizational Structure'. NCC is an adjunct of Indian Armed Forces that operates under the ambit of the Ministry of Defence through the Defence Secretary with Raksha Mantri as the political head.

'DGNCC'. Directorate General of NCC renders the command and administrative function of NCC. The executive head of NCC is Director General of NCC who is a Lt Gen rank officer from the Army.

'State NCC Directorate'. State NCC Directorates are directorates subordinate to DG NCC and render command and administrative control to NCC at State level and is headed by an Additional or Deputy Director General

'NCC Group HQ'. NCC Group HQs are subordinate to State Directorates and render command and administrative control to NCC at district or cluster of districts in a state and is headed by a Group Commander.

'NCC Units'. NCC Units are subordinate to Group HQs at the lowest rung of the command and administrative control exercised by military officers and is headed by a Commanding Officer or Officer Commanding. The NCC Units directly engage



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with educational institutions and ANOs and are primarily responsible for training of NCC in institutions under their jurisdiction.

'NCC Division/ Wing'. NCC Division/Wing are minor units of senior division/wing of NCC comprising of 160 senior cadets allotted to educational institutions. It can be further subdivided into NCC platoons of 53 to 54 cadets.

'NCC Troop'. NCC Troop are minor units of junior division/wing of NCC comprising of 100 junior cadets allotted to educational institutions. It can be further subdivided into NCC half troops comprising of 50 junior cadets.

RULE 2 : Admission and Other Provisions

The NCC Course under the CBCS as 'General Elective' shall be of three years (Six Semester) duration which may be completed in maximum duration of four year (8 semesters).

Students may complete NCC course minimum in Six semesters and maximum in eight semesters. Cadets may complete their 'B' Certificate in four semesters minimum and maximum six semesters. Cadets already having 'B' certificate may complete their 'C' certificate in minimum two semesters and maximum four semesters, and they may join NCC course 5 in first semester of college.

The intake to the course shall be decided according to the seats allotted to University/ college/ institution by DG NCC according to the availability of required infrastructure, faculty and resources.

The admission to the NCC Course under the CBCS as a 'General Elective' shall be governed by the provisions as laid down by the NCC Act 1948/ SNCCO 2020/ contemporary SNCCO and Academic council of parallel body of university. These rules and regulations may be modified from time to time (if needed) by the Academic body of the university in consultation with DG NCC or Act/ Ordinances prepared by DG NCC.

Students will be enrolled as NCC cadet as per existing Acts & Rules.

At the time of reporting for admission, the candidates are required to present medical & physical fitness documents as well as the admission proof of the university and submit these self-attested copies of aforesaid documents.

The admission of any candidate is liable to be cancelled without giving any further notice forthwith or at any time during the period of the course, if it is detected that the candidate has/had produced fake/forged certificate (s)/ document(s), indulged in any act of misconduct/indiscipline and has/had concealed any other relevant information at the time of admission.

The admission of the candidate to the course shall be subject to such ordinance, rules and regulations as may be framed from time to time by the university in consultation with DG NCC and NCC act 1948.

DG NCC shall have jurisdiction in case of any dispute relating to the provisional admission in the course.

RULE3 : For Eligibility, Medium of Instrs & Categories

Eligibility Conditions. Be governed by provisions of NCC Act and Rules and directions from DG NCC from time to time. These are readily available on DG NCC website www.nccindia.nic.in.

Standards for physical Fitness criteria for Male and Female Cadets/students shall be governed by provisions of NCC Act and Rules and policy documents released by DG NCC from time to time.

RULE4 : Medium of Instruction. English or Hindi. However, ANOs and training instructors are free to use vernacular language for helping students who are not fluent in Hindi or English.

RULE 5 : Course and Students. NCC course is unique, due to the nature of its military training content and component hence it is normally offered to students enrolled as NCC cadets only. This NCC Course is primarily designed for students enrolled as NCC cadets under provisions of NCC Act 1948. Institution allotted NCC will have the obligation to offer this course to all students from their institute enrolled as cadets as per vacancy allotted to the institution by DG NCC as also to those cadets enrolled under Open Quota seats.

RULES 6 :NCC Course for 'Cadet' Category

6.1. NCC Course for 'Cadet'

- (a) NCC course for Cadets comprises of total 24 credits (08 for theory, 06 for practical and 10 for camp component) over 6 semesters courses i.e., NCC course I to NCC course VI and NCC Camp I & NCC Camp II.
- (b) Cadets will not only earn the academic credits but also be given 'B', and 'C' Certificates after passing the exam conducted by DG NCC.
- (c) Students would be free to join NCC Course I or subsequent Courses in any semester, not necessarily Semester I or the designated Semester.
- (d) A student can opt for only one of the six Courses per semester and that too sequentially implying NCC Course II cannot be joined before completing NCC Course I and so on.
- (e) Under this category a fresh student/cadet will compulsorily have to opt for all six NCC Courses in minimum six Semesters. However, 'B' certificate holder may directly join NCC Course Number 5 in any semester. He will have to complete NCC Course Number 5 and NCC Course Number 6 for obtaining 'C' certificate and he will be awarded credit points only for NCC Course Number 5 and NCC Course Number 6.



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NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	1	1		2	
Semester - II	1	1		2	
Semester – III	1	1	5	7	Credits of 1 st Camp merged with 3 rd Sem
Semester – IV	2	1		3	
Semester – V	1	1	5	7	Credits of 2 nd Camp merged with 5 th Sem
Semester - VI	2	1		3	
Total	08	6	10	24	Twenty-Four Credits

RULE 7 :Mobility& Credit Bank

The mobility shall be permissible from the regular mode programme to the regular mode programme of learning only and cannot be replaced by open/distance/online programme.

It shall be the responsibility of the student to assess the feasibility and practicality of vertical mobility (across the Universities), as it doesn't entitle a student to be exempted or relaxed from any of the requisites (sessional, attendance, assignments, End-semester examinations and programme duration etc.) for completing the course.

After completing one semester/ one year cadet/student may pursue NCC course from any other institution/ University/ College having NCC and carry credits in credit bank as per NEP 2020. The NCC students/ Cadets of some other university shall in any case be admitted only at the beginning of the session to the fulfilment of the other requirements of the NCC Course (attendance, Formative assessment, Field-work, practical etc).

A student of NCC course availing inter-university mobility shall continue to be a bonafide student of the university where he/she initially got admission and as per the university/ Institutional rules for the inter-university mobility.

In case of inter-university mobility of NCC cadet for NCC Course is also the subject to availability of NCC for the cadets in that particular university/ institution and it shall be interpreted as inter-battalion migration (means another regimental no. shall be allotted to the cadet).

RULE 8 :Examination & Promotion

The examination of all the NCC courses shall be internal in nature and generally consisting of continuous internal assessment and End of semester Examination. For the preparation of final grade in a particular course, the continuous internal assessment (Formative in nature) and the End Semester Examination (Summative in nature) shall have the weightage as decided for other courses by the university as per the University norms for e.g., 25% internal assessment and 75% End of term exams or 30% internal assessment and 70% End of term exams etc.

For assigning the Grades and credit points to NCC Course Universities/ Institutions are free to use the same criteria which are decided by their academic bodies for providing the grades and credit points to the other courses

RULE 9 :Continuous Internal Assessment

The Continuous Internal Assessment of the NCC Cadets' and NCC students' learning and performance shall be carried out by the ANOs and PI staff.

Continuous Internal Assessment will be 100% Practical that includes Drill Square test, Map Reading, Weapon Training, Field craft & Battle craft.

CO of nominated NCC Unit will be deemed as Head of the Department and shall be responsible for approving the schedule and pattern of the continuous internal examination.



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ANO of the nominated institute shall maintain all the records related to attendance, teaching and assessment in a systematic manner, including award of final grade.

In case a student fails to appear in any Continuous Internal Assessment, they will be given a chance to reappear in retest and in case he/she fails to obtain 'P' grade he/she will be made to repeat the exam by carrying it forward for semester retest.

RULE 10 :Re-appear in the End Semester Examination for Improvement of Grades

If a student wishes to improve her/his grade(s) in NCC course(s), she/he can re-appear in the End Semester Examination in the subsequent odd/even semester(s), whenever the examination of the particular course(s) is held, on payment of fees in addition to the prescribed semester fee within the maximum permissible duration for the programme of study of the student/cadet.

A student may improve her/his points/grade by reappearing in the End Semester Examination of a course as per the provisions of reappearing mentioned above. In such cases points obtained by the student in the Continuous Internal Assessment of the particular course shall be carried forward to the subsequent End Semester Examination of the course. However, in such case, the points/grades obtained on the basis of latest appeared End Semester Examination shall be considered for calculation of final CGPA of the programme.

The re-appear examination of a course for improvement of grade shall be based on the syllabi of the course in force at the time of initial registration to the course.

A student who has got the Migration/Transfer Certificate issued from the University shall not be allowed to re-appear in any examination for improvement of grade.

RULE 11 :Repeating Courses

A student having attendance shortage in any course may repeat the course by taking re-admission in that course in subsequent odd/even semester(s), whenever the course is being offered, within the maximum permissible duration of the programme.

If a student repeats a course, she/he has to fulfil all the desired requirements afresh including attendance, Continuous Internal Assessment and the End Semester Examination. In such case the course content shall be based on the syllabi of the course in force at the time of repetition of the course.

RULE 12 : Promotion Rules

A student shall be declared as 'promoted' to the next semester when she/he earns 'P' Grade or above in the last concluded semester examination, maintaining the spirit and pattern of semester system and covering the mandatory components, such as Continuous Internal Assessment and End-Semester Examination in the NCC Courses.

A student shall be 'Provisionally Promoted' to the next semester if she/he secures less than 'P' grade but he /she has to pass all the courses of NCC course within permissible duration.

A cadet shall be eligible to attend the 'B' Certificate exam if he/she passed all the first four semester NCC course and completed one ATC/CATC. Similarly, cadet will be eligible to attend 'C' certificate examination if he/she has 'B' certificate and he /she has passed V, VI semester NCC course and attended one CATC/ATC after fourth semester and after having obtained 'B' certificate.

If a cadet/student is repeating a course in an academic session, whatever may be the reason, it shall not be counted in the total number of seats and shall not affect the fresh intake of cadets / student in that academic session.

RULE 13 :Computation of SGPA & CGPA

13. Computation of SGPA and CGPA . University may use their own criteria for giving the SGPA & CGPA which is prepared by the authorised academic body for the other courses.



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
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**Course Title: National Service Scheme
Semester III
Vocational skill development**

To enhance the employment potential and to set up small business enterprises skills of volunteers, a list of 12 to 15 vocational skills will be drawn up based on the local conditions and opportunities. Each volunteer will have the option to select two skill-areas out of this list

Issues related environment

Environmental conservation, enrichment and sustainability, climatic change, natural resource management (rain water harvesting, energy conservation, forestation, waste land development and soil conservations) and waste management

Disaster management

Introduction and classification of disaster, rehabilitation and management after disaster; role of NSS volunteers in disaster management.

Entrepreneurship development

Definition, meaning and quality of entrepreneur; steps in opening of an enterprise and role of financial and support service institution.

Formulation of production oriented project

Planning, implementation, management and impact assessment of project

Documentation and data reporting

Collection and analysis of data, documentation and dissemination of project reports

SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL
SUBJECT CODE – MC 3 (C)

CATEGORY – NNP

SUBJECT NAME – PHYSICAL EDUCATION

SEMESTER – III

ENGLISH I

UNIT-1

1. Vocabulary
2. I(1) Synonyms
- I.2 Antonyms

Common abbreviations in use

One words substitution

Words commonly Misspel

Idiosnatic Comparisions or similes

Word Formation by prefix and sutfix

UNIT-2

3. Common Errors & Transformations

Common Errors in sentences especially regarding number, gender, pronouns, prepositions, articles degrees etc.

Punctuations

Kinds of sentences- Assertive, Interrogative, Exclamatory, Imprative, Optative.

Transformation of sentences.

Tenses



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UNIT-III

Correspondence

1. Personal letter- To Mother, Father, Brother, Sister, Friends.
2. Application- For leave, For scholarship, for apology, for admission a ground or place to conduct Tournament or programming etc.

Invitations- Format and Informal, Invitation for chief guest, Invitation card,

Notice writing

Office correspondence.

UNIT - IV

General Knowledge

Complete Knowledge about National Flag.

Complete Knowledge about National Song.

Complete Knowledge about National Anthem

Name of states and their capitals of Hindustan.

UNIT - V

Text

The story of Maharani laxmibai

The story of Freedom fight since 1857 to 1947,

The Bharat Chodo Andolan.

The story of Prathvi Raj Chouhan

Namak Satyagrah

Story of Barkatullah Bhopali

Note: The questions related to unit V should be test Based.

METHODS IN PHYSICAL EDUCATION

UNIT-I

1. Introduction

Definitions and Functions of Education.

Types of Education-Formal, Informal and Non-formal education.

Process of Education.

Principles of teaching.

Importance of Devices of Teaching.

Various teaching devices: Exposition; and Explanation, Narration, Description, Assignment and Home work.

UNIT-II

Teaching Techniques in Physical education

Lecture Method.

Command Method.

Demonstration Method.

Imitation Method.

Project Method.

Discussion Method.

Group Directed Practice Method.

Teaching Procedures in Physical Education

Whole Method.

Whole-Part-Whole Method

Part-Whole Method

Stage-Whole Method.



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UNIT-III

Presentation Techniques-Meaning Importance and step of presentation.

Preparation-Personal and Technical Preparation.

Command-Its types, command used in different situations.

Formation-Its types, and situations for using different formations.

Class Management- Meaning and Principles.

Tournaments- Meaning and Types of Tournaments

Knockout or elimination Tournaments.

League or Round-Robin tournaments.

League cum knockout Tournament.

Challenge Tournament.

UNIT-IV

Lesson Planning

Meaning and objectives of lesson plan.

Values of lessons plan

Types of lesson plan

Principles of Lesson Plan.

Lessons and its parts along with their significance.

General and Specific Lesson Plan.

Coaching Lesson (A Brief Introduction).

Improvisation- Need, Ways & Means.

UNIT-V

Track Marking

Marking of 200 Meters track

Marking of 400 Meters track

Stagger marking for 200 meters and 400 meters run

Curve Marking- 800 meters run

Sector Marking for Throws- Shot-Put, Javelin, Discuss.

Marking for Jumps:- High Jump and Long Jump.

Marking of play Ground

Football

Hockey

Volleyball

Basketball

Badminton

Kabaddi

5.47 Kho-kho



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CRICKET

UNIT-I

1. Introduction of the game and historical development with special reference of India.
2. Important tournaments held at National and International levels and distinguished personalities related to the game.
3. Fundamental Skills.

Batting

Forward defensive stroke

Backward defensive stroke

Bowling

Simple bowling techniques.

Fielding

Defensive fielding - orthodox, Unorthodox.

3.2.2 Offensive fielding

Catching

High Catching

Slip Catching

Stopping and throwing techniques.

Wicket Keeping technique.

4. Advanced Skill

Batting

Forward defensive stroke

Backward defensive stroke

Forward off drive.

Forward on drive

Bowling

Simple bowling techniques.

Difference between pace.

Bowling and spin bowling : Off and leg spin bowling.

Fielding Different techniques of fielding and its importance.

Catching : Different types of catching, its technique and importance.

Stopping and throwing : Different techniques and its importance's.

4.7 Wicket keeping : Different techniques and their implications.

5. Rules and their interpretations and duties of Officials.

JUDO

- 1 Introduction of the game and historical development with special reference to India.
2. Important tournaments held at National and International levels and distinguished personalities related to the game.
3. Fundamental Skills.

Rei (salutation)

Ritsurei (Salutation in standing position).

Zarai (Salutation in the sitting position.)

How to wear Judo Costume.

Kumi Kata (Methods of holding judo costume).

Shisei (Posture in Judo).

Kuzushi (Act of disturbing the opponent posture).

Tsukuri and kake (Preparatory action for attack)

Ukemi (Break fall).

Urhiro Ukemi- (Rear break fall).

Yoko Ukemi (Side break fall).

Mae Ukemi (Front break fall).

Mae mawari Ukemi (Front rolling Break fall).

Shin Tai (Advance or Retreat foot Movement).

Suri-ashi (Gliding foot).

Tsugi-ashi (Following foot steps).



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Ayumi-ashi (Walking steps).

Tai Sabaki (Management of the body).

Nage-waze (Throwing Techniques).

Hiza Guruma (Knee wheel).

Sesae Tsurikomi- ashi (Drawing ankle throw).

De-ashi hari (Advance foot sweep).

O Goshi (Major Loin).

Seoi. nage (Shoulder throw)- Ippon scionage and Morote
Scionag.

Katama-waze (Grappling Techniques).

Kesa-gatame (Scaff hold).

Kata-gatma (Shoulder hold).

Kami-shiho gatama (Locking of upper four
quarters).

Method of escaping from each hold.

Lead-up games-

Break fall relay (maximum number of falls from standing
position in one minute duration).

Mae- mawriukemi relay (maximum number of falls in one
minute duration).

Maximum number of shoulder throw in one minute.

Maximum number of obstacles jumped while doing
maemawri-ukemi.

Rules and their interpretations and duties of officials.



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EE-41 - ELECTRICAL MACHINES – I

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 The Principle of Converting Electrical energy to Mechanical energy and Vice-versa through Electromagnetic field.
- 2 Understanding the effect of Armature reaction. Commutation and methods of improving commutation in D.C. Machines.
- 3 Understanding the Characteristics and Applications of DC motors.
- 4 The working principles and applications of 1- ϕ phase and 3- ϕ Transformers.

Unit-1

Transformer-I

Working principle, emf. equation, construction, phasor diagrams, equivalent circuit, voltage regulation, losses, separation of hysteresis and eddy current losses, efficiency, tests: open circuit and short circuit, load, Sumpner's test, Condition for maximum efficiency and regulation, Power and distribution transformer, all day efficiency, Excitation phenomenon, Autotransformer: working, advantages, its equivalent circuit and phasor diagram..

(12 HRS)

Unit -2

Transformer-II

Three phase transformer: its construction, groups and connections, their working and applications; Scott connection; Parallel operation of Transformers: application, advantages, requirement and load sharing; Tap changers, cooling, conservator and breather. Pulse and high frequency transformers.

(12 HRS)

Unit -3

Three phase Induction Motor-I

Working principle, construction, comparison of slip ring and squirrel cage motors, steady state analysis, phasor diagram and equivalent circuit, power flow diagram, torque-speed and power-speed characteristics, Losses and efficiency, No load and block rotor test, circle diagram.

(12 HRS)

Unit -4

Three phase Induction Motor-II

Starting of squirrel cage and slip ring motors, power factor control, Cogging & Crawling, Double cage & Deep bar Induction Motor, impact of unbalanced supply and harmonics on performance, speed control, braking, Induction Generator. Applications.

(12 HRS)



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IVTH SEMESTER

Unit -5

CATEGORY:-DC

Single Phase Motors:

Single Phase Induction motor; double revolving field theory, equivalent circuit and its determination, performance calculation, starting methods and types of single phase Induction



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motors: their working principle and applications, comparison with three phases Induction Motor. Single phase A.C. series motor, Servo motors, Linear Induction Motor.
(12 HRS)

COURSE OUTCOME:

After completion of this course, students will be able to

- 1 Describe the Principles of Electro Mechanical Energy Conversion.
- 2 Explain the effect of Armature reaction , Commutation and methods of improving commutation in D.C. Machines. & Solve Problems
- 3 Enumerate the Characteristics, speed control and Applications of D.C.Motors & Solve problems
- 4 I identify the Applications of 1-Phase and 3-Phase Transformers

Reference Books:

1. M. G. Say, 'Alternating Current Machines', (5th Ed.) ELBS, 1986.
2. V.Del Toro, "Electrical Machines & Power Systems", 1985, Prentice-Hall, Inc., Englewood Cliffs.
3. V.Del Toro, "Electromechanical Devices for Energy Conversion & Control Systems", PHI Pvt. Ltd., 1975. Text Books:
4. Electrical Machines by Nagrath and Kothari (TMH).
5. A.C. Machines by Langsdorf (McGraw-Hill)
6. Electrical Machines by Dr.P.S.Bimbhra (Khanna).
7. Electrical Machines by Ashfaq Hussain. (Dhanpat Rai).

List of Experiments (expandable)

1. Experiments can cover any of the above topics, following is a suggestive list:
2. Perform turn ratio and polarity test on 1-phase transformer
3. Perform load test on a 1-phase transformer and plot its load characteristic
4. Perform OC and SC tests on a 1-phase transformer and determine its equivalent circuit. Also find its efficiency and regulation at different load and power factor.
5. Perform OC and SC tests on a 3-phase transformer and determine its equivalent circuit. Also find its efficiency and regulation at different load and power factor.
6. Perform Sumpner's test on two 1-phase transformer and determine its efficiency at various load.
7. Perform No-load and block rotor test on a 3- phase IM and determine its equivalent circuit.
8. Perform load test on a 3- phase IM and plot its performance characteristics.
9. Study various types of starters used for 3- IMs.
10. Perform No-load and block rotor test on a 1- phase IM and determine its equivalent circuit.



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BHOPAL

IVTH SEMESTER

CATEGORY:-DC

EE-42 - POWER SYSTEM –I

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Conventional Energy Sources
- 2 Economics of Power Generation
- 3 Different Types of Insulators, Corona
- 4 Dunder Ground Cables & parameters of transmission lines

Unit -1

General consideration on various sources of energy, energy conversion employing steam, energy conversion using water gas turbine

- a) MHD generation
- b) Solar generation
- c) Wind power station
- d) Geothermal power generation.

(12 HRS)

Unit -2

Thermal, nuclear and gas power station:

Block diagram of thermal power station, selection of site. Different types of auxiliaries used in thermal power station. Nuclear Power Station: Different types of reactors and fuels, safety methods, waste disposal.

(12 HRS)

Unit- 3

Gas Power Station:

Block diagram, gas cycles, combined cycle power plants. Comparison between these power stations

(12 HRS)

Hydro Power Station:

Choice of site, block diagram including surge tank and penstock, Hydrographs, flow duration curve. Types of turbines, base load and peak load power station.

(12 HRS)

Unit -4

Economic aspects of power plant operations:

Definitions load factor, demand factor and Diversity factor. Calculation of cost of generation, fixed charges, interest and depreciations, Methods of Depreciation. Tariffs: Different types of tariffs, power factor improvement.

(12 HRS)



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Unit -5

Economic Scheduling of Power Stations:

Economic operation of power system, criteria of loading of power plants with and without transmission loss, load dispatching in power system, co-generation and coordination of power plants.

(12 HRS)

COURSE OUTCOME:

- 1 Describe the operation of conventional generating stations
- 2 Determine Different Types of Tariff's in power system
- 3 Design Distribution of voltage along the string insulators & Solve Problems
- 4 Discuss underground cables & circuit parameters of transmission lines & Solve Problems

Reference:

1. G.R.Nagpal,"Power Plant Engineering", Khanna Publisher
2. S.N. Singh Electric Power Generation. PHI.
3. M.V.Deshpandey,"Modern Design of Power Station"

List of Experiment

1. To study the Thermal Power Station.
2. To study the Hydro Power Station.
3. To study the Nuclear Power Station.
4. To study & draw Towers used in Transmission lines.
5. To study & draw the different types of insulator.
6. To study & design Electrical Power Transmission line.
7. Determination of Transmission Parameters of a transmission line.



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
BHOPAL**



IVTH SEMESTER

CATEGORY:-DC

EE-43 - DIGITAL ELECTRONICS & LOGIC DESIGN

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Switching algebra and various minimization techniques of switching functions.
- 2 Various combinational circuits and their applications.
- 3 Types of flip flops and their use in the design of sequential circuits.
- 4 Different logic family circuits and their performance.

Unit- 1

(A) Number System: Various number systems-decimal, Binary, Hex and Octal with mutual conversion, binary arithmetic in computers, addition, subtraction, multiplication and division.

(B) Binary Codes: Weighted, non-weighted codes, error detecting and correcting codes, alphanumeric codes, ASCII codes.

(12 HRS)

Unit- 2

Boolean Algebra & Logic Hardware

(A) Boolean Algebra: AND, OR, NOT, NAND, NOR, EXOR, operations and gates, laws of Boolean algebra, reduction of Boolean expression, logic diagram, universal building blocks, negative logic (B) Logic hardware “ Diode as switch, Bipolar transistor as switch FET as switch, MOSFET (Depletion and Enhancement mode) IC Technology, MSI, LSI, VLSI, logic specification, logic families (DTL, TTL, ECL, MOS, CMOS).

(12 HRS)

Unit- 3

Combinational circuits and system- (A) Combinational logic: Minterms and maxterms, Truth table and Karnaugh mapping, reduction of Boolean expression with SOP, POS and mixed terms, incompletely specified functions multiple output minimization, variable mapping, minimization by tabular/ Quine Mc cluskey method. (B) Encoders, Decoders, Multiplexers, Demultiplexers, code converters, Binary address Digital comparator, parity checker/ generator, programming logic Array (PLA);

(12 HRS)

Unit-4

Sequential circuits- (A) State tables and diagrams, flip flop and its various types- JK, RS, T, D, pulse and edge triggered flip flops transition and excitation tables, timing diagrams. (B) Shift registers: Series and parallel data transfer, ripple counters, synchronous counters, Modulo N counter design, Up down counters, Ring.

(12 HRS)



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Unit -5

Memory & A/D Conversion- (A) Semiconductor ROM, Bipolar and MOS RAM, organization of RAM memory subsystem. Timing circuit, clock circuit and IC Timer. (B) Analog/ Digital conversion: Digital to analog conversion, dual slope integration successive approximation, parallel and parallel/ series conversion, converter specifications.

(12 HRS)

COURSE OUTCOME :

After completion of this course, the students' will be able to

- 1 Apply various minimization techniques to obtain minimal SOP/POS forms of switching functions.
- 2 Design different combinational circuits and implement logic functions.
- 3 Explain the operation of flip flops and their application in the design of sequential circuits like counters, shift registers, sequence detectors etc.
- 4 Analyze the operation of various logic family circuits and compare their performance characteristics

Reference Books:

1. An Introduction to Digital Computer Design by V. rajaraman and T. Radhakrishnan, 3rd Edn. PHI.
2. Digital Principles and Applications by A.P. Malvino and B.P. Leach, 4th Edn. McGraw Hill.
3. Digital computer Fundamentals by T.C. Bratee, 6th Edn. McGraw Hill.
4. Pulse, Digital and switching circuits-Millman
5. Digital Electronics by WH Gothmann, 2nd Edn. PHI.

List of Experiments (Expandable):

1. To Verification of all Gates.
2. Draw of HALF ADDER circuit using EX-OR and AND gate and verification of its operation.
3. Construction of HALF ADDER circuit using EX-OR, AND and OR gate and verification of its operation.
4. Construction of HALF SUBTRACTOR circuit using EX-OR, AND and NOT gate and verification of its operation.
5. Construction of FULL SUBTRACTOR circuit using EX-OR, NOT AND and OR gates and verification of its operation
6. Analysis of operation COUNTER CONVERTER of D/A Converter circuit.
7. To testing the working of a MONOLITHIC CONVERTER of A/D Converter circuit.



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8. Functional verification of a WEIGHTED RESISTOR D/A converter The purpose of this experiment is to analyze the operation of weighted resistor D/A converter according to the binary system.
9. Functional verification of D/A converter with ladder network.
10. To Verification of ENCODER circuit.
11. To Verification of DCODER circuit.



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
BHOPAL**



IVTH SEMESTER

CATEGORY:-DC

EE-44 - ELECTRICAL & ENGINEERING MATERIAL

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Critical reasoning to identify fundamental issues and establish directions for investigation
- 2 Creative processes to define specific plans for problem solution
- 3 Analytical thought to interpret results and place them within a broader context.
- 4 Application of materials solutions to enhance or radically improve existing and future technology

Unit- 1

Conducting Material: Classification and main properties, High resistivity alloy: Constantan, Nichrome, Electrochemical, properties of copper, Aluminum, steel, tungsten, Molybdenum, Platinum, Tantalum, Niobium, Mercury, Nickel, Titanium, Carbon, Lead, transition metals, thermocouple, materials, specific resistance, conductance, variation of resistance with temperature, superconductors.

(12 HRS)

Unit -2

Semiconductor Materials: General conception, variation of electrical conductivity, Elements having semiconductor properties, general application, hall effect, energy levels, conduction in semiconductors, Intrinsic conduction, impurity conduction, P and N type impurities, electrical change, Neutrality, Drift, Mobility current flow in semiconductors P-N junction formation by alloying, Forward (forward and reverse) of P-n junction, Reverse saturation current, Zener effect, Junction, capacitance, hall defects and hall coefficient.

(12 HRS)

Unit- 3

Magnetic Materials: Details of magnetic materials, relationship between B.H. and μ , soft and hard magnetic materials. Diamagnetic, Paramagnetic and Ferromagnetic materials, electrical sheet steel, cast iron. Permanent magnetic materials. Dynamic and static hysteresis loop. Hysteresis loss, eddy current loss, Magnetization, magnetic susceptibility, coercive force, core temperature, rectangular hysteresis loop, Magnet core square loop core materials, iron silicon, Iron alloys.

(12 HRS)

Unit -4

Insulating Materials: General electrical mechanical and chemical properties of insulating material, Electrical characteristics volume and surface resistivity complex permittivity loss, and dielectric loss, equivalent circuits of an imperfect dielectric polarization and polarizability classification of dielectric.

(12 HRS)



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Unit- 5

Mechanical Properties: Classification insulating materials on the basis of temperature rise. General properties of transformer oil, commonly used varnishes, solidifying insulating materials, resins, bituminous waxes, drying oils, Fibrous insulating materials, wood, paper and cardboard, insulating textiles, varnished adhesive tapes, inorganic fibrous material and other insulating materials, such as mica, ceramic, Bakelite, ebonite, glass, PVC, rubber, other plastic molded materials.

(12 HRS)

COURSE OUTCOME:

After completion of this course, students will be able to

- 1 An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
- 2 An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- 3 An ability to communicate effectively with a range of audiences
- 4 Aan ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

References:

1. TTTI Madras; Electrical Engineering Materials; TMH.
2. Electrical Engineering Material s & Devices; John Allison ;TMH
3. Materials for Electrical Engineering: B.M. Tareev
4. Anderson; Di-Electrics :
5. Kortisky; Electrical Engineering Materials:
6. Indulkar and S. Thruvengadem; Electrical Engineering Materials; S. Chand



BHOPAL

IVTH SEMESTER

CATEGORY:-DC

EE-45 – ELECTRO MAGNETIC THEORY

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on-

- 1 Concepts of electric forces and fields for different configurations.
- 2 Fields in different materials and capacitance calculations.
- 3 Effects of magnetic fields in electrical systems, magnetic potentials and inductance.
- 4 Maxwell's equations, wave equation and Poynting vector

Unit-1

Cartesian, cylindrical & spherical co-ordinate systems, scalar & vector fields, gradient, divergence & curl of a vector field, Divergence theorem & Stokes's theorem, concept of vectors. Electrostatic Fields – Coulomb's law, electric field intensity due to different charge distribution viz. line charge, sheet charge, Field due to continuous volume – electric potential, properties of potential function, potential gradient equipotential surfaces, line of force, Gauss law, applications of Gauss law, Gauss law in point form, method of images.

(12 HRS)

Unit-2

Laplace's & Poisson's equations, solution of Laplace's equation, Electric dipole, dipole moment, potential & electric field intensity due to dipole, Behavior of conductors in an electric field. Conductor & insulator, electric field inside a dielectric, polarization, Boundary value conditions for electric Field, Capacitance & Capacitances of various types of capacitors, Energy stored and energy density in static electric field, Current density, conduction & convection current density ohms law in point form, equation of continuity.

(12 HRS)

Unit-3

Static Magnetic Field, Biot-Savart's law, Magnetic Field intensity due to straight current carrying filament, circular, square and solenoidal current carrying wire, Relationship between magnetic flux, flux density & magnetic Field intensity; Ampere's circuital law and its applications, magnetic Field intensity due to infinite sheet and various other configurations, Ampere's circuital law in point form, Magnetic force, moving charge in a magnetic field, Lorentz Force on straight and long current carrying conductors in magnetic field, force between two long & parallel current carrying conductors. Magnetic dipole & dipole moment, a differential current loop as dipole, torque on a current carrying loop in magnetic field, Magnetic Boundary conditions.

(12 HRS)

Unit-4

Scalar magnetic potential and its limitations, Vector magnetic potential and its properties, vector magnetic potential due to different simple configurations; Self and Mutual inductances, determination of self & mutual inductances, self inductance of solenoid, toroid coils, mutual inductance between a straight long wire & a square loop. Energy stored in magnetic Field & energy density, Faraday's Law, transformer & motional EMFs, Displacement current, Maxwell's equations as Generalization of circuit equations, Maxwell's equation in free space, Maxwell's equation for



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harmonically varying Field, static and steady fields, Maxwell's equations in differential & integral form.

(12 HRS)

Unit-5

Electro Magnetic Waves : Uniform plane wave in time domain in free space, Sinusoidally time varying uniform plane wave in free space, Wave equation and solution for material medium, Uniform plane wave in dielectrics and conductors, Pointing Vector theorem, instantaneous, average and complex poynting vector, power loss in a plane conductor, energy storage, Polarization of waves, Reflection by conductors and dielectric – Normal & Oblique incidence, Reflection at surface of a conducting medium, surface impedance, transmission line analogy.

(12 HRS)

COURSE OUTCOME :

After completion of this course, students will be able to

- 1 Realize electric field concepts and solve problems.
- 2 Examine the fields in different materials and capacitance concepts.
- 3 Use the magnetic field concepts and calculate the magnetic potentials and inductance for various cases.
- 4 Contrast Maxwell's equations, assess wave equation, Poynting vector and apply the concept of wave propagation.

Reference Books:

1. Mathew N.O Sadiku; Elements of Electromagnetic; Oxford.
2. P.V. Gupta; Electromagnetic Fields; Dhanpat Rai.
3. N.N. Rao; Element of Engineering Electromagnetic; PHI.
4. William H. Hayt; Engineering Electromagnetic; TMH.
5. John D. Kraus; Electromagnetic; TMH.
6. Jordan Balmian; Electromagnetic wave & Radiating System; PHI.
7. David K. Cheng; Fields and Wave Electromagnetic; Addison Wesley.
8. S.P. Seth; Electromagnetic Field ;Dhanpat Rai & Sons



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IVTH SEMESTER

CATEGORY:-DLC

BE-42 – MATLAB SIMULATION LAB

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 To familiarize the student in introducing and exploring MATLAB softwares.
- 2 To enable the student on how to approach for solving Engineering problems using simulation tools.
- 3 To prepare the students to use MATLAB in their project works.
- 4 To provide a foundation in use of this software's for real time applications.

LIST OF EXPERIMENT:-

1. To understand MATLAB basic features and built in functions available in MATLAB.
2. Generate various signals and sequences (periodic and aperiodic) (Ramp, Impulse, unit step, square, saw tooth, triangular sinusoidal and sine) using MATLAB software.
3. Perform the operations on signals and sequences such as addition, Multiplication, Scaling, shifting, folding and also compute energy and power.
4. Verify the Linearity of a given Discrete System.
5. Verification of Time Invariance of a Discrete System.
6. Compute the Unit Sample, Unit step and Sinusoidal Response of the given LTI System and Verifying its Stability.
7. Finding the Laplace transform & Inverse Laplace transform of some signals.
8. Compute the Unit Sample, Unit step and Sinusoidal Response of the given LTI System and Verifying its Stability.
9. Finding the Laplace transform & Inverse Laplace transform of some signals.
10. Perform the waveform synthesis using Laplace transform of a given signal.
11. Write the program for locating poles and zeros and plotting pole-zero maps in s-plane and z-plane for the given transfer function.
12. Verify the sampling theorem.

COURSE OUTCOME

After completion of this course, students will be able to

- 1 Ability to express programming & simulation for engineering problems.
- 2 Ability to find importance of this software for Lab Experimentation.
- 3 Articulate importance of software's in research by simulation work.
- 4 In-depth knowledge of providing virtual instruments on .Environment.
- 5 Ability to write basic mathematical ,electrical ,electronic problems in Matlab.



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REFERNCES :-

1. Shailandra Jain, Modeling and simulation using MATLAB/SIMULINK ,wiley
2. I.J.Nagrath,D.P. Kothari, Electrical machine,TMH
3. P.C. Sen ,Power Electronics, TMH



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**IVTH
SEMES
TER
CATEG
ORY: -
PDFS**

BE-44 PROFESSIONAL DEVELOPMENT

FINISHING SCHOOL LEVEL-II

TOTAL - 36 HOURS

OBJECTIVE

The students are to be groomed with respect to personality development. Further improvement in English, Aptitude and reasoning ability is desirable.

UNIT-I

Conversational English:

Grammar mainly Prepositions (550 small sentences of daily use related to day to day life

(18 –HRS)

UNIT-II

Aptitude / Reasoning:

Quantitative Aptitude and Logical Reasoning –

Level II Problem solving on,

Partnerships, Profit Loss and Discounts, Time and Distance.

Logical sequence of Figures, Cubes, Blood Relations, Data Sufficiency, Arrangement Problems.

(18 –HRS)

OUTCOME

Further improvement in English, Aptitude and reasoning ability is achieved.



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NATIONAL CADET CORPS



GROOMING TOMORROW'S LEADERS

**OFFERING NCC A GENERAL GENERIC ELECTIVE CREDIT
COURSE IN UNIVERSITIES UNDER CHOICE BASED CREDIT
SYSTEM TO ALIGN WITH NEW EDUCATION POLICY 2020**



SARVEPALLI RADHAKRISHNAN UNIVERSITY,



BHOPAL
CONTENTS

3. Section I : NCC Credit Course Design
4. Section II : NCC Credit Course Rules & Regulations aligned to UGC.

NATIONAL CADET CORPS

SECTION I: NCC CREDIT COURSE DESIGN DOCUMENT

UNDER CHOICE BASED CREDIT SYSTEM AS GENERAL ELECTIVE FOR SENIOR DIVISION / SENIOR WING

4. **Preamble.** The National Cadet Corps (NCC) is governed by NCC Act 1948 and attendant NCC Rules. It functions under the Ministry of Defence and is headed by DGNCC. It is organised into 17 State Directorates each headed by an Additional/Deputy Director General. The aims of NCC are:-

(a) To develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.

(b) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regard less of which career they choose.

(c) To provide a conducive environment to motivate young Indians to choose the Armed Forces as a career.

5. **Purpose.** Currently NCC training is imparted as extra-curricular activity to volunteer students from recognized schools and colleges who enroll as cadets. NCC as a Credit Course is designed with an intent to transform NCC training into a curricular activity from an extra-curricular thereby providing academic credits to students undergoing NCC training along with other attended advantages to the cadets in the college/ university.

6. **Introduction to NCC Credit Course Design.** Institutional Training is the mainstay of NCC training and it is conducted at colleges and universities by Associate NCC Officers and Armed Forces personnel. The application of knowledge gained through institutional training is further honed or developed to a higher degree in NCC Camps. The Institutional Training syllabus comprises Common Subjects and Specialised Subjects (military component). NCC Credit Course is designed to offer Institutional Training of

Senior Wing /Division is over six semesters (three years), comprising 300 periods (excluding Camp), of which 120 periods are meant for theory with 08 credits and 180 periods for practical with 6 credits. Each period is counted as hour. The ratio between theory and practical in terms of number of hours of training is 5:6, but in terms of credits is 5:3, since as per CBCS two hours of practical is counted towards one period of training as against one hour for theory. In addition two separate courses have been designed for two Camps normally referred to as Annual Training Camps (ATC).



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Training schedules planned for cadets ensure that the optimum benefits of the NCC organization reach maximum number of cadets. The main emphasis is on practical training which in consonance with theory is made to facilitate active participation of learner, better assimilation of knowledge, and proper development of various skills, strengthening of mind and body which is the bedrock of NCC training.

NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	01	01	-	02	
Semester - II	01	01	-	02	
Semester - III	01	01	05	07	Credits of 1 st Camp merged with 3 rd Sem
Semester - IV	02	01	-	03	
Semester - V	01	01	05	07	Credits of 2 nd Camp merged with 5 th Sem
Semester - VI	02	01	-	03	
Total	08	06	10	24	Twenty-Four Credits

INSTITUTIONAL TRG SYLLABUS

COMMON SUBJECTS				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	NCC General	06	-	06
2	National Integration	04		04
3	Drill	-	45	45
4	Weapon Training	-	25	25
5	Personality Development	25		25
6	Leadership	12	-	12
7	Disaster Management	13		13
8	Social Service & Community Development	08	39	47
9	Health & Hygiene	-	10	10
10	Adventure	01		01
11	Environmental awareness & conservation	03		03
12	Obstacle Training	-	09	09
13	General Awareness	04		04
14	Border & Coastal Areas	06		06
TOTAL HOURS COMMON SUBJECTS(a)		82	128	210

SPECIALISED SUBJECTS (ARMY)				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practical	
1	Armed Forces	09	-	09
2	Map Reading	-	24	24
3	Communications	03	03	06
4	Infantry Weapons	03	03	06
5	Field Craft & Battle Craft		22	22
6	Military History	23	-	23
Total Hours		38	52	90

**BHOPAL****SPECIALISED SUBJECTS (NAVY)**

Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	Naval Orientation	12	-	12
2	Naval Communication	02	18	20
3	Navigation	02	03	05
4	Seamanship	15	18	33
5	Fire Fighting and Damage Control	04	03	07
6	Ship and Boat Modelling	03	10	13
Total hours		38	52	90

SPECIALISED SUBJECTS (AIR FORCE)

Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	General Service Knowledge	08	-	08
2	Air Campaign	06	02	08
3	Principles of flight	06	06	12
4	Airmanship	01	07	08
5	Navigation	05	-	05
6	Aeroengines	06	-	06
7	Basic flight Instruments	03	03	06
8	Aero modelling	03	34	37
Total Hours		38	52	90

INSTITUTIONAL TRAINING: TOTAL HOURS & CREDITS

INSTITUTIONAL TRAINING: TOTAL HOURS & CREDITS			
ITEM	Periods (1 hour duration each)		Total
	Lectures/Tutorials	Practicals	
TOTAL HOURS COMMON SUBJECTS	82	128	210
TOTAL HOURS SPECIALISED SUBJECTS (ARMY/NAVY/AIR FORCE)	38	52	90
TOTAL HOURS INSTITUTIONAL TRAINING	120	180	300
TOTAL CREDITS INSTITUTIONAL TRAINING	08 CREDITS (15 HOUR THEORY = 1 CREDIT POINT)	6 CREDITS (30 HOURS PRACTICAL TRAINING = 1 CREDIT POINT)	



BHOPAL
NCC CAMP TRAINING SYLLABUS

COMMON SUBJECTS				
S No.	Subjects	Periods		Total
		L/T	P	
1.	Physical Training	-	18	18
2.	Drill	-	32	32
3.	Weapon Training	08	28	36
4.	National Integration and Awareness	08	-	08
5.	Personality Development	08	12	20
6.	Leadership	08	-	08
7.	Disaster Management	08	-	08
8. .	Social Service and Community Development	-	08	08
9.	Health & Hygiene	08	-	08
10.	Obstacle Training	-	04	04
11.	Military History	04	-	04
12.	Communication	04	-	04
13.	Games	-	18	18
14.	Culture	-	18	18
	TOTAL	56	138	194
<u>SPECIALISED SUBJECTS</u>				
1.	Map Reading	-	24	24
2.	Infantry Weapons	04	02	06
3.	Field Craft & Battle Craft	-	16	16
	TOTAL	04	42	46
	GRAND TOTAL	60 (4 credit)	180 (6 credit)	240 (10 credit)

NCC CAMP TRAINING SYLLABUS (FOR THEORY)

Ser No	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Weapon Training	-	-	04	-	04	-	08
2.	National Integration & Awareness	-	-	04	-	04	-	08
3.	Personality Development	-	-	04	-	04	-	08
4.	Leadership	-	-	04	-	04	-	08
5.	Disaster Management	-	-	04	-	04	-	08
6.	Health & Hygiene	-	-	04	-	04	-	08
7.	Military History			02		02		04
8.	Communication			02		02		04
9.	Infantry Weapons	-	-	02	-	02	-	04
	TOTAL	-	-	30	-	30	-	60
	TOTAL Credit	-	-	2	-	2	-	4

NCC CAMP TRAINING SYLLABUS (FOR PRACTICAL)

Ser No	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Physical Training	-	-	09	-	09	-	18
2.	Drill	-	-	16	-	16	-	32
3.	Weapon Training	-	-	14	-	14	-	28
4.	Personality Development	-	-	06	-	06	-	12
5.	Social Service and Community Development	-	-	04	-	04	-	08
6.	Obstacle Training	-	-	02	-	02	-	04
7.	Games			09		09		18
8.	Culture			09		09		18
9.	Map Reading	-	-	12	-	12	-	224
10.	Infantry Weapons	-	-	01	-	01	-	02
11.	Field Craft & Battle Craft	-	-	08	-	08	-	16
	TOTAL			90		90		180
	TOTAL CREDIT			03		03		06



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
BHOPAL**



SEMESTER WISE COURSE DESIGN ARMY CADETS

NATIONAL CADET CORPS

**INSTITUTIONAL TRAINING: SEMESTER WISE DISTRIBUTION OF NCC
SYLLABUS FOR THEORY (ARMY CADETS)**

S. NO.	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	NCC General	06	-	-	-	-	-	06
2.	National Integration	04	-	-	-	-	-	04
3.	Personality Development	02	05	05	04	06	04	25
4.	Leadership	-	05	04	03	-	-	12
5.	Disaster Management	-	-	03	10	-	-	13
6.	Social Service & Community Development	03	05	-	-	-	-	08
7.	Adventure	-	-	01	-	-	-	01
8.	Environmental Awareness & Conservation	-	-	-	03	-	-	03
9.	General Awareness	-	-	-	04	-	-	04
10.	Border & Coastal Areas	-	-	02	-	02	02	06
11.	Armed Forces	-	-	-	06	-	03	09
12.	Infantry Weapons	-	-	-	-	3	-	3
13.	Communication	-	-	-	-	-	03	03
14.	Military Hospital	-	-	-	-	04	19	23
	TOTAL	15	15	15	30	15	30	120
	TOTAL Credit	1	1	1	2	1	2	08

**INSTITUTIONAL TRAINING: SEMESTER WISE DISTRIBUTION OF NCC
SYLLABUS FOR PRACTICAL (ARMY CADETS)**

S. NO.	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Drill	12	12	08	07	03	03	45
2.	Field Craft & Battle Craft	03	04	04	04	04	03	22
3.	Map Reading	03	05	04	04	04	04	24
4.	Weapons Training	05	04	04	04	04	04	25
5.	Communication	-	-	-	-	-	03	03
6.	Infantry Weapons	-	-	-	-	-	03	03
7.	Social Service & Community Development	07	05	05	06	06	10	39
8.	Health & Hygiene	-	-	-	05	05	-	10
9.	Operation Training	-	-	05	-	04	-	09
	TOTAL	30	30	30	30	30	30	180
	TOTAL Credit	01	01	01	01	01	01	06



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INSTITUTIONAL TRAINING: SEMESTER WISE THEORY DETAILED SYLLABUS (ARMY CADETS)

<u>SEMESTER I</u>					
S.No	Subject	Periods	Chapter	Lesson	Hours
1	NCC General	6	NCC-I	Aims, Objectives and Org of NCC	1
			NCC-II	Incentives	2
			NCC-III	Duties of NCC Cadets	1
			NCC-IV	NCC Camps: Types and Conduct	2
2	National Integration and Awareness	4	NI-I	National Integration: Importance and Necessity	1
			NI-II	Factors affecting National Integration	1
			NI-III	Unity in Diversity	1
			NI-IV	Threats to National Security	1
3	Personality Development	2	PD - I	Factors Self-Awareness Empathy Critical and Creative Thinking Decision Making and Problem Solving	2
4	Social Service and Community Development	3	SSCD - I	Basics of Social Service Rural Development Programmes NGO's Contribution of Youth	3
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER II					
S.No	Subject	Periods	Chapter	Lesson	Hours
5	Personality Development	5	PD-II	Communication Skills	3
			PD-III	Group Discussion -Coping with Stress and Emotions	2
6	Leadership	5	L-I	Leadership Capsule Traits Indicators Motivation Moral Values Honour Code	3
			L-II	Case Studies Shivaji, Jhansi Ki Rani,	2
7	Social Service and Community Development	5	SS-IV	Protection of Children & Women Safety	1
			SS-V	Road/Rail Travel Safety	1
			SS-VI	New Initiatives	2
			SS-VII	Cyber and Mobile Security Awareness	1
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER III					
S.No	Subject	hours	Chapter	Lesson	HOURS
8	Personality Development	5	PD-III	Group Discussions - Change your Mindset	2
			PD-V	Public Speaking	3
9	Leadership	4	L-II	Case Studies – APJ Abdul Kalam, Deepa Malik, Maharana Pratap, N Narayan Murthy	4
10	Disaster Management	3	DM-I	Disaster Management Capsule Organisation Types of Disasters Essential Services Assistance Civil Defence Organisation	3
11	Adventure	1	AD-I	Adventure activities	1
12	Border & Coastal Areas	2	BCA-I	History, Geography & Topography of Border/ Coastal Areas	2
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER IV					
S.No	Subject	hours	Chapter	Lesson	HOURS
13	Personality Development	4	PD-III	Group Discussions - Time Management, Social Skills	4
14	Leadership	3	L-II	Case Studies – Ratan Tata, Rabindra Nath Tagore, Role of NCC cadets in 1965 war	3
15	Disaster Management	9	DM-II	Initiative Trg, Organising Skills, Dos and Don'ts Natural Disasters Man Made Disasters	9
		1	DM-III	Fire Services and Fire Fighting	1
16	Environmental Awareness	3	EA-I	Environmental Awareness and Conservation	3
17	General Awareness	4	GA-I	General Awareness	4
18	Armed Forces	6	AF-1	Army, Navy, Air Force and Central Armed Police Forces	6
TOTAL HOURS					30
TOTAL CREDITS					2



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SEMESTER V

S.No	Subject	hours	Chapter	Lesson	HOURS
19	Personality Development	6	PD-III	Group Discussions - Team Work	2
			PD-V	Public Speaking	4
20	Border & Coastal Areas	2	BCA-II	Security Setup and Border/Coastal management in the area	2
21	Introduction to Infantry Battalion and its Equipments	3	INF-1	Organisation of Infantry Battalion & its weapons	3
22	Military History	4	MH-3	Study of Battles of Indo-Pak Wars 1965 & 1971	4
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER VI

S.No	Subject	hours	Chapter	Lesson	HOURS
25	Personality Development	3	PD-IV	Career Counselling, SSB Procedure and Interview Skills	3
27	Border & Coastal Areas	2	BCA-III	Security Challenges & Role of cadets in Border management	2
28	Armed Forces	3	AF-2	Modes of Entry into Army, Police and CAPF	3
29	Military History	19	MH-1	Biographies of Renowned Generals	6
			MH-2	War Heroes : Param Veer Chakra Awardees	3
			MH-3	Study of Battles of Kargil	2
			MH-4	War Movies	8
30	Communication	3	C-1	Introduction to Communication & Latest Trends	3
TOTAL HOURS					30
TOTAL CREDITS					2

SIX SEMESTER NCC COURSE SYLLABUS

Training Objectives: Institutional Training

49. Institutional training includes basic military training of the cadets as part of the curriculum with its long-standing effort to mould young volunteers into disciplined and responsible citizens of India. NCC course is aimed to achieve following learning objectives:-

- (a) Develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.
- (b) To create interest in cadets by including and laying emphasis on those aspects of Institutional Training which attract young cadets into the NCC and provides them an element of thrill and excitement.
- (c) To inculcate defence Services work ethos that is characterized by hard work, sincerity of purpose, honesty, ideal of selfless service, dignity of labour, secular outlook, comradeship, spirit of adventure and sportsmanship.
- (d) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regardless of which career they choose.
- (e) To provide conducive environment to motivate young Indians to choose the Armed Forces as a career.



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SEMESTER I COURSE MODULE : NATIONAL CADET CORPS I

National Cadet Corps : Course Details			
Course Title: National Cadet Corps I			
Course Code	BNCC01GE03	Credits	1(Thr)+ 1(Pr) = 03
L /T + P	15+30	Course Duration	1 Semester
Semester	I (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e, 25% internal assessment and 75% end term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

50.

Course Objectives:

Cadets will be able to: -

- (a) Know about the history of NCC, its organization, and incentives of NCC for their career prospects.
- (b) Acquire knowledge of duties and conduct of ncc cadets.
- (c) Understand about different NCC camps and their conducts.
- (d) Understand the concept of national integration and its importance.
- (e) Understand the concept of self-awareness and emotional intelligence.
- (f) Understand the concept of critical & creative thinking.
- (g) Understand the process of decision making & problem solving.
- (h) Understand the concept of team and its functioning.
- (i) Understand the concept and importance of Social service.

51. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Imbibe the conduct of NCC cadets.
- (b) Respect the diversity of different Indian culture.
- (c) Practice togetherness and empathy in all walks of their life.
- (d) Do their own self analysis and will workout to overcome their weakness for better performance in all aspects of life.
- (e) Understand creative thinking & its components.
- (f) Think divergently and will try to break functional fixedness.
- (g) Make a team and will work together for achieving the common goals.
- (h) Do the social services on different occasions.

52. **Course Content Part (I) Theory**

- (a) **Unit 1- NCC General (N) (Contact Hrs. 06).** Introduction of NCC, History, Aims, Objective of NCC & NCC as Organization, Incentives of NCC, Duties of NCC Cadet. NCC Camps: Types & Conduct.
- (b) **Unit 2- National Integration & Awareness (NI) (Contact Hrs. 04) .** National Integration: Importance & Necessity, Factors Affecting National Integration, Unity in Diversity & Role of NCC in Nation Building, Threats to National Security.
- (c) **Unit 3- Personality Development (Contact Hrs. 3).** Intra & Interpersonal skills - Self-Awareness- & Analysis, Empathy, Critical & creative thinking, Decision making and problem solving.
- (d) **Unit 4- Social Service and Community Development (Contact Hrs. 02).** Basics of social service and its need, Types of social service activities, Objectives of rural development programs and its importance, NGO's and their contribution in social welfare, contribution of youth and NCC in Social welfare.

Course Content Part (II) Practical

53. **Course Objectives:** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of foot drill.
- (c) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.



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- (d) Develop awareness about different types of terrain and how it is used in battle craft.
- (e) Develop the concept of various markings on the map and how they are co-related to the ground features.
- (f) Understand the various social issues and their impact on social life.
- (g) Develop the sense of self-less social service for better social & community life.

54. **Expected Learning Outcomes:** After completing this course, the cadets will be able to: -

- (a) Perform foot drill and follow the different word of command.
- (b) Fire a weapon effectively with fair degree of marksmanship.
- (c) Undertake point to point navigation and take part in route marches by day and night.
- (d) Perform the social services on various occasions for better community & social life.

55. Course Content Part (II) Practical

- (a) **Unit 1. Drill (Contact Hrs. 12).** Foot Drill- Drill ki Aam Hidayaten, Word ki Command, Savdhan, Vishram, Aram Se, Murdha, Kadvar Sizing, Teen Line Banana, Khuli Line, Nikat Line, Khade Khade Salute Karna Parade Par, Visarjan, Line Tod, Tej Chal, Tham aur Dhire Chal, Tham.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 05).** Introduction & Characteristics of .22 rifle, Handling of .22 rifle.
- (c) **Unit 3. Map Reading (MR) (Contact Hrs. 03).** Definition of Map, Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Magnetic Variation and Grid Convergence.
- (d) **Unit 4. Field Craft & Battle Craft (FC & BC) (Contact Hrs. 03).** Introduction of Field Craft & Battle craft, Judging Distance, Method of Judging Distance.
- (e) **Unit 5. Social Service and Community Development (SSCD)(Contact Hrs.07).** Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc.

SEMESTER II COURSE MODULE : NATIONAL CADET CORPS II

<u>Course Title: National Cadet Corps II</u>			
Course Code	BNCC02GE03	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15+30	Course Duration	1 Semester
Semester	II (Even)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

56. **Course Objectives:** Cadets will be able to: -

- (a) Understand the thinking & reasoning process.
- (b) Understand the process to cope with Stress & emotions.
- (c) Understand the importance of improving communication skills.
- (d) Identify the leadership traits.
- (e) Admire the qualities of great leaders.
- (f) Know about different legal provisions for children & women safety and protection.
- (g) Understand the various rules & measures to be taken to ensure Road/Railsafety.
- (h) Understand & spread awareness about latest Government initiatives for welfare of citizens and contribute towards Nation building.
- (i) Understand concepts of cyber and mobile security.



57. **Expected Learning Outcomes.** After completing this course, the cadets will be able to:

- (a) Define thinking, reasoning, critical thinking and creative thinking.
- (b) To think critically about different life related issues.
- (c) Think divergently and will try to break functional fixedness.
- (d) Creatively in their real-life problems.
- (e) Understand the organizations related to disaster management and their functioning.
- (f) Appreciate the role of NCC cadets in disaster management.

58. **Course Content Part (I) Theory**

(a) **Unit 1. Personality Development (Contact Hrs.5)**

- (i) Thinking- Meaning and Concept of thinking, Reasoning, Process of thinking.
- (ii) Critical Thinking- Meaning & concept of critical thinking, Features of critical thinking, Process of critical thinking.
- (iii) Creative thinking- Meaning & concept of creative thinking, Features of creative thinking, Process of creative thinking, levels of Creativity, Characteristics of creative person.

(b) **Unit 2. Leadership Development (Contact Hrs.5)**

- (i) Leadership capsule.
- (ii) Important Leadership traits, Indicators of leadership and evaluation.
- (iii) Motivation- Meaning & concept, Types of motivation. Factors affecting motivation.
- (iv) Ethics and Honor codes.

(c) **Unit 3. Social Service and Community Development (Contact Hrs. 5)**

- (i) Protection of Children & Women Safety.
- (ii) Road/Rail Safety.
- (iii) New Government Initiatives.
- (iv) Cyber and mobile Security Awareness.

Course Content Part (II) Practical

59. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of foot drill.
- (c) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (d) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.

60. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform foot drill gracefully.
- (b) Give and follow the different word of command.
- (c) Fire a weapon effectively with fair degree of marksmanship.
- (d) Use of bearing and service protractor and locate the places and objects on the ground.
- (e) Do the social service and feel connected with social problems.

61. **Course Content Part (II) Practical**

(a) Unit 1. Drill (Contact Hrs. 12)

- (i) Foot Drill Dahine, Baen, Aageaur Piche Kadam Lena.
- (ii) Tej Chal se Murdna, Tej Chal se Salute Karna, Tej Kadam Taal aur Tham, Tej Kadam Taal se Kadam Badalna.
- (iii) Teeno Teen se Ek File aur ek file se Teeno Teen Banana

(b) Unit 2. Weapon Training (Contact Hrs. 04)

- (i) Range procedure & Theory of group.
- (ii) Short Range firing.

(c) Unit 3. Map Reading (Contact Hrs. 05)

- (i) Protractor Bearing and its conversion methods.
- (ii) Service protractor and its uses.
- (iii) Prismatic compass and its uses and GPS.
- (iv) Navigation by compass and GPS.



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(d) Unit 4. Field Craft & Battle Craft (Contact Hrs. 04)

- (i) Indications of landmarks and Targets.
- (ii) Intro, Definitions, Types of Ground, Indication of Landmarks, Methods of identification of targets, difficult targets.

(e) Unit 5. Social Service and Community Development (Contact Hrs. 05)

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc. as per the requirement and similar announced days- National and state level.

NATIONAL CADET CORPS

SEMESTER III COURSE MODULE : NATIONAL CADET CORPS III

<u>COURSE TITLE: NATIONAL CADET CORPS III</u>			
Course Code	BNCC03GE02	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15 +30	Course Duration	1 Semester
Semester	III (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

62. **Course Objectives.** Cadets will be able to: -

- (a) Understand the life history and leadership qualities of great leaders, sportspersons & entrepreneurs.
- (b) Understand the various aspects of types of mindset.
- (c) Understand public speaking methods & qualities.
- (d) Understand the organizations related to disaster management and their functioning.
- (e) Understand the role of NCC cadets in disaster management.
- (f) Understand the various types of adventure activities.
- (g) Understand the History, Geography & Topography of Border/ Coastal Areas.



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63. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Admire and get inspired from the accomplishments of leaders from various walks of life.
- (b) Develop public speaking skills.
- (c) Understand the importance of positive mindset and optimistic attitude in life.
- (d) Appreciate the need & requirement for disaster management and his role in disaster management activities.
- (e) Know the history & geographical peculiarity of our borders & coastal regions.

64. **Course Content Part (I) Theory**

(a) **Unit 1. Personality Development (Contact Hrs.5)**

- (i) Group Discussions - Change your Mindset
- (ii) Public Speaking.

(b) **Unit 2. Leadership Development (Contact Hrs.4).** Case Studies – APJ Abdul Kalam, Deepa Malik, Maharana Pratap, N Narayan Murthy.

(c) **Unit 3. Disaster management(Contact Hrs. 3)**

- (i) Disaster Management Capsule.
- (ii) Organisation.
- (iii) Types of Disasters.
- (iv) Essential Services.
- (v) Assistance.
- (vi) Civil Defence Organisation.

(d) **Adventure (Contact Hrs. 1).** Adventure activities.

(e) **Border & Coastal Areas(Contact Hrs. 2).** History, Geography & Topography of Border/ Coastal Areas.

Course Content Part (II) Practical

65. **Course Objectives.** Cadets will be able to : -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal
- (b) Appreciate grace and dignity in the performance of arm drill
- (c) Understand the concept and importance of social service.
- (d) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (e) Actively participate in social service and community development activities.

66. **Expected Learning Outcomes.** After completing this course, the cadets will be able to : -

- (a) Perform arm drill gracefully.
- (b) Give and follow the different word of command.
- (c) Fire a weapon effectively with fair degree of marksmanship.
- (d) Different positioning for fire and aiming.
- (e) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (f) Observe surroundings in better way.
- (g) Develop the qualities of patience and confidence and become better individuals.
- (h) Will develop physical as well as mental fitness.

67. **Course Content Part (II) Practical**

(a) **Unit 1. Drill (Contact Hrs. 08)**

- (i) Arm Drill.
- (ii) Rifle ke saath Savdhan, Vishram aur Aram se.
- (iii) Rifle ke saath Parade Par aur Saj, Rifle ke saath Visarjan, Line Tod.
- (iv) Bhumi Shastra aur Uthao Shastra, Bagal Shastra aur Baju Shastra.

(b) **Unit 2. Weapon Training (Contact Hrs. 04).** Short Range firing.

(c) **Unit 3. Map Reading (Contact Hrs. 04).**

- (i) Setting of Map.
- (ii) Findings North and Own Position.



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(d) Unit 4. Field Craft & Battle Craft (Contact Hrs. 04)

- (i) Observation.
- (ii) Camouflage.
- (iii) Concealment.

(e) Unit 5. Social Service and Community Development (Contact Hrs. 05).

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) Unit 6. Obstacle Training(Contact Hrs. 05)

- (i) Obstacle training - Introduction, Safety-measures, Benefits.
- (ii) Obstacle Course- Straight balance, Clear Jump, Gate Vault, Zig- ZagBalance, High Wall.

SEMESTER IV COURSE MODULE : NATIONAL CADET CORPS IV

<u>Course Title: National Cadet Corps IV</u>			
Course Code	BNCC04GE03	Credits	2(Thr)+ 1(Pr)=03
L /T + P	30+30	Course Duration	1 Semester
Semester	IV (Even)	Contact Hours	30(Thr)+30(Pr)=60Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70%end of term exams etc.		

Course Content Part (I) Theory

68. **Course Objectives.** Cadets will be able to: -

- (a) Develop a sense of time management and social skills.
- (b) Understand the life history & leadership qualities of personalities who have contributed in Nation Building and Literature.
- (c) Understand the role of NCC cadets as 2nd line Defence in 1965 War.
- (d) Develop awareness about various types of Natural and manmade disasters.
- (e) Know about life saving tips during disasters.
- (f) acquainted about Fire Services.
- (g) Understand importance of Environmental Awareness & conservation.
- (i) Understand importance of General Awareness.
- (j) Know about Armed Forces.



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(c) **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (i) Effectively Manage time.
- (ii) Develop the qualities of social skills.
- (iii) Imbibe leadership qualities.
- (iv) Do group discussions effectively.
- (v) Be motivated to serve the nation by joining Armed forces.
- (vi) Contribute in environmental awareness and conservation activities.
- (vii) Keep abreast of current affairs & general awareness.
- (viii) Effectively contribute in managing disaster relief tasks.

69. Course Content Part (I) Theory

(a) **Unit 1. Personality Development (Contact Hrs.4).** Group Discussions – Social Skills & Time management.

(b) **Unit 2. Leadership Development (Contact Hrs.3).** Case Studies – Case Studies – Ratan Tata, Rabindra Nath Tagore, Role of NCC cadets in 1965 war.

(c) **Unit 3. Disaster management(Contact Hrs. 10)**

- (i) Initiative Trg, Organising Skills.
- (ii) Dos and Don'ts.
- (iii) Natural Disasters.
- (iv) Man Made Disasters.
- (v) Fire Services and Fire Fighting.

(d) **Environmental Awareness (Contact Hrs. 3).** Adventure Environmental Awareness and Conservation.

(e) **General Awareness (Contact Hrs. 4).** General Awareness.

(f) **Armed Forces(Contact Hrs. 6).** Army, Navy, Air Force and Central Armed Police Forces.

Course Content Part (II) Practical

70. **Course Objectives.** Cadets will be able to: -
- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
 - (b) Understand various signals to convey messages in the army.
 - (c) Get acquainted various section formations.
 - (d) Understand the basics of personal and public hygiene.
 - (e) Get acquainted with the procedure to treat the wounds and fractures during emergencies.
71. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Perform weapon drill gracefully.
 - (b) Give and follow the different word of command.
 - (c) Appreciate grace and dignity in the performance of foot drill.
 - (d) Apply signals in there day to day functioning.
 - (e) Provide first aid during the emergencies.
 - (f) Navigate to the given location on ground using compass and GPS.
 - (g) Practice healthy practices for the personal sanitation and hygiene.
72. **Course Content Part (II) Practical**
- (a) **Unit 1. Drill (Contact Hrs. 08)**
 - (i) Arm Drill.
 - (ii) Salami Shastra.
 - (iii) Squad Drill with Arms.
 - (b) **Unit 2. Weapon Training (Contact Hrs. 04).** Short Range firing
 - (c) **Unit 3. Map Reading(Contact Hrs. 04)**
 - (i) Map to Ground.
 - (ii) Ground to Map.
 - (d) **Unit 4. Field Craft & Battle Craft(Contact Hrs. 04)**
 - (i) Fire and Move Capsule.
 - (ii) Field signal- with hand, with Weapons, Signal with Whistle.
 - (iii) Field signals as means of giving orders.



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(iv) Field signals by day, Field signals by night.

(v) Section Formation.

(e) **Unit 5. Social Service and Community Development(Contact Hrs. 05)**

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) **Unit 6. Health & Hygiene(Contact Hrs. 05)**

(i) Hygiene & Sanitation (Hygiene- Personal & Camp Hygiene).

(ii) First Aid in common medical emergencies.

(iii) Treatment & Care of Wounds.

NATIONAL CADET CORPS

SEMESTER V COURSE MODULE : NATIONAL CADET CORPS V

Course Title: National Cadet Corps V			
Course Code	BNCC05GE02	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15 +30	Course Duration	1 Semester
Semester	V (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exam setc.		

Course Content Part (I) Theory

73. **Course Objectives.** Cadets will be able to: -

- (a) Understand the concept of Team and its functioning.
- (b) Hone Public speaking skills.
- (c) Understand the security set up and management of Border/Coastal areas.
- (d) Acquire knowledge about an Infantry Battalion organisation and its weapons.
- (e) Acquire knowledge about Indo-Pak Wars fought in 1965 & 1971.

74. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Participate in team building exercise and value team work.
- (b) Improve communication skills by public speaking activities.
- (c) Understand the security mechanism and management of Border/Coastal areas.
- (d) Get motivated to join armed forces.



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75. **Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (Contact Hrs.6).**
 - (i) Group Discussions –Team work.
 - (ii) Public speaking.
- (b) **Unit 2. Border & Coastal Areas(Contact Hrs.2).** Security Setup and Border/Coastal management in the area.
- (c) **Unit 3. Introduction to Infantry Battalion and its Equipment(Contact Hrs. 3).** Organisation of Infantry Battalion & its weapons
- (d) **Military History(Contact Hrs. 4).** Study of Battles of Indo-Pak Wars 1965 & 1971.

Course Content Part (II) Practical

- 76. **Course Objectives.** Cadets will be able to: -
 - (a) Understand that drill as the foundation for discipline and to command a group for common goal.
 - (b) Appreciate grace and dignity in the performance of ceremonial drill.
 - (c) Use the compass and GPS to locate places on the ground and map.
- 77. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
 - (a) Perform ceremonial drill and follow the different word of command.
 - (b) Do the social service on various occasions and get connected with the community.
 - (c) Do all the asana and gain the physical& mental fitness.
- 78. **Course Content Part (II) practical**
 - (a) **Unit 1. Drill(Contact Hrs. 03)**
 - (i) Ceremonial Drill.
 - (ii) Guard Mounting.
 - (b) **Unit 2. Field Craft & Battle Craft(Contact Hrs. 04)**
 - (i) Fire control orders.
 - (ii) Types of fire control orders.

(iii) Fire and Movement- when to use fire and movements tactics, Basic considerations, Appreciation of ground cover, Types of cover, Dead ground, Common Mistakes, Map and air photography, Selection of Fire position and firecontrol.

(c) **Unit 3. Map Reading(Contact Hrs. 04).**Google Maps & applications

(d) **Unit 4. Weapon Training(Contact Hrs. 04).**Short Range firing

(e) **Unit 5. Social Service and Community Development (Contact Hrs. 05)**
Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc. as per the requirement and similar announced days- National and State level.

(f) **Unit 6. Health & Hygiene(Contact Hrs. 05)**

(i) Yoga- Introduction, Definition, Purpose, Benefits.

(ii) Asanas-Padamsana, Siddhasana, Gyan Mudra, Surya Namaskar, Shavasana, Vajrasana, Dhanurasana, Chakrasana, Sarvaangasana, Halasanaetc.

(g) **Unit 7. Obstacle Training(Contact Hrs. 05)**

(i) Obstacle training – Intro, Safety measures, Benefits.

(ii) Obstacle Course- Straight balance, Clear Jump, Gate Vault, Zig- ZagBalance, High Wall etc.



<u>Course Title: National Cadet Corps VI</u>			
Course Code	BNCC06GE03	Credits	2(Thr)+ 1(Pr)=03
L /T + P	30 +30	Course Duration	1 Semester
Semester	VI (Even)	Contact Hours	30(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

79. **Course Objectives.** Cadets will be able to: -

- (a) Get acquainted about counselling process its need and importance.
- (b) Know about SSB procedure and different tasks and tests.
- (c) Know about the conduction during the interview.
- (d) Understand the security challenges & role of cadets in Border Areas.
- (e) Know about the modes of entry in Armed forces, CAPF & police.
- (f) Understand the life history & leadership qualities of great generals.
- (g) Learn about 1999 Kargil war.
- (h) Acquire the knowledge about various wars and their heroes.
- (i) Know about various components of communication process.

80. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Get motivated to join Armed forces, police & CAPF.
- (b) Write their CV effective and appealing.
- (c) Face SSB interview effectively in their future.
- (d) Understand individual responsibilities & role in meeting the security challenges on Border/Coastal areas.
- (e) Imbibe the feeling of patriotism.
- (f) Communicate more effectively.

81. **Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (Contact Hrs.3).**
 - (i) Career Counselling.
 - (ii) SSB Procedure.
 - (iii) Interview Skills.
- (b) **Unit 2. Border & Coastal Areas (Contact Hrs.2).** Security Challenges & Role of cadets in Border management.
- (c) **Unit 3. Armed Forces (Contact Hrs. 3).** Modes of Entry into Army, Police and CAPF.
- (d) **Military History (Contact Hrs. 19).**
 - (i) Biographies of Renowned Generals.
 - (ii) War Heroes : Param Veer Chakra Awardees.
 - (iii) Study of Battles of Kargil.
 - (iv) War Movies.
- (e) **Communication (Contact Hrs. 3).** Introduction to Communication & Latest Trends.



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Course Content Part (II) Practical

82. **Course Objectives.** Cadets will be able to: -
- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
 - (b) Appreciate grace and dignity in the performance of ceremonial drill.
 - (c) Know about various knots and lashing used in soldiering.
 - (d) Acquire awareness about the basic weapon system in use in the Armed Forces.
83. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Perform foot drill and follow the different word of command.
 - (b) Aiming range and figure targets.
 - (c) Use the different knots and lashing in day-to-day life for different purposes.
 - (d) Develop the feeling of altruism.
84. **Course Content Part (II) Practical.**
- (a) **Unit 1. Drill (Contact Hrs. 03).**
 - (i) Ceremonial Drill.
 - (ii) Guard of Honour.
 - (b) **Unit 2. Weapon Training(WT) (Contact Hrs. 04).** Short Range firing.
 - (c) **Unit 3. Map Reading(MR) (Contact Hrs. 04).** Google maps and Applications.
 - (d) **Unit 4. Field Craft & Battle Craft(FCBC) (Contact Hrs. 03).** Knots, Lashing and Stretchers.
 - (e) **Unit 5. Social Service and Community Development(SSCD) (Contact Hrs. 05).** Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.
 - (f) **Unit 6 Introduction of Infantry Weapons & Equipment(INF) (Contact Hrs.03).** Characteristics of 5.56MM INSAS Rifle, Ammunition, Fire Power, Stripping, Assembling & Cleaning Practice.
 - (g) **Unit 7. Communication (COM) (Contact Hrs. 03).**
 - (i) Basic Radio Telephony (RT) Procedure.
 - (ii) Introduction, Advantages, Disadvantages, Need for standard procedures.

(iii) Types of Radio telephony communication.

(iv) Radio telephony procedure, Documentation.

NATIONAL CADET CORPS



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COURSE MODULE: NATIONAL CADET CORPS CAMP -I

Course Title: National Cadet Corps Camp I			
Course Code	BNCCCAMP03GE05	Credits	2(Thr)+ 3(Pr)=05
L /T + P	30+90	Course Duration	10 Days (24 hours each)
Semester	III (Odd)	Contact Hours	30(Thr)+90(Pr)=120Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work,self-study, individual and group tasks, team work, field-based assignments, Physical Training, endurance building and skill development practices		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

85. **Course Objectives**.Cadets will be able to: -

- (a) Acquire knowledge about the various aspects of personality development.
- (b) Understand the concept of leadership traits, moral values and character traits.
- (c) Develop awareness about the various types of natural disasters.
- (d) Develop sensitivity to the changing environment and understand the importance of conservation.
- (e) Understand the importance of hygiene and sanitation and common first aid procedures.
- (f) Acquire awareness about various types of weapon systems in the Armed Forces.

86. **Expected Learning Outcomes**. After completing this course, the cadets will be able to: -

- (a) Acquire adequate skill sets to overcome their weakness and reshape their personality.
- (b) Imbibe good moral values and character traits in their daily life.
- (c) Become useful members of the society and form part of disaster response team, if need arises.
- (d) Respect and make efforts to conserve natural resources
- (e) Follow good personal hygiene practices and provide first aid in emergencies.
- (f) Be motivated to join the armed forces.

87. **NCC Camp-I : Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (PD) (Contact Hrs. 04)**. Introduction to Personality Development, Factors influencing/shaping personality, Time Management and Interview Skills.
- (b) **Unit 2. Leadership (LDR) (Contact Hrs. 04)**. Leadership Traits, Moral Values and Character Traits.
- (c) **Unit 3. Disaster Management (DM) (Contact Hrs. 04)**. Assistance during natural disasters, Do's and Don'ts for NCC Cadets performing Disaster Management Duties
- (d) **Unit 4. National Integration and Awareness (NIA) (Contact Hrs. 04)**.
Water Conservation and Rain Harvesting, Waste Management and Energy Conservation
- (e) **Unit 5. Health and Hygiene (H&H) (Contact Hrs. 04)**. Hygiene and Sanitation, First Aid in Common Medical Emergencies.
- (f) **Unit 6. Infantry Weapons (IW) (Contact Hrs. 02)**. Characteristics of Company Support Weapons.
- (g) **Unit 7. Weapon Training (WT) (Contact Hrs. 04)**. Characteristics of Point 22 Rifle and its Ammunition, Range Procedure and Safety Precautions.
- (h) **Unit 8. Military History (MH) (Contact Hrs. 04)**. Guest lectures by War Veterans/decorated soldiers/veterans.
- (i) **Unit 9. Communication (COM) (Contact Hrs. 04)**. Basics of communication.



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NCC Camp-I : Course Content Part (II) Practical

88. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill is the foundation of discipline and command a group for a common goal.
- (b) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (c) Develop awareness about different types of terrain and how it is used in BattleCraft.
- (d) Develop the concept of various markings on the map and how they are co-related to the ground features.
- (e) Acquire awareness about the various types of weapon systems in the Armed Forces.
- (f) Understand the concept and importance of social service.
- (g) Understand the various nuances of Personality Development.
- (h) Understand the concept and importance of Physical Training in everyone's life.
- (i) Acquire skill sets about various games and understand the importance of teamwork.
- (j) Develop awareness about different cultures and different modes of its projection in artistic forms.

89. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform foot drill, arms drill, ceremonial drill and will be able to give out different words of command.
- (b) Fire a weapon effectively with fair degree of marksmanship.
- (c) Undertake point to point navigation and take part in route marches by day and night.
- (d) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (e) Be motivated to join the armed forces.
- (f) Acquire adequate skill sets to overcome their weakness and enhance their personality.
- (g) Gain adequate physical and mental endurance capabilities.
- (h) Play team games and be able to communicate and coordinate effectively in group events or situations.

- (i) Respect the diversity of Indian culture and develop pride by showcasing their own culture to others.

90. NCC Camp-I : Course Content Part (II) Practical

- (a) **Unit 1. Drill (Drill)(Contact Hrs. 16).** Drill ki Aam Hidayaten aur Words of Command, Savdhan, Vishram, Aram Se aur Mudna, Khuli Line aur Nikat Line mein march, Salute Karna Parade Par, Visarjan aur Line Tod, Tej Chal, Tham aur Dhire Chal, Tham, Dahine, Baen, Aage aur Piche Kadam lena, Tejchaal se Mudna, Tejchaalse Salute karna, Tej kadambaal aur Tham, Tej Kadambaal se kadam badhana, Teenon Teen se ek file Banana aur ek file se Teenon Teen Banana, Rifle Ke Saath Saavdhan, Vishram aur Aaram se, Rifle ke saath Parade par aur saaj, Rifle Ke saath visarjan aur line tod, Bhumi Sashtira aur Uthao Sashtira, Bagal Sashtira aur Baaju Shastra.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 14).** Stripping, Assembling, Cleaning of Point 22 rifle, Sight Setting and Sight Picture of Point 22 Rifle, Loading, Cocking and Unloading, Lying Position, Holding and Aiming of Point 22 rifle, Trigger Control and Firing of Shot, Theory of Group, Short-Range Aiming and Firing, Firing Practice I to VII.
- (c) **Unit 3. Field Craft & Battle Craft (FC/BC) (Contact Hrs. 06).** Introduction of Field Craft & Battle craft, Judging Distance, Indication of Landmarks and Targets, Observation, Camouflage and Concealment, Field Signals, Section formations.
- (d) **Unit 4. Map Reading (MR) (Contact Hrs. 12).** Introduction to Map and Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Types of Bearing and use of Service Protector, Prismatic Compass and its use, setting of a map, Finding North and own Position, Map to Ground and Ground to map, Point to Point march, Route March – I, Route March -II.
- (e) **Unit 5. Infantry Weapons (IW) (Contact Hrs. 01).** Characteristics of Battalion Support Weapons.
- (f) **Unit 6. Social Service and Community Development (SSCD) (Contact Hrs. 04).** Basics of Social Service and its need, Rural Development Programme, Civic Responsibilities: Cadets will participate in various activities throughout the camp e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc., Road /Rail Travel Safety
- (g) **Unit 7. Personality Development (PD) (Contact Hrs. 06).** Self-Awareness, Empathy, Critical and Creative Thinking, Decision making and problem Solving, Coping with Stress and Emotions, Time Management.
- (h) **Unit 8. Obstacle Training (OT) (Contact Hrs. 02).** OT Practice – I:- Untimed, Cadets will be familiarized with all the obstacles in the Obstacle Course and briefed about the correct method to do them, OT Practice -II: Timed practice for all the cadets and record to be maintained.



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- (i) **Unit 9. Physical Training (PT) (Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in morning hours. Training has to be progressive in degree of difficulty to improve individual stamina and endurance. Training to include warming up, running, exercises to strengthen upper body, lower body and core muscles. Two period each to be devoted to route march by day and night respectively and one period will be earmarked for trekking expedition as part of Adventure Activity.
- (j) **Unit 10. Games Training (G)(Contact Hrs. 09).** Games Training will be carried out on each day of the camp, except on last day, in evening hours. Training has to be progressive in degree of difficulty to improve individual skills, coordination, teamwork and desire to excel. Training to ensure that each and every boy and girl cadets participate in at least one game activity everyday.
- (k) **Unit 11. Cultural Activity (C)(Contact Hrs. 09).** Cultural Activity will be carried out on each day of the camp, except on last day, in evening hours. Cadets have to divided in Nine Groups consisting of a mix of boy and girl cadets and preferably belonging to the same geographical area. Each group has to present the unique culture, custom, tradition, folk lore, songs, drama, paintings and cuisine during one hour allotted. There will be a prize for the best group to encourage participation and to develop pride in their unique culture. This training activity should ensure that each and every boy and girl cadet participate in at least one game activity every-day.
- (l) **Unit 12. Spare (S)(Contact Hrs. 02).** Two periods in each camp will be earmarked as spare to cover disruptions in training activity due to weather or other administrative reasons.

COURSE MODULE : NATIONAL CADET CORPS CAMP – II

Course Title: National Cadet Corps Camp II			
Course Code	BNCCCAMP05GE05	Credits	2(Thr)+ 3(Pr)=05
L /T + P	30+90	Course Duration	10 Days (24 hours each)
Semester	V (Odd)	Contact Hours	30(Thr)+90(Pr)=120Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, individual and group tasks, team work, field-based assignments, Physical Training, endurance building and skill development practices		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (II) Theory

91. **Course Objectives.** Cadets will be able to: -

- (a) Acquire the concept self-awareness, emotional intelligence, critical and creative thinking, decision making and problem solving.
- (b) Learn about various indicators of good leadership and get an insight on principle of leadership and motivation.
- (c) Develop awareness about the various types of natural disasters and disaster management organization in our country.
- (d) Familiarize with natural resources, changing environment and understand the importance of conservation and waste management.
- (e) Value the importance of Physical and Mental health and understand how to deal with wounds of various types.
- (f) Acquire awareness about organization and role of an Infantry Battalion in the Armed Forces.



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92. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Develop a sense of responsibility, smartness in appearance and improve self-confidence, inculcate importance of empathizing with others, improve their deep-thinking ability and apply ideas and be able to face problems in a constructive manner with solutions.
- (b) Imbibe good leadership traits and apply them in practical life and appreciate the visible outcome of leadership and motivation.
- (c) Appreciate role of the org during emergency and become useful members of disaster response team, if need arises.
- (d) Learn about the various natural resources, their utilization and practice method of conservation of these resources in daily life.
- (e) Appreciate value of physical and mental health in daily life and spread awareness about treatment and care of wounds in their society.
- (f) Be motivated to join the armed forces.

93. **NCC Camp-II : Course Content Part (I) Theory.**

- (a) **Unit 1. Personality Development (PD) (Contact Hrs. 04).** Self-Awareness, Emotional intelligence, Critical and Creative Thinking, Decision-Making and Problem Solving.
- (b) **Unit 2. Leadership (LDR) (Contact Hrs. 02).** Indicators of Good Leadership, Leadership and Motivation.
- (c) **Unit 3. Disaster Management (DM) (Contact Hrs. 02).** Disaster Management Organization NDMA and NDRF, Types of Disasters.
- (d) **Unit 4. Environmental Awareness and Conservation (EAC) (Contact Hrs. 02).** Natural Resources, Conservation and Management, Water Conservation, Waste Management, Energy Conservation.
- (e) **Unit 5. Health and Hygiene (H&H) (Contact Hrs. 02).** Physical and Mental Health, Treatment and Care of Wounds.
- (f) **Unit 6. Infantry Weapons (IW) (Contact Hrs. 01).** Organization of Infantry Battalion.
- (g) **Unit 7. Weapon Training (WT) (Contact Hrs. 02).** Characteristics of Point 22 Rifle and its Ammunition, Range Procedure and Safety Precautions.
- (h) **Unit 8. Military History (MH) (Contact Hrs. 04).** Guest lectures by War Veterans/decorated soldiers/veterans.
- (i) **Unit 9. Communication (COM) (Contact Hrs. 04).** Latest trends in communication.

NCC Camp-II : Course Content Part (II) Practical

94. **Course Objectives.** Cadets will be able to: -

- (a) Inculcate spirit of discipline and follow command as a group for a common goal.
- (b) Fire a weapon with adequate safety precautions necessary for safe firing.
- (c) Understand the lay of the ground and use it skillfully towards own objective.
- (d) Understand and use the map, satellite imagery and GPS effectively.
- (e) Identify and be well versed with the primary weapon systems used in the Armed Forces.
- (f) Lead a life of selflessness and provide service towards society development and nation building.
- (g) Understand the importance of changing mindset, team work, social skills, etiquettes and manners, interview skills and importance of effective communication in daily life.
- (h) Learn the importance of physical fitness and nuances of physical training.
- (i) Inculcate esprit-de-corps through team games.
- (j) Have knowledge about cultural diversity of India and learn ways and means to adopt them.

95. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Practice problem solving, critical thinking in real life situations.
- (b) Practice leadership of small teams and groups under challenging environment.
- (c) Develop a positive attitude, have manners and etiquettes in social life, develop a sense of cooperation for group or team work, participate in an interview with confidence and inculcate verbal and non-verbal communication skills.
- (d) Develop adequate physical and mental endurance capabilities.
- (e) Fire a weapon effectively with fair degree of marksmanship.
- (f) Undertake point to point navigation and take part in endurance marches by day and night.
- (g) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (h) Be motivated to join the Armed Forces.
- (i) Play team games and be able to communicate and coordinate effectively in group events or situations.



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- (j) Perform foot drill, arms drill, ceremonial drill and will be able to take part in ceremonial parade and events.
- (k) Respect the diversity of indian culture and develop pride by showcasing their own culture to others.

96. NCC Camp-II : Course Content Part (II) Practical

- (a) **Unit 1. Drill (Drill) (Contact Hrs. 16).** Tejchaal se Mudna, Tejchaal se Salute karna, Tej kadamtaal aur Tham, Tej Kadamtaal se kadam badhana, Teenon Teen se ek file Banana aur ek file se Teenon Teen Banana, Rifle Ke Saath Saavdhan, Aaram se, Rifle ke saath Parade par aur saaj, Rifle Ke saath visarjanaur line tod, Bhumi Sashttra aur Uthao Sashttra, Bagal Sashttra aur Baaju Shastra, Salami Sashttra, Squad Drill, Guard Mounting, Guard of Honour, Platoon / Company Drill, Word of Command and Instructional Practice.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 14).** Stripping, Assembling, Cleaning of Point 22 rifle, Sight Setting and Sight Picture of Point 22 Rifle, Loading, Cocking and Unloading, Lying Position, Holding and Aiming of Point 22 rifle, Trigger Control and Firing of Shot, Theory of Group, Short-Range Aiming and Firing, Musketry Training, Firing Practice I to VII.
- (c) **Unit 3. Field Craft & Battle Craft (FC/BC) (Contact Hrs. 06).** Observation, Camouflage and Concealment, Field Signals, Section formations, Fire Control Orders, Fire and Movement, Knots and Lashings.
- (d) **Unit 4. Map Reading (MR) (Contact Hrs. 12).** Introduction to Map and Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Types of Bearing and use of Service Protector, Prismatic Compass and its use, setting of a map, Finding North and own Position, Map to Ground and Ground to map, Point to Point march, Endurance March – I (10 KM), Endurance March -II (20 KM).
- (e) **Unit 5. Infantry Weapons (IW) (Contact Hrs. 01).** Characteristics of Infantry Company support weapons and 5.56 MM INSAS Rifle.
- (f) **Unit 6. Social Service and Community Development (SSCD)(Contact Hrs. 04).** Contribution of Youth Towards Social Welfare: Cadets will participate in various activities throughout the camp e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc., Social Evils: Female Feticide, Dowry, Child Abuse, Trafficking and Corruption, Drug Abuse and Drug Trafficking, Protection of Children and POCSO Act 2012.
- (g) **Unit 7. Personality Development (PD)(Contact Hrs. 06).** Change Your Mindset, Team Work and Team Building, Social Skills, Etiquettes and Manners, Interview Skills, Communication Skills–I, Communication Skills -II

(h) **Unit 8. Obstacle Training (OT)(Contact Hrs. 02).** OT Practice – I: Untimed, Cadets will be familiarized with all the obstacles in the Obstacle Course and briefed about the correct method to do them, OT Practice -II: Timed practice for all the cadets and record to be maintained.

(i) **Unit 9. Physical Training (PT) (Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in morning hours. Training has to be progressive in degree of difficulty to improve individual stamina and endurance. Training to include warming up, running, exercises to strengthen upper body, lower body and core muscles. Two period each to be devoted to route march by day and night respectively and one period will be earmarked for trekking expedition as part of Adventure Activity.

(j) **Unit 10. Games Training (G)(Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in evening hours. Training has to be progressive in degree of difficulty to improve individual skills, coordination, teamwork and desire to excel. Training to ensure that each and every boy and girl cadets participate in at least one game activity everyday

(k) **Unit 11. Cultural Activity (C) (Contact Hrs. 09).** Cultural Activity will be carried out on each day of the camp, except on last day, in evening hours. Cadets have to be divided in Nine Groups consisting of a mix of boy and girl cadets and preferably belonging to the same geographical area. Each group has to present the unique culture, custom, tradition, folk lore, songs, drama, paintings and cuisine during one hour allotted. There will be a prize for the best group to encourage participation and to develop pride in their unique culture. This training activity should ensure that each and every boy and girl cadets participate in at least one game activity every day (Contact Hrs. 09)

(l) **Unit 12. Spare (S) (Contact Hrs. 02).** Two periods in each camp will be earmarked as spare to cover disruptions in training activity due to weather or other administrative reasons.



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SECTION II : RULES AND REGULATIONS

**GOVERNING NCC CREDIT COURSE UNDER CHOICE BASED CREDIT SYSTEM AS
GENERIC ELECTIVE FOR SENIOR DIVISION/WING**

RULE 1 :Definitions of Key Terms

General Definitions

‘Choice Based Credit System’ (CBCS).The CBCS provides choice for the student to select courses from the prescribed courses (Elective or Soft – Skill courses). It provides a ‘Cafeteria’ approach in which the students can take courses of their choice, learn at their own pace, study additional courses and acquire more than the minimum required credits, and adopt an inter-disciplinary approach.

‘Academic Year’.Two consecutive (one odd + one even) semesters shall constitute one academic year.

‘Credit Course’.Course, usually referred to as paper having specific title and code number, is a component of a programme. It consists of a list of topics/concepts/theories/principles/activities/tasks etc. which a student has to learn during the programme of study. Each course has some credits according to the nature and load of content. Each course should define the learning objectives/learning outcomes. A course may be designed to be delivered through lectures/tutorials/laboratory work/field work/out reach activities/project work / vocational training / physical training /viva / seminars /term papers / assignments / presentations / self-study work etc., or a combination of some of these.

‘Course Instructor/Teacher’.The course instructor generally will be a teaching faculty who has taken up the responsibility of teaching it and evaluating the performance of the students in that course. NCC course will be imparted by the ANO (Associate NCC Officer) and PI (Permanent Instructor) / Girl Cadet Instructor (GCI) staff together according to their area of specialization. Certain specific topics and training activity is imparted by Military Officers and Whole Time Lady (WTLO).

‘Credit’.A unit by which the course work is measured. It determines the number of hours of instructions required per week. One credit is equivalent to one hour of teaching (lecture or tutorial) or two hours of practical work per week. Thus, in each semester’s NCC course, credits are assigned on the basis of the number of lecture/tutorial/field work/physical training/excursions and other forms of learning required for completing the contents in a 15-18 week schedule. 2 hours of laboratory work/field work is generally considered equivalent to 1 hour of lecture.

- i. 1 credit = 1 hour of instruction per week (1 credit course = 15 contact hours of instruction per semester)

- ii. 4 credit = 4 hour of instruction per week (4 credit course = 60 contact hours of instruction per semester)
- iii. 1 credit = 2 hour of practical per week (1 credit course = 30 contact hours of instruction per semester)
- iv. 4 credit = 8 hour of practical per week (4 credit course = 120 contact hours of instruction per semester)

Number(s) of credit(s) assigned to a particular course are mentioned in the detailed syllabus of the courses.

‘Credit Point’. It is the product of the grade point and the number of credits for a course.

‘Letter Grade’. It is an index of the performance of students in a said course. Grades are denoted by letters O, A+, A, B+, B, C, P, and F. A letter grade is assigned to a student on the basis of evaluation of her/his performance in a course on a ten-point scale.

‘Grade Point’. It is a numerical weight allotted to each letter grade on a 10 -point scale.

Letter Grade	Grade Point
O	9-10
A+	8-9
A	7-8
B+	6-7
B	5-6
C	4-5
P	4
F	0
Ab	0

Note : University may use the above said criteria for providing the grades to the students or may adopt the same criteria which they are practicing for providing the letter grade and grade point for other subjects.

‘Programme’. An educational programme leading to the award of degree, Diploma or Certificate course. NCC course shall be offered only at under graduate level programmes for any stream or type of programme for example – Nonprofessional courses BA, B.Sc. B. Com etc. professional courses – B.A., LLB, B.A./B.Sc., B.Ed., BCA, BBA, B. Tech, MBBS etc.

‘Credit – Based Semester System (CBSS)’. Under the CBSS, the requirement of awarding a degree or diploma or certificate is prescribed in terms of number of credits to be completed by the students.



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'Semester' Each semester shall consist of 15 to 16 weeks of academic work equivalent to 90 actual teaching days. The odd semester may be scheduled from July to December and even semester from January to June. The Credit-based semester system provides flexibility in designing curriculum and assessing credits based on the course content and hrs of teaching.

'Semester Grade-Point Average (SGPA)' Semester Grade Point Average or SGPA, is an average grade point earned by the student at the end of an academic session i.e. semester at college. The formula for calculation of SGPA is the sum of all the credit points awarded for the subjects divided by total credits allotted to that semester. It shall be expressed up to two decimal places.

'Cumulative Grade Point (CGPA)' It is a measure of overall cumulative performance of a student over all semesters. The CGPA is the ratio of total credit points secured by a student in various courses in all the semesters. It shall be expressed up to two decimal places.

'Transcript/ Grade card or certificate' Based on the grades earned, a grade certificate shall be issued to all the registered students after every semester. The grade certificate will display the course details (code, title, number of credits, grade and / or marks secured) along with SGPA of semester. Overall Grade Certificate will be issued on completion of the course showing semester wise SGPA & CGPA.

'The University/ College/ Institution' The University/ College/ Institution in present document means the any recognized central/ state/ Deemed university or institution meant for higher education.

'NCC Course' In the present document 'NCC Course' means the course designed for imparting NCC curriculum in educational institutions as elaborated in this document under Choice Based Credit System as a General Elective Course for Senior Division/ Senior Wing.

Definitions Specific to NCC

'Institutional Training' Implies training conducted for NCC cadets as per Training Manuals and Cadet Hand Book issued by DG NCC, Ministry of Defence.

'Common Subjects' Implies those subjects specifically taught in NCC curriculum which are common to Army, Navy and Air Force and general training that can be imparted by Associate NCC Officers or Military staff or a suitably qualified person.

'Specialised Subjects' Implies subjects specifically taught in NCC curriculum by military instructors comprising specialised topics for Army, Navy and Air Force Cadets respectively.

'NCC Camps and Centralised Training Events' Collective training events conducted usually for 10 days with large number of cadets living under field conditions in selected places away from home. The training camp comprises of focused physical and mental training routines of different types as per syllabus and curriculum. Some training like route marches may happen overnight. Camps include, adventure camps, national integration camps,

Republic Day Parade Training Camps, ThalSainik, VayuSainik and NauSainik camps and other outdoor training activities as described in DG NCC Training Manuals.

NCC 'B' and 'C' Certificate Examinations. These are defined in Special National Cadet Corps Order 2020 issued by DG NCC, Ministry of Defence.

'Training Faculty'. Persons suitably trained & responsible for imparting training of different types and nature to students.

'Military Officers'. They are regular commissioned officers of Indian Armed Forces who serve in the NCC and render command, administrative and instructional functions for NCC.

'Whole Time Lady Officers (WTLO)'. They are women officers commissioned directly into the NCC.

'Associate NCC Officer (ANO)'. ANO will be a university/ college/ school faculty who are qualified in the PRCN (Pre-commission Course of NCC) conducted by DGNCC and are commissioned as Associate Officers in NCC as defined in NCC Act 1948 and NCC Rules. They have the eligibility to impart certain component of NCC Course and undertake training of cadets.

'Permanent Instructor (PI)'. PI Staff are Junior Commissioned Officers (JCO) and Non-Commissioned Officers (NCO) on deputation from Armed Forces to NCC as governed by NCC Act 1948. Retired PI Staff may be hired by a college as a substitute for ANO with prior concurrence of DGNCC.

'Girl Cadet Instructors (GCI)'. GCI are lady instructors' equivalent to PI Staff for specifically imparting instructions to women NCC cadets of Senior Wing.

'NCC Organizational Structure'. NCC is an adjunct of Indian Armed Forces that operates under the ambit of the Ministry of Defence through the Defence Secretary with Raksha Mantri as the political head.

'DGNCC'. Directorate General of NCC renders the command and administrative function of NCC. The executive head of NCC is Director General of NCC who is a Lt Gen rank officer from the Army.

'State NCC Directorate'. State NCC Directorates are directorates subordinate to DGNCC and render command and administrative control to NCC at State level and is headed by an Additional or Deputy Director General

'NCC Group HQ'. NCC Group HQs are subordinate to State Directorates and render command and administrative control to NCC at district or cluster of districts in a state and is headed by a Group Commander.

'NCC Units'. NCC Units are subordinate to Group HQs at the lowest rung of the command and administrative control exercised by military officers and is headed by a Commanding Officer or Officer Commanding. The NCC Units directly engage



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with educational institutions and ANOs and are primarily responsible for training of NCC in institutions under their jurisdiction.

‘NCC Division/ Wing’. NCC Division/Wing are minor units of senior division/wing of NCC comprising of 160 senior cadets allotted to educational institutions. It can be further subdivided into NCC platoons of 53 to 54 cadets.

‘NCC Troop’. NCC Troop are minor units of junior division/wing of NCC comprising of 100 junior cadets allotted to educational institutions. It can be further subdivided into NCC half troops comprising of 50 junior cadets.

RULE 2 : Admission and Other Provisions

The NCC Course under the CBCS as ‘General Elective’ shall be of three years (Six Semester) duration which may be completed in maximum duration of four year (8 semesters).

Students may complete NCC course minimum in Six semesters and maximum in eight semesters. Cadets may complete their ‘B’ Certificate in four semesters minimum and maximum six semesters. Cadets already having ‘B’ certificate may complete their ‘C’ certificate in minimum two semesters and maximum four semesters, and they may join NCC course 5 in first semester of college.

The intake to the course shall be decided according to the seats allotted to University/ college/ institution by DG NCC according to the availability of required infrastructure, faculty and resources.

The admission to the NCC Course under the CBCS as a ‘General Elective’ shall be governed by the provisions as laid down by the NCC Act 1948/ SNCCO 2020/ contemporary SNCCO and Academic council of parallel body of university. These rules and regulations may be modified from time to time (if needed) by the Academic body of the university in consultation with DG NCC or Act/ Ordinances prepared by DG NCC.

Students will be enrolled as NCC cadet as per existing Acts & Rules.

At the time of reporting for admission, the candidates are required to present medical & physical fitness documents as well as the admission proof of the university and submit the self-attested copies of aforesaid documents.

The admission of any candidate is liable to be cancelled without giving any further notice forthwith or at any time during the period of the course, if it is detected that the candidate has/had produced fake/forged certificate (s)/ document(s), indulged in any act of misconduct/indiscipline and has/had concealed any other relevant information at the time of admission.

The admission of the candidate to the course shall be subject to such ordinance, rules and regulations as may be framed from time to time by the university in consultation with DG NCC and NCC act 1948.

DG NCC shall have jurisdiction in case of any dispute relating to the provisional admission in the course.

RULE3 : For Eligibility, Medium of Instrs & Categories

Eligibility Conditions. Be governed by provisions of NCC Act and Rules and directions from DG NCC from time to time. These are readily available on DG NCC website www.nccindia.nic.in.

Standards for physical Fitness criteria for Male and Female Cadets/students shall be governed by provisions of NCC Act and Rules and policy documents released by DG NCC from time to time.

RULE4 : Medium of Instruction. English or Hindi. However, ANOs and training instructors are free to use vernacular language for helping students who are not fluent in Hindi or English.

RULE 5 : Course and Students. NCC course is unique, due to the nature of its military training content and component hence it is normally offered to students enrolled as NCC cadets only. This NCC Course is primarily designed for students enrolled as NCC cadets under provisions of NCC Act 1948. Institution allotted NCC will have the obligation to offer this course to all students from their institute enrolled as cadets as per vacancy allotted to the institution by DG NCC as also to those cadets enrolled under Open Quota seats.

RULES 6 :NCC Course for 'Cadet' Category

6.1. NCC Course for 'Cadet'

- (a) NCC course for Cadets comprises of total 24 credits (08 for theory, 06 for practical and 10 for camp component) over 6 semesters courses i.e., NCC course I to NCC course VI and NCC Camp I & NCC Camp II.
- (b) Cadets will not only earn the academic credits but also be given 'B', and 'C' Certificates after passing the exam conducted by DG NCC.
- (c) Students would be free to join NCC Course I or subsequent Courses in any semester, not necessarily Semester I or the designated Semester.
- (d) A student can opt for only one of the six Courses per semester and that too sequentially implying NCC Course II cannot be joined before completing NCC Course I and so on.
- (e) Under this category a fresh student/cadet will compulsorily have to opt for all six NCC Courses in minimum six Semesters. However, 'B' certificate holder may directly join NCC Course Number 5 in any semester. He will have to complete NCC Course Number 5 and NCC Course Number 6 for obtaining 'C' certificate and he will be awarded credit points only for NCC Course Number 5 and NCC Course Number 6.



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NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	1	1		2	
Semester - II	1	1		2	
Semester – III	1	1	5	7	Credits of 1 st Camp merged with 3 rd Sem
Semester – IV	2	1		3	
Semester – V	1	1	5	7	Credits of 2 nd Camp merged with 5 th Sem
Semester - VI	2	1		3	
Total	08	6	10	24	Twenty-Four Credits

RULE 7 :Mobility& Credit Bank

The mobility shall be permissible from the regular mode programme to the regular mode programme of learning only and cannot be replaced by open/distance/online programme.

It shall be the responsibility of the student to assess the feasibility and practicality of vertical mobility (across the Universities), as it doesn't entitle a student to be exempted or relaxed from any of the requisites (sessional, attendance, assignments, End-semester examinations and programme duration etc.) for completing the course.

After completing one semester/ one year cadet/student may pursue NCC course from any other institution/ University/ College having NCC and carry credits in credit bank as per NEP 2020. The NCC students/ Cadets of some other university shall in any case be admitted only at the beginning of the session to the fulfilment of the other requirements of the NCC Course (attendance, Formative assessment, Field-work, practical etc).

A student of NCC course availing inter-university mobility shall continue to be a bonafide student of the university where he/she initially got admission and as per the university/ Institutional rules for the inter-university mobility.

In case of inter-university mobility of NCC cadet for NCC Course is also the subject to availability of NCC for the cadets in that particular university/ institution and it shall be interpreted as inter-battalion migration (means another regimental no. shall be allotted to the cadet).

RULE 8 :Examination & Promotion

The examination of all the NCC courses shall be internal in nature and generally consisting of continuous internal assessment and End of semester Examination. For the preparation of final grade in a particular course, the continuous internal assessment (Formative in nature) and the End Semester Examination (Summative in nature) shall have the weightage as decided for other courses by the university as per the University norms fore.g., 25% internal assessment and 75% End of term exams or 30% internal assessment and 70% End of term exams etc.

For assigning the Grades and credit points to NCC Course Universities/ Institutions are free to use the same criteria which are decided by their academic bodies for providing the grades and credit points to the other courses

RULE 9 :Continuous Internal Assessment

The Continuous Internal Assessment of the NCC Cadets' and NCC students' learning and performance shall be carried out by the ANOs and PI staff.

Continuous Internal Assessment will be 100% Practical that includes Drill Square test, Map Reading, Weapon Training, Field craft & Battle craft.

CO of nominated NCC Unit will be deemed as Head of the Department and shall be responsible for approving the schedule and pattern of the continuous internal examination.



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ANO of the nominated institute shall maintain all the records related to attendance, teaching and assessment in a systematic manner, including award of final grade.

In case a student fails to appear in any Continuous Internal Assessment, they will be given a chance to reappear in retest and in case he/she fails to obtain 'P' grade he/she will be made to repeat the exam by carrying it forward for semester retest .

RULE 10 :Re-appear in the End Semester Examination for Improvement of Grades

If a student wishes to improve her/his grade(s) in NCC course(s), she/he can re-appear in the End Semester Examination in the subsequent odd/even semester(s), whenever the examination of the particular course(s) is held, on payment of fees in addition to the prescribed semester fee within the maximum permissible duration for the programme of study of the student/cadet.

A student may improve her/his points/grade by reappearing in the End Semester Examination of a course as per the provisions of reappearing mentioned above. In such cases points obtained by the student in the Continuous Internal Assessment of the particular course shall be carried forward to the subsequent End Semester Examination of the course. However, in such case, the points/grades obtained on the basis of latest appeared End Semester Examination shall be considered for calculation of final CGPA of the programme.

The re-appear examination of a course for improvement of grade shall be based on the syllabi of the course in force at the time of initial registration to the course.

A student who has got the Migration/Transfer Certificate issued from the University shall not be allowed to re-appear in any examination for improvement of grade.

RULE 11 :Repeating Courses

A student having attendance shortage in any course may repeat the course by taking re-admission in that course in subsequent odd/even semester(s), whenever the course is being offered, within the maximum permissible duration of the programme.

If a student repeats a course, she/he has to fulfil all the desired requirements afresh including attendance, Continuous Internal Assessment and the End Semester Examination. In such case the course content shall be based on the syllabi of the course in force at the time of repetition of the course.

RULE 12 : Promotion Rules

A student shall be declared as 'promoted' to the next semester when she/he earns 'P' Grade or above in the last concluded semester examination, maintaining the spirit and pattern of semester system and covering the mandatory components, such as Continuous Internal Assessment and End-Semester Examination in the NCC Courses.

A student shall be 'Provisionally Promoted' to the next semester if she/he secures less than 'P' grade but he /she has to pass all the courses of NCC course within permissible duration.

A cadet shall be eligible to attend the 'B' Certificate exam if he/she passed all the first four semester NCC course and completed one ATC/CATC. Similarly, cadet will be eligible to attend 'C' certificate examination if he/she has 'B' certificate and he /she has passed V, VI semester NCC course and attended one CATC/ATC after fourth semester and after having obtained 'B' certificate.

If a cadet/student is repeating a course in an academic session, whatever may be the reason, it shall not be counted in the total number of seats and shall not affect the fresh intake of cadets / student in that academic session.

RULE 13 :Computation of SGPA & CGPA

13. **Computation of SGPA and CGPA** . University may use their own criteria for giving the SGPA & CGPA which is prepared by the authorised academic body for the other courses.



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SUBJECT CODE – MC 4 (C)

CATEGORY – NNP

SUBJECT NAME – PHYSICAL EDUCATION

SEMESTER – IV

ENGLISH II

UNIT-I

1. Vocabulary

Homonyms

Foreign words & phrases commonly used

Proverbs

Idioms

One word substitution

UNIT-II

2. Grammar

Tenses

Direct Indirect

Active passive

Simple, Compound, Complex

Punctuation

UNIT-III

3. Correspondence

Business letters

Application for job

Resume, Curriculum Vita, Bio-data

Report writing on completion of events

Agenda

Minutes

UNIT-IV

4. Comprehension of unseen & seen passages.

Essay writing (500 words) about any sports tournament/Event.

News writing about any Match or sports event.

Interview- when how,types if interview.

UNIT-V

5. Text

Tales from freedom fights of India.

Binging of Freedom fight 1857 to 1900

Mahatma Gandhi and freedom Fight

Netaji Shubhas Chand Bose and Freedom fight.

Dissolve of various kingdom in India.

After 1947 Fights of India Short Stories freedom of Goa, India -China, India - Pakistan,

India - Pakistan 71, Kargilwar.



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EDUCATIONAL TECHNOLOGY

UNIT-I

1.1 Introduction to Educational Technology

Definition

Educative Process

The Teacher of Yesterday & Today

An outline of Teaching method used then and now

Use of sensory organ in the process of learning and remembering.

Communication:

Types of Communication

Communication Cycle

Communication in the Class room

UNIT-II**2. Teaching Aids:**

Importance of Teaching Aids.

Criteria for selecting Teaching Aid

Difference between Teaching Method and Teaching Aid.

Broad classification of Teaching Aids.

Audio Aids

Visual Aids

Audio-Visual Aids

Effectiveness of Edger Dale's cone classification

Advantage and suggestions for effective use of selected teaching Aids.

Verbal

Chock Board

Charts

Models

Slide Projector

Over Head Projector



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Motion Picture

Self Experiment and Projects.

3. New Teaching Techniques and Innovations-I:

Art of questioning and answering

Purpose of Questioning

Classification of Questioning

Techniques of asking questions

Programmed Learning

Concept of Programmed learning

Fundamental Principles of Programmed learning

Steps involved in preparation of Programme

Team Teaching

Meaning

Guiding principles of Team Teaching

Advantage of Team Teaching.

UNIT-IV

4. New Teaching Techniques and INNOVATIONS-II:

Micro Teaching

Concept and Features of Micro Teaching.

Micro Teaching Verses Traditional Teaching

Steps in Micro Teaching

Principles of Micro Teaching

Micro Teaching Skills

Limitation of Micro Teaching

Simulation Teaching

Meaning of Simulation

Types of activities in simulation

Steps in Simulation

Advantages of Simulation

Limitations of Simulation

System Approach

Definition of System

Components of an instructional system



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Flow Diagram for designing a system

Advantages of system approach

Teaching Machines

Meaning of teaching machines

Components of an instructional system

Flow Diagram for designing a system

Computer-Assisted Instruction.

WRESTLING

1. Introduction of the Wrestling and historical development with special reference to India.
2. Important tournaments held at National and International levels and distinguished personalities related to the game.
3. Fundamental Skills.

Learning and demonstrating fundamental skills involving drills and lead up games, if any, therein (Catch as can style).

Take downs: leg tackles, arm drag.

Counters for take downs: Cross face, whizzer series.

Escapes from under: Sitout-turn in triped.

Counters for escapes from under: Basic control, Back drop, counters for stand up.

Pinning combination: Nelson series, (Half Nelson, Half Nelson and bar arm) leg lift series, leg cradle series, Reverse double arm, chicken wing and half nelson.

Escapes from pinning: Wing lock series, Double arm lock roll, bridge.

Standing Wrestling: Head under arm series whizzer series.

Referees positions.

4. Rules and their interpretations and duties of officials.

TENNIS

1. Introduction of the game and Historical development with special reference to India.

2. Important Tournaments held at National and International levels and distinguished personalities related to the game.
3. Fundamental Skills

Grips

Eastern Forehand grip

Eastern Backhand grip

Western grip

3.1.5 Chopper grip

Stance and foot work.

Basic Ground strokes

Fore hand drive

Backhand drive

Basic Service

Basic Volley

Overhead Volley

Chop

Rules and their interpretations and duties of officials

VOLLEYBALL

- 1 Introduction of the game and historical development with special reference to India.



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2. Important tournaments help at National and International levels and distinguished personalities related to the game.
3. Fundamental Skills.

Player's stance- Receiving the ball & passing to the team mates.

The Volley (Over head pass)

The Dig (Under hand pass).

Service-

Under arm Service.

Side Arm Service.

Tennis Service.

Round Arm Service.

Lead up Games-

Three Volleys (These can be combined with service)

Three Digs (Receiving service using dig and setting and placing using volleying action)

Spike-

Straight Arm Spike.

Round Arm Spike.

Block- 3.5.1 Single Block.

4. Advanced Skills-

Pass-

Back Pass.

Back Roll Volley.

Back Roll Dig.

Jump and Pass.

Side Roll Dig.

Service.

Side Arm Floater.

Over head Floater.

Spike-

Spiking cross court.

Spiking down the line.

Block-

Double Block

Triple Black

Dive-

Dive combined with dig (Two handed).

Dive combined with dig (One handed).

5. Rules and their interpretations and duties of officials.



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EE-51 - UTILIZATION OF ELECTRICAL ENERGY

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Various electric traction systems with their performance.
- 2 Selection of motor for different industrial drives.
- 3 Electric heating and welding techniques.
- 4 Designing and selection of lamps for proper illumination

Unit-1

Illumination Engineering- Nature of light, units, sensitivity of the eye, luminous efficiency, glares. Production of Light; Incandescent lamps, arc lamps gas discharge lamps- fluorescent lamps polar curves, effect of voltage variation on efficiency and life of lamps, Distribution and control of light, lighting calculations, solid angle, inverse square and cosine laws, methods of calculations, factory lighting, flood lighting and street lighting, Direct diffused and mixed reflection & transmission factor, refractors, light fittings.

(12 HRS)

Unit-2

Heating, Welding And Electrolysis- Electrical heating-advantages, methods and applications, resistance heating, design of heating elements, efficiency and losses control. Induction heating: core type furnaces, core less furnaces and high frequency eddy current heating, dielectric heating: principle and special applications, arc furnaces: direct arc furnaces, Indirect arc furnaces, electrodes, design of heating elements, power supply and control. Different methods of electrical welding, resistance welding, arc welding, energy storage welding, laser welding, and electron beam welding, and electrical equipment for them. Arc furnaces transformer and welding transformers. Review of electrolytic principles, laws of electrolysis, electroplating, anodizing-electro-cleaning, extraction of refinery metals, power supply for electrolytic process, current and energy efficiency.

(12 HRS)

Unit-3

Traction - Special features of Traction motors, selection of Traction Motor, Different system of electric traction and their Advantages and disadvantages, Mechanics of train movement: simplified speed time curves for different services, average and schedule speed, tractive effort, specific energy consumption, factors affecting specific energy consumption, acceleration and braking retardation, adhesive weight and coefficient of adhesion,

(12 HRS)

Unit-4

Electric Drives - Individual and collective drives- electrical braking, plugging, rheostat and regenerative braking load equalization use of fly wheel criteria for selection of motors for various industrial drives, calculation of electrical loads for refrigeration and air-conditioning, intermittent loading and temperature rise curve.

(12 HRS)



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Unit-5

VTH SEMESTER

Introduction to Electric and Hybrid Vehicles - Configuration and performance of electrical vehicles, traction motor characteristics, motor selection, transmission requirement, vehicle performance and energy consumption.

CATEGORY: DG

(12 HRS)

COURSE OUTCOME:

After completion of this course, students will be able to

- 1 Choose the motor for different types of Electric traction systems.
- 2 Valuate the selection of a motor for different types of loads.
- 3 Use various heating and welding techniques for different applications.
- 4 Select and design the lamps for proper illumination. & Solve Problem.

References:

1. Open Shaw, Taylor, .Utilization of electrical energy., Orient Longmans, 1962.
2. H. Pratap, Art and Science of Utilization of Electrical Energy.
3. Gupta, J.B., Utilization of Elect. Energy ,Katariya and sons, New Delhi.
4. Garg, G.C., Utilization of Elect. Power and Elect. Traction.
5. N V Suryanarayan, Utilization of Elect. Power including Electric Drives and Elect. Traction, New Age International.
6. Hancock N N, Electric Power Utilisation, Wheeler Pub.
7. Mehrdad,Ehsani,Yimin Gao,Sabastien.E. Gay,Ali Emadi, “Modern electric, hybrid electric and fuel cell vehicles”, CRC Press.



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EE-52 - ELECTRICAL MACHINE-II

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Understanding the Construction, Working Principles and Performance characteristics of 3- Φ Induction Motors.
- 2 Knowing the Construction, Working Principles and Performance characteristics 3- Φ Synchronous Generators.
- 3 Understanding the Construction, Working Principles, Performance behavior and Applications of 3- Φ Synchronous Motors.
- 4 Knowing the Construction, Working principles and Various Applications of 1- Φ Induction Motors and Special Purpose Machines.

Unit-1

D.C. Machine-I- Basic construction of DC machines; types of DC machines and method of excitation; lap and wave windings; Emf equation; armature reaction and methods of limiting armature reaction; Commutation process and methods for improving commutation; Basic performance of DC generators and their performance characteristics; Metadyne and Amplidyne; permanent magnet DC motors; Brush less dc motors.

(12 HRS)

Unit-2

D.C. Machine-II - Basic operation of DC motors; Torque equation; Operating characteristics of DC motors, Starting of DC motors- 2point, 3 point and 4 point starters; speed control of DC motors; losses and efficiency of DC machines; testing of DC machines, direct testing, Swinburne's test and Hopkinson's test. Application of DC machines.

(12 HRS)

Unit-3

Synchronous Machine-I- Construction; types of prime movers; excitation system including brushless excitation; polyphase distributive winding, integral slot and fractional slot windings; emf equation, generation of harmonics and their elimination; armature reaction; synchronous reactance and impedance, equivalent circuit of alternator, relation between generated voltage and terminal voltage, voltage regulation of alternators using synchronous impedance, mmf, zpf and new A.S.A method.

(12 HRS)

Unit-4

Synchronous Machine-II - Salient pole machines; two reaction theory equivalent circuit model and phasor diagram; determination of X_d and X_q by slip test; SCR and its significance; regulation of salient pole alternator, power angle equation and characteristics; synchronizing of alternator with infinite busbar; parallel operation and load sharing; synchronizing current, synchronizing power and synchronising torque coefficient; synchroscopes and phase sequence indicator; effect of varying excitation and mechanical torque.

(12 HRS)



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VTH SEMESTER

Unit-5

Synchronous machine-III - Synchronous motor speed, starting and stopping of synchronous motor, pull in torque, motor under load power and torque, reluctance torque, effect of excitation, effect of armature reaction, power factor adjustment, V curves, inverted V curves, synchronous motors as power factor correcting device, super synchronous and sub synchronous motors, hunting and damper winding efficiency and losses. Analysis of short circuit oscillogram, determination of various transient, sub transient and steady reactances and time constants, expression of transient and sub transient reactances in terms of self and mutual inductances of various winding, short circuit current, equivalent circuit. Single phase synchronous motors- hysteresis motor, reluctance motor. Repulsion motor, stepper motor, switched reluctance.

(12 HRS)

COURSE OUTCOME:

After completion of this course, students will be able to

- 1 Describe the Performance Characteristics, starting methods and speed control of 3- Φ Induction motors.
- 2 Evaluate the Performance of 3- Φ Synchronous Generators. & Solve Problems
- 3 Identify Applications of 3- Φ Synchronous Motors & Solve Problems.
- 4 Explain the Various Applications of 1- Φ induction Motors and Special Purpose Machines.

List of Experiments (expandable)

Experiments can cover any of the above topics, following is a suggestive list:

1. To plot magnetization characteristic of a separately excited DC generator
2. To perform load test on DC generators.
3. To perform load test on DC series and shunt motor
4. To perform Swinburne's test on a DC machine and find out its efficiency under full load condition.
5. To conduct Hopkinson's test on a pair of DC shunt machine.
6. To perform OCC and SCC test on an alternator and determine its regulation.
7. To determine regulation of alternator using mmf and zpf methods.
8. To synchronise alternator with infinite bus bar.
9. To plot V and inverted V curves for a synchronous motor
10. To find X_d and X_q of salient pole synchronous machine by slip test.
11. To Determine negative sequence and zero sequence reactance of an alternator.
12. To determine subtransient direct axis and quadrature axis synchronous reactances of salient pole machine.



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Reference Books:

1. M.G. Say, Performance & design of AC machines, CBS publishers & distributors, Delhi, 3rd edition
2. A.E. Clayton & N.N. Nancock, The Performance & design of DC machines CBS publications & distributors, Delhi, 3rd edition
3. P.S. Bhimbra, Electrical Machinery, Khanna Pub.
4. 4. P.S. Bhimbra, Generalized theory of Electrical Machines, Khanna publishers, Delhi,
5. Ashfaq Husain, Electric Machines, Dhanpat Rai, New Delhi
6. I.J. Nagrath & D.P. Kothari, Electric Machines, Tata McGraw Hill, New Delhi,
7. 6. Syed A. Nasar, Electric Machines & Power Systems, Volume I, Tata McGraw Hill, New Delhi
8. 7. A. E. Fitzgerald, C. Kingsley & S.D. Umans, Electric Machinery Tata McGraw Hill, New Delhi, 5th edition



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VTH SEMESTER

CATEGORY:-DC

EE-53 - MICROPROCESSORS & MICROCONTROLLERS

(μ P & μ C)

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Architecture of 8085 microprocessor.
- 2 Assembly language programming.
- 3 Memory segmentation concept.
- 4: Architecture of 8051 microcontroller and its interfacing

Unit-1

Microprocessor 8086: Introduction to 16-bit 8086 microprocessors, architecture of 8086, Pin Configuration, interrupts, minimum mode and maximum mode, timing diagram, Memory interfacing, Comparative study of Salient features of 8086, 80286 and 80386.

(12 HRS)

Unit-2

Microprocessor 8086 programming: Instruction set of 8086, Addressing mode, Assembler directives & operations, assembly and machine language programming, subroutine call and returns, Concept of stack, Stack structure of 8086, timings and delays,

(12 HRS)

Unit-3

Input-Output interfacing: Memory Mapped I/O and Peripherals I/O. PPI 8255 Architecture and modes of operation, Interfacing to 16-bit microprocessor and programming, DMA controller (8257) Architecture, Programmable interval timer 8254, USART 8251, 8 bit ADC/DAC interfacing and programming.

(12 HRS)

Unit-4

Microcontroller 8051: Intel family of 8 bit microcontrollers, Architecture of 8051, Pin description, I/O configuration, interrupts; Interrupt structure and interrupt priorities, Port structure and operation, Accessing internal & external memories and different mode of operations, Memory organization, Addressing mode, instruction set of 8051 and programming.

(12 HRS)

Unit-5

8051 Interfacing, Applications and serial communication: 8051 interfacing to ADC and DAC, Stepper motor interfacing, Timer/ counter functions, 8051 based thyristors firing circuit, 8051 connections to RS-232, 8051 Serial communication, Serial communication modes, Serial communication programming, Serial port programming in C.

(12 HRS)



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COURSE OUTCOME:

After completion of this course, the students' will be able to

- 1 Explain the architecture of 8085.
- 2 Write assembly language programs by using 8085 microprocessor.
- 3 Discuss the various addressing modes of 8086 microprocessor.
- 4 Explain the architecture of 8051 microcontroller and its interfacing

Reference Books:

1. Hall Douglas V., Microprocessor and interfacing, revised second edition 2006, Macmillan, McGraw Hill.
2. A.K. Ray & K.M.Bhurchandi, Advanced Microprocessors and peripherals- Architecture, Programming and Interfacing, Tata McGraw – Hill, 2009 TMH reprint.
3. Kenneth J. Ayala, The 8086 microprocessor: programming and interfacing the PC, Indian - edition, CENGAGE Learning.
4. Muhammad Ali Mazidi and Janice Gillespie Mazidi, the 8051 Microcontroller and Embedded Systems, Pearson education, 2005.
5. Kenneth J. Ayala, the 8051 Microcontroller Architecture, III edition, CENGAGE Learning.
6. V.Udayashankara and M.S.Mallikarjunaswamy, 8051 Microcontroller: Hardware, Software & Applications, Tata McGraw – Hill, 2009.
7. McKinley, the 8051 Microcontroller and Embedded Systems – using assembly and C, PHI, 2006 / Pearson, 2006.
8. Microprocessor and Interfacing, I edition 2012, oxford press setnil kumar, Saravam Jeevanathan shah.

List of Experiment:

Introduction

1. Introduction to 8086 & 8051 kit, hardware features & modes of operation.
2. Introduction to 8086 & 8051 kit, hardware features & modes of operation.
3. Technique of programming & basic commands of kit.
4. Instruction set of 8086 & 8051.

Assembly language programming of 8086 & 8051.

1. Write a program to add two 8-bit numbers.
2. Write a program to add two 16-bit numbers.
3. Write a program for 8-bit decimal subtraction.
4. Write a program to find 1's complement and then 2's complement of a 16-bit numbers.
5. Write a program to find larger of two numbers.
6. Write a program to shift an 8-bit number left by 2-bits.
7. Write a program to multiply two 16-bit numbers.
8. Write a program for factorial of given number by recursion.



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9. Write a program to square of an 8-bit number.
10. Write a program to generate a square wave of 2 KHz Frequency on input pin.



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CATEGORY:-DC

EE-54 – POWER ELECTRONICS

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Characteristics and applications of basic power semiconductor switches
- 2 Performance of controlled rectifiers.
- 3 Performance of chopper, inverter operation
- 4 A.C voltage controllers & Cyclo converter operation and power electronic applications in industry.

Unit-1

Advantages and application of power electronic devices characteristics, Symbol & application of power diodes, power transistors, GTO, Triac, Diac, Power MOSFET, IGBT, LASCR, Fast recovery diode, Scotty diode MCTs. Principle of operation of SCR, Two transistor analogy, brief idea of construction of SCR, Static characteristics of SCR, Condition of turn on & off of SCR Gate characteristics, Method for turning on of SCR, Turnoff methods, different commutation techniques (Class A,B,C,D,E, & F Commutation) firing of SCR, Use of public transformer and opto isolator in firing, Resistance firing Ckt, Resistance capacitance firing circuit, UJT firing cut, and ramp triggering, firing for 3- Φ circuit. SCR rating & protection of SCR over voltage, Over current, Superior firing, Design of snubber circuit and protection of gate of SCR, heating, cooling & mounting of SCR series and parallel operation of SCR, String efficiency & problem associated with series and parallel operation of SCR.

(12 HRS)

Unit-2

Operation and analysis of single phase (Half wave & Full Wave) and multiphase (Three Phase) uncontrolled and controlled rectifier circuit with resistive, resistive & inductive load (continuous & non continuous conduction, Fw small & very large inductive loads) and RLE loads. Estimation of average load voltage and load current for above rectifier circuits active and reactive power input. Effect of free wheeling diode and source inductance on performance of these rectifier circuits. Comparison of mid point & Bridge rectifier circuits.

(12 HRS)

Unit-3

Series and parallel inverter, Voltage source & current source inverter, Single phase and three phase bridge inverter, Self cumulated inverters,, Mc- murray & MC murray bed ford inverters, Voltage control of single phase and three phase bridge inverter, Harmonics & their reduction techniques.

(12 HRS)

Unit-4

Principle of chopper operation, Various control strategies in chopper, Step up & step-up/step down choppers, chopper configuration (Type A,B, C,D, & E), Steady state analysis of chopper circuits, Current & voltage commutation of chopper circuits Jones & Morgens chopper.

(12 HRS)



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Unit-5

VTH SEMESTER

Single phase (mid point & bridge configuration) and three phase cyclo convertor configuration and operating principles. AC voltage controllers (using SCRs & Traic) single phase full wave controller with R and RL load, Estimation of RMS load voltage, RMS load current and input power factor, three phase AC voltage controller (Without analysis) Dual converter Switched mode voltage regulator buck, Boost, Buck & Boost, Ck regulators.

(12 HRS)

COURSE OUTCOME:

After completion of this course, students will be able to

- 1 Determine the power semiconductor switches characteristics and their applications & design of snubber circuit
- 2 Evaluate the performance of rectifiers. & Solve Problems
- 3 Aanalyze & describe the operation of inverters and choppers & Solve Problems
- 4 Evaluate the performance of AC voltage controllers and Cycloconverters

References:

1. M.H. Rashid, Power Electronics Circuits, Devices and Applications, Pearson Education, Singapore, 1993.
2. M Ramsmoorthy, An Introduction to transistor and their application, Affiliated East-West Press.
3. P.C. Sen, Power Electronics, TMH.
4. M.D. Singh, K.B. Khanchandani, Power Electronics, TMH, Delhi, 2001.
5. Chakravarti A., Fundamental of Power Electronics and Drives, Dhanpat Ray & Co.,

LIST OF EXPERIMENTS :-

1. V-I characteristics of SCR
2. V-I characteristics of DIAC
3. V-I characteristics of BJT
4. V-I characteristics of TRIAC
5. VI characteristics of MOSFET
6. Transfer characteristics of MOSFET
7. Output characteristics of IGBT
8. Transfer characteristics of IGBT
9. Single phase SCR half controlled converter with R load
10. 1- Φ SCR fully controlled converter with r-load
11. Study of 3 ϕ SCR Half controlled converter
12. Study of 3 ϕ SCR Fully controlled converter
13. Study of classes of commutation A,B,C,D,E & F.



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CATEGORY:-DE

EE-55 - ELECTRICAL ENERGY CONSERVATION AND AUDITING

TOTAL-60HOURS

COURSE OBJECTIVE:

To impart basic knowledge to the students about current energy scenario, energy conservation, audit & management. To inculcate among the students systematic knowledge & skill about assessing the energy efficiency, energy auditing & energy management.

Unit-1

General energy problem: Energy use patterns and scope for conservation. Energy audit: Energy monitoring, Energy accounting and analysis, Auditing and targeting. Energy conservation policy, Energy management & audit, Energy audit, Types of energy audit, energy management (audit), qualities and function of energy managers, language of an energy manager, Questionnaire, Check list for top management, Loss of energy in material flow, energy performance, Maximizing system efficiency, Optimizing, input energy requirements, Energy auditing instruments, Material load energy balance diagram..

(12 HRS)

Unit-2

Thermodynamics of Energy Conservation. Basic principle. Irreversibility and second law efficiency analysis of systems. Primary energy sources, optimum use of prime-movers, energy efficient house keeping, energy recovery in thermal systems, waste heat recovery techniques, thermal insulation. Thermal energy audit in heating, ventilation and air conditioning. Maintenance and Energy audit – friction, lubrication and tribo-logical innovations. Predictive and preventive maintenance.

(12 HRS)

Unit-3

Load curve analysis & load management DSM, Energy storage for power systems (Mechanical, Thermal, Electrical & Magnetic) Restructuring of electric tariff from energy conservation consideration, Economic analysis depreciation method, time value of money, Evaluation method of projects, replacement analysis, special problems inflation risk analysis. Pay back period, Energy economics, Cost Benefit Risk analysis, Pay back period..

(12 HRS)

Unit-4

Energy efficient electric drives, Energy efficient motors V.S.D. power factor improvement in power system. Energy Conservation in transportation system especially in electric vehicle. Energy flow networks, Simulation & modeling, formulation & Objective & constraints, alternative option, Matrix chart.

(12 HRS)

Unit-5

Energy conservation task before industry, Energy conservation equipments, Co-Generation, Energy conservation process, Industry Sugar, Textiles, Cement Industry etc Electrical Energy Conservation in building, heating and lighting. domestic gadgets .

(12 HRS)



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VTH SEMESTER

COURSE OUTCOME:

1. Students will be able to apply the knowledge of the subject to calculate the efficiency of various thermal utilities.
2. Students will be able to design suitable energy monitoring system to analyze and optimize the energy consumption in an organization.
3. Students will be able to improve the thermal efficiency by designing suitable systems for heat recovery and co-generation.
4. Students will be able to use the energy audit methods learnt to identify the areas deserving tighter control to save energy expenditure.
5. Students will be able to carry out the cost-benefit analysis of various investment alternatives for meeting the energy needs of the organization.

References:

1. Energy Management – W.R. Murphy & G. Mckey Butler worths. E
2. Energy Management Head Book- W.C. Turner, John Wiley
3. Energy Management Principles- Craig B. Smith, Pergamon Press
4. Energy Conservation- Paul O Callagan- Pergamon Press
5. Design & Management of energy conservation. Callaghan,
6. Elect, Energy Utilization & Conservation. Dr. Tripathi S.C.,



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CATEGORY:-DEL

EE-56 - INDUSTRIAL TRAINING (2 WEEKS)

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

1. To provide comprehensive learning platform to students where they can enhance their employ ability skills and become job ready along with real corporate exposure.
2. To enhance students' knowledge in one particular technology.
3. To Increase self-confidence of students and helps in finding their own proficiency
4. To cultivate student's leadership ability and responsibility to perform or execute the given task.
5. To provide learners hands on practice within a real job situation.

Concept and guideline

Student will undertake a small project which will pertain to live problems of Industry\Community. The project may be related to use of technology in industry or transfer of technology to introduce value addition for agriculture, improving health & hygienic, energy management & conservation, optimal use of local resources or in the new product areas. The student can undertake project singly or in a batch (of not more than five students). At the end of project student will submit a project report which will contain details of the problem identified and solution suggest for it.

COURSE OUTCOME:

After completion of this course, students will be able to

1. Become master in one's specialized technology
2. Become updated with all the latest changes in technological world.
3. Ability to communicate efficiently.
4. Knack to be a multi-skilled engineer with good technical knowledge, management, leadership and entrepreneurship skills.
5. Ability to identify, formulate and model problems and find engineering solution based on a systems approach.



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VTH SEMESTER

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CATEG

ORY: -

PDFS

BE-53 PROFESSIONAL DEVELOPMENT

FINISHING SCHOOL LEVEL-III

TOTAL - 36 HOURS

OBJECTIVE

The students are to be groomed with respect to personality development. Further improvement in English, Aptitude and reasoning ability is desirable.

UNIT-I

Conversational English:

Grammar mainly Active and Passive Voice, 250 sentences of daily use irrespective of any specific tenses.

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Conversational English:

100 sentences of daily use related to professional and formal environment Report Writing with necessary punctuations and with editor's eye, Thematic Apperception, Expression of Feelings 2- minutes Talk by the students, Smart Etiquettes and Tidiness .

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Aptitude/Reasoning

Quantitative Aptitude and Logical Reasoning

– Level III Problem solving on,

Average, Time work, percentage, Probability, Permutation and Combination.

Question- Statements, Theme Detection, Statement Assumptions,

Statement Argument. **(12 HRS)**



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VTH SEMESTER

OUTCOME

Further improvement in reading, writing and vocal English is achieved. Aptitude and reasoning aspect shows improvement.



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VTH SEMESTER

NATIONAL CADET CORPS



GROOMING TOMORROW'S LEADERS

**OFFERING NCC A GENERAL GENERIC ELECTIVE CREDIT
COURSE IN UNIVERSITIES UNDER CHOICE BASED CREDIT
SYSTEM TO ALIGN WITH NEW EDUCATION POLICY 2020**



SARVEPALLI RADHAKRISHNAN UNIVERSITY,



BHOPAL
CONTENTS
VTH SEMESTER

5. Section I : NCC Credit Course Design
6. Section II : NCC Credit Course Rules & Regulations aligned to UGC.

NATIONAL CADET CORPS

SECTION I: NCC CREDIT COURSE DESIGN DOCUMENT

UNDER CHOICE BASED CREDIT SYSTEM AS GENERAL ELECTIVE FOR SENIOR DIVISION / SENIOR WING

7. **Preamble.** The National Cadet Corps (NCC) is governed by NCC Act 1948 and attendant NCC Rules. It functions under the Ministry of Defence and is headed by DGNCC. It is organised into 17 State Directorates each headed by an Additional/Deputy Director General. The aims of NCC are:-

- (a) To develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.
- (b) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regard less of which career they choose.
- (c) To provide a conducive environment to motivate young Indians to choose the Armed Forces as a career.

8. **Purpose.** Currently NCC training is imparted as extra-curricular activity to volunteer students from recognized schools and colleges who enroll as cadets. NCC as a Credit Course is designed with an intent to transform NCC training into a curricular activity from an extra-curricular thereby providing academic credits to students undergoing NCC training along with other attended advantages to the cadets in the college/ university.

9. **Introduction to NCC Credit Course Design.** Institutional Training is the mainstay of NCC training and it is conducted at colleges and universities by Associate NCC Officers and Armed Forces personnel. The application of knowledge gained through institutional training is further honed or developed to a higher degree in NCC Camps. The Institutional Training syllabus comprises Common Subjects and Specialised Subjects (military component). NCC Credit Course is designed to offer Institutional Training of

Senior Wing /Division is over six semesters (three years), comprising ~~300~~ periods (excluding Camp), of which 120 periods are meant for theory with ~~08~~ credits and 180 periods for practical with 6 credits. Each period is counted as hour. The ratio between theory and practical in terms of number of hours of training is 5:6, but in terms of credits is 5:3, since as per CBCS two hours of practical is counted towards one period of training as against one hour for theory. In addition two separate courses have been designed for two Camps normally referred to as Annual Training Camps (ATC).

**BHOPAL****VTH SEMESTER**

Training schedules planned for cadets ensure that the optimum benefits of the NCC organization reach maximum number of cadets. The main emphasis is on practical training which in consonance with theory is made to facilitate active participation of learner, better assimilation of knowledge, and proper development of various skills, strengthening of mind and body which is the bedrock of NCC training.

NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	01	01	-	02	
Semester - II	01	01	-	02	
Semester - III	01	01	05	07	Credits of 1st Camp merged with 3rd Sem
Semester - IV	02	01	-	03	
Semester - V	01	01	05	07	Credits of 2nd Camp merged with 5th Sem
Semester - VI	02	01	-	03	
Total	08	06	10	24	Twenty-Four Credits

INSTITUTIONAL TRG SYLLABUS

COMMON SUBJECTS				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	NCC General	06	-	06
2	National Integration	04		04
3	Drill	-	45	45
4	Weapon Training	-	25	25
5	Personality Development	25		25
6	Leadership	12	-	12
7	Disaster Management	13		13
8	Social Service & Community Development	08	39	47
9	Health & Hygiene	-	10	10
10	Adventure	01		01
11	Environmental awareness & conservation	03		03
12	Obstacle Training	-	09	09
13	General Awareness	04		04
14	Border & Coastal Areas	06		06
TOTAL HOURS COMMON SUBJECTS(a)		82	128	210

SPECIALISED SUBJECTS (ARMY)				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practical	
1	Armed Forces	09	-	09
2	Map Reading	-	24	24
3	Communications	03	03	06
4	Infantry Weapons	03	03	06
5	Field Craft & Battle Craft		22	22
6	Military History	23	-	23
Total Hours		38	52	90

**BHOPAL****SPECIALISED SUBJECTS (NAVY)**

Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	Naval Orientation	12	-	12
2	Naval Communication	02	18	20
3	Navigation	02	03	05
4	Seamanship	15	18	33
5	Fire Fighting and Damage Control	04	03	07
6	Ship and Boat Modelling	03	10	13
Total hours		38	52	90

SPECIALISED SUBJECTS (AIR FORCE)

Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	General Service Knowledge	08	-	08
2	Air Campaign	06	02	08
3	Principles of flight	06	06	12
4	Airmanship	01	07	08
5	Navigation	05	-	05
6	Aeroengines	06	-	06
7	Basic flight Instruments	03	03	06
8	Aero modelling	03	34	37
Total Hours		38	52	90

INSTITUTIONAL TRAINING: TOTAL HOURS & CREDITS

INSTITUTIONAL TRAINING: TOTAL HOURS & CREDITS			
ITEM	Periods (1 hour duration each)		Total
	Lectures/Tutorials	Practicals	
TOTAL HOURS COMMON SUBJECTS	82	128	210
TOTAL HOURS SPECIALISED SUBJECTS (ARMY/NAVY/AIR FORCE)	38	52	90
TOTAL HOURS INSTITUTIONAL TRAINING	120	180	300
TOTAL CREDITS INSTITUTIONAL TRAINING	08 CREDITS (15 HOUR THEORY = 1 CREDIT POINT)	6 CREDITS (30 HOURS PRACTICAL TRAINING = 1 CREDIT POINT)	



BHOPAL
NCC CAMP TRAINING SYLLABUS
VTH SEMESTER

COMMON SUBJECTS				
S No.	Subjects	Periods		Total
		L/T	P	
1.	Physical Training	-	18	18
2.	Drill	-	32	32
3.	Weapon Training	08	28	36
4.	National Integration and Awareness	08	-	08
5.	Personality Development	08	12	20
6.	Leadership	08	-	08
7.	Disaster Management	08	-	08
8. .	Social Service and Community Development	-	08	08
9.	Health & Hygiene	08	-	08
10.	Obstacle Training	-	04	04
11.	Military History	04	-	04
12.	Communication	04	-	04
13.	Games	-	18	18
14.	Culture	-	18	18
	TOTAL	56	138	194
<u>SPECIALISED SUBJECTS</u>				
1.	Map Reading	-	24	24
2.	Infantry Weapons	04	02	06
3.	Field Craft & Battle Craft	-	16	16
	TOTAL	04	42	46
	GRAND TOTAL	60 (4 credit)	180 (6 credit)	240 (10 credit)

NCC CAMP TRAINING SYLLABUS (FOR THEORY)

Ser No	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Weapon Training	-	-	04	-	04	-	08
2.	National Integration & Awareness	-	-	04	-	04	-	08
3.	Personality Development	-	-	04	-	04	-	08
4.	Leadership	-	-	04	-	04	-	08
5.	Disaster Management	-	-	04	-	04	-	08
6.	Health & Hygiene	-	-	04	-	04	-	08
7.	Military History			02		02		04
8.	Communication			02		02		04
9.	Infantry Weapons	-	-	02	-	02	-	04
	TOTAL	-	-	30	-	30	-	60
	TOTAL Credit	-	-	2	-	2	-	4

NCC CAMP TRAINING SYLLABUS (FOR PRACTICAL)

Ser No	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Physical Training	-	-	09	-	09	-	18
2.	Drill	-	-	16	-	16	-	32
3.	Weapon Training	-	-	14	-	14	-	28
4.	Personality Development	-	-	06	-	06	-	12
5.	Social Service and Community Development	-	-	04	-	04	-	08
6.	Obstacle Training	-	-	02	-	02	-	04
7.	Games			09		09		18
8.	Culture			09		09		18
9.	Map Reading	-	-	12	-	12	-	224
10.	Infantry Weapons	-	-	01	-	01	-	02
11.	Field Craft & Battle Craft	-	-	08	-	08	-	16
	TOTAL			90		90		180
	TOTAL CREDIT			03		03		06



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BHOPAL**



VTH SEMESTER

SEMESTER WISE COURSE DESIGN ARMY CADETS

NATIONAL CADET CORPS

**INSTITUTIONAL TRAINING: SEMESTER WISE DISTRIBUTION OF NCC
SYLLABUS FOR THEORY (ARMY CADETS)**

S. NO.	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	NCC General	06	-	-	-	-	-	06
2.	National Integration	04	-	-	-	-	-	04
3.	Personality Development	02	05	05	04	06	04	25
4.	Leadership	-	05	04	03	-	-	12
5.	Disaster Management	-	-	03	10	-	-	13
6.	Social Service & Community Development	03	05	-	-	-	-	08
7.	Adventure	-	-	01	-	-	-	01
8.	Environmental Awareness & Conservation	-	-	-	03	-	-	03
9.	General Awareness	-	-	-	04	-	-	04
10.	Border & Coastal Areas	-	-	02	-	02	02	06
11.	Armed Forces	-	-	-	06	-	03	09
12.	Infantry Weapons	-	-	-	-	3	-	3
13.	Communication	-	-	-	-	-	03	03
14.	Military Hospital	-	-	-	-	04	19	23
	TOTAL	15	15	15	30	15	30	120
	TOTAL Credit	1	1	1	2	1	2	08

**INSTITUTIONAL TRAINING: SEMESTER WISE DISTRIBUTION OF NCC
SYLLABUS FOR PRACTICAL (ARMY CADETS)**

S. NO.	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Drill	12	12	08	07	03	03	45
2.	Field Craft & Battle Craft	03	04	04	04	04	03	22
3.	Map Reading	03	05	04	04	04	04	24
4.	Weapons Training	05	04	04	04	04	04	25
5.	Communication	-	-	-	-	-	03	03
6.	Infantry Weapons	-	-	-	-	-	03	03
7.	Social Service & Community Development	07	05	05	06	06	10	39
8.	Health & Hygiene	-	-	-	05	05	-	10
9.	Operation Training	-	-	05	-	04	-	09
	TOTAL	30	30	30	30	30	30	180
	TOTAL Credit	01	01	01	01	01	01	06



BHOPAL

INSTITUTIONAL TRAINING: SEMESTER WISE THEORY DETAILED SYLLABUS (ARMY CADETS)

<u>SEMESTER I</u>					
S.No	Subject	Periods	Chapter	Lesson	Hours
1	NCC General	6	NCC-I	Aims, Objectives and Org of NCC	1
			NCC-II	Incentives	2
			NCC-III	Duties of NCC Cadets	1
			NCC-IV	NCC Camps: Types and Conduct	2
2	National Integration and Awareness	4	NI-I	National Integration: Importance and Necessity	1
			NI-II	Factors affecting National Integration	1
			NI-III	Unity in Diversity	1
			NI-IV	Threats to National Security	1
3	Personality Development	2	PD - I	Factors Self-Awareness Empathy Critical and Creative Thinking Decision Making and Problem Solving	2
4	Social Service and Community Development	3	SSCD - I	Basics of Social Service Rural Development Programmes NGO's Contribution of Youth	3
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER II					
S.No	Subject	Periods	Chapter	Lesson	Hours
5	Personality Development	5	PD-II	Communication Skills	3
			PD-III	Group Discussion -Coping with Stress and Emotions	2
6	Leadership	5	L-I	Leadership Capsule Traits Indicators Motivation Moral Values Honour Code	3
			L-II	Case Studies Shivaji, Jhansi Ki Rani,	2
7	Social Service and Community Development	5	SS-IV	Protection of Children & Women Safety	1
			SS-V	Road/Rail Travel Safety	1
			SS-VI	New Initiatives	2
			SS-VII	Cyber and Mobile Security Awareness	1
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER III					
S.No	Subject	hours	Chapter	Lesson	HOURS
8	Personality Development	5	PD-III	Group Discussions - Change your Mindset	2
			PD-V	Public Speaking	3
9	Leadership	4	L-II	Case Studies – APJ Abdul Kalam, Deepa Malik, Maharana Pratap, N Narayan Murthy	4
10	Disaster Management	3	DM-I	Disaster Management Capsule Organisation Types of Disasters Essential Services Assistance Civil Defence Organisation	3
11	Adventure	1	AD-I	Adventure activities	1
12	Border & Coastal Areas	2	BCA-I	History, Geography & Topography of Border/ Coastal Areas	2
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER IV					
S.No	Subject	hours	Chapter	Lesson	HOURS
13	Personality Development	4	PD-III	Group Discussions - Time Management, Social Skills	4
14	Leadership	3	L-II	Case Studies – Ratan Tata, Rabindra Nath Tagore, Role of NCC cadets in 1965 war	3
15	Disaster Management	9	DM-II	Initiative Trg, Organising Skills, Dos and Don'ts Natural Disasters Man Made Disasters	9
		1	DM-III	Fire Services and Fire Fighting	1
16	Environmental Awareness	3	EA-I	Environmental Awareness and Conservation	3
17	General Awareness	4	GA-I	General Awareness	4
18	Armed Forces	6	AF-1	Army, Navy, Air Force and Central Armed Police Forces	6
TOTAL HOURS					30
TOTAL CREDITS					2



SEMESTER V

S.No	Subject	hours	Chapter	Lesson	HOURS
19	Personality Development	6	PD-III	Group Discussions - Team Work	2
			PD-V	Public Speaking	4
20	Border & Coastal Areas	2	BCA-II	Security Setup and Border/Coastal management in the area	2
21	Introduction to Infantry Battalion and its Equipments	3	INF-1	Organisation of Infantry Battalion & its weapons	3
22	Military History	4	MH-3	Study of Battles of Indo-Pak Wars 1965 & 1971	4
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER VI

S.No	Subject	hours	Chapter	Lesson	HOURS
25	Personality Development	3	PD-IV	Career Counselling, SSB Procedure and Interview Skills	3
27	Border & Coastal Areas	2	BCA-III	Security Challenges & Role of cadets in Border management	2
28	Armed Forces	3	AF-2	Modes of Entry into Army, Police and CAPF	3
29	Military History	19	MH-1	Biographies of Renowned Generals	6
			MH-2	War Heroes : Param Veer Chakra Awardees	3
			MH-3	Study of Battles of Kargil	2
			MH-4	War Movies	8
30	Communication	3	C-1	Introduction to Communication & Latest Trends	3
TOTAL HOURS					30
TOTAL CREDITS					2

SIX SEMESTER NCC COURSE SYLLABUS

Training Objectives: Institutional Training

97. Institutional training includes basic military training of the cadets as part of the curriculum with its long-standing effort to mould young volunteers into disciplined and responsible citizens of India. NCC course is aimed to achieve following learning objectives:-

- (a) Develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.
- (b) To create interest in cadets by including and laying emphasis on those aspects of Institutional Training which attract young cadets into the NCC and provides them an element of thrill and excitement.
- (c) To inculcate defence Services work ethos that is characterized by hard work, sincerity of purpose, honesty, ideal of selfless service, dignity of labour, secular outlook, comradeship, spirit of adventure and sportsmanship.
- (d) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regardless of which career they choose.
- (e) To provide conducive environment to motivate young Indians to choose the Armed Forces as a career.



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SEMESTER I COURSE MODULE : NATIONAL CADET CORPS I
VTH SEMESTER

National Cadet Corps : Course Details			
Course Title: National Cadet Corps I			
Course Code	BNCC01GE03	Credits	1(Thr)+ 1(Pr) = 03
L /T + P	15+30	Course Duration	1 Semester
Semester	I (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e, 25% internal assessment and 75% end term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

98.

Course Objectives:

Cadets will be able to: -

- (a) Know about the history of NCC, its organization, and incentives of NCC for their career prospects.
- (b) Acquire knowledge of duties and conduct of ncc cadets.
- (c) Understand about different NCC camps and their conducts.
- (d) Understand the concept of national integration and its importance.
- (e) Understand the concept of self-awareness and emotional intelligence.
- (f) Understand the concept of critical & creative thinking.
- (g) Understand the process of decision making & problem solving.
- (h) Understand the concept of team and its functioning.
- (i) Understand the concept and importance of Social service.

99. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Imbibe the conduct of NCC cadets.
- (b) Respect the diversity of different Indian culture.
- (c) Practice togetherness and empathy in all walks of their life.
- (d) Do their own self analysis and will workout to overcome their weakness for better performance in all aspects of life.
- (e) Understand creative thinking & its components.
- (f) Think divergently and will try to break functional fixedness.
- (g) Make a team and will work together for achieving the common goals.
- (h) Do the social services on different occasions.

100. **Course Content Part (I) Theory**

- (a) **Unit 1- NCC General (N) (Contact Hrs. 06).** Introduction of NCC, History, Aims, Objective of NCC & NCC as Organization, Incentives of NCC, Duties of NCC Cadet. NCC Camps: Types & Conduct.
- (b) **Unit 2- National Integration & Awareness (NI) (Contact Hrs. 04) .** National Integration: Importance & Necessity, Factors Affecting National Integration, Unity in Diversity & Role of NCC in Nation Building, Threats to National Security.
- (c) **Unit 3- Personality Development (Contact Hrs. 3).** Intra & Interpersonal skills - Self-Awareness- & Analysis, Empathy, Critical & creative thinking, Decision making and problem solving.
- (d) **Unit 4- Social Service and Community Development (Contact Hrs. 02).** Basics of social service and its need, Types of social service activities, Objectives of rural development programs and its importance, NGO's and their contribution in social welfare, contribution of youth and NCC in Social welfare.

Course Content Part (II) Practical

101. **Course Objectives:** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of foot drill.
- (c) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.



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- (d) Develop awareness about different types of terrain and how it is used in battle craft.
- (e) Develop the concept of various markings on the map and how they are co-related to the ground features.
- (f) Understand the various social issues and their impact on social life.
- (g) Develop the sense of self-less social service for better social & community life.

102. Expected Learning Outcomes: After completing this course, the cadets will be able to: -

- (a) Perform foot drill and follow the different word of command.
- (b) Fire a weapon effectively with fair degree of marksmanship.
- (c) Undertake point to point navigation and take part in route marches by day and night.
- (d) Perform the social services on various occasions for better community & social life.

103. Course Content Part (II) Practical

- (a) **Unit 1. Drill (Contact Hrs. 12).** Foot Drill- Drill ki Aam Hidayaten, Word ki Command, Savdhan, Vishram, Aram Se, Murdha, Kadvar Sizing, Teen Line Banana, Khuli Line, Nikat Line, Khade Khade Salute Karna Parade Par, Visarjan, Line Tod, Tej Chal, Tham aur Dhire Chal, Tham.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 05).** Introduction & Characteristics of .22 rifle, Handling of .22 rifle.
- (c) **Unit 3. Map Reading (MR) (Contact Hrs. 03).** Definition of Map, Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Magnetic Variation and Grid Convergence.
- (d) **Unit 4. Field Craft & Battle Craft (FC & BC) (Contact Hrs. 03).** Introduction of Field Craft & Battle craft, Judging Distance, Method of Judging Distance.
- (e) **Unit 5. Social Service and Community Development (SSCD)(Contact Hrs.07).** Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc.

SEMESTER II COURSE MODULE : NATIONAL CADET CORPS II

<u>Course Title: National Cadet Corps II</u>			
Course Code	BNCC02GE03	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15+30	Course Duration	1 Semester
Semester	II (Even)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

104. **Course Objectives:** Cadets will be able to: -

- (a) Understand the thinking & reasoning process.
- (b) Understand the process to cope with Stress & emotions.
- (c) Understand the importance of improving communication skills.
- (d) Identify the leadership traits.
- (e) Admire the qualities of great leaders.
- (f) Know about different legal provisions for children & women safety and protection.
- (g) Understand the various rules & measures to be taken to ensure Road/Railsafety.
- (h) Understand & spread awareness about latest Government initiatives for welfare of citizens and contribute towards Nation building.
- (i) Understand concepts of cyber and mobile security.



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105. **Expected Learning Outcomes.** After completing this course, the cadets will be able to:

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- (a) Define thinking, reasoning, critical thinking and creative thinking.
- (b) To think critically about different life related issues.
- (c) Think divergently and will try to break functional fixedness.
- (d) Creatively in their real-life problems.
- (e) Understand the organizations related to disaster management and their functioning.
- (f) Appreciate the role of NCC cadets in disaster management.

106. **Course Content Part (I) Theory**

(a) Unit 1. Personality Development (Contact Hrs.5)

- (i) Thinking- Meaning and Concept of thinking, Reasoning, Process of thinking.
- (ii) Critical Thinking- Meaning & concept of critical thinking, Features of critical thinking, Process of critical thinking.
- (iii) Creative thinking- Meaning & concept of creative thinking, Features of creative thinking, Process of creative thinking, levels of Creativity, Characteristics of creative person.

(b) Unit 2. Leadership Development (Contact Hrs.5)

- (i) Leadership capsule.
- (ii) Important Leadership traits, Indicators of leadership and evaluation.
- (iii) Motivation- Meaning & concept, Types of motivation. Factors affecting motivation.
- (iv) Ethics and Honor codes.

(c) Unit 3. Social Service and Community Development (Contact Hrs. 5)

- (i) Protection of Children & Women Safety.
- (ii) Road/Rail Safety.
- (iii) New Government Initiatives.
- (iv) Cyber and mobile Security Awareness.

Course Content Part (II) Practical

107. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of foot drill.
- (c) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (d) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.

108. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform foot drill gracefully.
- (b) Give and follow the different word of command.
- (c) Fire a weapon effectively with fair degree of marksmanship.
- (d) Use of bearing and service protractor and locate the places and objects on the ground.
- (e) Do the social service and feel connected with social problems.

109. **Course Content Part (II) Practical**

(a) Unit 1. Drill (Contact Hrs. 12)

- (i) Foot Drill Dahine, Baen, Aageaur Piche Kadam Lena.
- (ii) Tej Chal se Murdna, Tej Chal se Salute Karna, Tej Kadam Taal aur Tham, Tej Kadam Taal se Kadam Badalna.
- (iii) Teeno Teen se Ek File aur ek file se Teeno Teen Banana

(b) Unit 2. Weapon Training (Contact Hrs. 04)

- (i) Range procedure & Theory of group.
- (ii) Short Range firing.

(c) Unit 3. Map Reading (Contact Hrs. 05)

- (i) Protractor Bearing and its conversion methods.
- (ii) Service protractor and its uses.
- (iii) Prismatic compass and its uses and GPS.
- (iv) Navigation by compass and GPS.



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(d) Unit 4. Field Craft & Battle Craft (Contact Hrs. 04)

VTH SEMESTER

(i) Indications of landmarks and Targets.

(ii) Intro, Definitions, Types of Ground, Indication of Landmarks, Methods of identification of targets, difficult targets.

(e) Unit 5. Social Service and Community Development (Contact Hrs. 05)

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc. as per the requirement and similar announced days- National and state level.

NATIONAL CADET CORPS

SEMESTER III COURSE MODULE : NATIONAL CADET CORPS III

<u>COURSE TITLE: NATIONAL CADET CORPS III</u>			
Course Code	BNCC03GE02	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15 +30	Course Duration	1 Semester
Semester	III (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

110. **Course Objectives.** Cadets will be able to: -

- Understand the life history and leadership qualities of great leaders, sportspersons & entrepreneurs.
- Understand the various aspects of types of mindset.
- Understand public speaking methods & qualities.
- Understand the organizations related to disaster management and their functioning.
- Understand the role of NCC cadets in disaster management.
- Understand the various types of adventure activities.
- Understand the History, Geography & Topography of Border/ Coastal Areas.



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111. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: - **VTH SEMESTER**

- (a) Admire and get inspired from the accomplishments of leaders from various walks of life.
- (b) Develop public speaking skills.
- (c) Understand the importance of positive mindset and optimistic attitude in life.
- (d) Appreciate the need & requirement for disaster management and his role in disaster management activities.
- (e) Know the history & geographical peculiarity of our borders & coastal regions.

112. Course Content Part (I) Theory

(a) Unit 1. Personality Development (Contact Hrs.5)

- (i) Group Discussions - Change your Mindset
- (ii) Public Speaking.

(b) Unit 2. Leadership Development (Contact Hrs.4). Case Studies – APJ Abdul Kalam, Deepa Malik, Maharana Pratap, N Narayan Murthy.

(c) Unit 3. Disaster management(Contact Hrs. 3)

- (i) Disaster Management Capsule.
- (ii) Organisation.
- (iii) Types of Disasters.
- (iv) Essential Services.
- (v) Assistance.
- (vi) Civil Defence Organisation.

(d) Adventure (Contact Hrs. 1). Adventure activities.

(e) Border & Coastal Areas(Contact Hrs. 2). History, Geography & Topography of Border/ Coastal Areas.

Course Content Part (II) Practical

113. **Course Objectives.** Cadets will be able to : -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal
- (b) Appreciate grace and dignity in the performance of arm drill
- (c) Understand the concept and importance of social service.
- (d) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (e) Actively participate in social service and community development activities.

114. **Expected Learning Outcomes.** After completing this course, the cadets will be able to : -

- (a) Perform arm drill gracefully.
- (b) Give and follow the different word of command.
- (c) Fire a weapon effectively with fair degree of marksmanship.
- (d) Different positioning for fire and aiming.
- (e) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (f) Observe surroundings in better way.
- (g) Develop the qualities of patience and confidence and become better individuals.
- (h) Will develop physical as well as mental fitness.

115. **Course Content Part (II) Practical**

(a) **Unit 1. Drill (Contact Hrs. 08)**

- (i) Arm Drill.
- (ii) Rifle ke saath Savdhan, Vishram aur Aram se.
- (iii) Rifle ke saath Parade Par aur Saj, Rifle ke saath Visarjan, Line Tod.
- (iv) Bhumi Shastra aur Uthao Shastra, Bagal Shastra aur Baju Shastra.

(b) **Unit 2. Weapon Training (Contact Hrs. 04).** Short Range firing.

(c) **Unit 3. Map Reading (Contact Hrs. 04).**

- (i) Setting of Map.
- (ii) Findings North and Own Position.



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(d) Unit 4. Field Craft & Battle Craft (Contact Hrs. 04)

VTH SEMESTER

- (i) Observation.
- (ii) Camouflage.
- (iii) Concealment.

(e) Unit 5. Social Service and Community Development (Contact Hrs. 05).

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) Unit 6. Obstacle Training(Contact Hrs. 05)

- (i) Obstacle training - Introduction, Safety-measures, Benefits.
- (ii) Obstacle Course- Straight balance, Clear Jump, Gate Vault, Zig- ZagBalance, High Wall.

NATIONAL CADET CORPS

SEMESTER IV COURSE MODULE : NATIONAL CADET CORPS IV

<u>Course Title: National Cadet Corps IV</u>			
Course Code	BNCC04GE03	Credits	2(Thr)+ 1(Pr)=03
L /T + P	30+30	Course Duration	1 Semester
Semester	IV (Even)	Contact Hours	30(Thr)+30(Pr)=60Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70%end of term exams etc.		

Course Content Part (I) Theory

116. **Course Objectives.** Cadets will be able to: -

- (a) Develop a sense of time management and social skills.
- (b) Understand the life history & leadership qualities of personalities who have contributed in Nation Building and Literature.
- (c) Understand the role of NCC cadets as 2nd line Defence in 1965 War.
- (d) Develop awareness about various types of Natural and manmade disasters.
- (e) Know about life saving tips during disasters.
- (f) acquainted about Fire Services.
- (g) Understand importance of Environmental Awareness & conservation.
- (k) Understand importance of General Awareness.
- (l) Know about Armed Forces.



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(d) **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
VTH SEMESTER

- (i) Effectively Manage time.
- (ii) Develop the qualities of social skills.
- (iii) Imbibe leadership qualities.
- (iv) Do group discussions effectively.
- (v) Be motivated to serve the nation by joining Armed forces.
- (vi) Contribute in environmental awareness and conservation activities.
- (vii) Keep abreast of current affairs & general awareness.
- (viii) Effectively contribute in managing disaster relief tasks.

117. Course Content Part (I) Theory

- (a) **Unit 1. Personality Development (Contact Hrs.4).** Group Discussions – Social Skills & Time management.
- (b) **Unit 2. Leadership Development (Contact Hrs.3).** Case Studies – Case Studies – Ratan Tata, Rabindra Nath Tagore, Role of NCC cadets in 1965 war.
- (c) **Unit 3. Disaster management(Contact Hrs. 10)**
 - (i) Initiative Trg, Organising Skills.
 - (ii) Dos and Don'ts.
 - (iii) Natural Disasters.
 - (iv) Man Made Disasters.
 - (v) Fire Services and Fire Fighting.
- (d) **Environmental Awareness (Contact Hrs. 3).** Adventure Environmental Awareness and Conservation.
- (e) **General Awareness (Contact Hrs. 4).** General Awareness.
- (f) **Armed Forces(Contact Hrs. 6).** Army, Navy, Air Force and Central Armed Police Forces.

Course Content Part (II) Practical

118. **Course Objectives.** Cadets will be able to: -
- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
 - (b) Understand various signals to convey messages in the army.
 - (c) Get acquainted various section formations.
 - (d) Understand the basics of personal and public hygiene.
 - (e) Get acquainted with the procedure to treat the wounds and fractures during emergencies.
119. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Perform weapon drill gracefully.
 - (b) Give and follow the different word of command.
 - (c) Appreciate grace and dignity in the performance of foot drill.
 - (d) Apply signals in there day to day functioning.
 - (e) Provide first aid during the emergencies.
 - (f) Navigate to the given location on ground using compass and GPS.
 - (g) Practice healthy practices for the personal sanitation and hygiene.
120. **Course Content Part (II) Practical**
- (a) **Unit 1. Drill (Contact Hrs. 08)**
 - (i) Arm Drill.
 - (ii) Salami Shastra.
 - (iii) Squad Drill with Arms.
 - (b) **Unit 2. Weapon Training (Contact Hrs. 04).** Short Range firing
 - (c) **Unit 3. Map Reading(Contact Hrs. 04)**
 - (i) Map to Ground.
 - (ii) Ground to Map.
 - (d) **Unit 4. Field Craft & Battle Craft(Contact Hrs. 04)**
 - (i) Fire and Move Capsule.
 - (ii) Field signal- with hand, with Weapons, Signal with Whistle.
 - (iii) Field signals as means of giving orders.



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(iv) Field signals by day, Field signals by night.

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(v) Section Formation.

(e) **Unit 5. Social Service and Community Development(Contact Hrs. 05)**

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) **Unit 6. Health & Hygiene(Contact Hrs. 05)**

(i) Hygiene & Sanitation (Hygiene- Personal & Camp Hygiene).

(ii) First Aid in common medical emergencies.

(iii) Treatment & Care of Wounds.

NATIONAL CADET CORPS

SEMESTER V COURSE MODULE : NATIONAL CADET CORPS V

<u>Course Title: National Cadet Corps V</u>			
Course Code	BNCC05GE02	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15 +30	Course Duration	1 Semester
Semester	V (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exam setc.		

Course Content Part (I) Theory

121. **Course Objectives.** Cadets will be able to: -

- (a) Understand the concept of Team and its functioning.
- (b) Hone Public speaking skills.
- (c) Understand the security set up and management of Border/Coastal areas.
- (d) Acquire knowledge about an Infantry Battalion organisation and its weapons.
- (e) Acquire knowledge about Indo-Pak Wars fought in 1965 & 1971.

122. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Participate in team building exercise and value team work.
- (b) Improve communication skills by public speaking activities.
- (c) Understand the security mechanism and management of Border/Coastal areas.
- (d) Get motivated to join armed forces.



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123. **Course Content Part (I) Theory**

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(a) **Unit 1. Personality Development (Contact Hrs.6).**

(i) Group Discussions –Team work.

(ii) Public speaking.

(b) **Unit 2. Border & Coastal Areas(Contact Hrs.2).** Security Setup and Border/Coastal management in the area.

(c) **Unit 3. Introduction to Infantry Battalion and its Equipment(Contact Hrs. 3).** Organisation of Infantry Battalion & its weapons

(d) **Military History(Contact Hrs. 4).** Study of Battles of Indo-Pak Wars 1965 & 1971.

Course Content Part (II) Practical

124. **Course Objectives.** Cadets will be able to: -

(a) Understand that drill as the foundation for discipline and to command a group for common goal.

(b) Appreciate grace and dignity in the performance of ceremonial drill.

(c) Use the compass and GPS to locate places on the ground and map.

125. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

(a) Perform ceremonial drill and follow the different word of command.

(b) Do the social service on various occasions and get connected with the community.

(c) Do all the asana and gain the physical& mental fitness.

126. **Course Content Part (II) practical**

(a) **Unit 1. Drill(Contact Hrs. 03)**

(i) Ceremonial Drill.

(ii) Guard Mounting.

(b) **Unit 2. Field Craft & Battle Craft(Contact Hrs. 04)**

(i) Fire control orders.

(ii) Types of fire control orders.

(iii) Fire and Movement- when to use fire and movements tactics, Basic considerations, Appreciation of ground cover, Types of cover, Dead ground, Common Mistakes, Map and air photography, Selection of Fire position and firecontrol.

(c) **Unit 3. Map Reading(Contact Hrs. 04).**Google Maps & applications

(d) **Unit 4. Weapon Training(Contact Hrs. 04).**Short Range firing

(e) **Unit 5. Social Service and Community Development (Contact Hrs. 05)**
Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc. as per the requirement and similar announced days- National and State level.

(f) **Unit 6. Health & Hygiene(Contact Hrs. 05)**

(i) Yoga- Introduction, Definition, Purpose, Benefits.

(ii) Asanas-Padamsana, Siddhasana, Gyan Mudra, Surya Namaskar, Shavasana, Vajrasana, Dhanurasana, Chakrasana, Sarvaangasana, Halasanaetc.

(h) **Unit 7. Obstacle Training(Contact Hrs. 05)**

(i) Obstacle training – Intro, Safety measures, Benefits.

(ii) Obstacle Course- Straight balance, Clear Jump, Gate Vault, Zig- ZagBalance, High Wall etc.



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SEMESTER VI COURSE MODULE : NATIONAL CADET CORPS VI
VTH SEMESTER

<u>Course Title: National Cadet Corps VI</u>			
Course Code	BNCC06GE03	Credits	2(Thr)+ 1(Pr)=03
L /T + P	30 +30	Course Duration	1 Semester
Semester	VI (Even)	Contact Hours	30(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

127. **Course Objectives.** Cadets will be able to: -

- (a) Get acquainted about counselling process its need and importance.
- (b) Know about SSB procedure and different tasks and tests.
- (c) Know about the conduction during the interview.
- (d) Understand the security challenges & role of cadets in Border Areas.
- (e) Know about the modes of entry in Armed forces, CAPF & police.
- (f) Understand the life history & leadership qualities of great generals.
- (g) Learn about 1999 Kargil war.
- (h) Acquire the knowledge about various wars and their heroes.
- (i) Know about various components of communication process.

128. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Get motivated to join Armed forces, police & CAPF.
- (b) Write their CV effective and appealing.
- (c) Face SSB interview effectively in their future.
- (d) Understand individual responsibilities & role in meeting the security challenges on Border/Coastal areas.
- (e) Imbibe the feeling of patriotism.
- (f) Communicate more effectively.

129. **Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (Contact Hrs.3).**
 - (i) Career Counselling.
 - (ii) SSB Procedure.
 - (iii) Interview Skills.
- (b) **Unit 2. Border & Coastal Areas (Contact Hrs.2).** Security Challenges & Role of cadets in Border management.
- (c) **Unit 3. Armed Forces (Contact Hrs. 3).** Modes of Entry into Army, Police and CAPF.
- (d) **Military History (Contact Hrs. 19).**
 - (i) Biographies of Renowned Generals.
 - (ii) War Heroes : Param Veer Chakra Awardees.
 - (iii) Study of Battles of Kargil.
 - (iv) War Movies.
- (e) **Communication (Contact Hrs. 3).** Introduction to Communication & Latest Trends.



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Course Content Part (II) Practical

VTH SEMESTER

130. Course Objectives.

Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of ceremonial drill.
- (c) Know about various knots and lashing used in soldiering.
- (d) Acquire awareness about the basic weapon system in use in the Armed Forces.

131. Expected Learning Outcomes. After completing this course, the cadets will be able to: -

- (a) Perform foot drill and follow the different word of command.
- (b) Aiming range and figure targets.
- (c) Use the different knots and lashing in day-to-day life for different purposes.
- (d) Develop the feeling of altruism.

132. Course Content Part (II) Practical.

(a) Unit 1. Drill (Contact Hrs. 03).

- (i) Ceremonial Drill.
- (ii) Guard of Honour.

(b) Unit 2. Weapon Training(WT) (Contact Hrs. 04). Short Range firing.

(c) Unit 3. Map Reading(MR) (Contact Hrs. 04). Google maps and Applications.

(d) Unit 4. Field Craft & Battle Craft(FCBC) (Contact Hrs. 03). Knots, Lashing and Stretchers.

(e) Unit 5. Social Service and Community Development(SSCD) (Contact Hrs. 05). Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) Unit 6 Introduction of Infantry Weapons & Equipment(INF) (Contact Hrs.03). Characteristics of 5.56MM INSAS Rifle, Ammunition, Fire Power, Stripping, Assembling & Cleaning Practice.

(g) Unit 7. Communication (COM) (Contact Hrs. 03).

- (i) Basic Radio Telephony (RT) Procedure.
- (ii) Introduction, Advantages, Disadvantages, Need for standard procedures.

(iii) Types of Radio telephony communication.

(iv) Radio telephony procedure, Documentation.

NATIONAL CADET CORPS



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COURSE MODULE: NATIONAL CADET CORPS CAMP -I
VTH SEMESTER

Course Title: National Cadet Corps Camp I			
Course Code	BNCCCAMP03GE05	Credits	2(Thr)+ 3(Pr)=05
L /T + P	30+90	Course Duration	10 Days (24 hours each)
Semester	III (Odd)	Contact Hours	30(Thr)+90(Pr)=120Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work,self-study, individual and group tasks, team work, field-based assignments, Physical Training, endurance building and skill development practices		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of termexams etc.		

Course Content Part (I) Theory

133. **Course Objectives**.Cadets will be able to: -

- (a) Acquire knowledge about the various aspects of personality development.
- (b) Understand the concept of leadership traits, moral values and character traits.
- (c) Develop awareness about the various types of natural disasters.
- (d) Develop sensitivity to the changing environment and understand the importance of conservation.
- (e) Understand the importance of hygiene and sanitation and common first aid procedures.
- (f) Acquire awareness about various types of weapon systems in the Armed Forces.

134. **Expected Learning Outcomes**. After completing this course, the cadets will be able to: -

- (a) Acquire adequate skill sets to overcome their weakness and reshape their personality.
- (b) Imbibe good moral values and character traits in their daily life.
- (c) Become useful members of the society and form part of disaster response team, if need arises.
- (d) Respect and make efforts to conserve natural resources
- (e) Follow good personal hygiene practices and provide first aid in emergencies.
- (f) Be motivated to join the armed forces.

135. **NCC Camp-I : Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (PD) (Contact Hrs. 04)**. Introduction to Personality Development, Factors influencing/shaping personality, Time Management and Interview Skills.
- (b) **Unit 2. Leadership (LDR) (Contact Hrs. 04)**. Leadership Traits, Moral Values and Character Traits.
- (c) **Unit 3. Disaster Management (DM) (Contact Hrs. 04)**. Assistance during natural disasters, Do's and Don'ts for NCC Cadets performing Disaster Management Duties
- (d) **Unit 4. National Integration and Awareness (NIA) (Contact Hrs. 04)**.
Water Conservation and Rain Harvesting, Waste Management and Energy Conservation
- (e) **Unit 5. Health and Hygiene (H&H) (Contact Hrs. 04)**. Hygiene and Sanitation, First Aid in Common Medical Emergencies.
- (f) **Unit 6. Infantry Weapons (IW) (Contact Hrs. 02)**. Characteristics of Company Support Weapons.
- (g) **Unit 7. Weapon Training (WT) (Contact Hrs. 04)**. Characteristics of Point 22 Rifle and its Ammunition, Range Procedure and Safety Precautions.
- (h) **Unit 8. Military History (MH) (Contact Hrs. 04)**. Guest lectures by War Veterans/decorated soldiers/veterans.
- (i) **Unit 9. Communication (COM) (Contact Hrs. 04)**. Basics of communication.



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NCC Camp-I : Course Content Part (II) Practical

VTH SEMESTER

136. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill is the foundation of discipline and command a group for a common goal.
- (b) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (c) Develop awareness about different types of terrain and how it is used in BattleCraft.
- (d) Develop the concept of various markings on the map and how they are co-related to the ground features.
- (e) Acquire awareness about the various types of weapon systems in the Armed Forces.
- (f) Understand the concept and importance of social service.
- (g) Understand the various nuances of Personality Development.
- (h) Understand the concept and importance of Physical Training in everyone's life.
- (i) Acquire skill sets about various games and understand the importance of teamwork.
- (j) Develop awareness about different cultures and different modes of its projection in artistic forms.

137. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform foot drill, arms drill, ceremonial drill and will be able to give out different words of command.
- (b) Fire a weapon effectively with fair degree of marksmanship.
- (c) Undertake point to point navigation and take part in route marches by day and night.
- (d) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (e) Be motivated to join the armed forces.
- (f) Acquire adequate skill sets to overcome their weakness and enhance their personality.
- (g) Gain adequate physical and mental endurance capabilities.
- (h) Play team games and be able to communicate and coordinate effectively in group events or situations.

- (i) Respect the diversity of Indian culture and develop pride by showcasing their own culture to others.

138. NCC Camp-I : Course Content Part (II) Practical

- (a) **Unit 1. Drill (Drill)(Contact Hrs. 16).** Drill ki Aam Hidayaten aur Words of Command, Savdhan, Vishram, Aram Se aur Mudna, Khuli Line aur Nikat Line mein march, Salute Karna Parade Par, Visarjan aur Line Tod, Tej Chal, Tham aur Dhire Chal, Tham, Dahine, Baen, Aage aur Piche Kadam lena, Tejchaal se Mudna, Tejchaalse Salute karna, Tej kadambaal aur Tham, Tej Kadambaal se kadam badhana, Teenon Teen se ek file Banana aur ek file se Teenon Teen Banana, Rifle Ke Saath Saavdhan, Vishram aur Aaram se, Rifle ke saath Parade par aur saaj, Rifle Ke saath visarjan aur line tod, Bhumi Sashtira aur Uthao Sashtira, Bagal Sashtira aur Baaju Shastra.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 14).** Stripping, Assembling, Cleaning of Point 22 rifle, Sight Setting and Sight Picture of Point 22 Rifle, Loading, Cocking and Unloading, Lying Position, Holding and Aiming of Point 22 rifle, Trigger Control and Firing of Shot, Theory of Group, Short-Range Aiming and Firing, Firing Practice I to VII.
- (c) **Unit 3. Field Craft & Battle Craft (FC/BC) (Contact Hrs. 06).** Introduction of Field Craft & Battle craft, Judging Distance, Indication of Landmarks and Targets, Observation, Camouflage and Concealment, Field Signals, Section formations.
- (d) **Unit 4. Map Reading (MR) (Contact Hrs. 12).** Introduction to Map and Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Types of Bearing and use of Service Protector, Prismatic Compass and its use, setting of a map, Finding North and own Position, Map to Ground and Ground to map, Point to Point march, Route March – I, Route March -II.
- (e) **Unit 5. Infantry Weapons (IW) (Contact Hrs. 01).** Characteristics of Battalion Support Weapons.
- (f) **Unit 6. Social Service and Community Development (SSCD) (Contact Hrs. 04).** Basics of Social Service and its need, Rural Development Programme, Civic Responsibilities: Cadets will participate in various activities throughout the camp e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc., Road /Rail Travel Safety
- (g) **Unit 7. Personality Development (PD) (Contact Hrs. 06).** Self-Awareness, Empathy, Critical and Creative Thinking, Decision making and problem Solving, Coping with Stress and Emotions, Time Management.
- (h) **Unit 8. Obstacle Training (OT) (Contact Hrs. 02).** OT Practice – I:- Untimed, Cadets will be familiarized with all the obstacles in the Obstacle Course and briefed about the correct method to do them, OT Practice -II: Timed practice for all the cadets and record to be maintained.



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- (i) **Unit 9. Physical Training (PT) (Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in morning hours. Training has to be progressive in degree of difficulty to improve individual stamina and endurance. Training to include warming up, running, exercises to strengthen upper body, lower body and core muscles. Two period each to be devoted to route march by day and night respectively and one period will be earmarked for trekking expedition as part of Adventure Activity.
- (j) **Unit 10. Games Training (G)(Contact Hrs. 09).** Games Training will be carried out on each day of the camp, except on last day, in evening hours. Training has to be progressive in degree of difficulty to improve individual skills, coordination, teamwork and desire to excel. Training to ensure that each and every boy and girl cadets participate in at least one game activity everyday.
- (k) **Unit 11. Cultural Activity (C)(Contact Hrs. 09).** Cultural Activity will be carried out on each day of the camp, except on last day, in evening hours. Cadets have to divided in Nine Groups consisting of a mix of boy and girl cadets and preferably belonging to the same geographical area. Each group has to present the unique culture, custom, tradition, folk lore, songs, drama, paintings and cuisine during one hour allotted. There will be a prize for the best group to encourage participation and to develop pride in their unique culture. This training activity should ensure that each and every boy and girl cadet participate in at least one game activity every-day.
- (l) **Unit 12. Spare (S)(Contact Hrs. 02).** Two periods in each camp will be earmarked as spare to cover disruptions in training activity due to weather or other administrative reasons.

COURSE MODULE : NATIONAL CADET CORPS CAMP – II

Course Title: National Cadet Corps Camp II			
Course Code	BNCCCAMP05GE05	Credits	2(Thr)+ 3(Pr)=05
L /T + P	30+90	Course Duration	10 Days (24 hours each)
Semester	V (Odd)	Contact Hours	30(Thr)+90(Pr)=120Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, individual and group tasks, team work, field-based assignments, Physical Training, endurance building and skill development practices		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (II) Theory

139. **Course Objectives.** Cadets will be able to: -

- (a) Acquire the concept self-awareness, emotional intelligence, critical and creative thinking, decision making and problem solving.
- (b) Learn about various indicators of good leadership and get an insight on principle of leadership and motivation.
- (c) Develop awareness about the various types of natural disasters and disaster management organization in our country.
- (d) Familiarize with natural resources, changing environment and understand the importance of conservation and waste management.
- (e) Value the importance of Physical and Mental health and understand how to deal with wounds of various types.
- (f) Acquire awareness about organization and role of an Infantry Battalion in the Armed Forces.



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140. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
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- (a) Develop a sense of responsibility, smartness in appearance and improve self-confidence, inculcate importance of empathizing with others, improve their deep-thinking ability and apply ideas and be able to face problems in a constructive manner with solutions.
- (b) Imbibe good leadership traits and apply them in practical life and appreciate the visible outcome of leadership and motivation.
- (c) Appreciate role of the org during emergency and become useful members of disaster response team, if need arises.
- (d) Learn about the various natural resources, their utilization and practice method of conservation of these resources in daily life.
- (e) Appreciate value of physical and mental health in daily life and spread awareness about treatment and care of wounds in their society.
- (f) Be motivated to join the armed forces.

141. **NCC Camp-II : Course Content Part (I) Theory.**

- (a) **Unit 1. Personality Development (PD) (Contact Hrs. 04).** Self-Awareness, Emotional intelligence, Critical and Creative Thinking, Decision-Making and Problem Solving.
- (b) **Unit 2. Leadership (LDR) (Contact Hrs. 02).** Indicators of Good Leadership, Leadership and Motivation.
- (c) **Unit 3. Disaster Management (DM) (Contact Hrs. 02).** Disaster Management Organization NDMA and NDRF, Types of Disasters.
- (d) **Unit 4. Environmental Awareness and Conservation (EAC) (Contact Hrs. 02).** Natural Resources, Conservation and Management, Water Conservation, Waste Management, Energy Conservation.
- (e) **Unit 5. Health and Hygiene (H&H) (Contact Hrs. 02).** Physical and Mental Health, Treatment and Care of Wounds.
- (f) **Unit 6. Infantry Weapons (IW) (Contact Hrs. 01).** Organization of Infantry Battalion.
- (g) **Unit 7. Weapon Training (WT) (Contact Hrs. 02).** Characteristics of Point 22 Rifle and its Ammunition, Range Procedure and Safety Precautions.
- (h) **Unit 8. Military History (MH) (Contact Hrs. 04).** Guest lectures by War Veterans/decorated soldiers/veterans.
- (i) **Unit 9. Communication (COM) (Contact Hrs. 04).** Latest trends in communication.

NCC Camp-II : Course Content Part (II) Practical

142. **Course Objectives.** Cadets will be able to: -

- (a) Inculcate spirit of discipline and follow command as a group for a common goal.
- (b) Fire a weapon with adequate safety precautions necessary for safe firing.
- (c) Understand the lay of the ground and use it skillfully towards own objective.
- (d) Understand and use the map, satellite imagery and GPS effectively.
- (e) Identify and be well versed with the primary weapon systems used in the Armed Forces.
- (f) Lead a life of selflessness and provide service towards society development and nation building.
- (g) Understand the importance of changing mindset, team work, social skills, etiquettes and manners, interview skills and importance of effective communication in daily life.
- (h) Learn the importance of physical fitness and nuances of physical training.
- (i) Inculcate esprit-de-corps through team games.
- (j) Have knowledge about cultural diversity of India and learn ways and means to adopt them.

143. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Practice problem solving, critical thinking in real life situations.
- (b) Practice leadership of small teams and groups under challenging environment.
- (c) Develop a positive attitude, have manners and etiquettes in social life, develop a sense of cooperation for group or team work, participate in an interview with confidence and inculcate verbal and non-verbal communication skills.
- (d) Develop adequate physical and mental endurance capabilities.
- (e) Fire a weapon effectively with fair degree of marksmanship.
- (f) Undertake point to point navigation and take part in endurance marches by day and night.
- (g) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (h) Be motivated to join the Armed Forces.
- (i) Play team games and be able to communicate and coordinate effectively in group events or situations.



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(j) Perform foot drill, arms drill, ceremonial drill and will be able to take part in ceremonial parade and events. **VTH SEMESTER**

(k) Respect the diversity of indian culture and develop pride by showcasing their own culture to others.

144. NCC Camp-II : Course Content Part (II) Practical

(a) **Unit 1. Drill (Drill) (Contact Hrs. 16).** Tejchaal se Mudna, Tejchaal se Salute karna, Tej kadamtaal aur Tham, Tej Kadamtaal se kadam badhana, Teenon Teen se ek file Banana aur ek file se Teenon Teen Banana, Rifle Ke Saath Saavdhan, Aaram se, Rifle ke saath Parade par aur saaj, Rifle Ke saath visarjanaur line tod, Bhumi Sashttra aur Uthao Sashttra, Bagal Sashttra aur Baaju Shastra, Salami Sashttra, Squad Drill, Guard Mounting, Guard of Honour, Platoon / Company Drill, Word of Command and Instructional Practice.

(b) **Unit 2. Weapon Training (WT) (Contact Hrs. 14).** Stripping, Assembling, Cleaning of Point 22 rifle, Sight Setting and Sight Picture of Point 22 Rifle, Loading, Cocking and Unloading, Lying Position, Holding and Aiming of Point 22 rifle, Trigger Control and Firing of Shot, Theory of Group, Short-Range Aiming and Firing, Musketry Training, Firing Practice I to VII.

(c) **Unit 3. Field Craft & Battle Craft (FC/BC) (Contact Hrs. 06).** Observation, Camouflage and Concealment, Field Signals, Section formations, Fire Control Orders, Fire and Movement, Knots and Lashings.

(d) **Unit 4. Map Reading (MR) (Contact Hrs. 12).** Introduction to Map and Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Types of Bearing and use of Service Protector, Prismatic Compass and its use, setting of a map, Finding North and own Position, Map to Ground and Ground to map, Point to Point march, Endurance March – I (10 KM), Endurance March -II (20 KM).

(e) **Unit 5. Infantry Weapons (IW) (Contact Hrs. 01).** Characteristics of Infantry Company support weapons and 5.56 MM INSAS Rifle.

(f) **Unit 6. Social Service and Community Development (SSCD)(Contact Hrs. 04).** Contribution of Youth Towards Social Welfare: Cadets will participate in various activities throughout the camp e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc., Social Evils: Female Feticide, Dowry, Child Abuse, Trafficking and Corruption, Drug Abuse and Drug Trafficking, Protection of Children and POCSO Act 2012.

(g) **Unit 7. Personality Development (PD)(Contact Hrs. 06).** Change Your Mindset, Team Work and Team Building, Social Skills, Etiquettes and Manners, Interview Skills, Communication Skills–I, Communication Skills -II

(h) **Unit 8. Obstacle Training (OT)(Contact Hrs. 02).** OT Practice – I: Untimed, Cadets will be familiarized with all the obstacles in the Obstacle Course and briefed about the correct method to do them, OT Practice -II: Timed practice for all the cadets and record to be maintained.

(i) **Unit 9. Physical Training (PT) (Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in morning hours. Training has to be progressive in degree of difficulty to improve individual stamina and endurance. Training to include warming up, running, exercises to strengthen upper body, lower body and core muscles. Two period each to be devoted to route march by day and night respectively and one period will be earmarked for trekking expedition as part of Adventure Activity.

(j) **Unit 10. Games Training (G)(Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in evening hours. Training has to be progressive in degree of difficulty to improve individual skills, coordination, teamwork and desire to excel. Training to ensure that each and every boy and girl cadets participate in at least one game activity everyday

(k) **Unit 11. Cultural Activity (C) (Contact Hrs. 09).** Cultural Activity will be carried out on each day of the camp, except on last day, in evening hours. Cadets have to be divided in Nine Groups consisting of a mix of boy and girl cadets and preferably belonging to the same geographical area. Each group has to present the unique culture, custom, tradition, folk lore, songs, drama, paintings and cuisine during one hour allotted. There will be a prize for the best group to encourage participation and to develop pride in their unique culture. This training activity should ensure that each and every boy and girl cadets participate in at least one game activity every day (Contact Hrs. 09)

(l) **Unit 12. Spare (S) (Contact Hrs. 02).** Two periods in each camp will be earmarked as spare to cover disruptions in training activity due to weather or other administrative reasons.



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SECTION II : RULES AND REGULATIONS

**GOVERNING NCC CREDIT COURSE UNDER CHOICE BASED CREDIT SYSTEM AS
GENERIC ELECTIVE FOR SENIOR DIVISION/WING**

RULE 1 :Definitions of Key Terms

General Definitions

‘Choice Based Credit System’ (CBCS).The CBCS provides choice for the student to select courses from the prescribed courses (Elective or Soft – Skill courses). It provides a ‘Cafeteria’ approach in which the students can take courses of their choice, learn at their own pace, study additional courses and acquire more than the minimum required credits, and adopt an inter-disciplinary approach.

‘Academic Year’.Two consecutive (one odd + one even) semesters shall constitute one academic year.

‘Credit Course’.Course, usually referred to as paper having specific title and code number, is a component of a programme. It consists of a list of topics/concepts/theories/principles/activities/tasks etc. which a student has to learn during the programme of study. Each course has some credits according to the nature and load of content. Each course should define the learning objectives/learning outcomes. A course may be designed to be delivered through lectures/tutorials/laboratory work/field work/out reach activities/project work / vocational training / physical training /viva / seminars /term papers / assignments / presentations / self-study work etc., or a combination of some of these.

‘Course Instructor/Teacher’.The course instructor generally will be a teaching faculty who has taken up the responsibility of teaching it and evaluating the performance of the students in that course. NCC course will be imparted by the ANO (Associate NCC Officer) and PI (Permanent Instructor) / Girl Cadet Instructor (GCI) staff together according to their area of specialization. Certain specific topics and training activity is imparted by Military Officers and Whole Time Lady (WTLO).

‘Credit’.A unit by which the course work is measured. It determines the number of hours of instructions required per week. One credit is equivalent to one hour of teaching (lecture or tutorial) or two hours of practical work per week. Thus, in each semester’s NCC course, credits are assigned on the basis of the number of lecture/tutorial/field work/physical training/excursions and other forms of learning required for completing the contents in a 15-18 week schedule. 2 hours of laboratory work/field work is generally considered equivalent to 1 hour of lecture.

- i. 1 credit = 1 hour of instruction per week (1 credit course = 15 contact hours of instruction per semester)

- ii. 4 credit = 4 hour of instruction per week (4 credit course = 60 contact hours of instruction per semester)
- iii. 1 credit = 2 hour of practical per week (1 credit course = 30 contact hours of instruction per semester)
- iv. 4 credit = 8 hour of practical per week (4 credit course = 120 contact hours of instruction per semester)

Number(s) of credit(s) assigned to a particular course are mentioned in the detailed syllabus of the courses.

‘Credit Point’. It is the product of the grade point and the number of credits for a course.

‘Letter Grade’. It is an index of the performance of students in a said course. Grades are denoted by letters O, A+, A, B+, B, C, P, and F. A letter grade is assigned to a student on the basis of evaluation of her/his performance in a course on a ten-point scale.

‘Grade Point’. It is a numerical weight allotted to each letter grade on a 10 -point scale.

Letter Grade	Grade Point
O	9-10
A+	8-9
A	7-8
B+	6-7
B	5-6
C	4-5
P	4
F	0
Ab	0

Note : University may use the above said criteria for providing the grades to the students or may adopt the same criteria which they are practicing for providing the letter grade and grade point for other subjects.

‘Programme’. An educational programme leading to the award of degree, Diploma or Certificate course. NCC course shall be offered only at under graduate level programmes for any stream or type of programme for example – Nonprofessional courses BA, B.Sc. B. Com etc. professional courses – B.A., LLB, B.A./B.Sc., B.Ed., BCA, BBA, B. Tech, MBBS etc.

‘Credit – Based Semester System (CBSS)’. Under the CBSS, the requirement of awarding a degree or diploma or certificate is prescribed in terms of number of credits to be completed by the students.



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'Semester' Each semester shall consist of 15 to 16 weeks of academic work equivalent to 90 actual teaching days. The odd semester may be scheduled from July to December and even semester from January to June. The Credit-based semester system provides flexibility in designing curriculum and assessing credits based on the course content and hrs of teaching.

'Semester Grade-Point Average (SGPA)' Semester Grade Point Average or SGPA, is an average grade point earned by the student at the end of an academic session i.e. semester at college. The formula for calculation of SGPA is the sum of all the credit points awarded for the subjects divided by total credits allotted to that semester. It shall be expressed up to two decimal places.

'Cumulative Grade Point (CGPA)' It is a measure of overall cumulative performance of a student over all semesters. The CGPA is the ratio of total credit points secured by a student in various courses in all the semesters. It shall be expressed up to two decimal places.

'Transcript/ Grade card or certificate' Based on the grades earned, a grade certificate shall be issued to all the registered students after every semester. The grade certificate will display the course details (code, title, number of credits, grade and / or marks secured) along with SGPA of semester. Overall Grade Certificate will be issued on completion of the course showing semester wise SGPA & CGPA.

'The University/ College/ Institution' The University/ College/ Institution in present document means the any recognized central/ state/ Deemed university or institution meant for higher education.

'NCC Course' In the present document 'NCC Course' means the course designed for imparting NCC curriculum in educational institutions as elaborated in this document under Choice Based Credit System as a General Elective Course for Senior Division/ Senior Wing.

Definitions Specific to NCC

'Institutional Training' Implies training conducted for NCC cadets as per Training Manuals and Cadet Hand Book issued by DG NCC, Ministry of Defence.

'Common Subjects' Implies those subjects specifically taught in NCC curriculum which are common to Army, Navy and Air Force and general training that can be imparted by Associate NCC Officers or Military staff or a suitably qualified person.

'Specialised Subjects' Implies subjects specifically taught in NCC curriculum by military instructors comprising specialised topics for Army, Navy and Air Force Cadets respectively.

'NCC Camps and Centralised Training Events' Collective training events conducted usually for 10 days with large number of cadets living under field conditions in selected places away from home. The training camp comprises of focused physical and mental training routines of different types as per syllabus and curriculum. Some training like route marches may happen overnight. Camps include, adventure camps, national integration camps,

Republic Day Parade Training Camps, ThalSainik, VayuSainik and NauSainik camps and other outdoor training activities as described in DG NCC Training Manuals.

NCC 'B' and 'C' Certificate Examinations. These are defined in Special National Cadet Corps Order 2020 issued by DG NCC, Ministry of Defence.

'Training Faculty'. Persons suitably trained & responsible for imparting training of different types and nature to students.

'Military Officers'. They are regular commissioned officers of Indian Armed Forces who serve in the NCC and render command, administrative and instructional functions for NCC.

'Whole Time Lady Officers (WTLO)'. They are women officers commissioned directly into the NCC.

'Associate NCC Officer (ANO)'. ANO will be a university/ college/ school faculty who are qualified in the PRCN (Pre-commission Course of NCC) conducted by DGNCC and are commissioned as Associate Officers in NCC as defined in NCC Act 1948 and NCC Rules. They have the eligibility to impart certain component of NCC Course and undertake training of cadets.

'Permanent Instructor (PI)'. PI Staff are Junior Commissioned Officers (JCO) and Non-Commissioned Officers (NCO) on deputation from Armed Forces to NCC as governed by NCC Act 1948. Retired PI Staff may be hired by a college as a substitute for ANO with prior concurrence of DGNCC.

'Girl Cadet Instructors (GCI)'. GCI are lady instructors' equivalent to PI Staff for specifically imparting instructions to women NCC cadets of Senior Wing.

'NCC Organizational Structure'. NCC is an adjunct of Indian Armed Forces that operates under the ambit of the Ministry of Defence through the Defence Secretary with Raksha Mantri as the political head.

'DGNCC'. Directorate General of NCC renders the command and administrative function of NCC. The executive head of NCC is Director General of NCC who is a Lt Gen rank officer from the Army.

'State NCC Directorate'. State NCC Directorates are directorates subordinate to DGNCC and render command and administrative control to NCC at State level and is headed by an Additional or Deputy Director General

'NCC Group HQ'. NCC Group HQs are subordinate to State Directorates and render command and administrative control to NCC at district or cluster of districts in a state and is headed by a Group Commander.

'NCC Units'. NCC Units are subordinate to Group HQs at the lowest rung of the command and administrative control exercised by military officers and is headed by a Commanding Officer or Officer Commanding. The NCC Units directly engage



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with educational institutions and ANOs and are primarily responsible for training of NCC in institutions under their jurisdiction.

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'NCC Division/ Wing'. NCC Division/Wing are minor units of senior division/wing of NCC comprising of 160 senior cadets allotted to educational institutions. It can be further subdivided into NCC platoons of 53 to 54 cadets.

'NCC Troop'. NCC Troop are minor units of junior division/wing of NCC comprising of 100 junior cadets allotted to educational institutions. It can be further subdivided into NCC half troops comprising of 50 junior cadets.

RULE 2 : Admission and Other Provisions

The NCC Course under the CBCS as 'General Elective' shall be of three years (Six Semester) duration which may be completed in maximum duration of four year (8 semesters).

Students may complete NCC course minimum in Six semesters and maximum in eight semesters. Cadets may complete their 'B' Certificate in four semesters minimum and maximum six semesters. Cadets already having 'B' certificate may complete their 'C' certificate in minimum two semesters and maximum four semesters, and they may join NCC course 5 in first semester of college.

The intake to the course shall be decided according to the seats allotted to University/ college/ institution by DG NCC according to the availability of required infrastructure, faculty and resources.

The admission to the NCC Course under the CBCS as a 'General Elective' shall be governed by the provisions as laid down by the NCC Act 1948/ SNCCO 2020/ contemporary SNCCO and Academic council of parallel body of university. These rules and regulations may be modified from time to time (if needed) by the Academic body of the university in consultation with DG NCC or Act/ Ordinances prepared by DG NCC.

Students will be enrolled as NCC cadet as per existing Acts & Rules.

At the time of reporting for admission, the candidates are required to present medical & physical fitness documents as well as the admission proof of the university and submit the self-attested copies of aforesaid documents.

The admission of any candidate is liable to be cancelled without giving any further notice forthwith or at any time during the period of the course, if it is detected that the candidate has/had produced fake/forged certificate (s)/ document(s), indulged in any act of misconduct/indiscipline and has/had concealed any other relevant information at the time of admission.

The admission of the candidate to the course shall be subject to such ordinance, rules and regulations as may be framed from time to time by the university in consultation with DG NCC and NCC act 1948.

DG NCC shall have jurisdiction in case of any dispute relating to the provisional admission in the course.

RULE3 : For Eligibility, Medium of Instrs & Categories

Eligibility Conditions. Be governed by provisions of NCC Act and Rules and directions from DG NCC from time to time. These are readily available on DG NCC website www.nccindia.nic.in.

Standards for physical Fitness criteria for Male and Female Cadets/students shall be governed by provisions of NCC Act and Rules and policy documents released by DG NCC from time to time.

RULE4 : Medium of Instruction. English or Hindi. However, ANOs and training instructors are free to use vernacular language for helping students who are not fluent in Hindi or English.

RULE 5 : Course and Students. NCC course is unique, due to the nature of its military training content and component hence it is normally offered to students enrolled as NCC cadets only. This NCC Course is primarily designed for students enrolled as NCC cadets under provisions of NCC Act 1948. Institution allotted NCC will have the obligation to offer this course to all students from their institute enrolled as cadets as per vacancy allotted to the institution by DG NCC as also to those cadets enrolled under Open Quota seats.

RULES 6 :NCC Course for 'Cadet' Category

6.1. NCC Course for 'Cadet'

- (a) NCC course for Cadets comprises of total 24 credits (08 for theory, 06 for practical and 10 for camp component) over 6 semesters courses i.e., NCC course I to NCC course VI and NCC Camp I & NCC Camp II.
- (b) Cadets will not only earn the academic credits but also be given 'B', and 'C' Certificates after passing the exam conducted by DG NCC.
- (c) Students would be free to join NCC Course I or subsequent Courses in any semester, not necessarily Semester I or the designated Semester.
- (d) A student can opt for only one of the six Courses per semester and that too sequentially implying NCC Course II cannot be joined before completing NCC Course I and so on.
- (e) Under this category a fresh student/cadet will compulsorily have to opt for all six NCC Courses in minimum six Semesters. However, 'B' certificate holder may directly join NCC Course Number 5 in any semester. He will have to complete NCC Course Number 5 and NCC Course Number 6 for obtaining 'C' certificate and he will be awarded credit points only for NCC Course Number 5 and NCC Course Number 6.



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NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	1	1		2	
Semester - II	1	1		2	
Semester – III	1	1	5	7	Credits of 1 st Camp merged with 3 rd Sem
Semester – IV	2	1		3	
Semester – V	1	1	5	7	Credits of 2 nd Camp merged with 5 th Sem
Semester - VI	2	1		3	
Total	08	6	10	24	Twenty-Four Credits

RULE 7 :Mobility& Credit Bank

The mobility shall be permissible from the regular mode programme to the regular mode programme of learning only and cannot be replaced by open/distance/online programme.

It shall be the responsibility of the student to assess the feasibility and practicality of vertical mobility (across the Universities), as it doesn't entitle a student to be exempted or relaxed from any of the requisites (sessional, attendance, assignments, End-semester examinations and programme duration etc.) for completing the course.

After completing one semester/ one year cadet/student may pursue NCC course from any other institution/ University/ College having NCC and carry credits in credit bank as per NEP 2020. The NCC students/ Cadets of some other university shall in any case be admitted only at the beginning of the session to the fulfilment of the other requirements of the NCC Course (attendance, Formative assessment, Field-work, practical etc).

A student of NCC course availing inter-university mobility shall continue to be a bonafide student of the university where he/she initially got admission and as per the university/ Institutional rules for the inter-university mobility.

In case of inter-university mobility of NCC cadet for NCC Course is also the subject to availability of NCC for the cadets in that particular university/ institution and it shall be interpreted as inter-battalion migration (means another regimental no. shall be allotted to the cadet).

RULE 8 :Examination & Promotion

The examination of all the NCC courses shall be internal in nature and generally consisting of continuous internal assessment and End of semester Examination. For the preparation of final grade in a particular course, the continuous internal assessment (Formative in nature) and the End Semester Examination (Summative in nature) shall have the weightage as decided for other courses by the university as per the University norms fore.g., 25% internal assessment and 75% End of term exams or 30% internal assessment and 70% End of term exams etc.

For assigning the Grades and credit points to NCC Course Universities/ Institutions are free to use the same criteria which are decided by their academic bodies for providing the grades and credit points to the other courses

RULE 9 :Continuous Internal Assessment

The Continuous Internal Assessment of the NCC Cadets' and NCC students' learning and performance shall be carried out by the ANOs and PI staff.

Continuous Internal Assessment will be 100% Practical that includes Drill Square test, Map Reading, Weapon Training, Field craft & Battle craft.

CO of nominated NCC Unit will be deemed as Head of the Department and shall be responsible for approving the schedule and pattern of the continuous internal examination.



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ANO of the nominated institute shall maintain all the records related to attendance, teaching and assessment in a systematic manner, including award of final grade.

In case a student fails to appear in any Continuous Internal Assessment, they will be given a chance to reappear in retest and in case he/she fails to obtain 'P' grade he/she will be made to repeat the exam by carrying it forward for semester retest .

RULE 10 :Re-appear in the End Semester Examination for Improvement of Grades

If a student wishes to improve her/his grade(s) in NCC course(s), she/he can re-appear in the End Semester Examination in the subsequent odd/even semester(s), whenever the examination of the particular course(s) is held, on payment of fees in addition to the prescribed semester fee within the maximum permissible duration for the programme of study of the student/cadet.

A student may improve her/his points/grade by reappearing in the End Semester Examination of a course as per the provisions of reappearing mentioned above. In such cases points obtained by the student in the Continuous Internal Assessment of the particular course shall be carried forward to the subsequent End Semester Examination of the course. However, in such case, the points/grades obtained on the basis of latest appeared End Semester Examination shall be considered for calculation of final CGPA of the programme.

The re-appear examination of a course for improvement of grade shall be based on the syllabi of the course in force at the time of initial registration to the course.

A student who has got the Migration/Transfer Certificate issued from the University shall not be allowed to re-appear in any examination for improvement of grade.

RULE 11 :Repeating Courses

A student having attendance shortage in any course may repeat the course by taking re-admission in that course in subsequent odd/even semester(s), whenever the course is being offered, within the maximum permissible duration of the programme.

If a student repeats a course, she/he has to fulfil all the desired requirements afresh including attendance, Continuous Internal Assessment and the End Semester Examination. In such case the course content shall be based on the syllabi of the course in force at the time of repetition of the course.

RULE 12 : Promotion Rules

A student shall be declared as 'promoted' to the next semester when she/he earns 'P' Grade or above in the last concluded semester examination, maintaining the spirit and pattern of semester system and covering the mandatory components, such as Continuous Internal Assessment and End-Semester Examination in the NCC Courses.

A student shall be 'Provisionally Promoted' to the next semester if she/he secures less than 'P' grade but he /she has to pass all the courses of NCC course within permissible duration.

A cadet shall be eligible to attend the 'B' Certificate exam if he/she passed all the first four semester NCC course and completed one ATC/CATC. Similarly, cadet will be eligible to attend 'C' certificate examination if he/she has 'B' certificate and he /she has passed V, VI semester NCC course and attended one CATC/ATC after fourth semester and after having obtained 'B' certificate.

If a cadet/student is repeating a course in an academic session, whatever may be the reason, it shall not be counted in the total number of seats and shall not affect the fresh intake of cadets / student in that academic session.

RULE 13 :Computation of SGPA & CGPA

13. **Computation of SGPA and CGPA** . University may use their own criteria for giving the SGPA & CGPA which is prepared by the authorised academic body for the other courses.



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**VTH SEMESTER
SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL
SUBJECT CODE – MC 5 (C)
CATEGORY – NNP
SUBJECT NAME – PHYSICAL EDUCATION
SEMESTER – v**

ANATOMY AND FIRST AID

UNIT-1

1. Introduction and Structural Organization of Human body:

Introduction:

Meaning and Concept of Anatomy.

Need and Importance of the Knowledge of Anatomy in the Field of Physical Education.

Levels of Organization of Human Body:

Definition of Cell, Tissue, Organ and System.

Microscopic Structure, Composition and Function of Cell.

Essential Properties of living Organism.

Classification, Structure and Function of Tissues:

Epithelial tissue

Connective Tissue

Muscular Tissue

Nervous Tissue

UNIT-II

2. Musculo-skeletal System:

Skeleton

Different Parts of Human Skeleton.

Types of Bones.

Gross And Microscopic Structure and Function of Bones.

Joints

Joint's Nomenclature and Classification.

names of the Movements around Joints.

Structure of the Typical Synovial Joint.

Muscles

Structure and Function of Skeletal Muscles.

Names of Major Muscles of Different parts of Body.

UNIT-III

3. Cardio- respiratory, Digestive & Excretory system:

Circulatory System:

Anatomical Position and Gross Structure of the Heart.

Systemic and Pulmonary Circulation.

Blood Vessels- Artery, Vein Capillaries

Respiratory System:

Meaning of External and Internal Respiration

Organs of Respiration

Introduction to Mechanism of Respiration

Digestive System:

Parts of Digestive Tract, Structure and Function in brief

Steps of Digestion

Digestive Glands- Structure and Function

Excretory System:

Routes of Excretion from Human Body-

Organs of Urinary System.

Structure and Function of Kidney.

4. Neuro-Humoral & Reproductive System:

Nervous System:

Structural and functional Divisions of Nervous System.

Sensory Motor Nervous System.

Parts of Brain.

Structure and Functions of Spinal Cord.

Sense Organs- Eyes, Ears & Skin.

Endocrine system:

Names, Location and Functions of -

Pituitary Gland

Thyroid gland

Adrenal Gland

Pancreas

Reproductive System:

Primary and Secondary Male and Female Reproductive System.

Testes and Ovary.



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**UNIT V
VTH SEMESTER**

5. First - Aid:

Injuries:

Types of Wounds

Pressure Points

Role of Ice

Dressing and Bandages

Musculo-skeletal Injuries

Sprain

Strain

Dislocations/Subluxation

Types of fractures & their Management (Application) of
splints)/

Shock:

Common Causes

Signs and Symptoms

Immediate Management

Concept of Artificial Respiration.

Transportation of Severely Injured patients.

Snake Bite and Insect Bite.

HEALTH EDUCATION

UNIT-I

Health

- Concept of Health
- Dimensions of Health
- Spectrum of Health
- Positive Health
- Determinants of Health

Health Education

- Concept, aims objectives and scope
- Principles of Health Education
- Methods of Communication in Health Education

UNIT-II

Health Problems

- Communicable and non-communicable diseases
- Nutrition
- Environmental sanitation
- Medical care
- Population

An Introduction to Organization and Administrative set-up of Health System in India.

- National level
- State Level
- District Level

Steps of Planning of Health Education Programme.

UNIT-III

Personal and Environmental Hygiene

- Care of skin, mouth, nails, clothing, bathing etc.
- Importance of rest sleep and exercise.

Community Health

- Brief account of housing, water supply, sewage and refuse disposal

School Health Service

- Common Health Problems.
- Objectives of School health service



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Aspects of School Health Services

Health Appraisal.
Remedial Measures and Follow-up.
Prevention of Communicable Diseases.
Healthful School Environment
Nutritional Services.
First Aid and Emergency Care
Eye Health Service
Health Education
Education of Handicapped Children
School Health Records

UNIT-IV

4. Nutrition.

Classification of Foods
Proximate Principles and role of various nutrients
Balanced diet
Balanced diet of Indian School Children
Malnutrition Adulteration of Food.

UNIT-V

Family Planning

Concept, need and importance
Role of Health Education in Family.

Sex Education

Concept need and importance of sex education at school level.

National Health Programme in India (Breif description)

NMEP (National Malaria Eradication Programme)
DDCP (Diarrhea Diseases Control Programme)
National TB Control Programme
STD Control Programme.
NFPF National Family Planning Programme.

International Health Agencies.

WHO
UNICEF
UNDP

TAEKWONDO

1. Define Taekwondo, historical development of Taekwondo, Philosophy of Taekwondo.
2. Belt Promotion, Dress, oath, Tents of Taekwondo, equipments, and facilities used in Taekwondo.
3. Player Stance- walking, extending walking, L stance.
4. Fundamental Skills- Sitting stance punch, double punch, triple punch.

5. Bocks, Striks.
6. Punching Skill from sparring position- front- fist punch, rear fist punch, double punch, and four fist punch, double punch, and four combination punch.
7. Foot Techniques (Balgisul)- standing kick (soseochagi), Front kick (AP Chagi), Arc kick (Bandal Chagi), Side kick (Yeop Chagi), Turning kick (Dollyo Chagi), Back kick (Twit Chagi), Reverse turning kick (Bandae Dollyo Chagi), Jump kick (Twimyo Chagi).
8. Pumsae (Forms) - Jang, Yi Jang, Sam Jang, Sa Jang, O Jang, Yook Jang, Chil Jang, Pal Jang (Fundamental Movement- eye control, concentration of spirit, speed control, strength control, flexibility, balance, variety in techniques).
9. Sparring (gyeorugi)- One Step sparring (hand techniques, foot techniques, self defense techniques, (hosinsul), combination kicks and Free Sparring.
10. Board Breaking (gyeokpa)- eye control, balance, power control, speed, point of attack.
11. Competition Rules and their interpretation and duties of officials.
12. Stretching, Relaxation and Meditation exercises.

Athletics II

- 1 Introduction of Track & Field Athletics and Historical development of events with special reference to India.**
- 2 Organizational set-up of Track and Field Athletics at National Level.**
- 3 Important tournaments held at National and International levels.**
- 4 Fundamental Skills**

a. Field events.

- i. Technique of Long Jump (Sail technique, Hang technique) Approach run, take off, flight and landing.
- ii. Technique of Shot Put (O'Brien technique) Grip, Stance, Glide, Release and Reverse.
- iii. Technique of Triple Jump- Approach Run, Take-off, Hop, Step and Jump.
- iv. Technique of Discus throw
Grip, stance, swing, Release and Reverse.



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v. Technique of High Jump (Straddle roll)

Approach run, take off, Bar clearance and landing.

vi. Technique of Javelin Throw (Grip, Carry, Approach and Five Stride Rhythm)

vii. Technique of Pole-Vault (Grip, Approach, Take-off, Bar Clearance and Landing)

viii. Technique of Hammer Throw (Grip, Preliminary Swings, Turns, Release and Recovery)

4.2 Brief Introduction about Combined events (Heptathlon and Decathlon)

5 General Competition Rules of track and field events.

6 Marking for Track & Field Events.

GYMNASTICS

1. Introduction of the game and historical development with special reference to India.
2. Important tournaments held at National and International levels and distinguished personalities related to the game.
3. **Fundamental Skills.**

Floor exercise-

Forward roll.

Backward roll.

Sideward roll.

Cart Wheel.

Hand stand and forward roll,

Backward roll to hand stand.

Diving forward roll.

Side split.

Head stand.

Different kinds of scale.

Dive roll from beat board.

Round off.

Parallel Bars-

Mount from one bar.

Straddle walking on parallel bars.

Single and double step walk.

Perfect swing.

Shoulder stand on one bar and roll forward.

Roll side.

Shoulder stand.

Front on back vault to the side (dismount).

Vaulting Horse-

Approach run and jump from the spring board.

Cat vault.

Squat vault.

Straddle vault.

4. Rules of gymnastics and their interpretations and duties of officials.



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VITH SEMESTER

CATEGORY:-DC

EE-61-POWER SYSTEM PROTECTION

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1.Understand the different components of a protection system.
- 2.Evaluate fault current due to different types of fault in a network.
- 3.Understand the protection schemes for different power system components.
- 4.Understand the basic principles of digital protection.
- 5.Understand system protection schemes, & the use of wide-area measurements.

Unit I

Faults in power systems, single line diagram, equivalent impedance diagram, per unit reactances. Analysis (using matrices) of power systems by symmetrical components under:

- (a) Three phase short circuit.
- (b) Line to line fault.
- (c) Line to ground fault.
- (d) Double line to ground fault.

Sequence networks and their inter connections for different types of faults, effects of fault impedance. Current Limiting Reactors: Applications, types, construction and location of current limiting reactors, short circuit calculation using reactors.

(12 HRS)

Unit II Relays

General considerations, sensing of faults, construction of electro-magnetic attraction and induction types relays, Buchholz and negative sequence relay, concept of reset, pick up, inverse time and definite time characteristics, over current, over voltage, directional, differential and distance relays on R-X diagram. Static Relays: Introduction, advantage and limitation of static relays, static over current, directional, distance and differential relays.

(12 HRS)

Unit III Protection

Types & detection of faults and their effects, alternator protection scheme (stator, rotor, reverse power protection etc.). Power transformer protection (external and internal faults protection), generator-transformer unit protection scheme, bus bar protection. Transmission line protection (current/time grading, distance), Pilot relaying schemes, power line carrier protection.

(12 HRS)

Unit IV Switchgear

Theory of current interruption- energy balance and recovery rate theory, arc quenching, recovery and restriking voltages. Types of circuit breakers. bulk oil and minimum oil, air break and air blast, sulphur hexa fluoride (SF₆) and vacuum circuit breakers. Rating selection and testing of circuit breakers/operating mechanisms. LT switchgear, HRC fuses, types construction and applications.

(12 HRS)



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Unit V Modern Trends In Protection

Electronic relays, static relays functional circuits: comparators, level detectors, logic and training circuits, microprocessor and computer based protection schemes, software development for protection, security & reliability.

(12 HRS)

COURSE OUTCOME:

1. Student gains knowledge on different Protective Equipments or Power Systems
2. Know about various protective systems- how it works and where it works?
3. Different applications of the relays, circuit breakers, grounding for different elements of power system is also discussed in the subject.

Reference Books:

- Van A. R & Warrington C., “ Protective Relays : Their Theory and Practice”, Vol 1 & 2, Chapman and Hall.
- Paithankar Y. O.,” Transmission Network Protection: Theory and Practice”, Marcel Deicker, Inc.
- GEC Measurements,” Protective Relays : Application Guide”, GEC Measurements.
- Masson R.J., Art & Science of Protective Relaying.
- J & P Switchgear handbook Ravindra Nath B., and Chandar M., Power systems protection and switchgear
- Rao Sunil S, Switchgear and protection.
- Crane P.H.C., Switchgear Principle.
- The Elementary Council, “Power System Protection”, Vol.1,2 &3, Peter Peregrinus Ltd.
- Badriram & Vishwakarma, Power System Protection. Ravindranath & Chander, Power System Protection & Switchgear.



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VITH SEMESTER

CATEGORY:-DC

EE-62 - CONTROL SYSTEMS

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Transfer function representation & determination using block diagram Reduction & signal flow graphs of LTI Systems
- 2 Time domain analysis of LTI Systems and stability studies
- 3 Develop the concepts on stability analysis using frequency response.
- 4 Concepts of state space analysis & compensation techniques

Unit-1

Modeling of dynamic systems: Electrical, Mechanical and hydraulic systems, Concept of transfer function, Simulation of differential equations in analog computer, State space description of dynamic systems: Open and closed loop systems, Signal flow graph, Mason's formula, Components of control systems: Error detectors (Synchros & Potentiometer), Servomotors (AC & DC), techo generators, power amplifier, stepper motors.

(12 HRS)

Unit-2

Time – domain analysis of closed loop systems: Test signals, time response of first and second order systems, Time domain performance specifications, Steady state error & error constants Feedback control actions: Proportional, derivative and integral control. Solution of state equation: Eigen values & eigenvectors digitalization state transitive matrix, stability Routh-Hurwit stability analysis.

(12 HRS)

Unit-3

Characteristics equation of closed loop system root loci, construction of loci, Effect of adding, poles and Zeros on the loci, Stability by root loci.

(12 HRS)

Unit-4

Frequency, Domain analysis, Bode plots, Effect of adding, poles and Zeros, Polar plot, Nyquist stability analysis, Relative stability : Gain and phase margins.

(12 HRS)

Unit-5

Frequency- Domain compensation : lead lag, Lag-lead compensation, Design of compensating networks.

(12 HRS)



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COURSE OUTCOME:

After completion of this course, students will be able to

- 1 Determine the TF of a system using block diagram reduction technique & signal flow graphs of LTI system & Solve Problems.
- 2 Determine Transient and Steady State behavior of systems using standard test signals and stability in time domain & Solve Problems
- 3 Determine the stability of the LTI systems using frequency domain.
- 4 Analyze performance of state space analysis of a continuous system

References:

1. I.J. Nagrath and M. Gopal, "Control system Engineering", New Age International.
2. K. Ogata, Modern Control Engineering, PHI.
3. B.C. Kuo, Automatic Control systems, PHI
4. Gopal M., Control System : Principles & Design, TMH.
5. N.K. Sinha, Control Systems, New Age International
6. Stefani, Shahian, Savant, Hostetter – "Design of feed back control System's", Oxford.

List of Experiments :

1. Time response of second order system.
2. Characteristics of Synchrons.
3. Effect of feedback on servomotors.
4. Determination of transfer function of A-C servomotor.
5. Determination of transfer function of D-C motor.
6. Formulation of PI & PD controller and study of closed loop responses of 1st and 2nd order dynamic systems.
7. State space model for classical transfer function using MATLAB.
8. Simulation of transfer function using operational amplifier.
9. Design problem: Compensating Networks of lead and lag.
10. Temperature controller using PID.
11. Transfer function of a DC generator.
12. Characteristics of AC servomotor.
13. Use of MATLAB for root loci and Bode plots of type-1, type-2 systems.
14. Study of analog computer and simulation of 1st order and 2nd order dynamic equations.
15. Formulation of proportional control on 1st order and 2nd order dynamic systems.
16. Feed back control of 3rd order dynamic Systems
17. Study of lead and lag compensating networks.
18. Effect of adding poles & zeros on root loci and bode plots of type-1, type-2 systems through MATLAB.



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VITH SEMESTER

CATEGORY:-DC

EE-63 - POWER SYSTEM ENGINEERING-II

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Representation of transmission lines
- 2 Methods of voltage control
- 3 Symmetrical components and fault calculations
- 4 System neutral grounding and insulation co-ordination

Unit -1

Introduction: Problems associated with modern interconnected power Systems, deregulation, power systems restructuring, distributed generation, congestion, available transfer capacities, pricing of energy and transmission services.

(12 HRS)

Unit -2

Power flow studies: Formulation of static power flow equations and solutions using Gauss- Seidel, Newton Raphson and FDLF methods, comparison of these methods, Economic operation of power system - Economic dispatch, Emission dispatch, line loss, ITL, economic dispatch using lagrangian multiplier method.

(12 HRS)

Unit-3

MW Frequency control: Coherency, control area, modeling of speed control mechanism, load damping, block diagrammatic representation of single and two area interconnected system, static and dynamic response, .optimum parameter adjustment.

(12 HRS)

Unit-4

MVAR Voltage control Problem: Difference in control strategy over MW-f control, characteristics of an excitation system, DC AC and static excitation system, General block diagram representation of voltage regulators.

(12 HRS)

Unit-5

Power System Stability: Steady state, dynamic and transients stability, Swing equation, equal area criterion, solution of swing equation using step by step method modified Eulers method and Runge-Kutta method, methods of improving transient stability.

(12 HRS)



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COURSE OUTCOME:

After completion of this course, students will be able to

- 1 Analyze transmission line performance & Solve Problems
- 2 Explain the significance of per unit quantities.
- 3 Determine the fault currents for symmetrical and unbalanced faults
- 4 Realize the concept of Traveling waves on transmission line

Reference Books:

1. Modern Power System Analysis-by I.J. Nagrath & D.P. Kothari Tata Mc Graw – Hill Publication Company Ltd 2nd edition.
2. A Chakrawarti Power System Analysis:Operation and Control PHI Learning 3rd edition
3. Reactive power Control in Electric Systems-by T.J.E. Miller, John Wiley & Sons.
4. Electrical Power Systems-by C.L. Wadhwa New Age International (P) Limited Publishers, 2nd edition 1998.
5. Elgerd O.I., “Electric Energy Systems Theory”, TMH, New Delhi, Second Edition 1983.
6. Prabha Kundur, “Power system stability and control”, Mc-Graw Hill Inc, New York, 1993.
7. Taylor C.W., “Power System Voltage Stability”, Mc-Graw Hill Inc, New York, 1993.
8. Nagrath IJ, Kothari D.P., “Power System Engineering”, Tata Mc-Graw Hills, New Delhi 1994.
9. Weedy B.M. “Electric Power System” John Wiley and Sons, 3rd edition.
10. P.S.R. Murthy, “Power System Operation and Control”, B S Publication.
11. Power Generation, Operation and Control by A.J. wood and B.F. Wollenberg John Wiley & Sons Inc. 1984.
12. T.K. Nagsarkar, M.S. Sukhiza, -“Power System Analysis”, Oxford University Press.
13. Economic Operation of Power Systems- by L.K. Kirchmayer Wiley Eastern Ltd.

List of Experiments:

1. To develop a program in Mat-Lab for information of Y-bus matrix for N bus system.
2. Load flow solution for 3-bus system using Gauss- Seidel, Newton Raphson and FDLF methods up to 3 iteration.
3. Load flow solution for IEEE 6-bus and 30-bus system in Mat-Lab using Newton Raphson method.
4. Assessment of transient stability of a single machine system.
5. Effect of compensation on voltage profile of IEEE 6-bus system.
6. Study of any software tools (PSCAD,EDSA, Mi POWER, ETAP etc)



SARVEPALLI RADHAKRISHNAN UNIVERSITY,



BHOPAL

VITH SEMESTER

CATEGORY:-DE

EE-64(A) – SIGNAL & SYSTEMS

TOTAL-60HOURS

COURSE OBJECTIVE:

This course introduces students about the signals and systems mathematically and understands how to perform mathematical operations on them. **COURSE CONTENT**

Classification of signals and systems: Continuous time signals (CT signals), Discrete time signals (DT signals) -Step, ramp, pulse, impulse, sinusoidal and exponential signals, basic operations on signals, classifications of CT and DT signals-Periodic and aperiodic signals, energy and power signals, random signals, CT systems and DT systems, basic properties of systems, basic properties of systems, linear time invariant systems and properties.

Unit- 1

Analysis of continuous time signals: Time and frequency domain analysis, Fourier series analysis, spectrum of CT signals, Fourier transform and Laplace transform, region of convergence, wavelet transform..

(15 HRS)

Unit- 2

Linear time invariant continuous time systems: Differential equations representation, block diagram representation, state variable representation and matrix representation of systems, impulse response, step response, frequency response, realizability of systems, analog filters. Analysis of discrete time signals: Convolution sum and properties, sampling of CT signals and aliasing, DTFT and properties, Z transform and properties, inverse Z transform.

(15 HRS)

Unit- 3

Linear time invariant discrete time systems: Difference equations, block diagram representation, impulse response, analysis of DT LTI systems using DTFT and Z transform, state variable equations and matrix representation of systems, Digital filters. **COURSE OUTCOME** Student after successful completion of course must possess an Understanding of various signals and systems properties and be able to identify whether a given system exhibits these properties and its implication for practical systems.

(15 HRS)

Unit- 4

Evaluation will be continuous an integral part of the class as well through external assessment.

(15HRS)



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COURSE OUTCOME:

1. Knowledge about basic signal and system modeling concept and definitions Outcomes
2. Student understand continuous-time signals and discrete-time signals
3. Student understand linear time-invariant systems theory and applications
4. Student can perform mathematical and graphical convolution of signals and systems

REFERENCES

1. Alan V. Oppenheim, Alan S. Willsky, S Hamid Nawab, 'Signals and Systems', 2nd edition 2015 Pearson New International Edition
2. A. Anand Kumar, Signals and Systems, PHI, III edition, 2015
3. Mahmood Nahvi, Signals and Systems, McGraw Hill
4. Simon Haykins and Barry Van Veen, Signals and Systems, Wiley India
5. A. Nagoor Kani; 'Signals and Systems' McGraw Hill
6. Robert A. Gabel and Richard A. Roberts, Signals & Linear Systems, Wiley.
7. Rodger E. Ziemer, William H. Tranter, D. Ronald Fannin. Signals & systems, Pearson Education



SARVEPALLI RADHAKRISHNAN UNIVERSITY,



BHOPAL

VITH SEMESTER

CATEGORY:-DE

EE-64(B) – POWER PLANT ENGINEERING

TOTAL-60HOURS

COURSE OBJECTIVE

1. To introduce students to different aspects of power plant engineering.
2. To familiarize the students to the working of power plants based on different fuels.
3. To expose the students to the principles of safety and environmental issues.

Unit-1

Coal Based Thermal Power Plants

Rankine cycle – improvisations, Layout of modern coal power plant, Super Critical Boilers, FBC Boilers, Turbines, Condensers, Steam & Heat rate.

(12 HRS)

Unit-2

Subsystems of thermal power plants – Fuel and ash handling, Draught system, Feed water treatment. Binary Cycles and Cogeneration systems.

(12 HRS)

Unit-3

Diesel, Gas Turbine & Combined Cycle Power Plants

Otto, Diesel, Dual & Brayton Cycle – Analysis & Optimisation. Components of Diesel and Gas Turbine power plants. Combined Cycle Power Plants. Integrated Gasifier based Combined Cycle systems.

(12 HRS)

Unit-4

Nuclear Power Plants

Basics of Nuclear Engineering, Layout and subsystems of Nuclear Power Plants, Working of Nuclear Reactors : Boiling Water Reactor (BWR),

(12 HRS)

Unit-5

Pressurized Water Reactor (PWR), CANada Deuterium-Uranium reactor (CANDU), Breeder, Gas Cooled and Liquid Metal Cooled Reactors. Safety measures for Nuclear Power plants.

(12 HRS)



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COURSE OUTCOMES

1. Describe and analyze different types of sources and mathematical expressions related to thermodynamics and various terms and factors involved with power plant operation.
2. Analyze the working and layout of steam power plants and the different systems comprising the plant and discuss about its economic and safety impacts
3. Combine concepts of previously learnt courses to define the working principle of diesel power plant, its layout, safety principles and compare it with plants of other types.
4. Describe the working principle and basic components of the nuclear power plant and the economic and safety principles involved with it.

REFERENCES:

- Nag. P.K., “Power Plant Engineering”, Third Edition, Tata McGraw – Hill Publishing Company Ltd., 2008.
- El-Wakil. M.M., “Power Plant Technology”, Tata McGraw – Hill Publishing Company Ltd., 2010.
- Black & Veatch, Springer, “Power Plant Engineering”, 1996.
- Thomas C. Elliott, Kao Chen and Robert C. Swanekamp, “Power Plant Engineering”, Second Edition, Standard Handbook of McGraw – Hill, 1998.
- Godfrey Boyle, “Renewable energy”, Open University, Oxford University Press in association with the Open University, 2004.



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
BHOPAL**



VITH SEMESTER

CATEGORY:-DE

EE- 64(C) - GENERALISED THEORY OF ELECTRICAL MACHINES

TOTAL-60HOURS

COURSE OBJECTIVE

To impart knowledge on

1. The key principles in Analysis of electrical machines
2. The Generalized Representation and steady state analysis of Synchronous Machines
3. The generator and motor operation in steady state and transient conditions
4. The analysis of harmonics in Ac machines
5. The generalized representation of special machines

Unit-1

Review : Primitive machine, voltage and torque equation. Concept of transformation change of variables & m/c variables and transform variables. Application to D.C. machine for steady state and transient analysis, and equation of cross field commutator machine.

(12 HRS)

Unit-2

Induction Machine : Voltage, torque equation for steady state operation, Equivalent circuit, Dynamic performance during sudden changes in load torque and three phase fault at the machine terminals. Voltage & torque equation for steady state operation of 1-P induction motor & schrage motor.

(12 HRS)

Unit-3

Synchronous Machine : Transformation equations for rotating three phase windings, Voltage and power equation for salient and non salient alternator, their phasor diagrams, Simplified equations of a synchronous machine with two damper coils.

(12 HRS)

Unit-4

Operational Impedances and Time Constants of Synchronous Machines: Park's equations in operational form, operational impedances and $G(P)$ for a synchronous machine with four Rotor Windings, Standard synchronous machine Reactances, time constants, derived synchronous machine time constants, parameters from short circuit characteristics.

(12 HRS)

Unit-5

Approximate Methods for Generator & System Analysis : The problem of power system analysis, Equivalent circuit & vector diagrams for approximate calculations, Analysis of line to line short circuit, Application of approximate method to power system analysis.

(12 HRS)



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COURSE OUTCOME

Students will be able to describe

1. The Generalized Representation of machines and their analysis
2. The steady state analysis and transient analysis of various machines
3. The performance of special machines and their representation

References:

- 1 P.C.Krause, Analysis of Electric Machinery, Wiley India.
- 2 B.Adkins, The General theory of Electrical Machines.
- 3.B.Adkins & R.G.Harley, The General theory of AC Machines.
4. P.S.Bhimbra, Generalised theory of Electrical m/c
- 5.White & Woodson, Electro Mechanical Energy Conversion.



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
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VITH SEMESTER

CATEGORY:-OE

EE-65(A) - DIGITAL SIGNAL PROCESSING

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Continuous-time (CT) and discrete-time (DT) signals
- 2 Discrete Fourier transform (DFT), computational complexity of DFT and efficient implementation of DFT using fast Fourier transform (FFT)
- 3 Specifying characteristics of frequency selective filters, design of linear-phase FIR filters
- 4 Classical analog Butterworth & Chebyshev filters, converting analog filter into equivalent digital filter to design digital IIR filters

Unit- 1

Introduction to Digital Signal Processing: Discrete time signals & systems, linear shift invariant systems, stability and causality, Linear-constant coefficient difference equations, Frequency domain representation of discrete time signals and systems, properties of the Discrete Time Fourier transform (DTFT), Sampling and discrete time processing of continuous-time signals.

(12 HRS)

Unit- 2

Z-Transform: Applications of z-transforms, solution of difference equations of digital filters, System function, stability criterion, frequency response of stable systems, one-sided Z-transform and its applications.

(12 HRS)

Unit- 3

Discrete Fourier series: Properties of discrete Fourier series, DFS representation of periodic sequences. Discrete Fourier Transforms: Properties of DFT: Fast Fourier Transforms (FFT) - Radix-2 decimation in time and decimation in frequency FFT Algorithms. Inverse FFT.

(12 HRS)

Unit- 4

IIR DIGITAL FILTERS: Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters, Bilinear transformation method, step & impulse invariance techniques, Spectral Transformations, Realization of IIR digital filters - direct, canonic, cascade & parallel forms.

(12 HRS)

Unit- 5

FIR DIGITAL FILTERS: Characteristics of FIR Digital Filters frequency response, Design of FIR Digital Filters using Window Techniques. Comparison of IIR and FIR filters, Realization of FIR digital filters - direct, linear phase, cascade & parallel forms.

(12 HRS)



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COURSE OUTCOME :

After completion of this course, students will be able to

- 1 Explain Continuous-time (CT) and discrete-time (DT) signals
- 2 Find the DFT of a DT sequence, perform circular convolution using DFT & IDFT and compute 2, 4 & 8point DFT of a sequence using radix-2 DIT & DIF algorithms
- 3 Design a linear-phase FIR filter with a prescribed magnitude response using windowing & frequency- sampling methods
- 4 Design an IIR Butterworth/Chebyshev digital filter meeting the required specifications by performing impulse invariance/bilinear transformation

References:

1. Oppenheim & Schaffer, Digital Signal Processing, PHI.
2. J Cavacchi Digital Signal Processing Wiley India.
3. John G. Proakis Digital Signal Processing: Principles, Algorithms, and Applications, 4/E.
4. Ludeman Fundamental of Digital Signal Processing, wiley India.
5. A. Antoniou, Digital Filters Analysis & Design, TMH.
6. A. Anand Kumar Digital Signal Processing, PHI.
7. S.K. Mitra, Digital Signal Processing, TMH

List of Experiments :

1. To develop elementary signal function modules (m-files) for unit sample, unit step, exponential and unit ramp sequences.
2. To develop program for computing inverse Z-transform.
3. To develop program for computing discrete Fourier Transform (DFT) and inverse discrete Fourier Transform (IDFT).
4. To develop program for computing circular convolution.
5. To develop program for finding the response of the LTI system by difference equation.
6. To study important commands of MATLAB software.



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VITH SEMESTER

CATEGORY:-OE

EE-65(B) SCADA SYSTEMS AND APPLICATIONS

TOTAL-60HOURS

COURSE OBJECTIVE :

There are many objectives of SCADA System.

- 1.Improved overall System efficiency (capital & energy)
- 2.Increased penetration energy sources including renewable energy sources.
- 3.Reduced Energy Requirements in both the Transmission and Generation
- 4.Increased Relativity of sequence to essential loads.

Unit-1

Introduction to SCADA and PLC:SCADA: data acquisition system, evaluation of scada, communication technologies, monitoring and supervisory functions. plc: block diagram, rogramming languages, ladder diagram, functional block diagram, applications, interfacing of plc with scada.

(12 HRS)

Unit-2

SCADA system components: Schemes, Remote Terminal Unit, Intelligent Electronic Devices, Communication Network, SCADA server.

(12 HRS)

Unit-3

SCADA Architecture-Various SCADA Architectures, advantages and disadvantages of each system, single unified standard architecture IEC 61850 SCADA / HMI Systems.

(12 HRS)

Unit-4

SCADA Communication-Various industrial communication technologies- wired and wireless methods and fiber optics, open standard communication protocols.

(12 HRS)

Unit-5

Operation and control of interconnected power system-Automatic substation control, SCADA configuration, Energy management system, system operating states, system security, state estimation, SCADA applications Utility applications, transmission and distribution sector operation, monitoring analysis and improvement. Industries oil gas and water. Case studies, implementation, simulation exercises.

(12 HRS)



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Reference Books:

1. Stuart A Boyer: SCADA supervisory control and data acquisition.
2. Gordan Clark, Deem Reynders, Practical Modem SCADA Protocols.
3. Sunil S. Rao, Switchgear and Protections, Khanna Publication.



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
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VITH SEMESTER

CATEGORY:-OE

EE-65(C) INDUSTRIAL ELECTRONICS

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

1. Analyze the steady state and small signal AC response of simple electronic circuits containing diodes, transistors, and operational amplifiers
2. Apply performance criteria in the design of basic amplifier circuits and verify that the criteria were met.
3. Design and analyze circuits containing digital components and microprocessors.
4. Analyze and evaluate performance parameters of AC and DC motors.

Unit I

Advantages and application of power electronic devices characteristics, Symbol & application of power diodes, power transistors, GTO, Triac, Diac, Power MOSFET, IGBT, LASCR, Fast recovery diode, Scotty diode MCTs. Principle of operation of SCR, Two transistor analogy, brief idea of construction of SCR, Static characteristics of SCR, Condition of turn on & off of SCR Gate characteristics, Method for turning on of SCR, Turnoff methods, different commutation techniques (Class A,B,C,D,E, & F Commutation) firing of SCR, Use of pubic transformer and opto isolator in firing, Resistance firing Ckt, Resistance capacitance firing circuit, UJT firing cut, and ramp triggering, firing for 3- Φ circuit. SCR rating & protection of SCR over voltage, Over current, Suprior firing, Design of snubber circuit and protection of gate of SCR, heating, cooling & mounting of SCR series and parallel operation of SCR, String efficiency & problem associated with series and parallel operation of SCR.

(12 HRS)

Unit II

Operation and analysis of single phase (Half wave & Full Wave) and multiphase (Three Phase) uncontrolled and controlled rectifier circuit with resistive, resistive & inductive load (continuous & non continuous conduction, Fw small & very large inductive loads) and RLE loads. Estimation of average load voltage and load current for above rectifier circuits active and reactive power input. Effect of free wheeling diode and source inductance on performance of these rectifier circuits . Comparison of mid point & Bridge rectifier circuits.

(12 HRS)

Unit III

Series and parallel inverter, Voltage source & current source inverter, Single phase and three phase bridge inverter, Self cumulated inverters,, Mc- murray & MC murray bed ford inverters, Voltage control of single phase and three phase bridge inverter, Harmonics & their reduction techniques.

(12 HRS)

Unit IV

Principle of chopper operation, Various control strategies in chopper, Step up & step-up/step down choppers, chopper configuration (Type A,B, C,D, & E), Steady state analysis of chopper circuits, Current & voltage commutation of chopper circuits Jones & Morgens chopper.

(12 HRS)



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Unit V

Single phase (mid point & bridge configuration) and three phase cyclo convertor configuration and operating principles. AC voltage controllers (using SCRs & Traic) single phase full wave controller with R and RL load, Estimation of RMS load voltage, RMS load current and input power factor, three phase AC voltage controller (Without analysis) Dual converter Switched mode voltage regulator buck, Boost, Buck & Boost, Ck regulators.

(12 HRS)

COURSE OUTCOME:

After completion of this course, the students' will be able to

1. Awareness of general structure of industrial electronics
2. Impart the knowledge of generation of electricity based on conventional and nonconventional energy sources
3. To make students capable of analysis of mechanical and electrical design aspects of transmission system
4. Enable the students to do analysis of different types of distribution systems and its design.
5. Impart the knowledge of protective relays and circuit breakers.

List Of Experiments :-

Subject- Industrial Electronics

1. V-I characteristics of SCR
2. V-I characteristics of DIAC
3. V-I characteristics of BJT
4. V-I characteristics of TRIAC
5. VI characteristics of MOSFET
6. Transfer characteristics of MOSFET
7. Output characteristics of IGBT
8. Transfer characteristics of IGBT
9. Single phase SCR half controlled converter with R load
10. 1- Φ SCR fully controlled converter with r-load
11. Study of 3 ϕ SCR Half controlled converter
12. Study of 3 ϕ SCR Fully controlled converter
13. Study of classes of commutation A,B,C,D,E & F.

References:

1. M.H. Rashid, Power Electronics Circuits, Devices and Applications, Pearson Education, Singapore, 1993.
2. M Ramsmoorthy, An Introduction to transistor and their application, Affiliated East-



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West Press.

3. P.C. Sen, Power Electronics, TMH.

4. M.D. Singh, K.B. Khanchandani, Power Electronics, TMH, Delhi, 2001.

5. Chakravarti A., Fundamental of Power Electronics and Drives, Dhanpat Ray & Co.,



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VITH SEMESTER

CATEGORY:-P

EE-66 - ENGINEERING DESIGN PROJECT

COURSE OBJECTIVE:

1. Manufacturing and Design Engineering is an interdisciplinary field.
2. A balanced emphasis on the “design,” the “process,” and the “systems” view of manufacturing is essential.
3. Hands-on experience is required.

COURSE GUIDELINES

The Engineering Design Project Work provides students an opportunity to do something on their own and under the supervision of a guide. Each student shall work on an approved project, which may involve fabrication, design or investigation of a technical problem that may take design, experimental or analytical character or combine element of these areas. The project work involves sufficient work so that students get acquainted with different aspects of manufacture, design or analysis. The students also have to keep in mind that in final semester they would be required to implement whatever has been planned in the Major Project in this semester. It is possible that a work, which involves greater efforts and time may be taken up at this stage and finally completed in final semester, but partial completion report should be submitted in this semester and evaluated also at the end of the semester. At the end of semester, all students are required to submit a synopsis and be assessed by an external examiner.

COURSE OUTCOME:

1. Ability to apply mathematics, science, and engineering principles.
2. Ability to design and conduct experiments, analyze and interpret data.
3. Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
4. Ability to function on multidisciplinary teams.



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
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VITH SEMESTER CATEGORY: - PDFS

BE-61 PROFESSIONAL DEVELOPMENT

FINISHING SCHOOL (LEVEL-IV)

TOTAL - 36 HOURS

COURSE OBJECTIVE

The students are to be groomed with respect to personality development. Emphasis to be made in reading, writing and vocal English, quantitative aptitude and logical reasoning to be stressed.

UNIT-I

Final Finishing:

Final Preparation of CV. Final Compilation of Database of Students with Necessary Mapping, Mock Interviews, Group Discussions.

(18 HOURS)

UNIT-II

Aptitude / Reasoning:

Quantitative Aptitude and Logical Reasoning-

Level IV Problem solving on.

Ratio and Proportions, Solutions and Mixtures, Sets, Simple Interest and Compound Interest, Simple and Quadratic Equations.

(18 HOURS)

COURSE OUTCOME

Attainment of confidence the students to be able to face interviews, group discussion and presentation ability. Knowledge on basic mathematical ability attained. Hence forth a student become competent to face the challenges of the world after attainments of knowledge at college level



**SARVEPALLI RADHAKRISHNAN UNIVERSITY,
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NATIONAL CADET CORPS



GROOMING TOMORROW'S LEADERS

**OFFERING NCC A GENERAL GENERIC ELECTIVE CREDIT
COURSE IN UNIVERSITIES UNDER CHOICE BASED CREDIT
SYSTEM TO ALIGN WITH NEW EDUCATION POLICY 2020**



SARVEPALLI RADHAKRISHNAN UNIVERSITY,



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CONTENTS

7. Section I : NCC Credit Course Design
8. Section II : NCC Credit Course Rules & Regulations aligned to UGC.

NATIONAL CADET CORPS

SECTION I: NCC CREDIT COURSE DESIGN DOCUMENT

UNDER CHOICE BASED CREDIT SYSTEM AS GENERAL ELECTIVE FOR SENIOR DIVISION / SENIOR WING

10. **Preamble.** The National Cadet Corps (NCC) is governed by NCC Act 1948 and attendant NCC Rules. It functions under the Ministry of Defence and is headed by DGNCC. It is organised into 17 State Directorates each headed by an Additional/Deputy Director General. The aims of NCC are:-

- (a) To develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.
- (b) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regard less of which career they choose.
- (c) To provide a conducive environment to motivate young Indians to choose the Armed Forces as a career.

11. **Purpose.** Currently NCC training is imparted as extra-curricular activity to volunteer students from recognized schools and colleges who enroll as cadets. NCC as a Credit Course is designed with an intent to transform NCC training into a curricular activity from an extra-curricular thereby providing academic credits to students undergoing NCC training along with other attended advantages to the cadets in the college/ university.

12. **Introduction to NCC Credit Course Design.** Institutional Training is the mainstay of NCC training and it is conducted at colleges and universities by Associate NCC Officers and Armed Forces personnel. The application of knowledge gained through institutional training is further honed or developed to a higher degree in NCC Camps. The Institutional Training syllabus comprises Common Subjects and Specialised Subjects (military component). NCC Credit Course is designed to offer Institutional Training of Senior Wing /Division is over six semesters (three years), comprising 300 periods (excluding Camp), of which 120 periods are meant for theory with 08 credits and 180 periods for practical with 6 credits. Each period is counted as hour. The ratio between theory and practical in terms of number of hours of training is 5:6, but in terms of credits is 5:3, since as per CBCS two hours of practical is counted towards one period of training as against one hour for theory. In addition two separate courses have been designed for two Camps normally referred to as Annual Training Camps (ATC).



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Training schedules planned for cadets ensure that the optimum benefits of the NCC organization reach maximum number of cadets. The main emphasis is on practical training which in consonance with theory is made to facilitate active participation of learner, better assimilation of knowledge, and proper development of various skills, strengthening of mind and body which is the bedrock of NCC training.

NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	01	01	-	02	
Semester - II	01	01	-	02	
Semester – III	01	01	05	07	Credits of 1 st Camp merged with 3 rd Sem
Semester – IV	02	01	-	03	
Semester – V	01	01	05	07	Credits of 2 nd Camp merged with 5 th Sem
Semester - VI	02	01	-	03	
Total	08	06	10	24	Twenty-Four Credits

INSTITUTIONAL TRG SYLLABUS

COMMON SUBJECTS				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	NCC General	06	-	06
2	National Integration	04		04
3	Drill	-	45	45
4	Weapon Training	-	25	25
5	Personality Development	25		25
6	Leadership	12	-	12
7	Disaster Management	13		13
8	Social Service & Community Development	08	39	47
9	Health & Hygiene	-	10	10
10	Adventure	01		01
11	Environmental awareness & conservation	03		03
12	Obstacle Training	-	09	09
13	General Awareness	04		04
14	Border & Coastal Areas	06		06
TOTAL HOURS COMMON SUBJECTS(a)		82	128	210

SPECIALSED SUBJECTS (ARMY)				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practical	
1	Armed Forces	09	-	09
2	Map Reading	-	24	24
3	Communications	03	03	06
4	Infantry Weapons	03	03	06
5	Field Craft & Battle Craft		22	22
6	Military History	23	-	23
Total Hours		38	52	90

**BHOPAL****SPECIALISED SUBJECTS (NAVY)**

Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	Naval Orientation	12	-	12
2	Naval Communication	02	18	20
3	Navigation	02	03	05
4	Seamanship	15	18	33
5	Fire Fighting and Damage Control	04	03	07
6	Ship and Boat Modelling	03	10	13
Total hours		38	52	90

SPECIALISED SUBJECTS (AIR FORCE)

Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	General Service Knowledge	08	-	08
2	Air Campaign	06	02	08
3	Principles of flight	06	06	12
4	Airmanship	01	07	08
5	Navigation	05	-	05
6	Aeroengines	06	-	06
7	Basic flight Instruments	03	03	06
8	Aero modelling	03	34	37
Total Hours		38	52	90

INSTITUTIONAL TRAINING: TOTAL HOURS & CREDITS

INSTITUTIONAL TRAINING: TOTAL HOURS & CREDITS			
ITEM	Periods (1 hour duration each)		Total
	Lectures/Tutorials	Practicals	
TOTAL HOURS COMMON SUBJECTS	82	128	210
TOTAL HOURS SPECIALISED SUBJECTS (ARMY/NAVY/AIR FORCE)	38	52	90
TOTAL HOURS INSTITUTIONAL TRAINING	120	180	300
TOTAL CREDITS INSTITUTIONAL TRAINING	08 CREDITS (15 HOUR THEORY = 1 CREDIT POINT)	6 CREDITS (30 HOURS PRACTICAL TRAINING = 1 CREDIT POINT)	



BHOPAL
NCC CAMP TRAINING SYLLABUS

COMMON SUBJECTS				
S No.	Subjects	Periods		Total
		L/T	P	
1.	Physical Training	-	18	18
2.	Drill	-	32	32
3.	Weapon Training	08	28	36
4.	National Integration and Awareness	08	-	08
5.	Personality Development	08	12	20
6.	Leadership	08	-	08
7.	Disaster Management	08	-	08
8. .	Social Service and Community Development	-	08	08
9.	Health & Hygiene	08	-	08
10.	Obstacle Training	-	04	04
11.	Military History	04	-	04
12.	Communication	04	-	04
13.	Games	-	18	18
14.	Culture	-	18	18
	TOTAL	56	138	194
<u>SPECIALISED SUBJECTS</u>				
1.	Map Reading	-	24	24
2.	Infantry Weapons	04	02	06
3.	Field Craft & Battle Craft	-	16	16
	TOTAL	04	42	46
	GRAND TOTAL	60 (4 credit)	180 (6 credit)	240 (10 credit)

NCC CAMP TRAINING SYLLABUS (FOR THEORY)

Ser No	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Weapon Training	-	-	04	-	04	-	08
2.	National Integration & Awareness	-	-	04	-	04	-	08
3.	Personality Development	-	-	04	-	04	-	08
4.	Leadership	-	-	04	-	04	-	08
5.	Disaster Management	-	-	04	-	04	-	08
6.	Health & Hygiene	-	-	04	-	04	-	08
7.	Military History			02		02		04
8.	Communication			02		02		04
9.	Infantry Weapons	-	-	02	-	02	-	04
	TOTAL	-	-	30	-	30	-	60
	TOTAL Credit	-	-	2	-	2	-	4

NCC CAMP TRAINING SYLLABUS (FOR PRACTICAL)

Ser No	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Physical Training	-	-	09	-	09	-	18
2.	Drill	-	-	16	-	16	-	32
3.	Weapon Training	-	-	14	-	14	-	28
4.	Personality Development	-	-	06	-	06	-	12
5.	Social Service and Community Development	-	-	04	-	04	-	08
6.	Obstacle Training	-	-	02	-	02	-	04
7.	Games			09		09		18
8.	Culture			09		09		18
9.	Map Reading	-	-	12	-	12	-	224
10.	Infantry Weapons	-	-	01	-	01	-	02
11.	Field Craft & Battle Craft	-	-	08	-	08	-	16
	TOTAL			90		90		180
	TOTAL CREDIT			03		03		06



SARVEPALLI RADHAKRISHNAN UNIVERSITY,
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SEMESTER WISE COURSE DESIGN

ARMY CADETS

NATIONAL CADET CORPS

INSTITUTIONAL TRAINING: SEMESTER WISE DISTRIBUTION OF NCC SYLLABUS FOR THEORY (ARMY CADETS)

S. NO.	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	NCC General	06	-	-	-	-	-	06
2.	National Integration	04	-	-	-	-	-	04
3.	Personality Development	02	05	05	04	06	04	25
4.	Leadership	-	05	04	03	-	-	12
5.	Disaster Management	-	-	03	10	-	-	13
6.	Social Service & Community Development	03	05	-	-	-	-	08
7.	Adventure	-	-	01	-	-	-	01
8.	Environmental Awareness & Conservation	-	-	-	03	-	-	03
9.	General Awareness	-	-	-	04	-	-	04
10.	Border & Coastal Areas	-	-	02	-	02	02	06
11.	Armed Forces	-	-	-	06	-	03	09
12.	Infantry Weapons	-	-	-	-	3	-	3
13.	Communication	-	-	-	-	-	03	03
14.	Military Hospital	-	-	-	-	04	19	23
	TOTAL	15	15	15	30	15	30	120
	TOTAL Credit	1	1	1	2	1	2	08

INSTITUTIONAL TRAINING: SEMESTER WISE DISTRIBUTION OF NCC SYLLABUS FOR PRACTICAL (ARMY CADETS)

S. NO.	SUBJECT	I	II	III	IV	V	VI	TOTAL
1.	Drill	12	12	08	07	03	03	45
2.	Field Craft & Battle Craft	03	04	04	04	04	03	22
3.	Map Reading	03	05	04	04	04	04	24
4.	Weapons Training	05	04	04	04	04	04	25
5.	Communication	-	-	-	-	-	03	03
6.	Infantry Weapons	-	-	-	-	-	03	03
7.	Social Service & Community Development	07	05	05	06	06	10	39
8.	Health & Hygiene	-	-	-	05	05	-	10
9.	Operation Training	-	-	05	-	04	-	09
	TOTAL	30	30	30	30	30	30	180
	TOTAL Credit	01	01	01	01	01	01	06



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INSTITUTIONAL TRAINING: SEMESTER WISE THEORY DETAILED SYLLABUS (ARMY CADETS)

<u>SEMESTER I</u>					
S.No	Subject	Periods	Chapter	Lesson	Hours
1	NCC General	6	NCC-I	Aims, Objectives and Org of NCC	1
			NCC-II	Incentives	2
			NCC-III	Duties of NCC Cadets	1
			NCC-IV	NCC Camps: Types and Conduct	2
2	National Integration and Awareness	4	NI-I	National Integration: Importance and Necessity	1
			NI-II	Factors affecting National Integration	1
			NI-III	Unity in Diversity	1
			NI-IV	Threats to National Security	1
3	Personality Development	2	PD - I	Factors Self-Awareness Empathy Critical and Creative Thinking Decision Making and Problem Solving	2
4	Social Service and Community Development	3	SSCD - I	Basics of Social Service Rural Development Programmes NGO's Contribution of Youth	3
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER II					
S.No	Subject	Periods	Chapter	Lesson	Hours
5	Personality Development	5	PD-II	Communication Skills	3
			PD-III	Group Discussion -Coping with Stress and Emotions	2
6	Leadership	5	L-I	Leadership Capsule Traits Indicators Motivation Moral Values Honour Code	3
			L-II	Case Studies Shivaji, Jhansi Ki Rani,	2
7	Social Service and Community Development	5	SS-IV	Protection of Children & Women Safety	1
			SS-V	Road/Rail Travel Safety	1
			SS-VI	New Initiatives	2
			SS-VII	Cyber and Mobile Security Awareness	1
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER III					
S.No	Subject	hours	Chapter	Lesson	HOURS
8	Personality Development	5	PD-III	Group Discussions - Change your Mindset	2
			PD-V	Public Speaking	3
9	Leadership	4	L-II	Case Studies – APJ Abdul Kalam, Deepa Malik, Maharana Pratap, N Narayan Murthy	4
10	Disaster Management	3	DM-I	Disaster Management Capsule Organisation Types of Disasters Essential Services Assistance Civil Defence Organisation	3
11	Adventure	1	AD-I	Adventure activities	1
12	Border & Coastal Areas	2	BCA-I	History, Geography & Topography of Border/ Coastal Areas	2
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER IV					
S.No	Subject	hours	Chapter	Lesson	HOURS
13	Personality Development	4	PD-III	Group Discussions - Time Management, Social Skills	4
14	Leadership	3	L-II	Case Studies – Ratan Tata, Rabindra Nath Tagore, Role of NCC cadets in 1965 war	3
15	Disaster Management	9	DM-II	Initiative Trg, Organising Skills, Dos and Don'ts Natural Disasters Man Made Disasters	9
		1	DM-III	Fire Services and Fire Fighting	1
16	Environmental Awareness	3	EA-I	Environmental Awareness and Conservation	3
17	General Awareness	4	GA-I	General Awareness	4
18	Armed Forces	6	AF-1	Army, Navy, Air Force and Central Armed Police Forces	6
TOTAL HOURS					30
TOTAL CREDITS					2



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SEMESTER V

S.No	Subject	hours	Chapter	Lesson	HOURS
19	Personality Development	6	PD-III	Group Discussions - Team Work	2
			PD-V	Public Speaking	4
20	Border & Coastal Areas	2	BCA-II	Security Setup and Border/Coastal management in the area	2
21	Introduction to Infantry Battalion and its Equipments	3	INF-1	Organisation of Infantry Battalion & its weapons	3
22	Military History	4	MH-3	Study of Battles of Indo-Pak Wars 1965 & 1971	4
TOTAL HOURS					15
TOTAL CREDITS					1

SEMESTER VI

S.No	Subject	hours	Chapter	Lesson	HOURS
25	Personality Development	3	PD-IV	Career Counselling, SSB Procedure and Interview Skills	3
27	Border & Coastal Areas	2	BCA-III	Security Challenges & Role of cadets in Border management	2
28	Armed Forces	3	AF-2	Modes of Entry into Army, Police and CAPF	3
29	Military History	19	MH-1	Biographies of Renowned Generals	6
			MH-2	War Heroes : Param Veer Chakra Awardees	3
			MH-3	Study of Battles of Kargil	2
			MH-4	War Movies	8
30	Communication	3	C-1	Introduction to Communication & Latest Trends	3
TOTAL HOURS					30
TOTAL CREDITS					2

SIX SEMESTER NCC COURSE SYLLABUS

Training Objectives: Institutional Training

145. Institutional training includes basic military training of the cadets as part of the curriculum with its long-standing effort to mould young volunteers into disciplined and responsible citizens of India. NCC course is aimed to achieve following learning objectives:-

- (a) Develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.
- (b) To create interest in cadets by including and laying emphasis on those aspects of Institutional Training which attract young cadets into the NCC and provides them an element of thrill and excitement.
- (c) To inculcate defence Services work ethos that is characterized by hard work, sincerity of purpose, honesty, ideal of selfless service, dignity of labour, secular outlook, comradeship, spirit of adventure and sportsmanship.
- (d) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regardless of which career they choose.
- (e) To provide conducive environment to motivate young Indians to choose the Armed Forces as a career.



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SEMESTER I COURSE MODULE : NATIONAL CADET CORPS I

National Cadet Corps : Course Details			
Course Title: National Cadet Corps I			
Course Code	BNCC01GE03	Credits	1(Thr)+ 1(Pr) = 03
L / T + P	15+30	Course Duration	1 Semester
Semester	I (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e, 25% internal assessment and 75% end term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

146.

Course Objectives:

Cadets will be able to: -

- (a) Know about the history of NCC, its organization, and incentives of NCC for their career prospects.
- (b) Acquire knowledge of duties and conduct of ncc cadets.
- (c) Understand about different NCC camps and their conducts.
- (d) Understand the concept of national integration and its importance.
- (e) Understand the concept of self-awareness and emotional intelligence.
- (f) Understand the concept of critical & creative thinking.
- (g) Understand the process of decision making & problem solving.
- (h) Understand the concept of team and its functioning.
- (i) Understand the concept and importance of Social service.

147. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Imbibe the conduct of NCC cadets.
- (b) Respect the diversity of different Indian culture.
- (c) Practice togetherness and empathy in all walks of their life.
- (d) Do their own self analysis and will workout to overcome their weakness for better performance in all aspects of life.
- (e) Understand creative thinking & its components.
- (f) Think divergently and will try to break functional fixedness.
- (g) Make a team and will work together for achieving the common goals.
- (h) Do the social services on different occasions.

148. **Course Content Part (I) Theory**

- (a) **Unit 1- NCC General (N) (Contact Hrs. 06).** Introduction of NCC, History, Aims, Objective of NCC & NCC as Organization, Incentives of NCC, Duties of NCC Cadet. NCC Camps: Types & Conduct.
- (b) **Unit 2- National Integration & Awareness (NI) (Contact Hrs. 04) .** National Integration: Importance & Necessity, Factors Affecting National Integration, Unity in Diversity & Role of NCC in Nation Building, Threats to National Security.
- (c) **Unit 3- Personality Development (Contact Hrs. 3).** Intra & Interpersonal skills - Self-Awareness- & Analysis, Empathy, Critical & creative thinking, Decision making and problem solving.
- (d) **Unit 4- Social Service and Community Development(Contact Hrs. 02).** Basics of social service and its need, Types of social service activities, Objectives of rural development programs and its importance, NGO's and their contribution in social welfare, contribution of youth and NCC in Social welfare.

Course Content Part (II) Practical

149. **Course Objectives:** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of foot drill.
- (c) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.



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- (d) Develop awareness about different types of terrain and how it is used in battle craft.
- (e) Develop the concept of various markings on the map and how they are co-related to the ground features.
- (f) Understand the various social issues and their impact on social life.
- (g) Develop the sense of self-less social service for better social & community life.

150. **Expected Learning Outcomes:** After completing this course, the cadets will be able to: -

- (a) Perform foot drill and follow the different word of command.
- (b) Fire a weapon effectively with fair degree of marksmanship.
- (c) Undertake point to point navigation and take part in route marches by day and night.
- (d) Perform the social services on various occasions for better community & social life.

151. Course Content Part (II) Practical

- (a) **Unit 1. Drill (Contact Hrs. 12).** Foot Drill- Drill ki Aam Hidayaten, Word ki Command, Savdhan, Vishram, Aram Se, Murdna, Kadvar Sizing, Teen Line Banana, Khuli Line, Nikat Line, Khade Khade Salute Karna Parade Par, Visarjan, Line Tod, Tej Chal, Tham aur Dhire Chal, Tham.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 05).** Introduction & Characteristics of .22 rifle, Handling of .22 rifle.
- (c) **Unit 3. Map Reading (MR) (Contact Hrs. 03).** Definition of Map, Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Magnetic Variation and Grid Convergence.
- (d) **Unit 4. Field Craft & Battle Craft (FC & BC) (Contact Hrs. 03).** Introduction of Field Craft & Battle craft, Judging Distance, Method of Judging Distance.
- (e) **Unit 5. Social Service and Community Development (SSCD)(Contact Hrs.07).** Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc.

SEMESTER II COURSE MODULE : NATIONAL CADET CORPS II

Course Title: National Cadet Corps II			
Course Code	BNCC02GE03	Credits	1(Thr)+ 1(Pr)=02
L / T + P	15+30	Course Duration	1 Semester
Semester	II (Even)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

152.

Course Objectives: Cadets will be able to: -

- (a) Understand the thinking & reasoning process.
- (b) Understand the process to cope with Stress & emotions.
- (c) Understand the importance of improving communication skills.
- (d) Identify the leadership traits.
- (e) Admire the qualities of great leaders.
- (f) Know about different legal provisions for children & women safety and protection.
- (g) Understand the various rules & measures to be taken to ensure Road/Rail safety.
- (h) Understand & spread awareness about latest Government initiatives for welfare of citizens and contribute towards Nation building.
- (i) Understand concepts of cyber and mobile security.



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153. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Define thinking, reasoning, critical thinking and creative thinking.
- (b) To think critically about different life related issues.
- (c) Think divergently and will try to break functional fixedness.
- (d) Creatively in their real-life problems.
- (e) Understand the organizations related to disaster management and their functioning.
- (f) Appreciate the role of NCC cadets in disaster management.

154. **Course Content Part (I) Theory**

(a) Unit 1. Personality Development (Contact Hrs.5)

- (i) Thinking- Meaning and Concept of thinking, Reasoning, Process of thinking.
- (ii) Critical Thinking- Meaning & concept of critical thinking, Features of critical thinking, Process of critical thinking.
- (iii) Creative thinking- Meaning & concept of creative thinking, Features of creative thinking, Process of creative thinking, levels of Creativity, Characteristics of creative person.

(b) Unit 2. Leadership Development (Contact Hrs.5)

- (i) Leadership capsule.
- (ii) Important Leadership traits, Indicators of leadership and evaluation.
- (iii) Motivation- Meaning & concept, Types of motivation. Factors affecting motivation.
- (iv) Ethics and Honor codes.

(c) Unit 3. Social Service and Community Development (Contact Hrs. 5)

- (i) Protection of Children & Women Safety.
- (ii) Road/Rail Safety.
- (iii) New Government Initiatives.
- (iv) Cyber and mobile Security Awareness.

Course Content Part (II) Practical

155. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of foot drill.
- (c) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (d) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.

156. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform foot drill gracefully.
- (b) Give and follow the different word of command.
- (c) Fire a weapon effectively with fair degree of marksmanship.
- (d) Use of bearing and service protractor and locate the places and objects on the ground.
- (e) Do the social service and feel connected with social problems.

157. **Course Content Part (II) Practical**

(a) Unit 1. Drill (Contact Hrs. 12)

- (i) Foot Drill Dahine, Baen, Aageaur Piche Kadam Lena.
- (ii) Tej Chal se Murdna, Tej Chal se Salute Karna, Tej Kadam Taal aur Tham, Tej Kadam Taal se Kadam Badalna.
- (iii) Teeno Teen se Ek File aur ek file se Teeno Teen Banana

(b) Unit 2. Weapon Training (Contact Hrs. 04)

- (i) Range procedure & Theory of group.
- (ii) Short Range firing.

(c) Unit 3. Map Reading (Contact Hrs. 05)

- (i) Protractor Bearing and its conversion methods.
- (ii) Service protractor and its uses.
- (iii) Prismatic compass and its uses and GPS.
- (iv) Navigation by compass and GPS.



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(d) Unit 4. Field Craft & Battle Craft (Contact Hrs. 04)

- (i) Indications of landmarks and Targets.
- (ii) Intro, Definitions, Types of Ground, Indication of Landmarks, Methods of identification of targets, difficult targets.

(e) Unit 5. Social Service and Community Development (Contact Hrs. 05)

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc. as per the requirement and similar announced days- National and state level.

NATIONAL CADET CORPS

SEMESTER III COURSE MODULE : NATIONAL CADET CORPS III

<u>COURSE TITLE: NATIONAL CADET CORPS III</u>			
Course Code	BNCC03GE02	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15 +30	Course Duration	1 Semester
Semester	III (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

158. **Course Objectives.** Cadets will be able to: -

- (a) Understand the life history and leadership qualities of great leaders, sportspersons & entrepreneurs.
- (b) Understand the various aspects of types of mindset.
- (c) Understand public speaking methods & qualities.
- (d) Understand the organizations related to disaster management and their functioning.
- (e) Understand the role of NCC cadets in disaster management.
- (f) Understand the various types of adventure activities.
- (g) Understand the History, Geography & Topography of Border/ Coastal Areas.



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159. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Admire and get inspired from the accomplishments of leaders from various walks of life.
- (b) Develop public speaking skills.
- (c) Understand the importance of positive mindset and optimistic attitude in life.
- (d) Appreciate the need & requirement for disaster management and his role in disaster management activities.
- (e) Know the history & geographical peculiarity of our borders & coastal regions.

160. Course Content Part (I) Theory

(a) Unit 1. Personality Development (Contact Hrs.5)

- (i) Group Discussions - Change your Mindset
- (ii) Public Speaking.

(b) Unit 2. Leadership Development (Contact Hrs.4). Case Studies – APJ Abdul Kalam, Deepa Malik, Maharana Pratap, N Narayan Murthy.

(c) Unit 3. Disaster management(Contact Hrs. 3)

- (i) Disaster Management Capsule.
- (ii) Organisation.
- (iii) Types of Disasters.
- (iv) Essential Services.
- (v) Assistance.
- (vi) Civil Defence Organisation.

(d) Adventure (Contact Hrs. 1). Adventure activities.

(e) Border & Coastal Areas(Contact Hrs. 2). History, Geography & Topography of Border/ Coastal Areas.

Course Content Part (II) Practical

161. **Course Objectives.** Cadets will be able to : -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal
- (b) Appreciate grace and dignity in the performance of arm drill
- (c) Understand the concept and importance of social service.
- (d) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (e) Actively participate in social service and community development activities.

162. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform arm drill gracefully.
- (b) Give and follow the different word of command.
- (c) Fire a weapon effectively with fair degree of marksmanship.
- (d) Different positioning for fire and aiming.
- (e) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (f) Observe surroundings in better way.
- (g) Develop the qualities of patience and confidence and become better individuals.
- (h) Will develop physical as well as mental fitness.

163. **Course Content Part (II) Practical**

(a) **Unit 1. Drill(Contact Hrs. 08)**

- (i) Arm Drill.
- (ii) Rifle ke saath Savdhan, Vishram aur Aram se.
- (iii) Rifle ke saath Parade Par aur Saj, Rifle ke saath Visarjan, Line Tod.
- (iv) Bhumi Shastra aur Uthao Shastra, Bagal Shastra aur Baju Shastra.

(b) **Unit 2. Weapon Training(Contact Hrs. 04).** Short Range firing.

(c) **Unit 3. Map Reading (Contact Hrs. 04).**

- (i) Setting of Map.
- (ii) Findings North and Own Position.



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(d) Unit 4. Field Craft & Battle Craft (Contact Hrs. 04)

- (i) Observation.
- (ii) Camouflage.
- (iii) Concealment.

(e) Unit 5. Social Service and Community Development (Contact Hrs. 05).

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) Unit 6. Obstacle Training(Contact Hrs. 05)

- (i) Obstacle training - Introduction, Safety-measures, Benefits.
- (ii) Obstacle Course- Straight balance, Clear Jump, Gate Vault, Zig- Zag Balance, High Wall.

NATIONAL CADET CORPS

SEMESTER IV COURSE MODULE : NATIONAL CADET CORPS IV

Course Title: National Cadet Corps IV			
Course Code	BNCC04GE03	Credits	2(Thr)+ 1(Pr)=03
L /T + P	30+30	Course Duration	1 Semester
Semester	IV (Even)	Contact Hours	30(Thr)+30(Pr)=60Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

164. **Course Objectives.** Cadets will be able to: -

- (a) Develop a sense of time management and social skills.
- (b) Understand the life history & leadership qualities of personalities who have contributed in Nation Building and Literature.
- (c) Understand the role of NCC cadets as 2nd line Defence in 1965 War.
- (d) Develop awareness about various types of Natural and manmade disasters.
- (e) Know about life saving tips during disasters.
- (f) acquainted about Fire Services.
- (g) Understand importance of Environmental Awareness & conservation.
- (m) Understand importance of General Awareness.
- (n) Know about Armed Forces.



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(e) **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (i) Effectively Manage time.
- (ii) Develop the qualities of social skills.
- (iii) Imbibe leadership qualities.
- (iv) Do group discussions effectively.
- (v) Be motivated to serve the nation by joining Armed forces.
- (vi) Contribute in environmental awareness and conservation activities.
- (vii) Keep abreast of current affairs & general awareness.
- (viii) Effectively contribute in managing disaster relief tasks.

165. Course Content Part (I) Theory

- (a) **Unit 1. Personality Development (Contact Hrs.4).** Group Discussions – Social Skills & Time management.
- (b) **Unit 2. Leadership Development (Contact Hrs.3).** Case Studies – Case Studies – Ratan Tata, Rabindra Nath Tagore, Role of NCC cadets in 1965 war.
- (c) **Unit 3. Disaster management(Contact Hrs. 10)**
 - (i) Initiative Trg, Organising Skills.
 - (ii) Dos and Don'ts.
 - (iii) Natural Disasters.
 - (iv) Man Made Disasters.
 - (v) Fire Services and Fire Fighting.
- (d) **Environmental Awareness (Contact Hrs. 3).** Adventure Environmental Awareness and Conservation.
- (e) **General Awareness (Contact Hrs. 4).** General Awareness.
- (f) **Armed Forces(Contact Hrs. 6).** Army, Navy, Air Force and Central Armed Police Forces.

Course Content Part (II) Practical

166. **Course Objectives.** Cadets will be able to: -
- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
 - (b) Understand various signals to convey messages in the army.
 - (c) Get acquainted various section formations.
 - (d) Understand the basics of personal and public hygiene.
 - (e) Get acquainted with the procedure to treat the wounds and fractures during emergencies.
167. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -
- (a) Perform weapon drill gracefully.
 - (b) Give and follow the different word of command.
 - (c) Appreciate grace and dignity in the performance of foot drill.
 - (d) Apply signals in there day to day functioning.
 - (e) Provide first aid during the emergencies.
 - (f) Navigate to the given location on ground using compass and GPS.
 - (g) Practice healthy practices for the personal sanitation and hygiene.
168. **Course Content Part (II) Practical**
- (a) **Unit 1. Drill (Contact Hrs. 08)**
 - (i) Arm Drill.
 - (ii) Salami Shastra.
 - (iii) Squad Drill with Arms.
 - (b) **Unit 2. Weapon Training (Contact Hrs. 04).** Short Range firing
 - (c) **Unit 3. Map Reading(Contact Hrs. 04)**
 - (i) Map to Ground.
 - (ii) Ground to Map.
 - (d) **Unit 4. Field Craft & Battle Craft(Contact Hrs. 04)**
 - (i) Fire and Move Capsule.
 - (ii) Field signal- with hand, with Weapons, Signal with Whistle.
 - (iii) Field signals as means of giving orders.



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(iv) Field signals by day, Field signals by night.

(v) Section Formation.

(e) **Unit 5. Social Service and Community Development(Contact Hrs. 05)**

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) **Unit 6. Health & Hygiene(Contact Hrs. 05)**

(i) Hygiene & Sanitation (Hygiene- Personal & Camp Hygiene).

(ii) First Aid in common medical emergencies.

(iii) Treatment & Care of Wounds.

NATIONAL CADET CORPS

SEMESTER V COURSE MODULE : NATIONAL CADET CORPS V

Course Title: National Cadet Corps V			
Course Code	BNCC05GE02	Credits	1(Thr)+ 1(Pr)=02
L /T + P	15 +30	Course Duration	1 Semester
Semester	V (Odd)	Contact Hours	15(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

169. **Course Objectives.** Cadets will be able to: -

- (a) Understand the concept of Team and its functioning.
- (b) Hone Public speaking skills.
- (c) Understand the security set up and management of Border/Coastal areas.
- (d) Acquire knowledge about an Infantry Battalion organisation and its weapons.
- (e) Acquire knowledge about Indo-Pak Wars fought in 1965 & 1971.

170. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Participate in team building exercise and value team work.
- (b) Improve communication skills by public speaking activities.
- (c) Understand the security mechanism and management of Border/Coastal areas.
- (d) Get motivated to join armed forces.



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171. **Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (Contact Hrs.6).**
 - (i) Group Discussions –Team work.
 - (ii) Public speaking.
- (b) **Unit 2. Border & Coastal Areas(Contact Hrs.2).** Security Setup and Border/Coastal management in the area.
- (c) **Unit 3. Introduction to Infantry Battalion and its Equipment(Contact Hrs. 3).** Organisation of Infantry Battalion & its weapons
- (d) **Military History(Contact Hrs. 4).** Study of Battles of Indo-Pak Wars 1965 & 1971.

Course Content Part (II) Practical

172. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of ceremonial drill.
- (c) Use the compass and GPS to locate places on the ground and map.

173. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform ceremonial drill and follow the different word of command.
- (b) Do the social service on various occasions and get connected with the community.
- (c) Do all the asana and gain the physical& mental fitness.

174. **Course Content Part (II) practical**

- (a) **Unit 1. Drill(Contact Hrs. 03)**
 - (i) Ceremonial Drill.
 - (ii) Guard Mounting.
- (b) **Unit 2. Field Craft & Battle Craft(Contact Hrs. 04)**
 - (i) Fire control orders.
 - (ii) Types of fire control orders.

(iii) Fire and Movement- when to use fire and movements tactics, Basic considerations, Appreciation of ground cover, Types of cover, Dead ground, Common Mistakes, Map and air photography, Selection of Fire position and fire control.

(c) **Unit 3. Map Reading(Contact Hrs. 04)**.Google Maps & applications

(d) **Unit 4. Weapon Training(Contact Hrs. 04)**.Short Range firing

(e) **Unit 5. Social Service and Community Development (Contact Hrs. 05)**

Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc. as per the requirement and similar announced days- National and State level.

(f) **Unit 6. Health & Hygiene(Contact Hrs. 05)**

(i) Yoga- Introduction, Definition, Purpose, Benefits.

(ii) Asanas-Padamsana, Siddhasana, Gyan Mudra, Surya Namaskar, Shavasana, Vajrasana, Dhanurasana, Chakrasana, Sarvaangasana, Halasana etc.

(i) **Unit 7. Obstacle Training(Contact Hrs. 05)**

(i) Obstacle training – Intro, Safety measures, Benefits.

(ii) Obstacle Course- Straight balance, Clear Jump, Gate Vault, Zig- Zag Balance, High Wall etc.



Course Title: National Cadet Corps VI			
Course Code	BNCC06GE03	Credits	2(Thr)+ 1(Pr)=03
L / T + P	30 +30	Course Duration	1 Semester
Semester	VI (Even)	Contact Hours	30(Thr)+30(Pr)=45Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, Seminar presentations by students, individual and group drills, group and individual field-based assignments, Educational Excursion		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

175. **Course Objectives.** Cadets will be able to: -

- (a) Get acquainted about counselling process its need and importance.
- (b) Know about SSB procedure and different tasks and tests.
- (c) Know about the conduction during the interview.
- (d) Understand the security challenges & role of cadets in Border Areas.
- (e) Know about the modes of entry in Armed forces, CAPF & police.
- (f) Understand the life history & leadership qualities of great generals.
- (g) Learn about 1999 Kargil war.
- (h) Acquire the knowledge about various wars and their heroes.
- (i) Know about various components of communication process.

176. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Get motivated to join Armed forces, police & CAPF.
- (b) Write their CV effective and appealing.
- (c) Face SSB interview effectively in their future.
- (d) Understand individual responsibilities & role in meeting the security challenges on Border/Coastal areas.
- (e) Imbibe the feeling of patriotism.
- (f) Communicate more effectively.

177. **Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (Contact Hrs.3).**
 - (i) Career Counselling.
 - (ii) SSB Procedure.
 - (iii) Interview Skills.
- (b) **Unit 2. Border & Coastal Areas(Contact Hrs.2).** Security Challenges & Role of cadets in Border management.
- (c) **Unit 3. Armed Forces(Contact Hrs. 3).** Modes of Entry into Army, Police and CAPF.
- (d) **Military History(Contact Hrs. 19).**
 - (i) Biographies of Renowned Generals.
 - (ii) War Heroes : Param Veer Chakra Awardees.
 - (iii) Study of Battles of Kargil.
 - (iv) War Movies.
- (e) **Communication(Contact Hrs. 3).** Introduction to Communication & Latest Trends.



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Course Content Part (II) Practical

178. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill as the foundation for discipline and to command a group for common goal.
- (b) Appreciate grace and dignity in the performance of ceremonial drill.
- (c) Know about various knots and lashing used in soldiering.
- (d) Acquire awareness about the basic weapon system in use in the Armed Forces.

179. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform foot drill and follow the different word of command.
- (b) Aiming range and figure targets.
- (c) Use the different knots and lashing in day-to-day life for different purposes.
- (d) Develop the feeling of altruism.

180. **Course Content Part (II) Practical.**

(a) **Unit 1. Drill (Contact Hrs. 03).**

- (i) Ceremonial Drill.
- (ii) Guard of Honour.

(b) **Unit 2. Weapon Training(WT) (Contact Hrs. 04).** Short Range firing.

(c) **Unit 3. Map Reading(MR) (Contact Hrs. 04).** Google maps and Applications.

(d) **Unit 4. Field Craft & Battle Craft(FCBC) (Contact Hrs. 03).** Knots, Lashing and Stretchers.

(e) **Unit 5. Social Service and Community Development(SSCD) (Contact Hrs. 05).** Cadets will participate in various activities throughout the semester e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc as per the requirement and similar announced days- National and State level.

(f) **Unit 6 Introduction of Infantry Weapons & Equipment(INF) (Contact Hrs.03).** Characteristics of 5.56MM INSAS Rifle, Ammunition, Fire Power, Stripping, Assembling & Cleaning Practice.

(g) **Unit 7. Communication (COM) (Contact Hrs. 03).**

- (i) Basic Radio Telephony (RT) Procedure.
- (ii) Introduction, Advantages, Disadvantages, Need for standard procedures.

(iii) Types of Radio telephony communication.

(iv) Radio telephony procedure, Documentation.

NATIONAL CADET CORPS



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COURSE MODULE: NATIONAL CADET CORPS CAMP -I

Course Title: National Cadet Corps Camp I			
Course Code	BNCCCAMP03GE05	Credits	2(Thr)+ 3(Pr)=05
L /T + P	30+90	Course Duration	10 Days (24 hours each)
Semester	III (Odd)	Contact Hours	30(Thr)+90(Pr)=120Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, individual and group tasks, team work, field-based assignments, Physical Training, endurance building and skill development practices		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (I) Theory

181. **Course Objectives**. Cadets will be able to: -

- (a) Acquire knowledge about the various aspects of personality development.
- (b) Understand the concept of leadership traits, moral values and character traits.
- (c) Develop awareness about the various types of natural disasters.
- (d) Develop sensitivity to the changing environment and understand the importance of conservation.
- (e) Understand the importance of hygiene and sanitation and common first aid procedures.
- (f) Acquire awareness about various types of weapon systems in the Armed Forces.

182. **Expected Learning Outcomes**. After completing this course, the cadets will be able to: -

- (a) Acquire adequate skill sets to overcome their weakness and reshape their personality.
- (b) Imbibe good moral values and character traits in their daily life.
- (c) Become useful members of the society and form part of disaster response team, if need arises.
- (d) Respect and make efforts to conserve natural resources
- (e) Follow good personal hygiene practices and provide first aid in emergencies.
- (f) Be motivated to join the armed forces.

183. **NCC Camp-I : Course Content Part (I) Theory**

- (a) **Unit 1. Personality Development (PD) (Contact Hrs. 04)**. Introduction to Personality Development, Factors influencing/shaping personality, Time Management and Interview Skills.
- (b) **Unit 2. Leadership (LDR) (Contact Hrs. 04)**. Leadership Traits, Moral Values and Character Traits.
- (c) **Unit 3. Disaster Management (DM) (Contact Hrs. 04)**. Assistance during natural disasters, Do's and Don'ts for NCC Cadets performing Disaster Management Duties
- (d) **Unit 4. National Integration and Awareness (NIA)(Contact Hrs. 04)**.
Water Conservation and Rain Harvesting, Waste Management and Energy Conservation
- (e) **Unit 5. Health and Hygiene (H&H)(Contact Hrs. 04)**. Hygiene and Sanitation, First Aid in Common Medical Emergencies.
- (f) **Unit 6. Infantry Weapons (IW) (Contact Hrs. 02)**. Characteristics of Company Support Weapons.
- (g) **Unit 7. Weapon Training (WT) (Contact Hrs. 04)**. Characteristics of Point 22 Rifle and its Ammunition, Range Procedure and Safety Precautions.
- (h) **Unit 8. Military History (MH) (Contact Hrs. 04)**. Guest lectures by War Veterans/decorated soldiers/veterans.
- (i) **Unit 9. Communication (COM) (Contact Hrs. 04)**. Basics of communication.



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NCC Camp-I : Course Content Part (II) Practical

184. **Course Objectives.** Cadets will be able to: -

- (a) Understand that drill is the foundation of discipline and command a group for a common goal.
- (b) Understand the importance of a weapon its detailed safety precautions necessary for prevention of accidents.
- (c) Develop awareness about different types of terrain and how it is used in Battle Craft.
- (d) Develop the concept of various markings on the map and how they are co-related to the ground features.
- (e) Acquire awareness about the various types of weapon systems in the Armed Forces.
- (f) Understand the concept and importance of social service.
- (g) Understand the various nuances of Personality Development.
- (h) Understand the concept and importance of Physical Training in everyone's life.
- (i) Acquire skill sets about various games and understand the importance of team work.
- (j) Develop awareness about different cultures and different modes of its projection in artistic forms.

185. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Perform foot drill, arms drill, ceremonial drill and will be able to give out different words of command.
- (b) Fire a weapon effectively with fair degree of marksmanship.
- (c) Undertake point to point navigation and take part in route marches by day and night.
- (d) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (e) Be motivated to join the armed forces.
- (f) Acquire adequate skill sets to overcome their weakness and enhance their personality.
- (g) Gain adequate physical and mental endurance capabilities.
- (h) Play team games and be able to communicate and coordinate effectively in group events or situations.

- (i) Respect the diversity of Indian culture and develop pride by showcasing their own culture to others.

186. NCC Camp-I : Course Content Part (II) Practical

- (a) **Unit 1. Drill (Drill)(Contact Hrs. 16).** Drill ki Aam Hidayaten aur Words of Command, Savdhan, Vishram, Aram Se aur Mudna, Khuli Line aur Nikat Line mein march, Salute Karna Parade Par, Visarjan aur Line Tod, Tej Chal, Tham aur Dhire Chal, Tham, Dahine, Baen, Aage aur Piche Kadam lena, Tejchaal se Mudna, Tejchaal se Salute karna, Tej kadamtaal aur Tham, Tej Kadamtaal se kadam badhana, Teenon Teen se ek file Banana aur ek file se Teenon Teen Banana, Rifle Ke Saath Saavdhan, Vishram aur Aaram se, Rifle ke saath Parade par aur saaj, Rifle Ke saath visarjan aur line tod, Bhumi Sashttra aur Uthao Sashttra, Bagal Sashttra aur Baaju Shastra.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 14).** Stripping, Assembling, Cleaning of Point 22 rifle, Sight Setting and Sight Picture of Point 22 Rifle, Loading, Cocking and Unloading, Lying Position, Holding and Aiming of Point 22 rifle, Trigger Control and Firing of Shot, Theory of Group, Short-Range Aiming and Firing, Firing Practice I to VII.
- (c) **Unit 3. Field Craft & Battle Craft (FC/BC) (Contact Hrs. 06).** Introduction of Field Craft & Battle craft, Judging Distance, Indication of Landmarks and Targets, Observation, Camouflage and Concealment, Field Signals, Section formations.
- (d) **Unit 4. Map Reading (MR) (Contact Hrs. 12).** Introduction to Map and Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Types of Bearing and use of Service Protector, Prismatic Compass and its use, setting of a map, Finding North and own Position, Map to Ground and Ground to map, Point to Point march, Route March – I, Route March -II.
- (e) **Unit 5. Infantry Weapons (IW) (Contact Hrs. 01).** Characteristics of Battalion Support Weapons.
- (f) **Unit 6. Social Service and Community Development (SSCD) (Contact Hrs. 04).** Basics of Social Service and its need, Rural Development Programme, Civic Responsibilities: Cadets will participate in various activities throughout the camp e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc., Road /Rail Travel Safety
- (g) **Unit 7. Personality Development (PD) (Contact Hrs. 06).** Self-Awareness, Empathy, Critical and Creative Thinking, Decision making and problem Solving, Coping with Stress and Emotions, Time Management.
- (h) **Unit 8. Obstacle Training (OT) (Contact Hrs. 02).** OT Practice – I:- Untimed, Cadets will be familiarized with all the obstacles in the Obstacle Course and briefed about the correct method to do them, OT Practice -II: Timed practice for all the cadets and record to be maintained.



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- (i) **Unit 9. Physical Training (PT) (Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in morning hours. Training has to be progressive in degree of difficulty to improve individual stamina and endurance. Training to include warming up, running, exercises to strengthen upper body, lower body and core muscles. Two period each to be devoted to route march by day and night respectively and one period will be earmarked for trekking expedition as part of Adventure Activity.
- (j) **Unit 10. Games Training (G)(Contact Hrs. 09).** Games Training will be carried out on each day of the camp, except on last day, in evening hours. Training has to be progressive in degree of difficulty to improve individual skills, coordination, team work and desire to excel. Training to ensure that each and every boy and girl cadets participate in at least one game activity everyday.
- (k) **Unit 11. Cultural Activity (C)(Contact Hrs. 09).** Cultural Activity will be carried out on each day of the camp, except on last day, in evening hours. Cadets have to divided in Nine Groups consisting of a mix of boy and girl cadets and preferably belonging to the same geographical area. Each group has to present the unique culture, custom, tradition, folk lore, songs, drama, paintings and cuisine during one hour allotted. There will be a prize for the best group to encourage participation and to develop pride in their unique culture. This training activity should ensure that each and every boy and girl cadet participate in at least one game activity every-day.
- (l) **Unit 12. Spare (S)(Contact Hrs. 02).** Two periods in each camp will be earmarked as spare to cover disruptions in training activity due to weather or other administrative reasons.

COURSE MODULE : NATIONAL CADET CORPS CAMP – II

Course Title: National Cadet Corps Camp II			
Course Code	BNCCCAMP05GE05	Credits	2(Thr)+ 3(Pr)=05
L /T + P	30+90	Course Duration	10 Days (24 hours each)
Semester	V (Odd)	Contact Hours	30(Thr)+90(Pr)=120Hours
Methods of Content Interaction	Lecture, Tutorials, Group discussion, Collaborative work, self-study, individual and group tasks, team work, field-based assignments, Physical Training, endurance building and skill development practices		
Assessment and Evaluation	As per the University norms i.e. 25% internal assessment and 75% End of term exams , or 30% internal assessment and 70% end of term exams etc.		

Course Content Part (II) Theory

187. **Course Objectives.** Cadets will be able to: -

- (a) Acquire the concept self-awareness, emotional intelligence, critical and creative thinking, decision making and problem solving.
- (b) Learn about various indicators of good leadership and get an insight on principle of leadership and motivation.
- (c) Develop awareness about the various types of natural disasters and disaster management organization in our country.
- (d) Familiarize with natural resources, changing environment and understand the importance of conservation and waste management.
- (e) Value the importance of Physical and Mental health and understand how to deal with wounds of various types.
- (f) Acquire awareness about organization and role of an Infantry Battalion in the Armed Forces.



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188. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Develop a sense of responsibility, smartness in appearance and improve self-confidence, inculcate importance of empathizing with others, improve their deep-thinking ability and apply ideas and be able to face problems in a constructive manner with solutions.
- (b) Imbibe good leadership traits and apply them in practical life and appreciate the visible outcome of leadership and motivation.
- (c) Appreciate role of the org during emergency and become useful members of disaster response team, if need arises.
- (d) Learn about the various natural resources, their utilization and practice method of conservation of these resources in daily life.
- (e) Appreciate value of physical and mental health in daily life and spread awareness about treatment and care of wounds in their society.
- (f) Be motivated to join the armed forces.

189. **NCC Camp-II : Course Content Part (I) Theory.**

- (a) **Unit 1. Personality Development (PD) (Contact Hrs. 04).** Self-Awareness, Emotional intelligence, Critical and Creative Thinking, Decision-Making and Problem Solving.
- (b) **Unit 2. Leadership (LDR) (Contact Hrs. 02).** Indicators of Good Leadership, Leadership and Motivation.
- (c) **Unit 3. Disaster Management (DM) (Contact Hrs. 02).** Disaster Management Organization NDMA and NDRF, Types of Disasters.
- (d) **Unit 4. Environmental Awareness and Conservation (EAC) (Contact Hrs. 02).** Natural Resources, Conservation and Management, Water Conservation, Waste Management, Energy Conservation.
- (e) **Unit 5. Health and Hygiene (H&H) (Contact Hrs. 02).** Physical and Mental Health, Treatment and Care of Wounds.
- (f) **Unit 6. Infantry Weapons (IW) (Contact Hrs. 01).** Organization of Infantry Battalion.
- (g) **Unit 7. Weapon Training (WT) (Contact Hrs. 02).** Characteristics of Point 22 Rifle and its Ammunition, Range Procedure and Safety Precautions.
- (h) **Unit 8. Military History (MH) (Contact Hrs. 04).** Guest lectures by War Veterans/decorated soldiers/veterans.
- (i) **Unit 9. Communication (COM) (Contact Hrs. 04).** Latest trends in communication.

NCC Camp-II : Course Content Part (II) Practical

190. **Course Objectives.** Cadets will be able to: -

- (a) Inculcate spirit of discipline and follow command as a group for a common goal.
- (b) Fire a weapon with adequate safety precautions necessary for safe firing.
- (c) Understand the lay of the ground and use it skillfully towards own objective.
- (d) Understand and use the map, satellite imagery and GPS effectively.
- (e) Identify and be well versed with the primary weapon systems used in the Armed Forces.
- (f) Lead a life of selflessness and provide service towards society development and nation building.
- (g) Understand the importance of changing mindset, team work, social skills etiquettes and manners, interview skills and importance of effective communication in daily life.
- (h) Learn the importance of physical fitness and nuances of physical training.
- (i) Inculcate esprit-de-corps through team games.
- (j) Have knowledge about cultural diversity of India and learn ways and means to adopt them.

191. **Expected Learning Outcomes.** After completing this course, the cadets will be able to: -

- (a) Practice problem solving, critical thinking in real life situations.
- (b) Practice leadership of small teams and groups under challenging environment.
- (c) Develop a positive attitude, have manners and etiquettes in social life, develop a sense of cooperation for group or team work, participate in an interview with confidence and inculcate verbal and non-verbal communication skills.
- (d) Develop adequate physical and mental endurance capabilities.
- (e) Fire a weapon effectively with fair degree of marksmanship.
- (f) Undertake point to point navigation and take part in endurance marches by day and night.
- (g) Use terrain effectively for concealment, camouflage, indicate landmarks and give field signals.
- (h) Be motivated to join the Armed Forces.
- (i) Play team games and be able to communicate and coordinate effectively in group events or situations.



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- (j) Perform foot drill, arms drill, ceremonial drill and will be able to take part in ceremonial parade and events.
- (k) Respect the diversity of indian culture and develop pride by showcasing their own culture to others.

192. NCC Camp-II : Course Content Part (II) Practical

- (a) **Unit 1. Drill (Drill) (Contact Hrs. 16).** Tejchaal se Mudna, Tejchaal se Salute karna, Tej kadamtaal aur Tham, Tej Kadamtaal se kadam badhana, Teenon Teen se ek file Banana aur ek file se Teenon Teen Banana, Rifle Ke Saath Saavdhan, Aaram se, Rifle ke saath Parade par aur saaj, Rifle Ke saath visarjanaur line tod, Bhumi Sashttra aur Uthao Sashttra, Bagal Sashttra aur Baaju Shastra, Salami Sashttra, Squad Drill, Guard Mounting, Guard of Honour, Platoon / Company Drill, Word of Command and Instructional Practice.
- (b) **Unit 2. Weapon Training (WT) (Contact Hrs. 14).** Stripping, Assembling, Cleaning of Point 22 rifle, Sight Setting and Sight Picture of Point 22 Rifle, Loading, Cocking and Unloading, Lying Position, Holding and Aiming of Point 22 rifle, Trigger Control and Firing of Shot, Theory of Group, Short-Range Aiming and Firing, Musketry Training, Firing Practice I to VII.
- (c) **Unit 3. Field Craft & Battle Craft (FC/BC) (Contact Hrs. 06).** Observation, Camouflage and Concealment, Field Signals, Section formations, Fire Control Orders, Fire and Movement, Knots and Lashings.
- (d) **Unit 4. Map Reading (MR) (Contact Hrs. 12).** Introduction to Map and Conventional signs, Scale and Grid System, Topographical forms and technical terms, Relief, Contours and gradients, Cardinal points and types of North, Types of Bearing and use of Service Protector, Prismatic Compass and its use, setting of a map, Finding North and own Position, Map to Ground and Ground to map, Point to Point march, Endurance March – I (10 KM), Endurance March -II (20 KM).
- (e) **Unit 5. Infantry Weapons (IW) (Contact Hrs. 01).** Characteristics of Infantry Company support weapons and 5.56 MM INSAS Rifle.
- (f) **Unit 6. Social Service and Community Development (SSCD)(Contact Hrs. 04).** Contribution of Youth Towards Social Welfare: Cadets will participate in various activities throughout the camp e.g., Blood donation Camp, Swachhata Abhiyan, Constitution Day, Jan Jeevan Hariyali Abhiyan, Beti Bachao Beti Padhao etc., Social Evils: Female Feticide, Dowry, Child Abuse, Trafficking and Corruption, Drug Abuse and Drug Trafficking, Protection of Children and POCSO Act 2012.
- (g) **Unit 7. Personality Development (PD)(Contact Hrs. 06).** Change Your Mindset, Team Work and Team Building, Social Skills, Etiquettes and Manners, Interview Skills, Communication Skills–I, Communication Skills -II

(h) **Unit 8. Obstacle Training (OT)(Contact Hrs. 02).** OT Practice – I: Untimed, Cadets will be familiarized with all the obstacles in the Obstacle Course and briefed about the correct method to do them, OT Practice -II: Timed practice for all the cadets and record to be maintained.

(i) **Unit 9. Physical Training (PT) (Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in morning hours. Training has to be progressive in degree of difficulty to improve individual stamina and endurance. Training to include warming up, running, exercises to strengthen upper body, lower body and core muscles. Two period each to be devoted to route march by day and night respectively and one period will be earmarked for trekking expedition as part of Adventure Activity.

(j) **Unit 10. Games Training (G)(Contact Hrs. 09).** Physical Training will be carried out on each day of the camp, except on last day, in evening hours. Training has to be progressive in degree of difficulty to improve individual skills, coordination, team work and desire to excel. Training to ensure that each and every boy and girl cadets participate in at least one game activity everyday

(k) **Unit 11. Cultural Activity (C) (Contact Hrs. 09).** Cultural Activity will be carried out on each day of the camp, except on last day, in evening hours. Cadets have to divided in Nine Groups consisting of a mix of boy and girl cadets and preferably belonging to the same geographical area. Each group has to present the unique culture, custom, tradition, folk lore, songs, drama, paintings and cuisine during one hour allotted. There will be a prize for the best group to encourage participation and to develop pride in their unique culture. This training activity should ensure that each and every boy and girl cadets participate in at least one game activity every day (Contact Hrs. 09)

(l) **Unit 12. Spare (S) (Contact Hrs. 02).** Two periods in each camp will be earmarked as spare to cover disruptions in training activity due to weather or other administrative reasons.



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SECTION II : RULES AND REGULATIONS

**GOVERNING NCC CREDIT COURSE UNDER CHOICE BASED CREDIT SYSTEM AS
GENERIC ELECTIVE FOR SENIOR DIVISION/WING**

RULE 1 :Definitions of Key Terms

General Definitions

‘Choice Based Credit System’ (CBCS). The CBCS provides choice for the student to select courses from the prescribed courses (Elective or Soft – Skill courses). It provides a ‘Cafeteria’ approach in which the students can take courses of their choice, learn at their own pace, study additional courses and acquire more than the minimum required credits, and adopt an inter-disciplinary approach.

‘Academic Year’. Two consecutive (one odd + one even) semesters shall constitute one academic year.

‘Credit Course’. Course, usually referred to as paper having specific title and code number, is a component of a programme. It consists of a list of topics/concepts/theories/principles/activities/tasks etc. which a student has to learn during the programme of study. Each course has some credits according to the nature and load of content. Each course should define the learning objectives/learning outcomes. A course may be designed to be delivered through lectures/tutorials/laboratory work/field work/out reach activities/project work / vocational training / physical training / viva / seminars / term papers / assignments / presentations / self-study work etc., or a combination of some of these.

‘Course Instructor/Teacher’. The course instructor generally will be a teaching faculty who has taken up the responsibility of teaching it and evaluating the performance of the students in that course. NCC course will be imparted by the ANO (Associate NCC Officer) and PI (Permanent Instructor) / Girl Cadet Instructor (GCI) staff together according to their area of specialization. Certain specific topics and training activity is imparted by Military Officers and Whole Time Lady (WTLO).

‘Credit’. A unit by which the course work is measured. It determines the number of hours of instructions required per week. One credit is equivalent to one hour of teaching (lecture or tutorial) or two hours of practical work per week. Thus, in each semester’s NCC course, credits are assigned on the basis of the number of lecture/tutorial/field work/physical training/excursions and other forms of learning required for completing the contents in a 15-18 week schedule. 2 hours of laboratory work/field work is generally considered equivalent to 1 hour of lecture.

- i. 1 credit = 1 hour of instruction per week (1 credit course = 15 contact hours of instruction per semester)

- ii. 4 credit = 4 hour of instruction per week (4 credit course = 60 contact hours of instruction per semester)
- iii. 1 credit = 2 hour of practical per week (1 credit course = 30 contact hours of instruction per semester)
- iv. 4 credit = 8 hour of practical per week (4 credit course = 120 contact hours of instruction per semester)

Number(s) of credit(s) assigned to a particular course are mentioned in the detailed syllabus of the courses.

‘Credit Point’. It is the product of the grade point and the number of credits for a course.

‘Letter Grade’. It is an index of the performance of students in a said course. Grades are denoted by letters O, A+, A, B+, B, C, P, and F. A letter grade is assigned to a student on the basis of evaluation of her/his performance in a course on a ten-point scale.

‘Grade Point’. It is a numerical weight allotted to each letter grade on a 10 -point scale.

Letter Grade	Grade Point
O	9-10
A+	8-9
A	7-8
B+	6-7
B	5-6
C	4-5
P	4
F	0
Ab	0

Note : University may use the above said criteria for providing the grades to the students or may adopt the same criteria which they are practicing for providing the letter grade and grade point for other subjects.

‘Programme’. An educational programme leading to the award of degree, Diploma or Certificate course. NCC course shall be offered only at under graduate level programmes for any stream or type of programme for example – Nonprofessional courses BA, B.Sc. B. Com etc. professional courses – B.A., LLB, B.A./B.Sc., B.Ed., BCA, BBA, B. Tech, MBBS etc.

‘Credit – Based Semester System (CBSS)’. Under the CBSS, the requirement of awarding a degree or diploma or certificate is prescribed in terms of number of credits to be completed by the students.



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'Semester' Each semester shall consist of 15 to 16 weeks of academic work equivalent to 90 actual teaching days. The odd semester may be scheduled from July to December and even semester from January to June. The Credit-based semester system provides flexibility in designing curriculum and assessing credits based on the course content and hrs of teaching.

'Semester Grade-Point Average (SGPA)' Semester Grade Point Average or SGPA, is an average grade point earned by the student at the end of an academic session i.e. semester at college. The formula for calculation of SGPA is the sum of all the credit points awarded for the subjects divided by total credits allotted to that semester. It shall be expressed up to two decimal places.

'Cumulative Grade Point (CGPA)' It is a measure of overall cumulative performance of a student over all semesters. The CGPA is the ratio of total credit points secured by a student in various courses in all the semesters. It shall be expressed up to two decimal places.

'Transcript/ Grade card or certificate' Based on the grades earned, a grade certificate shall be issued to all the registered students after every semester. The grade certificate will display the course details (code, title, number of credits, grade and / or marks secured) along with SGPA of semester. Overall Grade Certificate will be issued on completion of the course showing semester wise SGPA & CGPA.

'The University/ College/ Institution' The University/ College/ Institution in present document means the any recognized central/ state/ Deemed university or institution meant for higher education.

'NCC Course' In the present document 'NCC Course' means the course designed for imparting NCC curriculum in educational institutions as elaborated in this document under Choice Based Credit System as a General Elective Course for Senior Division/ Senior Wing.

Definitions Specific to NCC

'Institutional Training' Implies training conducted for NCC cadets as per Training Manuals and Cadet Hand Book issued by DG NCC, Ministry of Defence.

'Common Subjects' Implies those subjects specifically taught in NCC curriculum which are common to Army, Navy and Air Force and general training that can be imparted by Associate NCC Officers or Military staff or a suitably qualified person.

'Specialised Subjects' Implies subjects specifically taught in NCC curriculum by military instructors comprising specialised topics for Army, Navy and Air Force Cadets respectively.

'NCC Camps and Centralised Training Events' Collective training events conducted usually for 10 days with large number of cadets living under field conditions in selected places away from home. The training camp comprises of focused physical and mental training routines of different types as per syllabus and curriculum. Some training like route marches may happen overnight. Camps include, adventure camps, national integration camps,

Republic Day Parade Training Camps, ThalSainik, VayuSainik and NauSainik camps and other outdoor training activities as described in DG NCC Training Manuals.

NCC 'B' and 'C' Certificate Examinations. These are defined in Special National Cadet Corps Order 2020 issued by DG NCC, Ministry of Defence.

'Training Faculty'. Persons suitably trained & responsible for imparting training of different types and nature to students.

'Military Officers'. They are regular commissioned officers of Indian Armed Forces who serve in the NCC and render command, administrative and instructional functions for NCC.

'Whole Time Lady Officers (WTLO)'. They are women officers commissioned directly into the NCC.

'Associate NCC Officer (ANO)'. ANO will be a university/ college/ school faculty who are qualified in the PRCN (Pre-commission Course of NCC) conducted by DGNCC and are commissioned as Associate Officers in NCC as defined in NCC Act 1948 and NCC Rules. They have the eligibility to impart certain component of NCC Course and undertake training of cadets.

'Permanent Instructor (PI)'. PI Staff are Junior Commissioned Officers (JCO) and Non-Commissioned Officers (NCO) on deputation from Armed Forces to NCC as governed by NCC Act 1948. Retired PI Staff may be hired by a college as a substitute for ANO with prior concurrence of DGNCC.

'Girl Cadet Instructors (GCI)'. GCI are lady instructors' equivalent to PI Staff for specifically imparting instructions to women NCC cadets of Senior Wing.

'NCC Organizational Structure'. NCC is an adjunct of Indian Armed Forces that operates under the ambit of the Ministry of Defence through the Defence Secretary with Raksha Mantri as the political head.

'DGNCC'. Directorate General of NCC renders the command and administrative function of NCC. The executive head of NCC is Director General of NCC who is a Lt Gen rank officer from the Army.

'State NCC Directorate'. State NCC Directorates are directorates subordinate to DG NCC and render command and administrative control to NCC at State level and is headed by an Additional or Deputy Director General

'NCC Group HQ'. NCC Group HQs are subordinate to State Directorates and render command and administrative control to NCC at district or cluster of districts in a state and is headed by a Group Commander.

'NCC Units'. NCC Units are subordinate to Group HQs at the lowest rung of the command and administrative control exercised by military officers and is headed by a Commanding Officer or Officer Commanding. The NCC Units directly engage



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with educational institutions and ANOs and are primarily responsible for training of NCC in institutions under their jurisdiction.

'NCC Division/ Wing'. NCC Division/Wing are minor units of senior division/wing of NCC comprising of 160 senior cadets allotted to educational institutions. It can be further subdivided into NCC platoons of 53 to 54 cadets.

'NCC Troop'. NCC Troop are minor units of junior division/wing of NCC comprising of 100 junior cadets allotted to educational institutions. It can be further subdivided into NCC half troops comprising of 50 junior cadets.

RULE 2 : Admission and Other Provisions

The NCC Course under the CBCS as 'General Elective' shall be of three years (Six Semester) duration which may be completed in maximum duration of four year (8 semesters).

Students may complete NCC course minimum in Six semesters and maximum in eight semesters. Cadets may complete their 'B' Certificate in four semesters minimum and maximum six semesters. Cadets already having 'B' certificate may complete their 'C' certificate in minimum two semesters and maximum four semesters, and they may join NCC course 5 in first semester of college.

The intake to the course shall be decided according to the seats allotted to University/ college/ institution by DG NCC according to the availability of required infrastructure, faculty and resources.

The admission to the NCC Course under the CBCS as a 'General Elective' shall be governed by the provisions as laid down by the NCC Act 1948/ SNCCO 2020/ contemporary SNCCO and Academic council of parallel body of university. These rules and regulations may be modified from time to time (if needed) by the Academic body of the university in consultation with DG NCC or Act/ Ordinances prepared by DG NCC.

Students will be enrolled as NCC cadet as per existing Acts & Rules.

At the time of reporting for admission, the candidates are required to present medical & physical fitness documents as well as the admission proof of the university and submit these self-attested copies of aforesaid documents.

The admission of any candidate is liable to be cancelled without giving any further notice forthwith or at any time during the period of the course, if it is detected that the candidate has/had produced fake/forged certificate (s)/ document(s), indulged in any act of misconduct/indiscipline and has/had concealed any other relevant information at the time of admission.

The admission of the candidate to the course shall be subject to such ordinance, rules and regulations as may be framed from time to time by the university in consultation with DG NCC and NCC act 1948.

DG NCC shall have jurisdiction in case of any dispute relating to the provisional admission in the course.

RULE3 : For Eligibility, Medium of Instrs & Categories

Eligibility Conditions. Be governed by provisions of NCC Act and Rules and directions from DG NCC from time to time. These are readily available on DG NCC website www.nccindia.nic.in.

Standards for physical Fitness criteria for Male and Female Cadets/students shall be governed by provisions of NCC Act and Rules and policy documents released by DG NCC from time to time.

RULE4 : Medium of Instruction. English or Hindi. However, ANOs and training instructors are free to use vernacular language for helping students who are not fluent in Hindi or English.

RULE 5 : Course and Students. NCC course is unique, due to the nature of its military training content and component hence it is normally offered to students enrolled as NCC cadets only. This NCC Course is primarily designed for students enrolled as NCC cadets under provisions of NCC Act 1948. Institution allotted NCC will have the obligation to offer this course to all students from their institute enrolled as cadets as per vacancy allotted to the institution by DG NCC as also to those cadets enrolled under Open Quota seats.

RULES 6 :NCC Course for ‘Cadet’ Category

6.1. NCC Course for ‘Cadet’

- (a) NCC course for Cadets comprises of total 24 credits (08 for theory, 06 for practical and 10 for camp component) over 6 semesters courses i.e., NCC course I to NCC course VI and NCC Camp I & NCC Camp II.
- (b) Cadets will not only earn the academic credits but also be given ‘B’, and ‘C’ Certificates after passing the exam conducted by DG NCC.
- (c) Students would be free to join NCC Course I or subsequent Courses in any semester, not necessarily Semester I or the designated Semester.
- (d) A student can opt for only one of the six Courses per semester and that too sequentially implying NCC Course II cannot be joined before completing NCC Course I and so on.
- (e) Under this category a fresh student/cadet will compulsorily have to opt for all six NCC Courses in minimum six Semesters. However, ‘B’ certificate holder may directly join NCC Course Number 5 in any semester. He will have to complete NCC Course Number 5 and NCC Course Number 6 for obtaining ‘C’ certificate and he will be awarded credit points only for NCC Course Number 5 and NCC Course Number 6.



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NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	1	1		2	
Semester - II	1	1		2	
Semester – III	1	1	5	7	Credits of 1 st Camp merged with 3 rd Sem
Semester – IV	2	1		3	
Semester – V	1	1	5	7	Credits of 2 nd Camp merged with 5 th Sem
Semester - VI	2	1		3	
Total	08	6	10	24	Twenty-Four Credits

RULE 7 :Mobility& Credit Bank

The mobility shall be permissible from the regular mode programme to the regular mode programme of learning only and cannot be replaced by open/distance/online programme.

It shall be the responsibility of the student to assess the feasibility and practicality of vertical mobility (across the Universities), as it doesn't entitle a student to be exempted or relaxed from any of the requisites (sessional, attendance, assignments, End-semester examinations and programme duration etc.) for completing the course.

After completing one semester/ one year cadet/student may pursue NCC course from any other institution/ University/ College having NCC and carry credits in credit bank as per NEP 2020. The NCC students/ Cadets of some other university shall in any case be admitted only at the beginning of the session to the fulfilment of the other requirements of the NCC Course (attendance, Formative assessment, Field-work, practical etc).

A student of NCC course availing inter-university mobility shall continue to be a bonafide student of the university where he/she initially got admission and as per the university/ Institutional rules for the inter-university mobility.

In case of inter-university mobility of NCC cadet for NCC Course is also the subject to availability of NCC for the cadets in that particular university/ institution and it shall be interpreted as inter-battalion migration (means another regimental no. shall be allotted to the cadet).

RULE 8 :Examination & Promotion

The examination of all the NCC courses shall be internal in nature and generally consisting of continuous internal assessment and End of semester Examination. For the preparation of final grade in a particular course, the continuous internal assessment (Formative in nature) and the End Semester Examination (Summative in nature) shall have the weightage as decided for other courses by the university as per the University norms for e.g., 25% internal assessment and 75% End of term exams or 30% internal assessment and 70% End of term exams etc.

For assigning the Grades and credit points to NCC Course Universities/ Institutions are free to use the same criteria which are decided by their academic bodies for providing the grades and credit points to the other courses

RULE 9 :Continuous Internal Assessment

The Continuous Internal Assessment of the NCC Cadets' and NCC students' learning and performance shall be carried out by the ANOs and PI staff.

Continuous Internal Assessment will be 100% Practical that includes Drill Square test, Map Reading, Weapon Training, Field craft & Battle craft.

CO of nominated NCC Unit will be deemed as Head of the Department and shall be responsible for approving the schedule and pattern of the continuous internal examination.



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ANO of the nominated institute shall maintain all the records related to attendance, teaching and assessment in a systematic manner, including award of final grade.

In case a student fails to appear in any Continuous Internal Assessment, they will be given a chance to reappear in retest and in case he/she fails to obtain 'P' grade he/she will be made to repeat the exam by carrying it forward for semester retest.

RULE 10 :Re-appear in the End Semester Examination for Improvement of Grades

If a student wishes to improve her/his grade(s) in NCC course(s), she/he can re-appear in the End Semester Examination in the subsequent odd/even semester(s), whenever the examination of the particular course(s) is held, on payment of fees in addition to the prescribed semester fee within the maximum permissible duration for the programme of study of the student/cadet.

A student may improve her/his points/grade by reappearing in the End Semester Examination of a course as per the provisions of reappearing mentioned above. In such cases points obtained by the student in the Continuous Internal Assessment of the particular course shall be carried forward to the subsequent End Semester Examination of the course. However, in such case, the points/grades obtained on the basis of latest appeared End Semester Examination shall be considered for calculation of final CGPA of the programme.

The re-appear examination of a course for improvement of grade shall be based on the syllabi of the course in force at the time of initial registration to the course.

A student who has got the Migration/Transfer Certificate issued from the University shall not be allowed to re-appear in any examination for improvement of grade.

RULE 11 :Repeating Courses

A student having attendance shortage in any course may repeat the course by taking re-admission in that course in subsequent odd/even semester(s), whenever the course is being offered, within the maximum permissible duration of the programme.

If a student repeats a course, she/he has to fulfil all the desired requirements afresh including attendance, Continuous Internal Assessment and the End Semester Examination. In such case the course content shall be based on the syllabi of the course in force at the time of repetition of the course.

RULE 12 : Promotion Rules

A student shall be declared as 'promoted' to the next semester when she/he earns 'P' Grade or above in the last concluded semester examination, maintaining the spirit and pattern of semester system and covering the mandatory components, such as Continuous Internal Assessment and End-Semester Examination in the NCC Courses.

A student shall be 'Provisionally Promoted' to the next semester if she/he secures less than 'P' grade but he /she has to pass all the courses of NCC course within permissible duration.

A cadet shall be eligible to attend the 'B' Certificate exam if he/she passed all the first four semester NCC course and completed one ATC/CATC. Similarly, cadet will be eligible to attend 'C' certificate examination if he/she has 'B' certificate and he /she has passed V, VI semester NCC course and attended one CATC/ATC after fourth semester and after having obtained 'B' certificate.

If a cadet/student is repeating a course in an academic session, whatever may be the reason, it shall not be counted in the total number of seats and shall not affect the fresh intake of cadets / student in that academic session.

RULE 13 :Computation of SGPA & CGPA

13. Computation of SGPA and CGPA . University may use their own criteria for giving the SGPA & CGPA which is prepared by the authorised academic body for the other courses.



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SUBJECT CODE – MC 6 (C)

CATEGORY – NNP

SUBJECT NAME – PHYSICAL EDUCATION

SEMESTER – VI

Fitness Management

UNIT-I

1 Introduction

- Concept of fitness and wellness, components and their significance in Modern fitness.
- Types of fitness and its testing.
- Factors affecting physical fitness and values of physical fitness.
- Concept of fitness management in modern era.
- Precaution before, during and after exercise.
- Prominent health problem associated with inactivity.
- 1.7 Concept of weight training and principles of weight training.
- Brief introduction of various fitness centre equipments/gadgets.
 - Cardiac machines
 - Strength training machines.
 - Swiss ball, terra bands etc.
 - Sauna & Steam bath, Jacuzzi etc.

UNIT-II

2 Exercise for fitness & designing of the programme.

- Understanding suitability and form of exercise for fitness.
 - Aerobic and anaerobic exercise.
 - Calisthenics exercise.
 - Other forms of exercise - Dance, aquatic, skating, skipping rope etc.
- Calculate target heart rate zones for various exercise intensities.
- Women & weight training, pregnancy, and post natal exercise, exercise and aging.
- Concept of free weight Vs. Machine, sets & repetitions, variation of sets & repetitions for weight training.
- Concept of designing different fitness training programme for different age & life style people.
- Concept of designing a weight reduction training programme.

UNIT-III

3 Establishment & Management of Fitness Centre.

- Environmental considerations for fitness centre.
- Fitness center - operations.
 - Evolutions of fitness center culture.
 - Principle of starting a fitness center - location policy, offer of programmes, record keeping, code of ethics.
- Different exercise equipment & their management.
- Personal outfits & gadgets required for the members of the fitness

centre.

Safety measures, prevention & Management of injury in a fitness centre.

Membership for fitness centre and its type. Fitness centre agreement form.

UNIT-IV

4 Carrier option & business of fitness centre.

Scope of a fitness trainer.

Qualification & Qualities for a fitness trainer.

Management Skills needed for a successful fitness manger.

Instructor's duties for handling beginners.

Individualized/group grooming programmes and reports.

4.6. Personal training - Various aspects.

4.7. Formalities prior to conducting fitness programme for various categories of clients.

Basic marketing & Promoting business

Budgeting.

Marketing & Clientage.

Record keeping.

Public relations.

Exercise & music

UNIT-V

5 Nutrition & Weight Management

Nutrition - daily caloric requirement and daily energy expenditure.

Exercise & energy expenditure and body weight.

Macronutrients concerns before, during and after exercise.

Obesity & its assessment, body mass index, body composition & determination of desirable weight.

Effects of excess weight & appraisal of body fat.

Dieting versus exercise for weight control.

Fat- Dissolving criteria & spot reduction.



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PHYSIOLOGY UNIT-I

Introduction

Concept of Physiology

Essential properties of life

Passage of water and soluble across cell membrane

Cardio-Vascular System and Blood

Composition and functions of blood

Cardiac Cycle

Blood pressure and its maintenance

Cardiac output and its regulation

UNIT-II

Respiratory System

Mechanism of Respiration

Pulmonary ventilation and its regulation

Secretion and functions of Digestive Juices

Functions of Liver

Absorption of Food

Digestive system

Metabolism of Food (Carbohydrates, Fat & Protein)

Temperature regulation.

UNIT-III

Nervous System

Functions of important parts of system (cerebrum, cerebellum, medulla oblongata and spinal cord.)

3.2.2 Functions of autonomic nervous system

Sensory System

General sensations- Coetaneous and Kinesthetic

Visual and Auditory senses.

UNIT-IV**Excretory system**

Excretion of water Through Skin, Kidney and Gastrointestinal Tract.

Endocrine system

Secretion and function of Endocrine Glands- Pituitary, Thyroid, Adrenal and Pancreas.

Reproductive System

Introduction to Physiology of human reproduction.

Transmissions of hereditary characteristics.

UNIT-V**5. Physiology of Exercise**

Physiological concept of Health and Fitness.

Effect of exercise on- Circulatory, Respiratory and Muscular Systems.

Changes during Muscular Contraction.

Nervous control of Muscular activity.

Training, Conditioning and Warm-up.

Oxygen Debt, Second Wind, Stitch and Cramp.



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HANDBALL

UNIT-I

1. Introduction of the game and historical development with special reference to India.
2. Important tournaments held at National and International Levels.
3. Rules and their interpretations.
4. Duties of Officials.
5. Fundamental Skills.

Passing and receiving techniques.

Overhand Pass

Push Pass

Wrist Pass

Bounce Pass

Behind the back Pass

Ball Reception Techniques : -

Catching at Chest Level

Catching below waist

Catching at sides

Catching at head height

Catching in the air

Dribbling :

High Dribbling

Low Dribbling

Shooting :

Set Shot

Jump Shot Long

Jump Shot High

Wing Shot

6. Positional Play in attack & defense.
7. Drills and Lead-up games.

FUNDAMENTAL OF COMPUTER & INFORMATION TECHNOLOGY

UNIT-I

1. Introduction to computers

Brief history development of computers

Generations of computers.

Types of PCs-Desktop, Laptop, Notebook, Laptop, Workstations
etc.

Basic components of a computer system

Memory- Ram, Rom, and other types of memory

Operating system

Need of Software, Types of Software

Types of Virus, virus detection and prevention.

Binary Number system

UNIT-III

2. Introduction to windows

Using Mouse and moving icons on the screen

My Computer, Recycle Bin, Status Bar

Start-menu selection, running an application,

Window Explorer to view files, folders and directories, creating and
renaming of files and folders,

Operating and closing of different windows, Minimize, Restore and
Maximize forms of windows,

Basic components of a window: Desktop, Frame, Title Bar, Menu
Bar, Status Bar, Scroll Bars, Using right button of the Mouse,

Creating shortcut, Basic Windows Accessories: Power Point.
Presentation, Notepad, Paint, Colculator, Word pad, using



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UNIT-III

3. Introduction to MS Office & Word Processor

Types of Word Processor

Creating and Saving a documents, Editing and Formatting a Document including Changing colour, Size Font, alignment of text, Formatting paragraphs with line or paragraph spacing, adding headers and footers, numbering pages.

Using grammar and spell check utilities, etc., printing a document.

Inserting word art, clipart and Pictures,

Page setting, Bullet and Numbering, Borders, Shading Format painter find and replace.

Inserting Tables, Mail Merge.

UNIT-IV

4. Introduction to information & Communication Technology

Concept, Importance, Meaning & Nature of Information & Communication Technology

Need of Information & Communication Technology in Physical Education.

Scope of ICT in Education & Physical Education.

Teaching Learning Process, Publication, Evaluation, Research Administration.

Paradigm shift in education due to ICT content with special reference to curriculum.

Role of Teacher, Methods of Teaching, Classroom Environment, Evaluation procedure.

UNIT-V

5. INTRODUCTION TO INTERNET BROWSING

Internet: Evolution, Protocols, Interlace Concepts,

Internet Vs Intranet, Growth of Internet, ISP.

Connectivity-Dial-up, Leased line, VSAT etc., URLs, Domains names,

Application, E-mail: Concepts, POP and WEB Based E-mail, merits, address, Basics of sending & Receiving, E-mail protocols, Mailing List, Free E-mail services.

Telnet Concept, Remote Logging, Protocols, Terminal Emulation. Message Board, Internet Chatting voice chat text chat.

World Wide web (www)- History, working web browsers, Its functions, concept of search Engines, Searching the Web, HTTP, URLs, Web Servers web; Protocols.

Web publishing concepts, Domain name Registration, place on Host server for web sites HTML, Design tools HTML editors, Image editors, Image editor, Issues in Web site creations & maintained, FTP software for upload web site.

Concepts of Hypertext, Versions of HTML, Elements of HTML syntax, Head & Body Section, Building HTML documents, Inserting tests, Images, Hyperlinks, Backgrounds and Colour controls, Different HTML, tags, Table layout and presentation, Use of font size & Attributes.

Practicals:

The practicals will be conducted based on the syllabus.



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EE-71 - ELECTRICAL MACHINE DESIGN

TOTAL-60HOURS

COURSE OBJECTIVE:

1. To study mmf calculation & thermal rating of various types of electrical machines.
2. To study armature & field systems of D.C machines.
3. To design core, yoke, windings & cooling system of transformers.
4. To design stator & rotor of induction machines.
5. To design stator & rotor of synchronous machines & study their thermal behavior.

Unit-1

Introduction: Design problem -Mathematical programming methods, computer aided design Mathematical formulation of the problem. Programming techniques (LP & NLP only), Methods of solution, Unconstrained optimization problems, constrained optimization problems.

(12 HRS)

Unit-2

Optimal design of DC machine: Design of armature, Windings and field systems, Selection of variables for optimal design, Formulations of design equations, Objective function, Constraint functions, Algorithms for optimal design.

(12 HRS)

Unit-3

Optimal design of power transformer: Design of magnetic circuit, Design of windings, Selection of variables for optimal design, Formulation of design equations, Objective function, Constraint functions, Algorithms for optimal design.

(12 HRS)

Unit-4

Optimal design for 3-phase alternator: Design of stator, windings, Design of Field systems for salient pole and non-salient pole machines, Selection of variables for optimal design.

(12 HRS)

Unit-5

Optimal design of 3-phase induction motor: Design of stator, Windings Design of squirrel cage rotor, Design of slip ring rotor, Selection of variables for optimal design, Formulation of design equations, Objective functions Constraint functions, Algorithms for optimal design.

(12 HRS)



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VIITH SEMESTER

COURSE OUTCOME:

CATEGORY:-DC

Ability to model & analyze electrical apparatus & their application to power system.

References Books:

1. Design and Testing of Electrical Machines, M.V. Deshpandey, PHI Learning
2. Computer- Aided Design of Electrical Equipment by Dr. M. Ramamurthy Affiliated East-West press Pvt. Ltd. New Delhi.
3. Electrical Machine Design by A.K. Sawhney, Dhanpat Rai & Sons.
4. Principles of Electrical Machine Design with Computer Programmes by S.K. Sen, Oxford & IBH Publishing Co.
5. Performance and Design of A.C. Machines -M.G. Say, Affiliated East West Press Pvt. Ltd., New Delhi.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL



EE-72 – ELECTRICAL DRIVES

TOTAL-60HOURS

COURSE OBJECTIVE:

1. To learn basics of electric drive analysis
2. To be able to analyze and design systems with electric drives
3. To learn to work in teams while working on engineering problems
4. To be able to write reports on the technical analysis performed
5. To learn to search the literature for more information on electric drives and report back on what was found in writing and orally

Unit-1

Review of electric motors & Solid state converters: Speed control techniques of DC, Induction & synchronous motor, Converters, inverters, chopper and cyclo converter operation, Effects of power electronic equipments on load side & supply side.

(12 HRS)

Unit-2

Review of closed loop controllers, sensors & transducers: PI, PID, Variable structure. AC, DC & Pulse tacho- generators.

(12 HRS)

Unit-3

DC Drives: Converter & chopper fed DC drive, Reversing, Starting, Regenerative Four quadrant operation, High power application.

(12 HRS)

Unit-4

AC Drive: Inverter & cyclo converter fed drive, Vector control, Sensor less operation, Linear Electrical motor concept, Synchronous motor Drive.

(12 HRS)

Unit-5

Special Drives: Switched reluctance & permanent magnet brushless DC Operation, Converters Characteristics & Control, PLC based drives.

(12 HRS)

COURSE OUTCOME:

This course will give the students a basic understanding of various methods of controlling electric machines for use in variable speed and positioning applications. The student will learn to analyze the steady state behavior of electric machine and drive systems.



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REFERENCE

VIITH SEMESTER

- CATEGORY-DC**
- 1 Ned Mohan, T.M. Undeland, W.P. Robbins, "Power Electronics design", John Wiley & Sons.
 - 2 J.M.D. Murphy, F.O. Turnbull, "Power Electronic Control of AC motors", Pergamon Press.
 3. P.C. Sen, D.C. drive, Pergamon Press.
 4. B.K. Bose, Power Electronics & AC drive prentice Hall.
 5. Dubey G.K. "Power semi Conductor controller drives, Prentice Hall.
 6. Vedam Subramanyam, "Electrical Drives".
 7. T.J.E. Miller, Switched Reluctance & P.M. B.L. DC motor, Pergamon Press.
 8. P.V. Rao, "Power semiconductor Drives", BS Publications.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL



EE-73- HIGH VOLTAGE ENGINEERING

TOTAL-60HOURS

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Breakdown mechanisms in solid, liquid and gaseous dielectrics.
- 2 High voltage and current generation circuits and the operation of each circuit.
- 3 Different techniques employed in high voltage and current measurements.
- 4 Impulse and power frequency tests on the power system components.

Unit-1

Introduction:-Introduction to HV technology, advantages of transmitting electrical power at high voltages, need for generating high voltages in laboratory. Important applications of high voltage.

(12 HRS)

Unit-2

Breakdown phenomena:-Classification of HV insulating media, Properties of important HV insulating media. Gaseous dielectrics: Ionizations: primary and secondary ionization processes. Criteria for gaseous insulation breakdown based on Townsend's theory, Limitations of Townsend's Streamer's theory breakdown in non uniform fields. Corona discharges. Paschen's law and its significance. Time lags of Breakdown. Breakdown in solid dielectrics: Intrinsic Breakdown avalanche breakdown, thermal breakdown, and electro mechanic breakdown, Breakdown of liquids dielectric dielectrics: Suspended particle theory, electronic Breakdown, cavity breakdown (bubble's theory), electro convection breakdown..

(12 HRS)

Unit-3

Generation of HV AC DC and Impulse Voltage and current:HV AC-HV transformer; Need for cascade connection and working of transformers units connected in cascade, Series resonant circuit principle of operation and advantages. Tesla coil.HV DC- voltage doubler circuit, Cockcroft-Walton type high voltage DC set, Introduction to standard lightning and switching impulse voltages. Analysis of single stage impulse generator expression for Output impulse voltage, multistage impulse, generator Components of multistage, impulse generator. Triggering of impulse generator by three electrode gap arrangement.Triggering gap and oscillograph time sweep circuits. Generation of switching impulse voltage.Generation of high impulse Current.

(12 HRS)

Unit-4

Measurement of high voltages: Electrostatic voltmeter-principle, construction and limitation. Generating voltmeter- Principle,construction. Series resistance micro ammeter for HV DC measurements. Standard sphere gap measurements of HV AC, HV DC, and impulse voltage. Factors affecting the measurements.Potential dividers-resistance dividers capacitance mixed RC potential dividers. Surge current measurement.

(12 HRS)



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VIITH SEMESTER

Unit-5

High voltage tests on electrical apparatus: Applications of technologies, tests on isolators, circuit breakers, cables insulators and transformers.

(12 HRS)

COURSE OUTCOME:

After completion of this course, students will be able to:

- 1 Illustrate the different breakdown mechanisms in solid, liquid and gaseous dielectrics.
- 2 Design and describe the various circuits for generation of high voltages and currents.
- 3 Design and describe the various circuits for measurement of high voltages and currents.
- 4 Illustrate the power frequency and impulse tests on the power system components

Reference books:

1. E. Kuffel and W.S. Zaengl, "High voltage engineering fundamentals", 2nd edition, Elsevier, Press, 2005.
2. M.S.Naidu and Kamaraju, "High Voltage Engineering", 3rd edition, THM, 2007.
3. L. L. Alston, "High Voltage technology", BSB Publication, 2007..
4. Rakosh Das Begamudre, Extra High voltage AC transmission engineering, Wiley 1987.
5. Transmission and distribution reference book -Westing House.C.L.Wadhwa, High voltage Engineering, New Age International Private limited, 1995.



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DE-74(A) -EHV A.C. AND D.C. TRANSMISSION

TOTAL-60HOURS

COURSE OBJECTIVE:

1. To Provide In-depth understanding of different aspects of Extra High Voltage AC transmission system design and Analysis.
2. To Calculate the Value of Line Inductance and Capacitance of EHV transmission Line
3. To understand the concept of Voltage gradients of conductors.
4. To develop the empirical formula to determine the Corona loss occurring in EHV AC transmission Line.
5. To determine the interference caused by Corona and to measure its magnitude

Unit-1

Constitution of EHV a.c. and d.c. links, Kind of d.c. links, Limitations and Advantages of a.c. and d.c. transmission, Principal application of a.c. and d.c. transmission, Trends in EHV a.c. and d.c. transmission, Power handling control, Converter analysis garetz circuit, firing angle control overlapping.

(12 HRS)

Unit-2

FACTS devices, basic types of controller, series controller, static synchronous series compensator (SSSC), thyristor-controlled series capacitor (TCSC), thyristor-controlled series reactor (TCSR), shunt controller (STATCOM), static VAR controller (SVC), series- series, combined series-shunt controller, unified power flow controller (UPFC), thyristor controlled phase shifting transformer (TCPST).

(12 HRS)

Unit-3

Components of EHV d.c. system, converter circuits, rectifier and inverter valves, Reactive power requirements, harmonics generation, Adverse effects, Classification, Remedial measures to suppress, filters, Ground return. Converter faults & protection harmonics misoperation, Commutation failure, Multiterminal D.C. lines.

(12 HRS)

Unit-4

Control of EHV d.c. system desired features of control, control characteristics, Constant current control, Constant extinction angle control. Ignition Angle control. Parallel operation of HVAC & DC system. Problems & advantages.

(12 HRS)

Unit-5

Travelling waves on transmission systems, their shape, Attenuation and distortion, effect of junction and termination on propagation of traveling waves. Over voltages in transmission



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system. Lightning, switching and temporary overvoltages. Control of lighting and switching over voltages.

(12 HRS)

CATEGORY:-DE

COURSE OUTCOME:

1. Qualitative comparison of AC and DC transmission system with all aspects
2. Understand the need of EHV AC transmission and various issues related with it
3. Reactive power management, Stability of AC and DC systems
4. In depth converter analysis, faults, protections, harmonic considerations, grounding system

Reference:

1. S. Rao,- “EHV AC & DC Transmission” Khanna pub.
2. Kimbark,-” HVDC Transmission” john willy & sons pub.
3. Arrillaga,- “HVDC Transmission” 2nd Edition ,IEE london pub.
4. Padiyar, -“HVDC Transmission” 1st Edition ,New age international pub.
5. T.K. Nagsarkar,M.S. Sukhiza, -“Power System Analysis”, Oxford University
6. Narain.G. Hingorani, I. Gyugyi,-”Understanding of FACTS concept and technology”, John Wiley & sons pub.
7. P.Kundur- “H.V.D.C. Transmission” McGraw Hill Pub.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL



DE-74(B)- RENEWABLE ENERGY SOURCE

TOTAL-60HOURS

COURSE OBJECTIVE :

The course should enable the students to :

1. Understand the various forms of conventional energy resources.
2. Learn the present energy scenario and the need for energy conservation
3. Explain the concept of various forms of renewable energy
4. Outline division aspects and utilization of renewable energy sources for both domestics and industrial application
5. Analyse the environmental aspects of renewable energy resources.

Unit – 1

Renewable Energy Systems: Energy Sources, Comparison of Conventional and non-conventional, non-renewable sources. Statistics of world resources and data on different sources globally and in Indian context. Significance of renewable sources and their exploitation. Energy planning, Energy efficiency and management.

(12 HRS)

Unit – 2

Wind Energy System: Wind Energy, Wind Mills, Grid connected systems. System configuration, working principles, limitations. Effects of wind speed and grid conditions. Grid independent systems wind-battery, wind diesel, wind-hydro biomass etc. wind operated pumps, controller for energy balance. Small Hydro System Grid connected system, system configuration, working principles, limitations. Effect of hydro potential and grid condition. Synchronous versus Induction Generator for standalone systems. Use of electronic load controllers and self excited induction generators. Wave Energy System: System configuration: grid connected and hybrid system..

(12 HRS)

Unit – 3

Solar Radiation: Extraterrestrial solar radiation, terrestrial solar radiation, Solar thermal conversion, Solar Photo tonic System Solar cell, Solar cell materials, efficiency, Characteristics of PV panels under varying insulation. PV operated lighting and water pumps, characteristics of motors and pumps connected to PV panels Energy System, Biomass System configuration, Biomass engine driven stand-alone or hybrid modes, Biomass engine drive generators, feeding loads in stand-alone or hybrid modes, Biomass energy and their characteristics.

(12 HRS)

Unit – IV

Energy from oceans: Ocean temperature difference, Principles of OTEC, plant operations, Geothermal Energy, Electric Energy from gaseous cells, Magneto from nuclear wastes, Possibilities of other modern non-conventional energy sources.

(12 HRS)



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Unit – V

VIITH SEMESTER

Electric Energy Conservation: Energy efficient motors and other equipment. Energy saving in Power Electronic controlled drives. **CATEGORY-DE** pumps, air-conditioning, power plants, process industries, illumination etc. Methods of Energy Audit.

Measurements systems; efficiency measurements. Energy regulation, typical case studies, various measuring devices analog and digital, use of thyristers.

(12 HRS)

COURSE OUTCOME:

Upon completion of the course, the student will be able to:

- 1 Aanalyze the Renewable Sources of Energy and Distributed Generation (DG) & DSM options, Energy storage and Interconnection technologies
- 2 Eexplain the concepts of Generation of Electricity by Photovoltaic Effect & its applications.
- 3 Ddescribe the Principle of Generation of Electricity by Wind power plants, Various wind turbines & controller for wind Applications.
- 4 Eexplain construction & working of Energy Storage Parameters-Lead–Acid Batteries & issues involved in Power Injection & Interconnection Technologies with Grid.

References:

1. John Twidell & Toney Weir, Renewable Energy Resources, E & F N Spon.
2. El-Wiki, Power Plant Technology, McGraw Hill.
3. Rai G D, Non-conventional Energy Resources, Khanna.
4. F Howard E. Jordan, "Energy Efficient Electric Motor & their Application-II", Plenum Press, New York, USA.
5. Anna Mani, "Wind Energy Resource Survey in India-III", Allied Publishers Ltd., New Delhi.
6. S.P. Sukhatme: Solar Energy, TMH-4e.
7. Dr. A. Ramachandran, Prof B.V Sreekantan & M F.C. Kohli etc, "TERI Energy Data Directory & Year book 1994-95", Teri Tata Energy Research Institute, New Delhi.



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DE-74(C)-FLEXIBLE AC TRANSMISSION SYSTEM

TOTAL-60HOURS

COURSE OBJECTIVE :

To impart the students with various flexible ac transmission system devices which are used for proper operation of existing AC system more flexible in normal and abnormal conditions.

Unit-1

Basic Issues Involved in Bulk Power Transmission, Review of basics of power transmission networks-control of power flow in AC transmission line- Analysis of uncompensated AC Transmission line- Passive reactive power compensation, Principle of Transmission system compensation, Need for FACTS controllers - types of FACTS controllers and Benefits.

(12 HRS)

Unit-2

STATIC VAR COMPENSATOR (SVC) AND PURPOSE: Voltage control by SVC – Advantages of slope in dynamic characteristics- Influence of SVC on system voltage, Design of SVC voltage regulator, Modeling of SVC for power flow and stability studies, Applications- Enhancement of transient stability, Steady state power transfer, Enhancement of Power system damping, Prevention of voltage instability.

(12 HRS)

Unit-3

THYRISTOR AND GTO THYRISTOR CONTROLLED SERIES CAPACITORS (TCSC and GCSC): Concepts of Controlled Series Compensation–Analysis of TCSC-GCSC , Different modes of operation, Modeling of TCSC and GCSC for load flow studies- modeling TCSC and GCSC for stability studies- Applications of TCSC and GCSC, SSR mitigation.

(12 HRS)

Unit-4

VOLTAGE SOURCE CONVERTER BASED FACTS CONTROLLERS: Static synchronous compensator(STATCOM)- Static synchronous series compensator (SSSC) -Power flow control with STATCOM and SSSC-Modeling of STATCOM and SSSC for power flow studies- operation of Unified and Interline power flow Controllers (UPFC and IPFC).

(12 HRS)

Unit-5

CONTROLLERS AND THEIR –CO-ORDINATION: FACTS Controller interactions – SVC– SVC interaction - co-ordination of multiple controllers using linear control techniques – Quantitative treatment of control coordination.

(12 HRS)



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VIITH SEMESTER

After completion of the course, the student will be able to

1. Understand the operations of different flexible ac transmission system devices.
2. Select the controllers for different Contingencies.
3. Analyze the different flexible ac transmission system devices in different stability conditions.
4. Select an appropriate flexible ac transmission system device for a particular application.

References-

1. Mohan Mathur, R., Rajiv. K. Varma, Thyristor– Based FACTS Controllers for Transmission Systems, IEEE press and John Wiley & Sons, Inc, 2002.
2. K.R.Padiyar, FACTS Controllers in Power Transmission and Distribution, New Age International (P) Ltd., Publishers, New Delhi, Reprint, 2008.



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COURSE OUTCOME:

EE-75(A)- COMPUTER APPLICATION TO POWER SYSTEMS

TOTAL-60HOURS

COURSE OBJECTIVE:

The objective of this course is to give an overview of control and operation of power systems in a wide sense, including generation, transmission and distribution of electric power.

Unit-1

Models of power system components, network model using graph theory, formation of Z bus transmission line models, regulating transformer, line load ability, capability curves of alternator.

(12 HRS)

Unit-2

Control of load bus voltage using reactive power control variable, SVC & SVS, Regulated shunt Compensation, series and shunt compensation, Uniform series and shunt compensation and effect on load ability of transmission lines.

(12 HRS)

Unit-3

Sensitivity analysis: General sensitivity ,relations, generation shift distribution factors, line outage distribution factors, compensated shift factors, sensitivity associated with voltage VAR sensitivities relating load bus voltage changes in terms of PV bus voltage changes, sensitivity relating changes in reactive power generation for changes in PV Bus Voltage.

(12 HRS)

Unit-4

Power system security :Security functions, Security level, contingency analysis, security Control, economic dispatch using LP formulation,pre-contingency and post-contingency corrective rescheduling.

(12 HRS)

Unit-5

Voltage stability :Difference between voltage and angle stability, PV Curve for voltage stability Assessment, proximity and mechanism, modal analysis using reduced Jacobian, participation factor, effect of series and shunt compensation on voltage stability,effect of load models.

(12 HRS)



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VIITH SEMESTER

~~CATEGORY: -OE~~ CATEGORY: -OE

1. Perform basic design and evaluation of SCADA system architectures including local systems, communication infrastructure and central systems.
2. Describe basic power system instrumentation technologies and principles
3. Describe basic power system protection technologies and schemes
4. Analyse and evaluate current processes and technologies employed for control and operation of transmission grids

References:

1. Computer Modeling of Electrical Power Systems, Arrillaga J. Watson N R Wiley India.
2. A Chakrawarti Power System Analysis: Operation and Control PHI Learning 3rd edition
3. Power Generation, Operation and Control by A.J.wood and B.F.Wollenberg John Wiley & Sons Inc. 1984.
4. Computer Techniques in Power Systems Analysis- Pai M.A. Tata Mc Graw Hill.
5. Computer Aided Power Systems Analysis Kusic G.L. 2nd Edition, CRC Press
6. Modern Power Systems Analysis Nagrath I.J. and Kothari D.P. Tata Mc Graw Hill.
7. Power System Analysis Grainger J.J. & Stevnson W.D. Mc Graw Hill.
8. Power System Stability and control-P Kundur, IEEE Press 1994.
9. Advance Power Systems Analysis and Dynamics Singh L.P.John Wiley.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL



COURSE OUTCOME:

At the end of the course, a student will be able to:

75(B) – ADVANCED ELECTRICAL DRIVES

TOTAL-60HOURS

COURSE OBJECTIVE :

This course will develop students' knowledge in/on

1. To impart knowledge on Performance of the fundamental control practices associated with AC and DC machines (starting, reversing, braking, plugging, etc.) using power electronics
2. To impart industry oriented learning
3. To evaluate the use of computer-based analysis tools to review the major classes of machines and their physical basis for operation

Unit-1

Review of electric motors & Solid state converters: Speed control techniques of DC, Induction & synchronous motor, Converters, inverters, chopper and cyclo converter operation, Effects of power electronic equipments on load side & supply side.

(12 HRS)

Unit-2

Review of closed loop controllers, sensors & transducers: PI, PID, Variable structure. AC, DC & Pulse tacho- generators.

(12 HRS)

Unit-3

DC Drives: Converter & chopper fed DC drive, Reversing, Starting, Regenerative braking, Four quadrant operation, High power application.

(12 HRS)

Unit-4

AC Drive: Inverter & cyclo converter fed drive, Vector control, Sensor less operation, Linear Electrical motor concept, Synchronous motor Drive.

(12 HRS)

Unit-5

Special Drives: Switched reluctance & permanent magnet brushless DC Operation, Converters, Characteristics & Control, PLC based drives.

(12 HRS)



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VIITH SEMESTER

CATEGORY- OE

1. Identify relevant information to supplement the Electric Drives (EE 701) course.
2. Set up control strategies to synthesize the voltages in dc and ac motor drives.
3. Develop testing and experimental procedures applying basic knowledge in electronics, electrical circuit analysis, electrical machines, microprocessors, and programmable logic controllers.
4. An ability to use standard methods to determine accurate modeling/simulation parameters for various general-purpose electrical machines and power electronics devices required for designing a system and solve drives related problems

REFERENCE

- 1 Ned Mohan, T.M. Undeland, W.P. Robbins, Power Electronics -Converters, Applications and design", John Wiley & Sons.
- 2 .J.M.D.Murphy, F.O.Turnbull, "Power Electronic Control of AC motors", Pergamon Press.
3. P.C. Sen, D.C. drive, Pergamon Press
4. B.K. Bose, Power Electronics & AC drive prentice Hall.
5. Dubey G.K. "Power semi Conductor controller drives, Prentice Hall.
6. Vedam Subramanyam, "Electrical Drives".
7. T.J.E. Miller, Switched Reluctance & P.M. B.L. DC motor, Pergamon Press
8. P.V. Rao, "Power semiconductor Drives", BS Publications.



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COURSE OUTCOME:

At the end of the course, a student will be able to:

75(C)-WIND AND SOLAR ENERGY SYSTEMS

TOTAL-60HOURS

COURSE OBJECTIVE:

The course should enable the students to :

1. Understand the various forms of conventional energy resources.
2. Learn the present energy scenario and the need for energy conservation
3. Explain the concept of various forms of renewable energy
4. Outline division aspects and utilization of renewable energy sources for both domestic and industrial application
5. Analyse the environmental aspects of renewable energy resources

Unit-1

Physics of Wind Power: History of wind power, Indian and Global statistics, Wind physics, Betz limit, Tip speed ratio, stall and pitch control, Wind speed statistics-probability distributions, Wind speed and power-cumulative distribution functions.

(12 HRS)

Unit-2

Wind generator topologies: Review of modern wind turbine technologies, Fixed and Variable speed wind turbines, Induction Generators, Doubly-Fed Induction Generators and their characteristics, Permanent Magnet Synchronous Generators, Power electronics converters. Generator-Converter configurations, Converter Control.

(12 HRS)

Unit-3

The Solar Resource Introduction, solar radiation spectra, solar geometry, Earth Sun angles, observer Sun angles, solar day length, Estimation of solar energy availability.

(12 HRS)

Unit-4

Solar photovoltaic: Technologies-Amorphous, mono crystalline, polycrystalline; V-I characteristics of a PV cell, PV module, array, Power Electronic Converters for Solar Systems, Maximum Power Point Tracking (MPPT) algorithms. Converter Control.

(12 HRS)

Unit-5

Network Integration Issues and Solar thermal power generation: Overview of grid code technical requirements. Fault ride-through for wind farms - real and reactive power regulation, voltage and frequency operating limits, solar PV and wind farm behavior during grid disturbances. Power quality issues. Power system interconnection experiences in the world. Hybrid and isolated operations of solar PV and wind systems. Technologies, Parabolic trough, central receivers, parabolic dish, Fresnel, solar pond, elementary analysis.

(12 HRS)



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COURSE OUTCOMES: VIITH SEMESTER

1. Understand the energy scenario & the consequent growth of the power generation from renewable energy sources.
2. Understand the basic physics of wind & solar power generation.
3. Understand the power electronic interfaces for wind & solar generation.
4. Understand the issues related to the grid-integration of solar & wind energy systems.

References:

1. T. Ackermann, “Wind Power in Power Systems”, John Wiley and Sons Ltd., 2005.
2. G. M. Masters, “Renewable and Efficient Electric Power Systems”, John Wiley and Sons, 2004.
3. S. P. Sukhatme, “Solar Energy: Principles of Thermal Collection and Storage”, McGraw Hill, 1984.
4. H. Siegfried and R. Waddington, “Grid integration of wind energy conversion systems” John Wiley and Sons Ltd., 2006.
5. G. N. Tiwari and M. K. Ghosal, “Renewable Energy Applications”, Narosa Publications, 2004



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CATEGORY:-P

EE-76-MAJOR PROJECT-I

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Problem based & project based learning
- 2 Mmajor project design in one of the selected areas of specialization with substantial multi-disciplinary component
- 3 Aanalytical and research skills
- 4 Team work, leadership and interpersonal skills

COURSE GUIDELINES

The objectives of the course 'Major Project' are to provide students with a comprehensive experience for applying the knowledge gained so far by studying various courses. To develop an inquiring aptitude and build confidence among students by working on solutions of small industrial problems. To give students an opportunity to do something creative and to assimilate real life work situation in institution. To adapt students for latest developments and to handle independently new situations. To develop good expressions power and presentation abilities in students. The focus of the Major Project is on preparing a working system or some design or understanding of a complex system using system analysis tools and submit it the same in the form of a write-up i.e. detail project report. The student should select some real life problems for their project and maintain proper documentation of different stages of project such as need analysis, market analysis, concept evaluation, requirement specification, objectives, work plan, analysis, design, implementation and test plan. Each student is required to prepare a project report and present the same at the final examination with a demonstration. The faculty and student should work according to following schedule: i) Each student undertakes substantial project in an approved area of the subject and supervised by a member of staff. ii) The student must submit outline and action plan for the project execution (time schedule) and the same be approved by the concerned faculty. iii) At all the steps of the project, students must submit a written report of the same.

COURSE OUTCOME:

After completion of this course, the students will be able to

- 1 Demonstrate creativity in the design of components, systems or processes of their program of study
- 2 Ddesign an innovative product by applying current knowledge and adopt to emerging applications of engineering & technology
- 3 Work cooperatively with others to achieve shared goal by motivating team-mates with a clear sense of direction, values and ethics
- 4 Write concisely & convey meaning in a manner appropriate to different readers and verbally express ideas easily understood by others who are unfamiliar with the topic



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VIITH SEMESTER

CATEGORY:-P

EE-81-MAJOR PROJECT-II

COURSE OBJECTIVE:

This course will develop students' knowledge in/on

- 1 Problem based & project based learning
- 2 Major project design in one of the selected areas of specialization with substantial multi-disciplinary component
- 3 Analytical and research skills
- 4 Team work, leadership and interpersonal skills

COURSE GUIDELINES

The objectives of the course 'Major Project' are to provide students with a comprehensive experience for applying the knowledge gained so far by studying various courses. To develop an inquiring aptitude and build confidence among students by working on solutions of small industrial problems. To give students an opportunity to do something creative and to assimilate real life work situation in institution. To adapt students for latest developments and to handle independently new situations. To develop good expressions power and presentation abilities in students. The focus of the Major Project is on preparing a working system or some design or understanding of a complex system using system analysis tools and submit it the same in the form of a write-up i.e. detail project report. The student should select some real life problems for their project and maintain proper documentation of different stages of project such as need analysis, market analysis, concept evaluation, requirement specification, objectives, work plan, analysis, design, implementation and test plan. Each student is required to prepare a project report and present the same at the final examination with a demonstration. The faculty and student should work according to following schedule:

- i) Each student undertakes substantial project in an approved area of the subject and supervised by a member of staff.
- ii) The student must submit outline and action plan for the project execution (time schedule) and the same be approved by the concerned faculty.
- iii) At all the steps of the project, students must submit a written report of the same.



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COURSE OUTCOME:

After completion of this course, the students will be able to

- 1 Work cooperatively with others to achieve shared goal by motivating team-mates with a clear sense of direction, values and ethics
- 2 Write concisely & convey meaning in a manner appropriate to different readers and verbally express ideas easily understood by others who are unfamiliar with the topic
- 3 Design an innovative product by applying current knowledge and adopt to emerging applications of engineering & technology



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